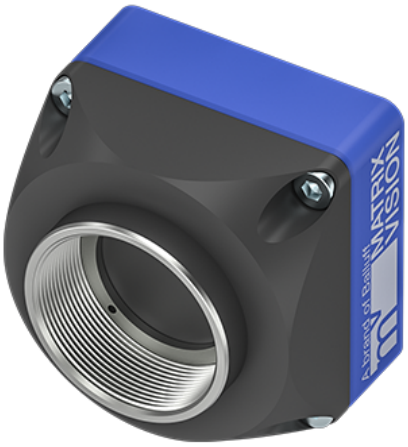
BALLUFF

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mvBlueNAOS

Technical Manual

English - Version 2.14

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1.1 About this manual

1.1.1 Goal of the manual

This manual gives you an overview of the mvBlueNAOS, MATRIX VISION's compact PCI Express camera family compliant to GenICam GenTL Producer, its technical data and basic operation of the mvBlueNAOS. Programming the device is detailed in a separate documentation, which will be available in an online format.

1.1.2 Contents of the manual

At the beginning of the manual, you will get an **introduction** (p. 16) to the possible usages of the camera. The following chapters contain general information about the camera including:

- **Quickstart** (p. 21) followed by
- **Technical Data** (p. 36)
- **Sensor Overview** (p. 54)
- **Filters** (p. 63)

The general information is followed by the description of the

- **Software tools for the camera** (p. 65) including the tools
- **GenICam and advanced features** (p. 66) introduces the GenICam and the advanced features of the camera (the cameras are GenICam compliant devices).
- **Developing applications using the mvIMPACT Acquire SDK** (p. 86)
- **DirectShow developers** (p. 87) documents MATRIX VISION's mvIMPACT Acquire to DirectShow interface(↩ DirectShow_acquire).
- **Troubleshooting** (p. 92) shows how to detect damages and other inconveniences.
- **Use Cases** (p. 122) describes solutions for general tasks and
- A **Glossary** (p. 113) explains abbreviations and technical terms.
- **Appendix A. Specific Camera / Sensor Data** (p. 212) contains all data of the Pregius CMOS sensors like timings, details of operation, etc.
- **Appendix C. Tested ARM platforms** (p. 273) contains a list of ARM platforms tested with this product and information on how to setup these systems for achieving optimal results

1.2 Imprint

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2020

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1.3 Legal notice

1.3.1 Introduction

The firmware running on MATRIX VISION devices make use of a couple of third party software packages that come with various licenses. This section is meant to list all these packages and to give credit to those whose code helped in the creation of this software:

The mvBlueNAOS Kernel module was released under the terms of GPL v2.0 and is available as source code in the installer:

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```

1.3.2 cJSON

A slightly modified version of cJSON is used inside some of the modules that eventually build up the firmware.

```
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THE SOFTWARE.

1.3.3 Unity

A slightly modified version of Unity (https://github.com/ThrowTheSwitch/Unity) is used for unit testing various modules that eventually build up the firmware.

1.4 Revisions

Date	Rev.	Author	Description	Driver / Firmware version
24. May 2022	V2.14	LAN	Add **mvBlueNAOS4** (p. 19) .	
9. May 2022	V2.13	LAN	New **sensors for mvBlueNAOS4** (p. 57) added. Added **OCuLink** (p. 45)	
31. March 2022	V2.12	LAN	Added temperatures **Summary of components** (p. 52) .	
10. March 2022	V2.11	LAN	Updated **Troubleshooting** (p. 92) .	
07. March 2022	V2.10	LAN	Added note about distribution of electricity consumption: **CMOS sensors** (p. 57) .	
28. January 2022	V2.09	LAN	Added **BVS CA-BN4-0124BC (12.4 Mpix [4128 x 3008])** (p. 263) .	
26. January 2022	V2.08	LAN	Added **mvBlueNAOS4** (p. 38) information and **BVS CA-BN4-0124BG (12.4 Mpix [4128 x 3008])** (p. 233) .	
22. September 2021	V2.07	LAN	Added **Add-in cards for the NAOS for Embedded (N4e) Interface** (p. 43) .	
17. September 2021	V2.06	LAN	New control **mv Auto Feature Control** (p. 84) .	FW Revision 2.40.2546.0
19. August 2021	V2.05	LAN	Updated power consumption information.	
11. August 2021	V2.04	LAN	Introduced **"NAOS for Embedded (N4e)"** for the PCIe Embedded interface.	
21. July 2021	V2.03	LAN	Updated **Order code nomenclature** (p. 18) . Added **dimensional drawings** (p. 36). Added **accessories** (p. 20).	
16. July 2021	V2.02	LAN	Updated sensor data **Appendix A. Specific Camera / Sensor Data** (p. 212) .	
01. July 2021	V2.01	LAN	Added use case **Low latency triggering** (p. 176).	
03. May 2021	V2.00	LAN	Corrected **Symbols and Conventions** (p. 10).	
13. January 2021	V1.00	LAN	Seperated **GUI tools** (p. 65).	

1.5 Symbols and Conventions

Note

This symbol indicates general notes.

1.5.1 Explanation of the warnings

Always observe the warnings in these instructions and the measures described to avoid hazards. The warnings used here contain various signal words and are structured as follows:

Attention

SIGNAL WORD
"Type and source of the hazard"

Consequences if not complied with

→ Measures to avoid hazards.

The individual signal words mean:

Attention

Indicates a **danger** that can lead to **damage** or **destruction** of the product.

All due care and attention has been taken in preparing this manual. In view of our policy of continuous product improvement, however, we can accept no liability for completeness and correctness of the information contained in this manual. We make every effort to provide you with a flawless product.

In the context of the applicable statutory regulations, we shall accept no liability for direct damage, indirect damage or third-party damage resulting from the acquisition or operation of a MATRIX VISION product. Our liability for intent and gross negligence is unaffected. In any case, the extend of our liability shall be limited to the purchase price.

1.6 Important Information



1.6.1 Important Safety Instructions

- We cannot and do not take any responsibility for the damage caused to you or to any other equipment connected to a MATRIX VISION device. Similarly, warranty will be void, if a damage is caused by not following the manual.
- Handle your MATRIX VISION device with care. Do not misuse it. Avoid shaking, striking, etc.. Your MATRIX VISION device could be damaged by faulty handling or shortage.
- Do not use accessories not recommended by the product manufacturer as they may cause hazards.
- The product should be situated away from heat sources such as radiators, heat registers, stoves, or other products (including amplifiers) that produce heat.
- Using the board-level version:
  - Provide sufficient cooling because single components can reach high temperatures.
  - Handle with care and avoid damage of electrical components by electrostatic discharge (ESD):
    - Discharge body static (contact a grounded surface and maintain contact).
    - Avoid all plastic, vinyl, and Styrofoam (except anti-static versions) around printed circuit boards.
    - Do not touch components on the printed circuit board with your hands or with conductive devices.

1.6.2 Operating considerations

1.6.2.1 Important Safety Notes

- Handle with care and avoid damage of electrical components by electrostatic discharge (ESD):
  - Discharge body static (contact a grounded surface and maintain contact).
  - Avoid all plastic, vinyl, and Styrofoam (except anti-static versions) around printed circuit boards.
  - Do not touch components on the printed circuit board with your hands or with conductive devices.

Be careful when bending the flex cable. The minimum bending radius is approx. 3 mm.

- Provide sufficient cooling because single components can reach high temperatures. Inadequate and incorrect cooling invalidate the guarantee. For heat dissipation, we recommend to use a thermal conduction plate.
- Camera case temperature should not exceed 65 °C.

Attention

"Overheating"

The mainboards temperature may not exceed 80 °C otherwise the device can be damaged.

1.6.2.2 Handling And Cleaning

- Do not attempt to disassemble camera.
- When installing or removing a lens, take care that water or dust does not enter the inside of the camera.

1.6.2.3 Installing

Avoid installing or storing the camera in the following environments:

- Environments exposed to direct sunlight, rain or snow.
- Environments where combustible or corrosive gas exists.
- Excessively warm or cold environment (Operating ambient temperature: 0 to 45 ℃)
- Humid or dusty environment.
- Place subjected to excessive vibration or shock.
- Environment exposed to strong electric or magnetic field.
- It is recommended to mount the camera on a thermoconducting surface such as aluminum or other metals rather than plastic or wood.
- Please contact manufacturer or local distributor if you want to use additional enclosures for higher ingress protection.
- Do not aim the camera at the sun or other very strong light sources, otherwise you can destroy the image sensor.

1.6.2.4 Optimizing performance and life time

When the camera is used continuously for long time under high ambient temperature, the inside electrical parts may deteriorate, resulting in shorter life span. Additional cooling by e.g. air convection is recommended.

1.6.2.5 Connectors

Confirm the power is off before connecting or disconnecting a signal cable. Grap connectors by the body, not the attached wires.

1.6.2.6 Cleaning

- Use a blower or a lens brush to remove dust on the lens or the optical filter.
- Do not disassemble front flange.
- Clean case with dry soft cloth. Use neutral detergent liquid if needed; wipe the cover with dry cloth.
- Do not use benzene, thinner, alcohol, liquid cleaner or spray-type cleaner.

1.6.2.7 Adjusting the C-mount

The mvBlueNAOS does not support back focus adjustment in any way. There might be future versions with different lensholders but for now, the back focus is fixed.

1.6.3 Conformity

CE	The mvBlueNAOS is in conformity with all applicable essential requirements necessary for CE marking.
UKCA	
	MATRIX VISION corresponds to the EU guideline WEEE 2002/96/EG on waste electrical and electronic equipment and is registered under WEEE-Reg.-No. DE 25244305.

A brand of Balluff

MATRIX VISION

EG-Konformitätserklärung
Declaration of conformity

Der Hersteller
The Manufacturer

MATRIX VISION GmbH
Talstraße 16
71570 Oppenweiler
Germany

erklärt hiermit, dass sein Produkt
herewith declares, that his product

Typbezeichnung: mvBlueNAOS2 (BVS CA-BN2-xxxxxx-xxxx1x)
Type: mvBlueNAOS2 (BVS CA-BN2-xxxxxx-xxxx1x)

mit den Bestimmungen folgender Europäischer Richtlinien übereinstimmt:
complies with the provisions of the following European Directives:

2014/30/EU EMC
2011/65/EU RoHS II

auf Grundlage folgender harmonisierter Normen:
based on the following harmonized standards:

Elektromagnetische Verträglichkeit (EMV) / *Electromagnetic compatibility (EMC)*
Störaussendung / *Interference emission*
EN 61000-6-3:2007 + A1:2011

Störfestigkeit / *Interference immunity*
EN61000-6-2:2019

Beschränkung gefährlicher Stoffe / Restriction of certain Hazardous Substances
EN IEC 63000:2018

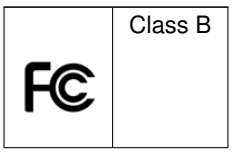
Oppenweiler, 11.08.2021

Uwe Hagmaier
Vice President R & D

Bitte beachten: Um die Einhaltung obiger Normen sicherzustellen, ist die Verwendung eines geeigneten EMV-Gehäuses unbedingt erforderlich.
Please note: In order to fulfill the above standards, the use of a suitable EMC-case is required.

1.6.4 Additional notices

1.6.4.1 For customers in the U.S.A.



This equipment has been tested and found to comply with the limits for a Class B digital device, pursuant to Part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference when the equipment is operated in a residential environment. This equipment generates, uses, and can radiate radio frequency energy and, if not installed and used in accordance with the instruction manual, may cause harmful interference to radio communications. However there is no guarantee that interferences will not occur in a particular installation. If the equipment does cause harmful interference to radio or television reception, the user is encouraged to try to correct the interference by one or more of the following measures:

- Reorient or relocate the receiving antenna.
- Increase the distance between the equipment and the receiver.
- Use a different line outlet for the receiver.
- Consult a radio or TV technician for help.

You are cautioned that any changes or modifications not expressly approved in this manual could void your authority to operate this equipment. The shielded interface cable recommended in this manual must be used with this equipment in order to comply with the limits for a computing device pursuant to Subpart B of Part 15 of FCC Rules.

1.6.4.2 For customers in Canada

This apparatus complies with the Class B limits for radio noise emissions set out in the Radio Interference Regulations.

1.6.4.3 Pour utilisateurs au Canada

Cet appareil est conforme aux normes classe B pour bruits radioélectriques, spécifiées dans le Règlement sur le brouillage radioélectrique.

1.7 Introduction

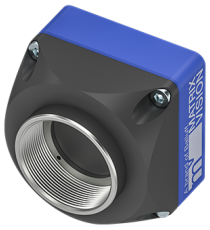


Figure 1: mvBlueNAOS

The mvBlueNAOS is a PCI Express board-level camera family compliant to the **GenICam GenTL Producer** (p. 113) standard. The camera family combines the Sony IMX sensor with up to 16 LVDS lanes with a 4 lane PCI Express interface including Generation 2 performance. Furthermore, the mvBlueNAOS offers

- a wide range of sensors and resolutions with high frame rates,
- I/Os suitable for industrial applications, and
- a modular design.

The PCI Express interface enables large sensors with high frame rates to be fully exploited at full speed. This and the modular concept makes the mvBlueNAOS ideally suited for applications in the medicine, reverse vending, traffic surveillance, 3D measurement, packaging, logistics, and robotics area.

1.7.1 Software concept

The mvBlueNAOS is a **GenICam** (p. 113) compliant device, using an XML-file describing the device capabilities. Within this XML-file it uses the names, units, data types etc. recommended by the **SFNC** (p. 121) to describe the devices features. Whenever custom features have been added to the mvBlueNAOS firmware these can clearly be spotted by the leading 'mv' in the features name.

The following figure shows the software concept of MATRIX VISION's camera devices:

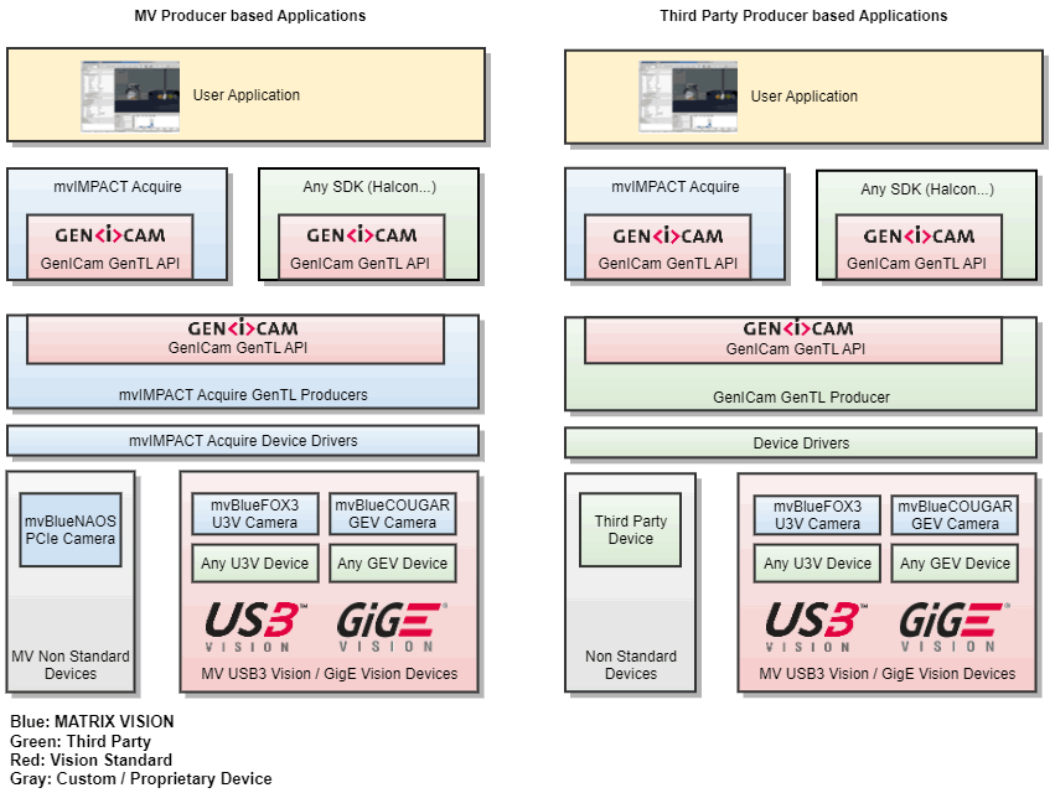


Figure 2: Software concept

As shown in figure 2, for the mvBlueNAOS the **mvIMPACT_Acquire** (p. 119) interface is stacked on the **Genicam** (p. 113) layers. The **mvIMPACT_Acquire** (p. 119) interface internally uses the **GenICam** (p. 113) runtime libraries, so that it can be considered as a user application written with the **GenICam** (p. 113) interface.

1.7.2 Order code nomenclature

1.7.2.1 mvBlueNAOS2

The mvBlueNAOS2 nomenclature scheme is as follows:

Sensor	Model name
1.6 Mpix, 1456 x 1088, 1/2.9", IMX273, CMOS	BVS CA-BN2-0016ZG/C
3.2 Mpix, 2064 x 1544, 1/1.8", IMX252, CMOS	BVS CA-BN2-0032ZG/C
5.1 Mpix, 2464 x 2056, 2/3", IMX250, CMOS	BVS CA-BN2-0051ZG/C
8.9 Mpix, 4112 x 2176 (G) | 4096 x 2176 (C), 1", IMX255, CMOS	BVS CA-BN2-0089ZG/C
12.4 Mpix, 4112 x 3008 (G) | 4096 x 3008 (C), 1.1", IMX253, CMOS	BVS CA-BN2-0124ZG/C
24.6 Mpix, 5328 x 4608, 1.2", IMX540, CMOS	BVS CA-BN2-0246AG/C

	Series	Sensor	Color		HW Variant		CUP/STD
Model name	BVS CA-BN2-	xxxxx	G/C	-	see legend below	-	see legend below

Legend	Product name
HW Variant	(1)(2)(3)(4)(5)(6) (1): Handling **1**: Standard handling (2): Lensholder **0**: None **1**: C-Mount, Type2, with high durability and factory-set backfocus, fixed square filter **2**: C-Mount, Type1, with adjustable backfocus, round filter with screw ring, Ø23.2 mm **5**: C-Mount, Type3, low cost, factory-set backfocus, D23.2mm Filter **A:** C-Mount, Type2, with high durability and factory-set backfocus, fixed square filter, adjustable center position **B:** C-Mount Type2 + D23.2mm Filter with screwring **Y:** S-Mount, 13 mm length for sensors with image circule <= 2/3 inch (3): Filter **0**: None **1**: IR-CUT (standard) **2**: Glass **3**: Daylight cut filter **6**: Dualband filter DB850 **8**: Customized filter type (4): Housing **0**: None (standard) **1**: Color blue **5**: Mounting and thermal conduction plate for board level camera (5): I/O **1**: NAOS for Embedded (N4e) interface (standard) **H:** OCuLink for PCIe (6): Software Adjustments **0**: None
CUP/STD	**001**

1.7.2.2 mvBlueNAOS4

The mvBlueNAOS4 nomenclature scheme is as follows:

Sensor	Model name
5.1 Mpix, 2464 x 2064, 1/1.8", IMX537, CMOS	BVS CA-BN4-0051BG/C
8.1 Mpix, 2848 x 2848, 2/3", IMX536, CMOS	BVS CA-BN4-0081ZG/C
12.4 Mpix, 4128 x 3008, 1/1.1", IMX535, CMOS	BVS CA-BN4-0124BG/C
16.2 Mpix, 5328 x 3040, 1.1", IMX532, CMOS	BVS CA-BN4-0162ZG/C
20.4 Mpix, 4512 x 4512, 1.1", IMX531, CMOS	BVS CA-BN4-0204ZG/C
24.6 Mpix, 5328 x 4608, 1.2", IMX530, CMOS	BVS CA-BN4-0246ZG/C

	Series	Sensor	Color		HW Variant		CUP/STD
Model name	BVS CA-BN4-	xxxxx	G/C	-	see legend below	-	see legend below

Legend	Product name
HW Variant	(1)(2)(3)(4)(5)(6) (1): Handling **1**: Standard handling (2): Lensholder **0**: None **1**: C-Mount, Type2, with high durability and factory-set backfocus, fixed square filter **2**: C-Mount, Type1, with adjustable backfocus, round filter with screw ring, Ø23.2 mm **5**: C-Mount, Type3, low cost, factory-set backfocus, D23.2mm Filter **A:** C-Mount, Type2, with high durability and factory-set backfocus, fixed square filter, adjustable center position **B:** C-Mount Type2 + D23.2mm Filter with screwring **Y:** S-Mount, 13 mm length for sensors with image circule <= 2/3 inch (3): Filter **0**: None **1**: IR-CUT (standard) **2**: Glass **3**: Daylight cut filter **6**: Dualband filter DB850 **8**: Customized filter type (4): Housing **0**: None **1**: Color blue (standard) **5**: Mounting and thermal conduction plate for board level camera (5): I/O **1**: NAOS for Embedded (N4e) interface (standard) **H:** OCuLink for PCIe (6): Software Adjustments **0**: None
CUP/STD	**001**

1.7.3 What's inside and accessories

Due to the varying fields of application the mvBlueNAOS is shipped without accessories. The package contents:

- **mvBlueNAOS**

For the first use of the mvBlueNAOS we recommend the following accessories to get the camera up and running:

- a **PCI Express adapter card** or a **M.2 Key M adapter** (if you want to connect the mvBlueNAOS directly)
- a **Flex extension cable**,
- a **lens**.

Accessories for the mvBlueNAOS:

Part code	Description
BNE-IF-PCIeX4-N4e-IO14	Adapter board for PCIe x4 interface to camera interface PCIe Embedded
BNE-IF-M2M-N4e-IO14	Adapter board for the M.2M interface to the camera interface PCIe Embedded
BNE-FLEX-300	Connection and extension cable for the PCIe Embedded interface of the mvBlue↩ NAOS, length 0.3 m
KS-10TFM-OPEN LS 01.0	Cable for the mvBlueNAOS to connect Power / IOs, connector female TFM 10 pins to open end, length 1.0 m

1.8 Quickstart

- **System Requirements** (p. 21)
- **Installing The mvGenTL-Acquire Package** (p. 22)
- **Connecting The Camera** (p. 30)
- **Driver concept** (p. 33)
- **Relationship Between Driver, Firmware And SDK** (p. 35)

1.8.1 System Requirements

1.8.1.1 Host System

The mvBlueNAOS is a high-performance camera which requires a high data throughput on the host system, for example, when processing large amount of image data and high CPU resources if processing color images on the host PC.

For this reason we recommend the following components:

Component	Recommendation
RAM	8 GB in 64-bit OS
Mainboard	Latest PC architecture and chipset
PCIe	x4 slot, preferably if available the x16 graphic card slot

1.8.1.2 Supported Operating Systems

The following operating system are supported officially:

1.8.1.2.1 Windows

- Microsoft Windows 7 (64-bit)
- Microsoft Windows 8.1 (64-bit)
- Microsoft Windows 10 (64-bit)
- Microsoft Windows 11

Other Windows versions might work as well but will not be tested on a regular basis.

Note

Since mvIMPACT Acquire version 2.8.0 it could be possible that you have to update your Windows installer at least using Windows XP. The necessary packages are available from Microsoft's website: `http://www.microsoft.com/en-US/download/details.aspx?id=8483`

1.8.1.2.2 Linux

This following description is only valid for 64-bit ARM and Intel architectures.

Please check the the 'Support' section of the MATRIX VISION website for the availability of the latest Linux driver package.

See also

`https://matrix-vision.com/software-drivers-en.html`

Currently supported Kernel versions are:

- Kernel 4.9.x or greater (64-bit Linux systems)

In case the target system runs an older Linux kernel, **it is absolutely necessary to update the kernel to at least version 4.9.0** . Please refer to the documentation of your Linux distribution for information on how to update your system's Linux kernel.

Following Kernels have been tested and verified by MATRIX VISION. They work flawless with mvBlueNAOS devices:

- Kernel 4.9.140
- Kernel 4.15.0
- Kernel 5.3.0

1.8.2 Installing The mvGenTL-Acquire Package

All necessary drivers are available from the MATRIX VISION website at `https://www.matrix-vision.de`, section "`Products -> Cameras -> your interface -> your product -> Downloads`".

1.8.2.1 Installing the hardware

Attention

"Electro Static Discharge (ESD)"

Please take all proper Electro Static Discharge (ESD) precautions during the installation of your new hardware!

Before starting the installation, turn off your computer and all peripheral devices. Disconnect the computer from the power supply and all necessary components.

Note

To avoid doing damage to the hardware, discharge yourself of static charge by touching e.g. the casing. Beware of touching contacts of the PCIe adapter card, the mvBlueNAOS or of the computer.

Attention

"Short circuit"

Since the connector of the rigid-flex extension cable can be connected in both ways you have to check if it is connected the right way and that is:

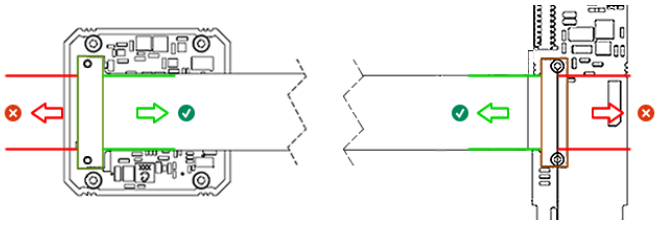


Figure 1: Connecting the rigid-flex extension cable

- Select a free PCIe slot (preferably the x16 graphic card slot). Remove the slot's cover at the back of the computer and keep the screw.
- Carefully insert the PCIe adapter card into the slot by holding the board at the top and gently pushing both ends into the slot at the same time. Press onto the upper edge of the board to make sure it is seated in the slot firmly.
- Do not force the PCIe adapter card into the slot! You run the risk of bending the contacts. If the board does not fit easily, pull it back out, and try again.
- Connect the flex cable extension to the PCIe adapter card.
- Lead out the flex cable extension from the case.
- Put the cover back on the computer and reconnect the peripheral devices.
- Connect the flex cable extension to the mvBlueNAOS.

  Note

  Please note **pin 1** (p. 36) of the PCIe-Embedded interface.

- Start the computer.

Attention

"Short-circuit"

If you want to connect or disconnect the device, please be sure that the PC is switched off! Otherwise, during the connection, the device could be short-circuited and possibly destroyed!

→ Switch off the PC if you want to connect or disconnect the device.

1.8.2.2 Windows

Starting the installer application

- **mvGenTL_Acquire-x86_64-n.n.n.msi** (for 64-bit systems): will display the following dialog:

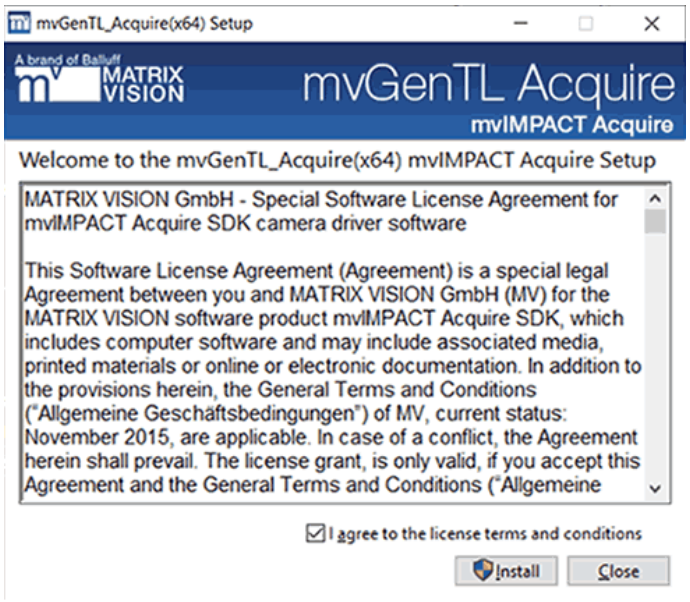


Figure 2: Driver installation - Start Window

- Now, follow the instructions of installation program and adjust the settings to your needs:

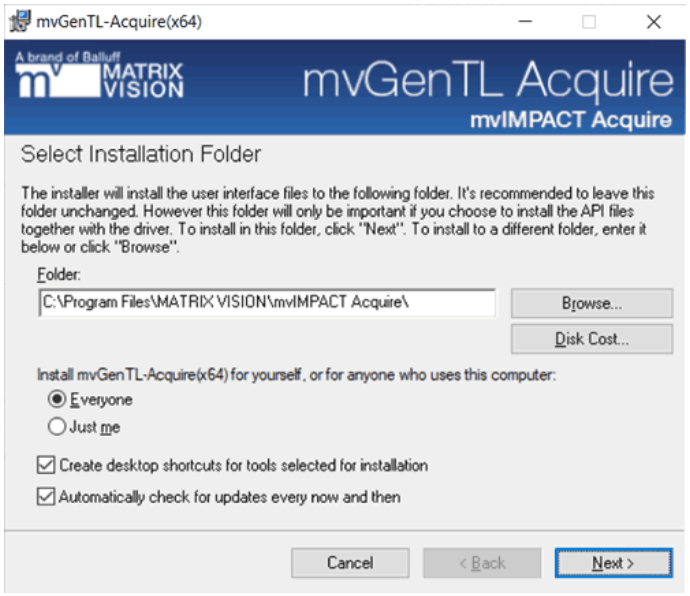


Figure 3: Driver installation - Select Installation Folder

Since

mvIMPACT Acquire 2.25.0

wxPropView (p. 65) is able to check if new driver version became available on a weekly basis. Deactivate the check box if wxPropView should not check for updates. You can activate this again in **wxPropView** (p. 65) via the help menu.

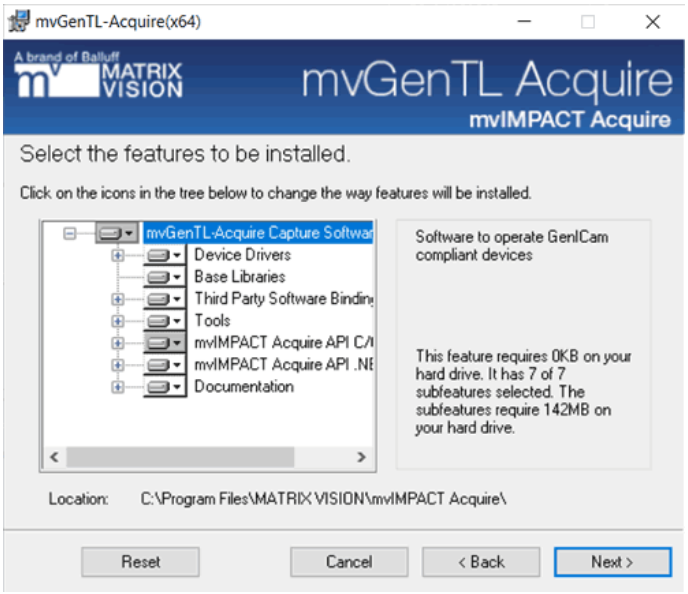


Figure 4: Driver installation - Select Features

- After confirmation, the installation will start and copy files and install device drivers.

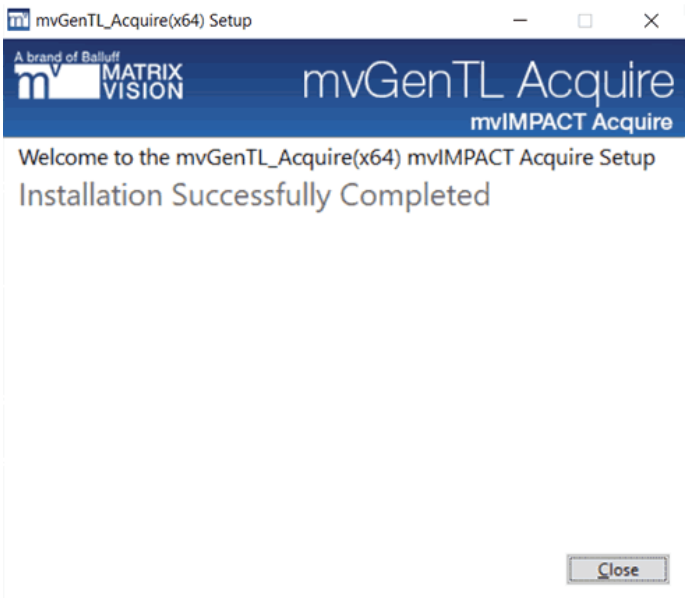


Figure 5: Driver installation - Installation Completed

You will find all tools like

- **wxPropView** (p. 65) and
- **mvDeviceConfigure** (p. 65)

either as shortcuts on the desktop or in the Windows start menu under **"MATRIX VISION ->** **mvIMPACT Acquire"**.

Note

You can ignore the other tools mvIPConfigure and mvGigEConfigure, because they are only necessary in combination with GigE Vision™ devices like the mvBlueCOUGAR-X.

Afterwards, you can use **mvDeviceConfigure** (p. 65) to update the firmware if needed. The current firmware version can be read out using **wxPropView** (p. 65).

1.8.2.3 Linux

The following (additional) packages will be needed to use all features of mvIMPACT Acquire:

- libwxbase3.0-0v5
- libwxbase3.0-dev
- libwxgtk3.0-gtk3-0v5
- libwxgtk3.0-gtk3-dev
- libwxgtk-webview3.0-gtk3-0v5
- libwxgtk-webview3.0-gtk3-dev
- wx3.0-headers
- build-essential (meta package)
- libgtk2.0-dev
- gcc 4.8.5 (4.9.4 for ARM versions) environment or newer

Note

The names of the packages as mentioned above are the Ubuntu specific ones. Other distributions (e.g. Debian, Arch, Redhat, ...) might use different names.

The installation script will ask if the packages should be downloaded during the installation process. If some of the packages are not installed some features might not be available. If the e.g. wxWidgets related packages are missing on the target system then all GUI application coming as part of the mvIMPACT Acquire installation won't be available.

To use a (camera) device in Linux (capture images from it and change its settings), a driver is needed, consisting of several libraries and several configuration files. These files are required during run time.

To develop applications that can use the device an API is needed, containing header files, makefiles, samples, and a few libraries. These files are required at compile time.

Both file collections are distributed in a single package which is available in the `Support` section of the MATRIX VISION website. In addition to that an installation script is provided which can be downloaded from the same location. Using this script makes installing the driver package a lot easier.

Note

The following table shows the supported platforms and the corresponding package and installation script name:

Architecture	Package	Installation Script
ARM64	mvGenTL_Acquire-ARM64_gnu-n.n.n.tgz	install_mvGenTL_Acquire_ARM.sh
x86_64	mvGenTL_Acquire-x86_64_ABI2-n.n.n.tgz	install_mvGenTL_Acquire.sh

The following example explains the installation process for the x86_64 package. **The installation process for other packages will work almost identical except different names as mentioned in the previous table.**

- Please start a console and change into the directory where the installation script and the installation package are located e.g. `/home/username/Downloads` :

```
cd /home/username/Downloads
```

> **Note**
>
> If root permissions are needed, the script will ask for the permissions. **There is no need to call it with root permissions.**

- You might need to enable the execute flag with:

```
chmod a+x install_mvGenTL_Acquire.sh
```

- Run the install script:

```
./install_mvGenTL_Acquire.sh
```

During installation the script will ask, if it should build all tools and samples.

> **Note**
>
> The installation scripts is developed for Ubuntu/Debian, SUSE Linux and Red Hat Linux based distributions. On other distributions some features of the installation script may or may not work. Get in touch with us if you encounter any problems!

The installation script checks for package dependencies described above and installs them with the respective standard package manager (e.g. **apt-get**) if necessary. So an Internet connection is recommended.

> **Note**
>
> The installation script (`install_mvGenTL_Acquire.sh`) and the archive (`mvGenTL_Acquire-x86_64_ABI2-n.n.n.tgz`) must reside in the same directory. Nothing is written to this directory during script execution, so no write access to the directory is needed in order to execute the script.

You need Internet access in case one or more of the packages on which the **GenICam** (p. 113) libs depend are not yet installed on your system. In this case, the script will install these packages, and for that, Internet access is required.

The script supports various arguments, which allow to customize the installation, the desired functionalities and the installation process itself. All arguments are optional:

Argument	Function
`-h` or `-help`	Display the help.
`-p` or `-path`	Define a custom installation directory.
`-gev` or `-gev_support`	Install the GigE Vision related features of the driver. Default: **yes**
`-u3v` or `-u3v_support`	Install the USB3 Vision related features of the driver. Default: **yes**
`-pcie` or `-pcie_support`	Install the PCI Express related features of the driver. Default: **yes**
`-ogev` or `-only_gev`	Install only the GigE Vision related features of the driver (deprecated, favor flavors without the 'only' instead). Default: **no**
`-ou3v` or `-only_u3v`	Install only the USB3 Vision related features of the driver (deprecated, favor flavors without the 'only' instead). Default: **no**
`-onaos` or `-only_naos`	Install only the PCI Express related features of the driver (deprecated, favor flavors without the 'only' instead). Default: **no**
`-u` or `-unattended`	Unattended installation with default settings. By using this parameter you explicitly accept the EULA.
`-m` or `-minimal`	Minimal installation. No tools or samples will be built, and no automatic configuration and/or optimizations will be done. By using this parameter you explicitly accept the EULA.

The target directory name specifies where to place the driver. If the directory does not yet exist, it will be created. The path can be either absolute or relative; i.e. the name may but need not start with / .

If no path is specified, the package will be installed to /**opt/mvIMPACT_Acquire**.

1.8.2.3.1 UEFI SecureBoot with Linux on x86_64 platforms UEFI SecureBoot is available on most modern x86 hardware and is used to ensure that only validated operating system binaries can be used. If you are using the mvBlueNAOS camera on an ARM-based system please ignore the following section.

SecureBoot requires the Linux kernel and all loadable kernel modules to be signed using a digital signature saved on the system. Since the MATRIX VISION mvBlueNAOS PCIe camera makes use of a kernel module which has been individually built for the system, this kernel module must also be signed in order to be loadable.

If you do not need UEFI SecureBoot protection it is advisable to turn it off in the BIOS settings for your PC. For details please see the BIOS documentation for your PC. In this case you can ignore the following section, since the MATRIX VISION kernel module will not need to be signed.

However, if you are unable to turn off SecureBoot or would like to continue using it then the kernel module will need to be signed after it has been built. The install script will try to do this using appropriate digital keys it finds on the system. In some cases you may need to install extra software on the PC and reinstall mvIMPACT Acquire using the install script or you may want to use your own digital keys to sign the kernel module.

In all cases you will have to register the key used with each PC individually, unless this has already been done with a key pair for which you have the private key (see below).

The following section describes how to install and use keys supplied by Debian-based distributions such as Ubuntu. Other Linux distributions may have similar packages that contain keys - please refer to your distribution's documentation.

Using keys supplied by Debian / Ubuntu

The MATRIX VISION install script attempts to use keys found in the directory "/var/lib/shim-signed/mok". If this directory or the keys **"MOK.priv"** and **"MOK.der"** do not exist on your system please try installing this package: **"shim-signed"**

Afterwards, reinstall mvIMPACT Acquire using the install script. The tool called **"kmodsign"** and the keys **"MOK.priv"** and **"MOK.der"** will be used automatically after building the kernel module.

Using your own keys or signing by hand Alternatively you can sign the kernel module yourself, by hand, using your own keys or those supplied by **"shim-signed"**. An example of signing a kernel-module is shown below. Substitute your own keys and their locations, if required.

```
cd /lib/modules/$(uname -r)/kernel/misc
sudo kmodsign sha512 /var/lib/shim-signed/mok/MOK.priv /var/lib/shim-signed/mok/MOK.der mvpci.ko
```

Registering validation keys with the system Debian and Ubuntu use a tool called **"mokutil"** to import a key for registration on a system. This action is only needed once per key on each PC.

The command to use looks like this (substitute your own key and its location, if required):

```
sudo mokutil --import /var/lib/shim-signed/mok/MOK.der
```

You will be asked to provide a one-time password. Choose anything you like, but take note of what you have chosen!

The next time your PC is restarted and before the Linux kernel boots a program called **"MokManager"** will be started. You will be presented with a blue box asking if you want to administer the keys on the system. Choose the option to do this and then follow the instructions on the screen to register your key. You will be asked to provide your one-off password that you used with **"mokutil"!**

Once the key has been successfully registered the kernel will boot and it will be possible to load any kernel modules that have been signed using this key.

References

- https://wiki.ubuntu.com/UEFI/SecureBoot
- https://ubuntu.com/blog/how-to-sign-things-for-secure-boot

1.8.3 Connecting The Camera

To start the device please accomplish following steps:

1. **Power down** the **host system**
2. **Connect the mvBlueNAOS** to the host system using the appropriate adapter
   - PCIe or M.2 Key M and
   - cable
3. **Power up the host** system.
   The mvBlueNAOS will boot immediately.

1.8.3.1 Communicating With The Camera

You can communicate with the camera the following way

1. **Via wxPropView** (p. 65) :
   Since driver version 2.11.3, starting **wxPropView** (p. 65) the first time, the so called Quick Setup Wizard will be launched.

   Read more about how to make optimal use of it in the mvIMPACT Acquire GUI manual: `https://www.matrix-vision.com/manuals/SDK_GUI_Tools/index.html`

1.8.3.2 Setting Up The Camera

MATRIX VISION offers several **GUI tools** (p. 65) to work with the camera. Please have a look at the specific chapter.

1.8.3.3 About Settings

A setting contains all parameters that are needed to configure the device to a state it was in when the setting was created. Every image can be captured with a completely different set of parameters. In almost every case, these parameters are accessible via a property offered by the device driver. A setting e.g. might contain

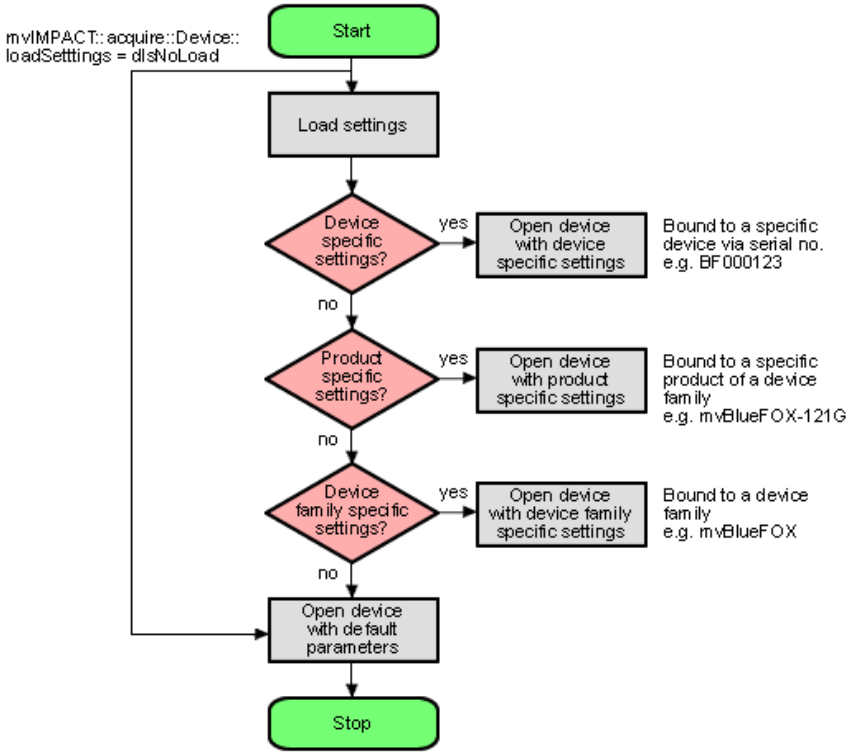
- The gain to be applied to the analog to digital conversion process for analog video sources or
- The AOI to be captured from the incoming image data.

So for the user a setting is the one and only place where all the necessary modifications can be applied to achieve the desired data acquisition mode. There is however an important difference in behaviour between different interface layouts. Please have a look at the **"mvIMPACT Acquire SDK GUI Applications"** chapter **"wxPropView ->** **Device Configuration ->** **General Device Configuration ->** **Changing The Interface Layout To GenICam Or Device Specific"** to find out how to modify the interface layout or check in the API documentation for the **interfaceLayout** property of the class **Device**.

- When working with the **DeviceSpecific** interface layout, each frame will be captured with the settings as present when requesting the image. Every parameter can be modified at any time. When requesting another image the settings valid at that moment will be used to fill this buffer with data

- For the **GenICam** interface layout all device properties modified during a continuous acquisition will be applied at once so might affect this or the next image transmitted by the device. Depending on various parameters (the number of buffer already captured but not collected by the application, the way the device internally operates(e.g. has already captured a couple of images that await transmission), etc.) this will have impact on that captured images somewhere in the near future thus when a precise moment to change settings is needed, continuous acquisition must be stopped and then restarted after modifying the features. Certain features (typically those affecting the buffer layout/size) cannot be changed while a continuous acquisition is running in GenICam interface layout anyway.

Now, whenever a device is opened, the driver will execute following procedure:



wxPropView - Device setting start procedure

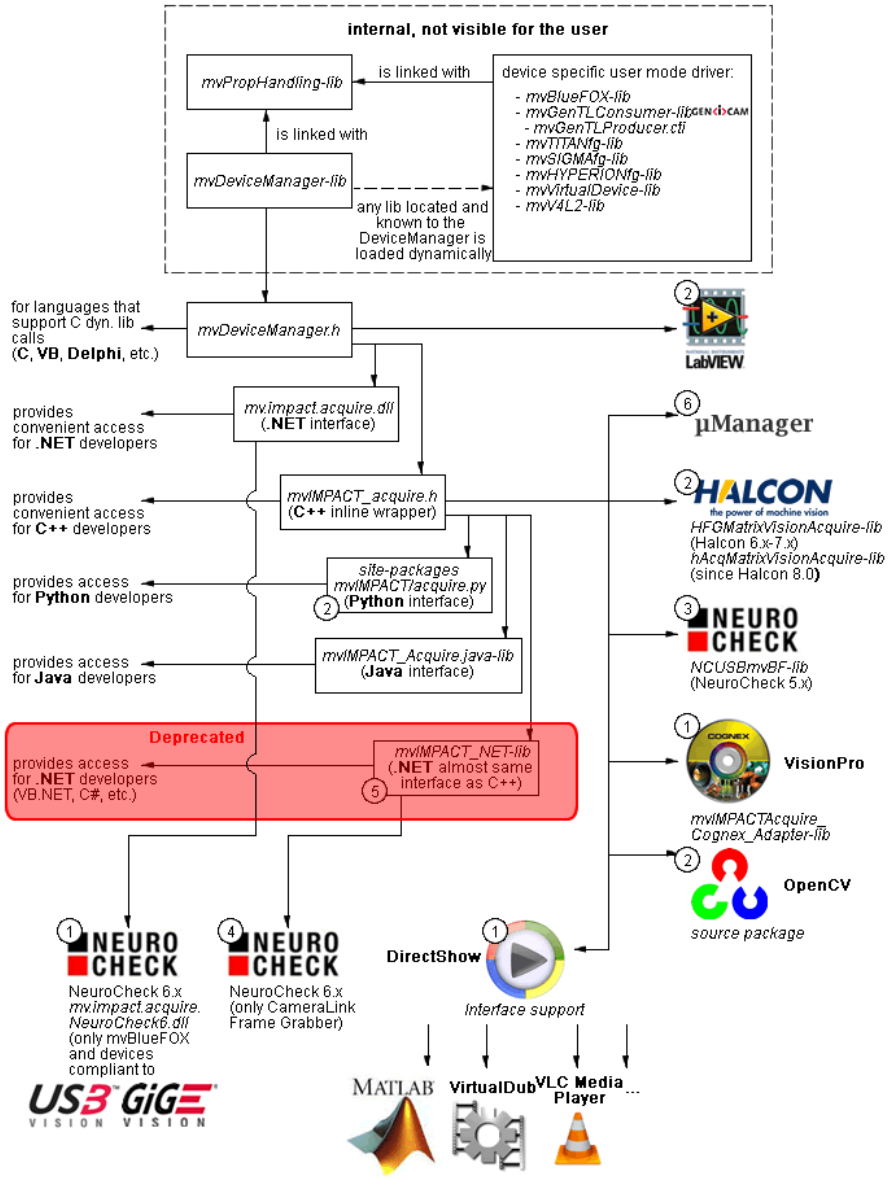
- Please note that each setting location step in the figure from above internally contains two search steps. First the framework will try to locate a setting with user scope and if this can't be located, the same setting will be searched with global (system-wide) scope. On Windows this e.g. will access either the **HKEY_CURRENT_USER** or (in the second step) the **HKEY_LOCAL_MACHINE** branch in the Registry.
- Whenever storing a product specific setting, the device specific setting of the device used for storing will be deleted (if existing). E.g. you have a device "VD000001" which belongs to the product group "VirtualDevice" with a setting exclusively for "VD000001". As soon as you store a product specific setting **using THIS device**, the (device specific) setting for "VD000001" will be deleted. Otherwise a product specific setting would never be loaded as a device specific setting will always be found first. Storing a product specific setting with a different device belonging to the same family however will **NOT** delete device specific settings for other devices.
- The very same thing will also happen when opening a device from any other application! mvPropView does not behave in a special way but only acts as an arbitrary user application.
- Whenever storing a device family specific setting, the device specific or product specific setting of the device used for storing will be deleted (if existing). See above to find out why.

- On Windows the driver will **not** look for a matching XML file during start-up automatically as the native storage location for settings is the Windows Registry. This must be loaded explicitly by the user by using the appropriate API function offered by the SDK. However, under **Linux** XML files are the only setting formats understood by the driver framework thus here the driver will also look for them at start-up. The device specific setting will be an XML file with the serial number of the device as the file name, the product specific setting will be an XML file with the product string as the filename, the device family specific setting will be an XML file with the device family name as the file name. All other XML files containing settings will be ignored!
- Restoring of settings previously stored works in a similar way. After a device has been opened the settings will be loaded automatically as described above.
- A detailed description of the individual properties offered by a device will **not** be provided here but can be found in the C++ API reference, where descriptions for all properties relevant for the user (grouped together in classes sorted by topic) can be found. As mvPropView doesn't introduce new functionality but simply evaluates the list of features offered by the device driver and lists them any modification made using the GUI controls just calls the underlying function needed to write to the selected component. mvPropView also doesn't know about the type of component or e.g. the list of allowed values for a property. This again is information delivered by the driver and therefore can be queried by the user as well without the need to have special *inside information*. One version of the tool will always be delivered in source so it can be used as a reference to find out how to get the desired information from the device driver.

1.8.4 Driver concept

The driver supplied with the MATRIX VISION product represents the port between the programmer and the hardware. The driver concept of MATRIX VISION provides a standardized programming interface to all image processing products made by MATRIX VISION GmbH.
The advantage of this concept for the programmer is that a developed application runs without the need for any major modifications to the various image processing products made by MATRIX VISION GmbH. You can also incorporate new driver versions, which are available for download free of charge on our website:
`https://www.matrix-vision.com`.

The following diagram shows a schematic structure of the driver concept:



Driver concept

- **1** Part of any mvIMPACT Acquire driver installation package (Windows).
- **2** Separately available for 32 bit and 64 bit. Requires at least one installed driver package.
- **3** See 2, but requires an installed version of the mvBlueFOX driver.

- **4** Part of the NeuroCheck installer but requires at least one installed frame grabber driver.
- **5** Part of the mvIMPACT SDK installation. However, new designs should use the .NET libs that are now part of mvIMPACT Acquire (`"mv.impact.acquire.dll"`). The namespace `"mv.impact.acquire"` of `"mv.impact.acquire.dll"` provides a more natural and more efficient access to the same features as contained in the namespace `"mvIMPACT_NET.acquire"` of `"mvIMPACT_NET.dll"`, which is why the latter one should only be used for backward compatibility but **NOT** when developing a new application.
- **6** Part of Micro-Manager.

1.8.4.1 VisionPro Support

Every *mvIMPACT Acquire* driver package on Windows comes with an adapter to VisionPro from Cognex. The installation order does not matter. After the driver package and VisionPro has been installed, the next time VisionPro is started it will allow selecting the *mvIMPACT Acquire* device. No additional steps are needed.

MATRIX VISION devices that also comply with the GigE Vision or USB3 Vision standard don't need any software at all, but can also use VisionPro's built-in GigE Vision or USB3 Vision support.

1.8.4.2 HALCON Support

HALCON comes with built-in support for *mvIMPACT Acquire* compliant devices, so once a device driver has been installed for the *mvIMPACT Acquire* device, it can also be operated from a HALCON environment using the corresponding acquisition interface. No additional steps are needed.

MATRIX VISION devices that also comply with the GigE Vision standard don't need any software at all, but can also use HALCON's built-in GigE Vision support.

As some *mvIMPACT Acquire* device driver packages also come with a GenTL compliant interface, these can also be operated through HALCON's built-in GenTL acquisition interface.

1.8.4.3 LabVIEW Support

Every *mvIMPACT Acquire* compliant device can be operated under LabVIEW through an additional set of VIs which is shipped by MATRIX VISION as a separate installation (`"mvLabVIEW Acquire"`).

MATRIX VISION devices that also comply with the GigE Vision or USB3 Vision standard don't need any additional software at all, but can also be operated through LabVIEW's GigE Vision or USB3 Vision driver packages.

1.8.4.4 DirectShow Support

Every *mvIMPACT Acquire* compliant device driver package comes with an interface to DirectShow. In order to be usable from a DirectShow compliant application, devices must first be registered for DirectShow support. How to this is explained **here** (p. 87).

1.8.4.5 Micro-Manager Support

Every *mvIMPACT Acquire* compliant device can be operated under `https://micro-manager.org` when using mvIMPACT Acquire 2.18.0 or later and at least *Micro-Manager* 1.4.23 build **AFTER** 15.12.2016. The adapter needed is part of the *Micro-Manager* release. Additional information can be found here: `https://micro-manager.org/wiki/MatrixVision`.

1.8.5 Relationship Between Driver, Firmware And SDK

In order to operate an mvBlueNAOS, not only the hardware but also the corresponding software is required. This can be installed via the mvIMPACT Acquire GenTL-Acquire package and consists of the following different parts.

1. Firmware - runs on the camera and controls the hardware there. The camera must be ready for operation before or while the host is booting so that it can be found by the OS. The firmware required for this is therefore stored on the camera. If the firmware and driver do not match, an error message is displayed when attempting to use the camera with the request to update the firmware. The MATRIX VISION tool mvDeviceConfigure can be used for this. Only after that the camera can be used.
2. Kernel Module - is used for communication between the HW and the OS. The module is specific for each OS and has to match the next layer, i.e. the GenICam producer, see below
3. GenICam Producer - provides the standardized GenTL interface and is specific for the mvBlueNAOS series. All SDKs that have a corresponding GenTL consumer can communicate via this interface not only the MATRIX VISION SDK. Also possible are for example HALCON, MIL, CVB, MATLAB and many other third party packages.
4. MATRIX VISION GenICam Consumer - communicates with the camera via the GenTL interface and with the application via mvIMPACT Acquire functions.

1.9 Technical Data

1.9.1 mvBlueNAOS2

1.9.1.1 Dimensions

1.9.1.1.1 BVS CA-BN2-xxxxx-x5x110 (Standard housing with NAOS for Embedded (N4e) interface)

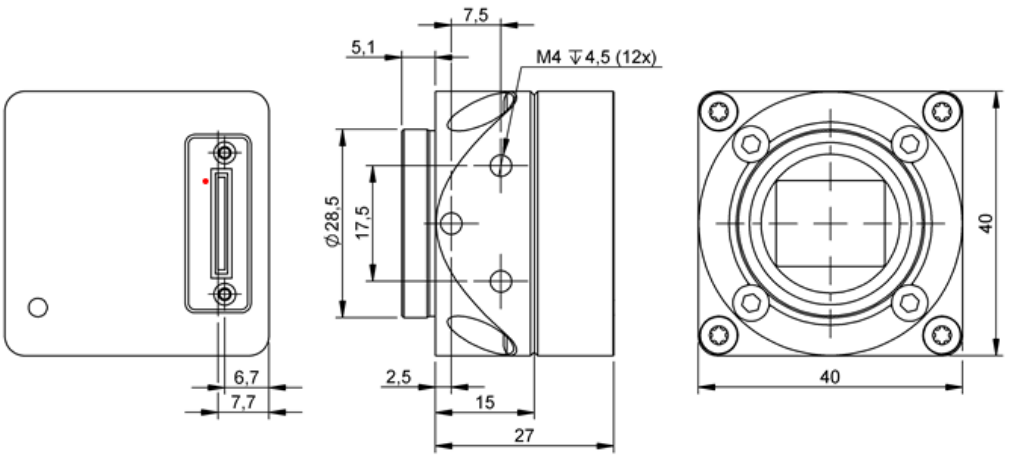


Figure 1: Connectors BVS CA-BN2-xxxxx-x5x110

mvBlueNAOS2	
Size of body (w x h x l)	40 x 40 x 27 mm

1.9.1.1.2 BVS CA-BN2-xxxxx-x00010 (Board-level)

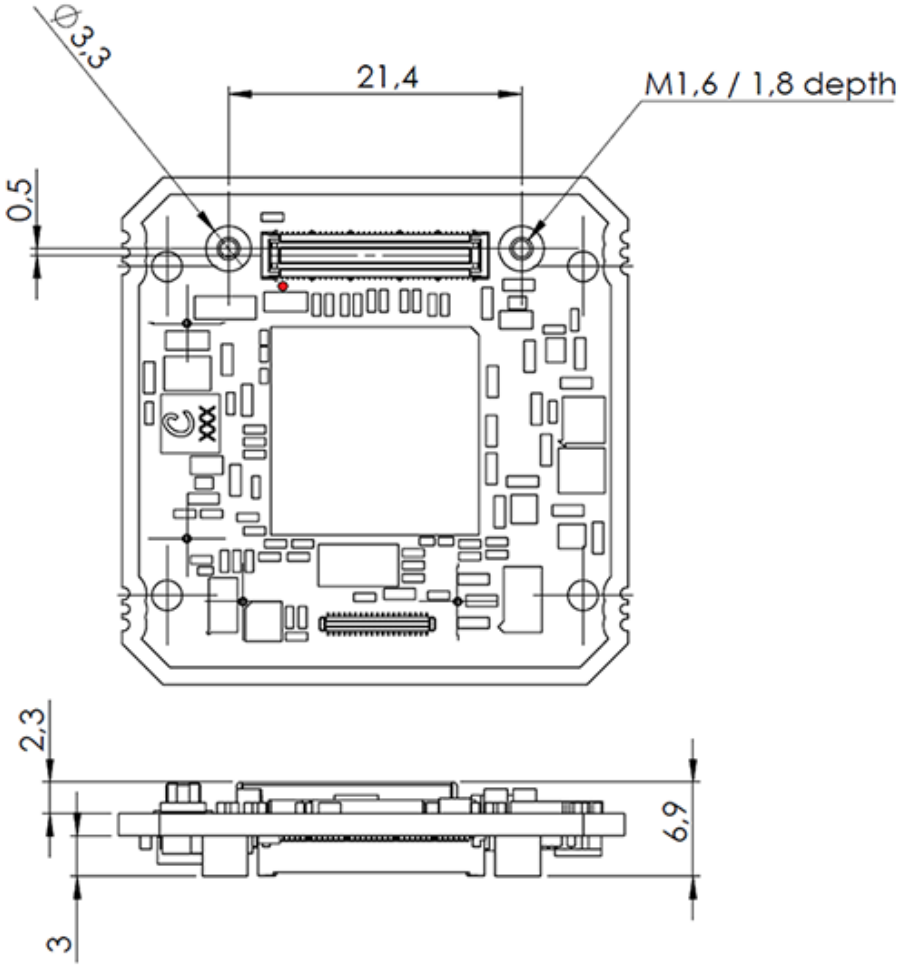


Figure 2: Connectors BVS CA-BN2-xxxxx-x00010

mvBlueNAOS	
Size of body (w x h x l)	37 x 37 x 6.9 mm

1.9.2 mvBlueNAOS4

1.9.2.1 Dimensions

1.9.2.1.1 BVS CA-BN4-xxxxx-x5x510 (Board-level)

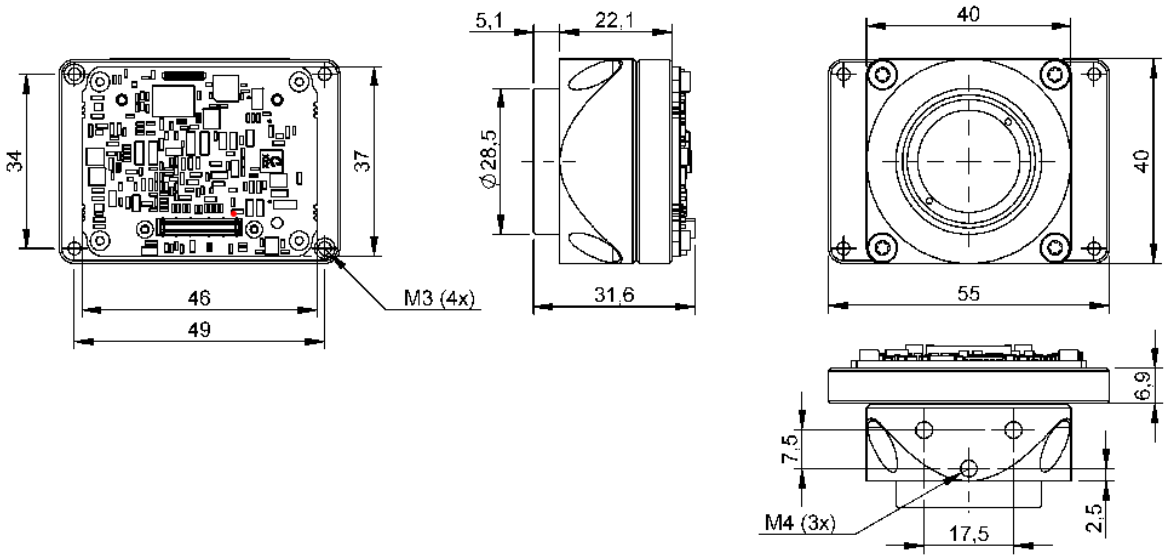


Figure 3: Connectors BVS CA-BN4-xxxxx-x5x510

mvBlueNAOS4	
Size of body (w x h x l)	55 x 40 x 26.5 mm

1.9.3 NAOS for Embedded (N4e) Interface

The NAOS for Embedded (N4e) interface provides PCIe Gen.2 and various I/O functionality on a single 70-pin Board-to-Board connector. The main features of the user I/O interface are

- 4 digital inputs,
- 4 digital outputs,
- a UART interface for serial communication, and
- a I2C two wire serial interface.

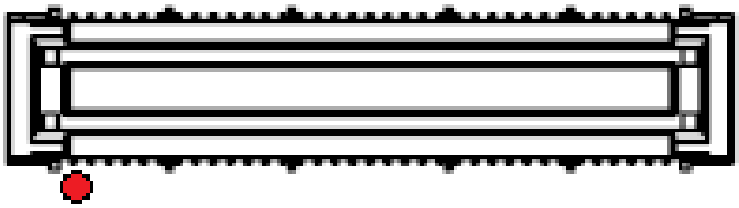


Figure 4: NAOS for Embedded (N4e) interface

"Camera connector"

- Hirose DF40GB(3.0)-70DS-0.4V

"Mating Connector"

Used for accessory IO Boards or customer implementation.

- Hirose DF40GB-70DP-0.4V

"Mechanical characteristics"

- 70-pin, 0.4mm pitch, shielded, stacking height: 3.0mm
- Steel spacers (height 3mm). Screws M1.6, max. screw depth 1.8mm.

1.9.3.1 Pin assignment

Note

The red dot marks pin 1 of Hirose DF40GB(3.0)-70DS-0.4V.

Pin	Signal	Description
1	VCC_IN	Input Voltage (5V-12V)
2	VCC3V3	Power Output, max. 10mA for customer use
3	VCC_IN	Input Voltage (5V-12V)
4	I2C_SCL_USER	I2C_USER (see details below) LVCMOS 3.3V IO level
5	VCC_IN	Input Voltage (5V-12V)
6	I2C_SDA_USER	I2C_USER (see details below) LVCMOS 3.3V IO level
7	DigIn0	Digital input (with level shifter) Voltage reference VCC_IO
8	UART_RX	LVCMOS 3.3V IO level 100 Ohm series resistor placed on mainboard
9	DigIn1	Digital input (with level shifter) Voltage reference VCC_IO
10	UART_TX	LVCMOS 3.3V IO level 100 Ohm series resistor placed on mainboard
11	GND	Ground
12	GND	Ground
13	GND	Ground
14	GND	Ground
15	PCIe_CLK_P	PCIE_CLK + (differential)
16	-	Do not connect, internal use
17	PCIe_CLK_N	PCIE_CLK - (differential)
18	-	Do not connect, internal use
19	GND	Ground
20	GND	Ground
21	PCIe_TX0_P	PCIe Transmitter (differential)
22	-	Do not connect, internal use
23	PCIe_TX0_N	PCIe Transmitter (differential)
24	-	Do not connect, internal use
25	GND	Ground
26	GND	Ground
27	GND	Ground
28	GND	Ground
29	DigIn2	Digital input (with level shifter) Voltage reference VCC_IO
30	GND	Ground
31	DigIn3	Digital input (with level shifter) Voltage reference VCC_IO
32	PCIe_RX3_P	PCIe Receiver (differential)
33	GND	Ground
34	PCIe_RX3_N	PCIe Receiver (differential)
35	PCIe_TX1_P	PCIe Transmitter (differential)
36	GND	Ground
37	PCIe_TX1_N	PCIe Transmitter (differential)
38	PCIe_RX0_P	PCIe Receiver (differential)

39	GND	Ground
40	PCIe_RX0_N	PCIe Receiver (differential)
41	PCIe_PERST↩ _N	Reset
42	GND	Ground
43	GND	Ground
44	GND	Ground
45	GND	Ground
46	GND	Ground
47	PCIe_TX2_P	PCIe Transmitter (differential)
48	PCIe_RX2_P	PCIe Receiver (differential)
49	PCIe_TX2_N	PCIe Transmitter (differential)
50	PCIe_RX2_N	PCIe Receiver (differential)
51	GND	Ground
52	GND	Ground
53	PCIe_TX3_P	PCIe Transmitter (differential)
54	PCIe_RX1_P	PCIe Receiver (differential)
55	PCIe_TX3_N	PCIe Transmitter (differential)
56	PCIe_RX1_N	PCIe Receiver (differential)
57	GND	Ground
58	GND	Ground
59	GND	Ground
60	GND	Ground
61	DigOut0	Digital output (with level shifter) Voltage reference VCC_IO
62	-	Do not connect, internal use
63	DigOut1	Digital output (with level shifter) Voltage reference VCC_IO
64	-	Do not connect, internal use
65	DigOut2	Digital output (with level shifter) Voltage reference VCC_IO
66	VCC_IN	Input Voltage (5V-12V)
67	DigOut3	Digital output (with level shifter) Voltage reference VCC_IO
68	VCC_IN	Input Voltage (5V-12V)
69	VCC_IO	Voltage reference for digital inputs / outputs.
70	VCC_IN	Input Voltage (5V-12V)

Note

Additional shield pins are connected to GND.

1.9.3.1.1 Electrical characteristics of signals

Input Voltage

Parameter	Description	min	nom	max	Unit
VCC_IN		5	12	17	V
I_{VCC_IN}				1.8	A
I per pin				0.3	A

VCC_IO / DIGIN / DIGOUT

Parameter	Description	min	nom	max	Unit
VCC_IO	I/O voltage power	1.8	3.3	5.5	V
$U_{DIG_IN_LOW}$	Low-level input voltage			0.8	V
$U_{DIG_IN_HIGH}$	High-level input voltage	0.7 x VCC_IO			V
I_{OH}	Output current for VCC_IO = 3.3 V			-24	mA
I_{OL}	Output current for VCC_IO = 3.3 V			24	mA
$U_{DIG_OUT_HIGH}$	Output voltage for VCC_IO = 3.3 V, I_{OUT}= -24 mA	2.4			V
$U_{DIG_OUT_LOW}$	Output voltage for VCC_IO = 3.3 V, I_{OUT}= 24 mA			0.55	V

Note

Digital I/Os include Level Shifters which allow customized I/O levels. Output characteristics given for VCC_IO = 3.3 V. For other values see datasheet SN74LVC2T45

Serial Interface, I2C and Power Down

Signal	Parameters / Properties	min	nom	max	Unit
PCIe_PERST_N	Reset		3.3		
	PCIe_PERST_N Input Low Voltage	-0.5		0.8	V
	PCIe_PERST_N Input High Voltage	2		3.8	V
UART_RX / UART_TX	Voltage level		3.3	3.4	V
	The camera provides a UART Interface for serial communication (internally protected by 100Ohm series resistor)				
I2C_SCL / I2C_SDA	USER_I2C for customer applications		3.3		V

1.9.3.2 Add-in cards for the NAOS for Embedded (N4e) Interface

1.9.3.2.1 BNE-IF-PCIeX4-N4e-IO14 Add-in card for connecting the camera to the PCIe x4 slot of a host board. The board includes

- the N4e interface (CON2)
- a 14-pin I/O connector (CON3) and
- the PCIe x4 edge (CON1).

The add-in card is compliant with PCIe 2.0 mainboard or higher (up to 1600 MB/s). Dimensions: 37.5 mm x 46 mm.

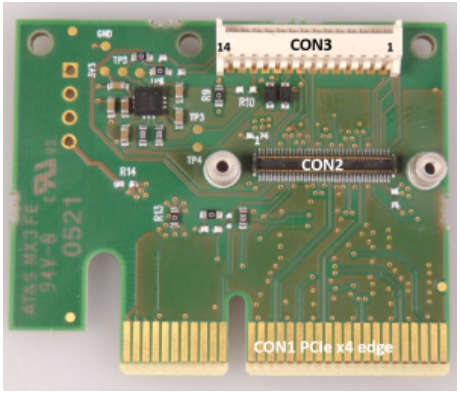


Figure 5: PCI Express Add-in card

1.9.3.2.2 BNE-IF-M2M-N4e-IO14 M.2-M-Key add-in card for connecting the camera to the PCIe x4 slot of a host board. The board includes

- the N4e interface (CON2)
- a 14-pin I/O connector (CON3) and
- the PCIe x4 edge (CON1).

The add-in card is compliant with PCIe 2.0 mainboard or higher (up to 1600 MB/s). Dimensions: 22 mm x 80 mm (or 60 mm).

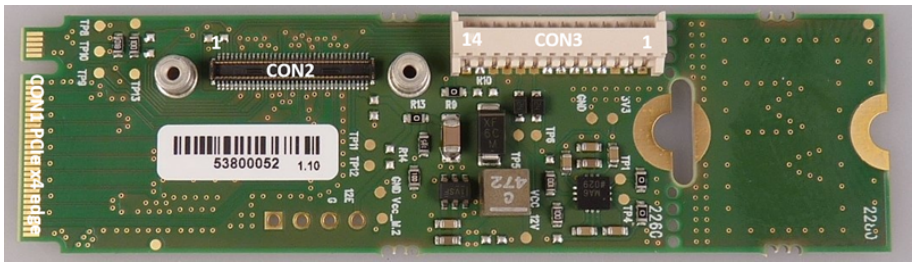


Figure 6: M.2-M-Key add-in card

1.9.3.2.3 Pin assignments 12-pin I/O connector (CON3)

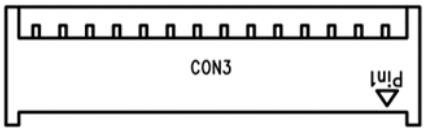


Figure 7: 14-pin I/O connector (CON3

- Part type:
  - Wire to Bord Connector 1.25mm MOLEX PicoBlade 53048_1410
- Connecting part:
  - Molex 0510211400, 1.25 Wire to Board Connection Receptacle Crimp Housing, Applicable Terminal 50058-8000, 50079-8000 Series

Pin	I/O	Signal	Description
1	Out	DigOut0	Digital Output
2	Out	DigOut1	Digital Output
3	Out	DigOut2	Digital Output
4	Out	DigOut3	Digital Output
5	In	DigIn0	Digital Input
6	In	DigIn1	Digital Input
7	In	DigIn2	Digital Input
8	In	DigIn3	Digital Input
9	Out	RS232_TXD	Serial Interface RS232_TXD
10	In	RS232_RXD	Serial Interface RS232_RXD
11	GND	GND	Ground
12	POWER	VCC_IO	Not connected
13	Out	I2C_SCL_USER	I2C Clock
14	BI	I2C_SDA_USER	I2C Data

"Electrical characteristics of signals"

"VCC_IO / DIGIN / DIGOUT"

Parameter	Description	min	nom	max	Unit
VCC_IO	I/O voltage power (connected to 3.3 V on-board)		3.3		V
$U_{DIG_IN_LOW}$	Low-level input voltage	0		0.8	V
$U_{DIG_IN_HIGH}$	High-level input voltage	2		3.5	V
I_{OH}	High-level output current			-10	mA
I_{OL}	Low-level output current			10	mA
$U_{DIG_OUT_HIGH}$	High-level output voltage	2.↩ 4	3.3		V
$U_{DIG_OUT_LOW}$	High-level output voltage		0.0	0.55	V

Note

Optional user defined VCC_IO voltage 1.8 V .. 5.5 V possible. For further information please contact MATRIX VISION.

"Serial interface RS232"

Parameter	Description	min	nom	max	Unit
EIA/TIA-232E Input Voltage Range*		-30		+30	V
EIA/TIA-232E Input Threshold Low		0.6	1.3		V
EIA/TIA-232E Input Threshold High			1.6	2.4	V
EIA/TIA-232E Input Hysteresis			0.4		mA
EIA/TIA-232E Input Resistance		3	5	7	V
Output Voltage Swing		±5.↩ 0	±5.↩ 7		V
RS-232 Output Short-Circuit Current			±15		mA
ESD PROTECTION	Human body model air discharge Human body model contact discharge		±15 ±8		kV kV
TIMING CHARACTERISTICS	Maximum Data Rate			115	kbps

"I2C_USER"

Signal	Parameter	min	nom	max	Unit
U_{I2C} - I2C_SCL_USER / I2C_SDA_USER			3.3		V
$U_{_IN_LOW}$	V_{IL} (low level input voltage)	0		0.15	V
$U_{_IN_HIGH}$	V_{IH} (high level input voltage)	2.↩ 9		3.3	V
$U_{_OUT_HIGH}$	depending on user pullup (default 2k)	2.↩ 2			V
$U_{_OUT_LOW}$	Digital output (I_{OUT}=1mA)			0.4	V
Frequency	Interface clock rate		100	400	kHz

Note

10K pull up resistor to 3.3 V is intergrated. For 400 kHz operation additional 2K pull up is recommended.

1.9.4 OCuLink

The OCuLink interface provides PCIe Gen.2 functionality on a single 42-pin connector.

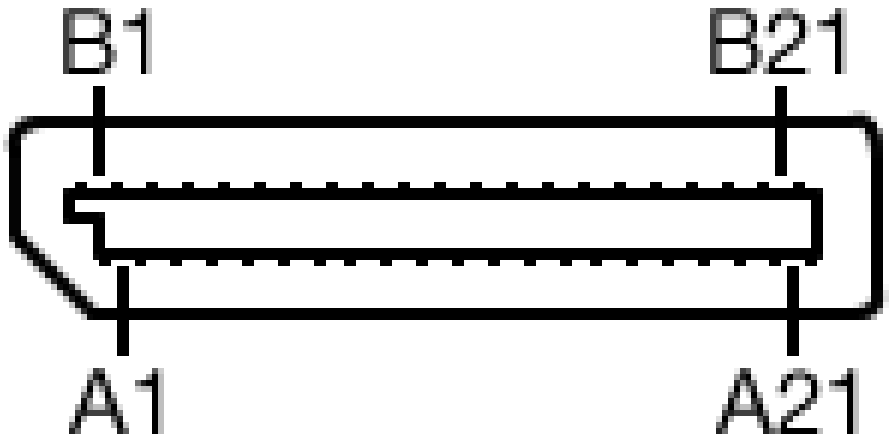


Figure 8: OCuLink interface

"Camera connector"

- Amphenol type G14A42211912HRV

"Matching cable"

Used for accessory IO Boards or customer implementation.

- OCuLink standard cable, for example Molex 2060611151/2021431003 or similar from other suppliers

1.9.4.1 Pin assignment

Note

The red dot marks pin 1.

Pin	Signal	Description
A1	-	Do not connect
A2	GND	Ground
A3	PCIe_RX0_P	PCIe Receiver IN (differential)
A4	PCIe_RX0_N	PCIe Receiver IN (differential)
A5	GND	Ground
A6	PCIe_RX1_P	PCIe Receiver IN (differential)
A7	PCIe_RX1_N	PCIe Receiver IN (differential)
A8	GND	Ground
A9	-	Do not connect
A10	WAKE_PCIE	IN CWAKE
A11	GND	Ground
A12	PCIe_CLK_P	IN PCIE_CLK +
A13	PCIe_CLK_N	IN PCIE_CLK +
A14	GND	Ground
A15	PCIe_RX2_P	PCIe Receiver IN (differential)
A16	PCIe_RX2_N	PCIe Receiver IN (differential)
A17	GND	Ground
A18	PCIe_RX3_P	PCIe Receiver IN (differential)
A19	PCIe_RX3_N	PCIe Receiver IN (differential)
A20	GND	Ground
A21	**PWR_IN** (p. 52)	Power-In
B1	**PWR_IN** (p. 52)	Power-In
B2	GND	Ground
B3	PCIe_TX0_P	PCIe Transmitter OUT (differential)
B4	PCIe_TX0_N	PCIe Transmitter OUT (differential)
B5	GND	Ground
B6	PCIe_TX1_P	PCIe Transmitter OUT (differential)
B7	PCIe_TX1_N	PCIe Transmitter OUT (differential)
B8	GND	Ground
B9	-	Do not connect
B10	-	Do not connect
B11	GND	Ground
B12	PCIe_PERST↩_N	IN Reset
B13	CPRSNT	IN PRESENT
B14	GND	Ground
B15	PCIe_TX2_P	PCIe Transmitter OUT (differential)
B16	PCIe_TX2_N	PCIe Transmitter OUT (differential)
B17	GND	Ground
B18	PCIe_TX3_P	PCIe Transmitter OUT (differential)
B19	PCIe_TX3_N	PCIe Transmitter OUT (differential)
B20	GND	Ground
B21	-	Do not connect

Directions of the PCIe signals (Transmitter/Receiver) are from the FPGA to the add-in card.

1.9.4.1.1 Electrical characteristics of signals

Input Voltage

Signal	min	nom	max	Unit
V_IN	-	5	-	V
I_{VCC_IN}			2	A
I per pin			0.5	A

Power Down

Signal	Properties	min	nom	max	Unit
PCIe_PERST_N	Voltage level	-	3.3	-	V
	V_{IL} (low level input voltage)	-0.5	-	0.8	V
	V_{IH} (high level input voltage)	2	-	3.8	V
WAKE_PCIe	Voltage level	-	3.3	-	V
	V_{IL} (low level input voltage)	-0.5	-	0.8	V
	V_{IH} (high level input voltage)	1	-	3.8	V
	Current WAKE_PCIe	-	-	4	mA

1.9.4.2 Power/IO connector (for OCuLink model)

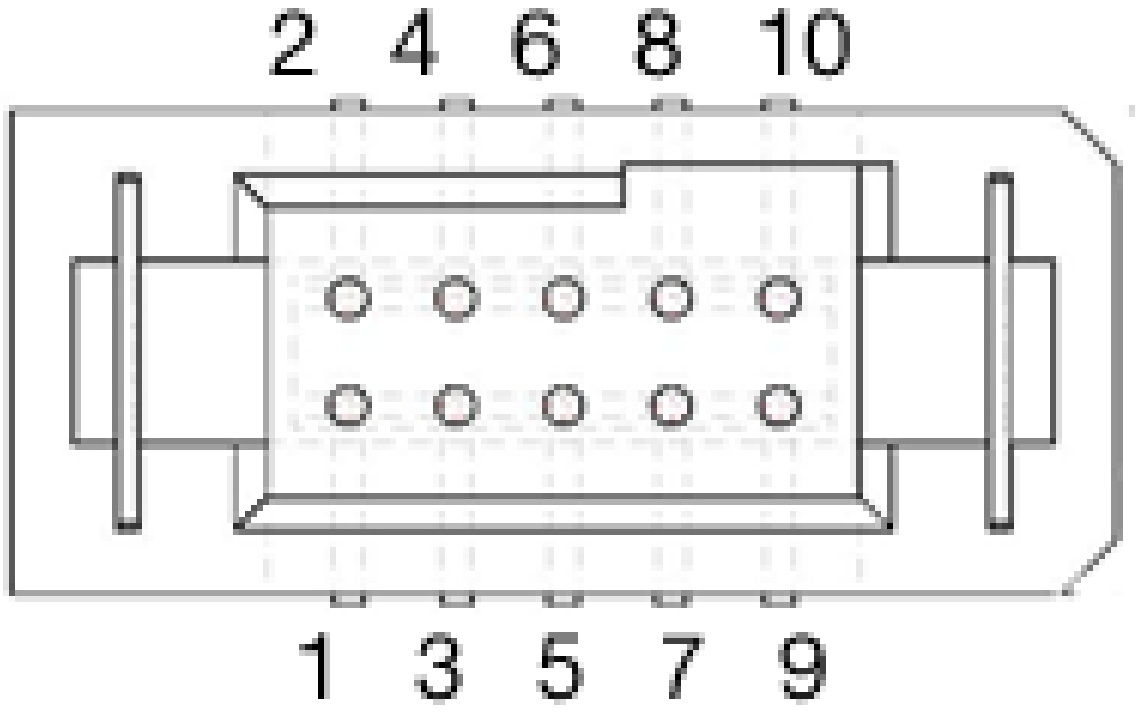


Figure 9: Power/IO connector (for OCuLink model)

"Camera connector"

- 10-Pin Samtec Connector
- Part type: TFM-105-02-L-D-WT
- 1.27 mm pitch Samtec Tiger Eye

"Matching cable"

- Alysium A65-3210 or **KS-10TFM-OPEN LS 01.0** (p. 20)
- Samtec SFSD series or similar from other suppliers

Pin	Signal	Description
1	GND (for PWR_IN)	ground for Power-In
2	**PWR_IN** (p. 52)	power-In
3	Opto DigIn_GND	ground for opto-coupler (In)
4	DigOut_PWR_IN	power-In for the opto-coupler outputs
5	Opto DigIn0	digital input, opto-coupler
6	Opto DigOut0	digital output, opto-coupler
7	Opto DigIn1	digital input, opto-coupler
8	Opto DigOut1	digital output, opto-coupler
9	DigOut2	digital output, TTL level
10	DigIn2	digital input, TTL level

1.9.4.2.1 Pin assignment

- Opto DigIn 0 & 1 and Opto DigOut 0 & 1 are galvanically isolated from the camera electronic
  - The isolation distance is specified by the connector (0.5mm Pitch $\rightarrow$ 1kV)
- As an option pin 10 can be used as CASE_GND instead of the digital input

Characteristics of input voltage

Characteristics	min.	nom.	max.	Unit
Camera power supply	5	12	17	V
Camera current supply		1	1.8	A

Characteristics of the digital inputs

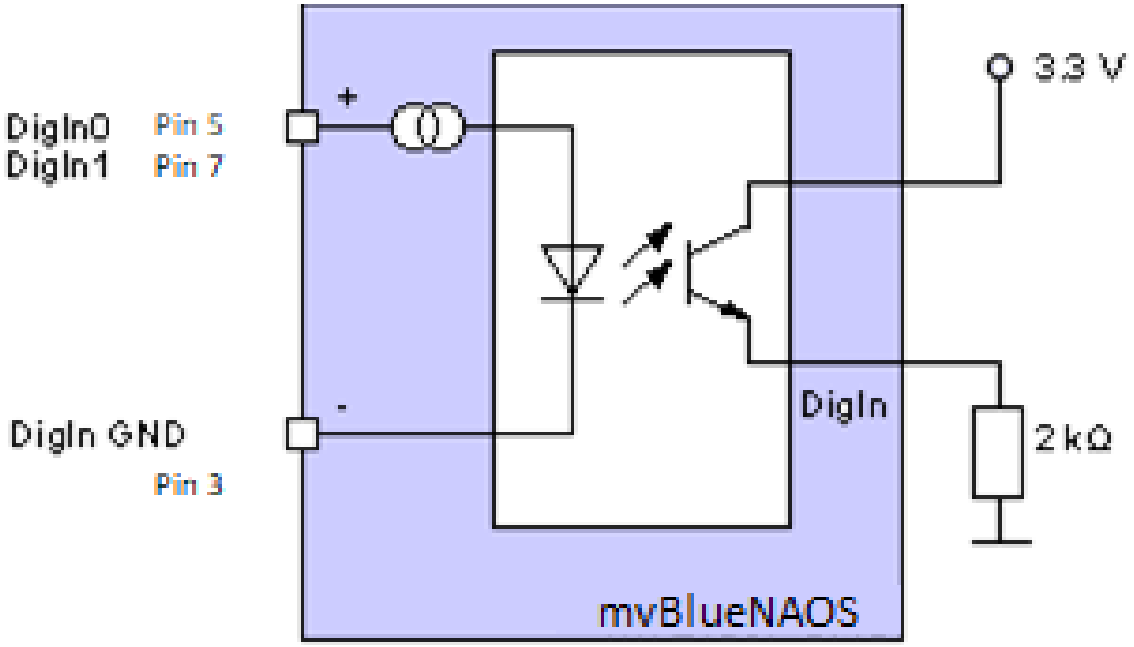


Figure 10: DigIn

Opto DigIn				
Characteristics	min.	nom.	max.	Unit
$U_{Opto\ DigIn_LOW}$ / V_{IL} (low level input voltage)	0		1	V
$U_{Opto\ DigIn_HIGH}$ / V_{IH} (high level input voltage)	2.5		28	V
I_{max}			8	mA
Switching Frequency			5	MHz

Characteristics of the digital outputs

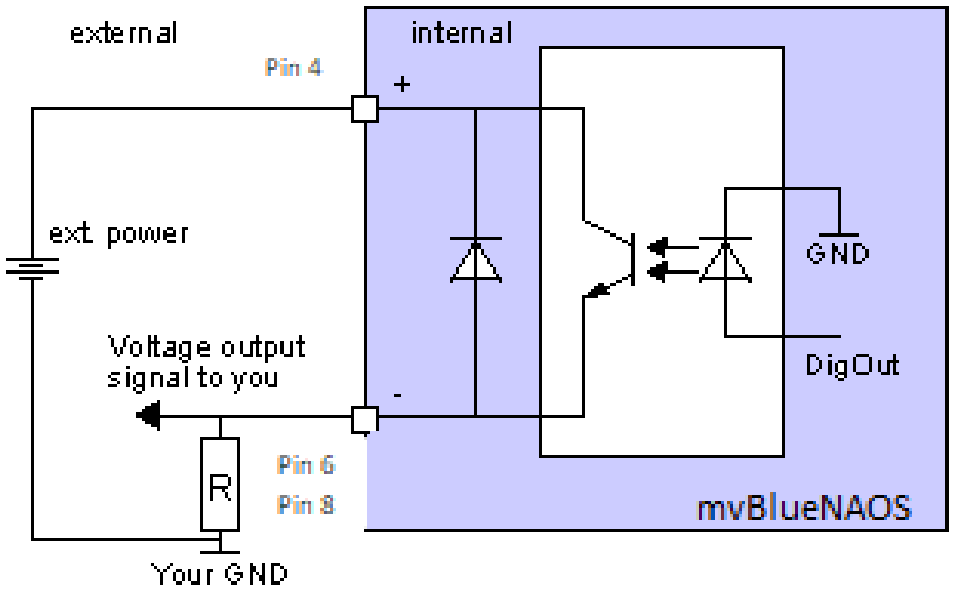


Figure 11: DigOut

Opto DigOut				
Characteristics	min.	nom.	max.	Unit
$U_{Opto\ DIG_OUT_LOW}$ / V_{IL} (low level input voltage)	0		1	V
$U_{Opto\ DIG_OUT_HIGH}$ / V_{IH} (high level input voltage)	3		30	V
$U_{CE(sat)}$ at I_C = 7 mA		0.4		V
I_C Load current			15	mA

Switching characteristics			
Characteristics	Test Conditions	Typ.	Unit
t_{ON} Turn-On time	R_L = 100 Ohm, V_{CC} 10 V, I_C = 2 mA	3	us
t_S Storage time		3	us
t_{OFF} Turn-Off time		3	us
t_{ON} Turn-On time	R_L = 1.9 kOhm, V_{CC} 5 V, I_C = 16 mA	2	us
t_S Storage time		25	us
t_{OFF} Turn-Off time		40	us

VCC_IO / DIGIN / DIGOUT

Signal	Parameter	min	nom	max	Unit
VCC_IO	I/O voltage power		3.3		V
$U_{DIG_IN_LOW}$	V_{IL} (low level input voltage)			0.8	V
$U_{DIG_IN_HIGH}$	V_{IH} (high level input voltage)	2		3.5	V
I_{OH}	High-level output current VCC_IO: 3.3 V			-24	mA
I_{OL}	Low-level output current VCC_IO: 3.3 V			24	mA
$U_{DIG_OUT_HIGH}$	Digital output (I_{OUT}= -24 mA)	2.↩ 4			V
$U_{DIG_OUT_LOW}$	Digital output (I_{OUT}= 24 mA)			0.55	V

1.9.5 Signal LED

The mvBlueNAOS features a RGB LED. There are following states:

State	LED color
FPGA loaded	White on
Starting **wxPropView** (p. 65)	
Waiting for client	Blue on
Connected, streaming off	Yellow on
Streaming on	Green on
Error	
	Red on

1.9.6 Summary of components

Features		mvBlueNAOS2	mvBlueNAOS4
Image Memory		-	
Flash (FPGA Configuration, Device Specific Data)		16 MByte	
PCIe Interface		x4 Gen2	
PCIe Bandwidth max.		16 Gbit/s	
Max. effective PCIe Bandwidth		12.8 Gbit/s	
Bandwidth FPGA Pipeline		800 MPixel/s	1300 MPixel/s
SLVS Lanes		16	-
SLVS-EC Lanes		-	4
Inputs		4	
Outputs		4	
RS232		Controlling e.g. varioptic liquid lens, lights, etc.	
Environment			
	Ambient Temperature		
	Operation	0..45 deg C / 30 to 80% RH	
	Storage	-20..60 deg C / 20 to 90% RH	
	Protection class [1]	IP30	
Weight without lens		approx. 71 g	approx. 81.6 g
Power supply (PWR_IN)			
	P_{typ}	4 W	6 W

[1] not evaluated by UL

1.9.7 Summary of advanced features

The following table shows an excerpt of the advanced features.

Advanced features		mvBlueNAOS2	mvBlueNAOS4
MultiFrame / **SingleFrame** / **Continuous** (p. 69)		X	
Auto Exposure Control (AEC) (p. 69)		X	
Auto Gain Control (AGC) (p. 69)		X	

Auto white balance (AWB)		X	
Binning / **Decimation** (p. 68)		X	
Chunk Data (p. 79)		X	
Gamma (p. 75)	on the fly on the camera	X	
LUT	on the fly on the camera	-	
Flat Field Correction (FFC)		-	
"Defective Pixel Correction"		-	
Sequencer		X	-
Digital I/O (p. 80)		X	
Trigger Overlap (p. 69)		X	
Enhanced I/O functionality acc. to SFNC (counter, timer) (p. 73)		X	
Events (p. 78)		X	
User Sets (p. 82)		X	

1.10 Sensor Overview

1.10.1 Image data flow

The following block diagrams show the data flow of the image data after being read from the sensor chip in the camera. The transfer latency (stream controller + resend buffer + packet generator) of the data is less than 100 us.

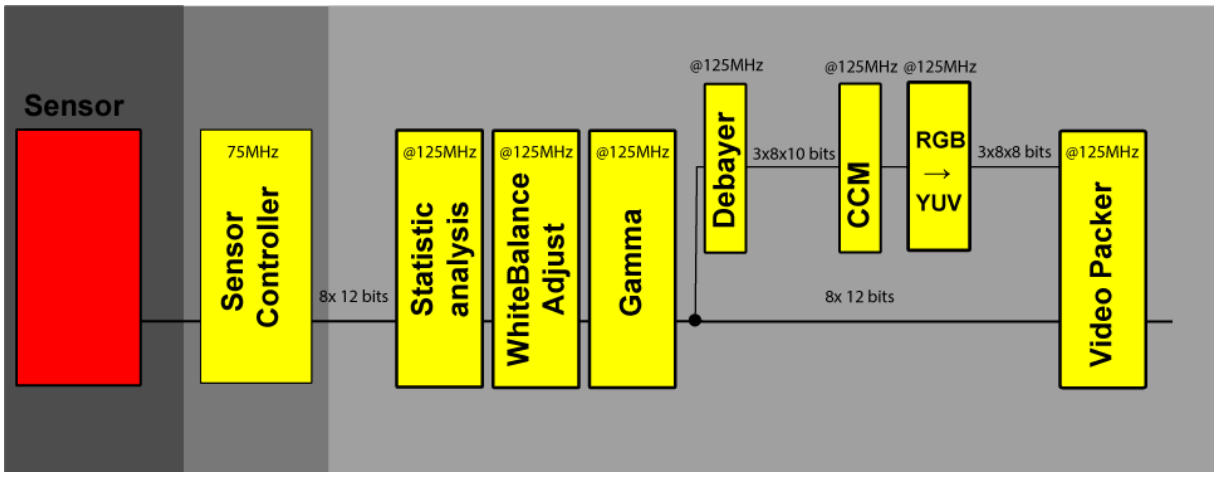


Figure 1: Block diagram

[1] Measured accord. to EMVA1288 with gray scale version of the camera

1.10.2 Output sequence of color sensors (RGB Bayer)

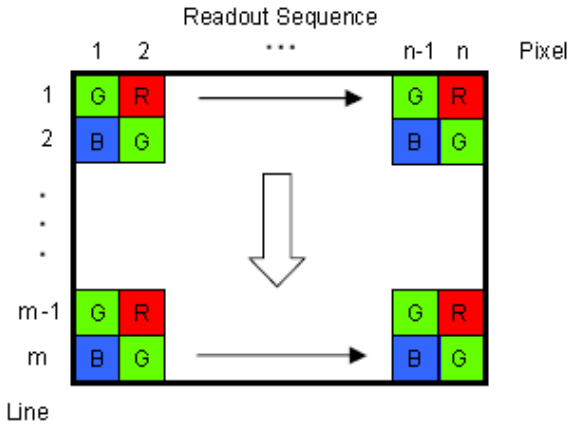


Figure 2: Output sequence of RAW data

1.10.3 Debayering of color sensors (RGB Bayer)

For Bayer demosaicing in the camera, we use our **"Adaptive Edge Sensing Plus"** method. I.e. **"Adaptive Edge Sensing Plus"** detects edges, filters and sharpens them depending on the location. It tries to reconstruct the green channel based on a gradient. Afterwards, it creates the differences between red and green, and blue and green, interpolates the values and adds the green values again. The second path below adds a high path filter, an amplifier, and a threhold to the Bayer data.

The advantages of the **"Adaptive Edge Sensing Plus"** method are:

- Very few false colors
- No alias effects
- Sharp edges
- Low noise

On the host side, the **"Adaptive Edge Sensing Plus"** method needs 20 % CPU load.

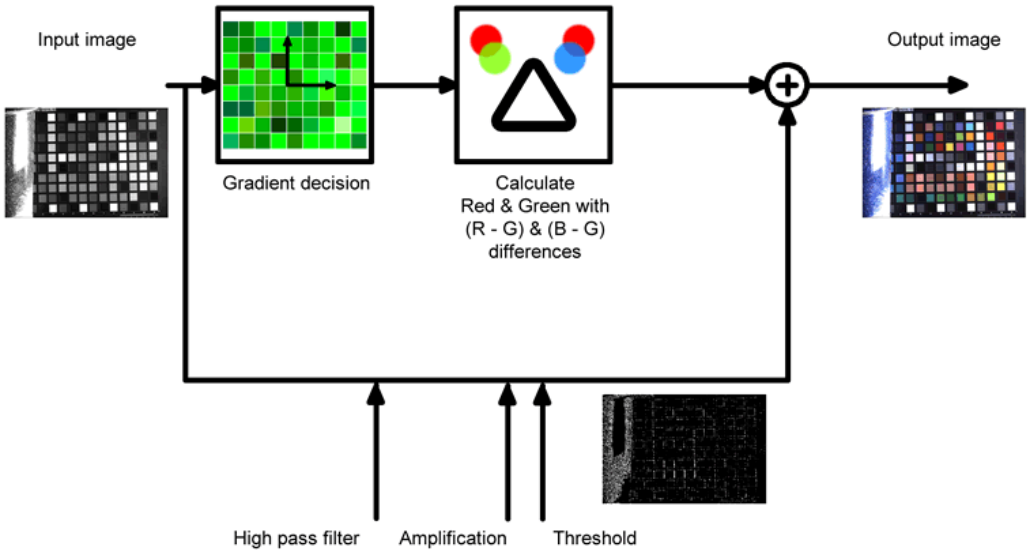


Figure 3: Debayering with Adaptive Edge Sensing Plus

1.10.4 Supported image formats

It depends on the sensor model which image formats are actually supported.

The parameter **PixelFormat** in the **ImageFormatControl** (p. 68) lists all the supported image formats of the specific camera model. In **wxPropView** (p. 65) you can find the **PixelFormat** parameter in *"Setting -> Base -> Camera -> GenICam -> Image Format Control -> Pixel Format"*.

Gray scale version	Color version
Mono8	
Mono10	
Mono12Packed	
Mono16	
	BayerGR8 (GR, RG, GB, BG depends on camera type)
	BayerGR10 (GR, RG, GB, BG depends on camera type)
	BayerGR12p (GR, RG, GB, BG depends on camera type)
	BayerGR16 (GR, RG, GB, BG depends on camera type)
	BayerGR12Packed (GR, RG, GB, BG depends on camera type)
	RGB8Packed
	BGR8Packed
	BGR10V2Packed
	YUV422Packed
	YUV422_YUYVPacked
	YUV444Packed

See also

For more details about the image formats, please have a look at the enums `TImageDestination↩ PixelFormat (C++)` and `TImageBufferPixelFormat (C++)` in the `mvIMPACT Acquire API manuals`.
An example application about the pixel formats called **ContinuousAllFormats** is also available.

1.10.5 CMOS sensors

Approximately 33 % of power is consumed by the sensor board and the rest by the main board. The main consumers are the FPGA and the sensor itself. Approximately 15 % of consumption is spread over various voltage regulators.

Summary of CMOS sensors (p. 57)

1.10.6 Summary of CMOS sensors

Sensor Type	Color	Resolution	Sensor size	Max. FPS	FPS exact	ADC	SNR_{max} [dB][1]	DR [dB][1]	Shutter	Trigger (HW / SW)	Pix↩ Size in um	Watt	Model
IMX250 (Sony)	Color	2464 x 2056	2/3	140	+	12	40,2	71,2	Global	X / X	3.45	4.0	**BVS CA-↩ BN2-0051↩ ZC (5.1 Mpix [2464 x 2056])** (p. 248)
IMX250 (Sony)	Gray	2464 x 2056	2/3	140	+	12	40,3	71,3	Global	X / X	3.45	4.0	**BVS CA-↩ BN2-0051↩ ZG (5.1 Mpix [2464 x 2056])** (p. 218)
IMX252 (Sony)	Color	2064 x 1544	1/1.↩ 8	191.↩ 5	+	12	40,3	71,4	Global	X / X	3.45	4.0	**BVS CA-↩ BN2-0032↩ ZC (3.2 Mpix [2064 x 1544])** (p. 246)

IMX252 (Sony)	Gray	2064 x 1544	1/1.↩ 8	191.↩ 5	+	12	40,2	71,3	Global	X / X	3.45	4.0	**BVS CA-↩ BN2-0032↩ ZG (3.2 Mpix [2064 x 1544])** (p. 215)
IMX253 (Sony)	Color	4096 x 3008	1.1	64.↩ 6	+	12	40,2	70,9	Global	X / X	3.45	4.5	**BVS CA-↩ BN2-0124↩ ZC (12.↩ 3 Mpix [4096 x 3008])** (p. 253)
IMX253 (Sony)	Gray	4112 x 3008	1.1	64.↩ 6	+	12	40,1	71,1	Global	X / X	3.45	4.5	**BVS CA-↩ BN2-0124↩ ZG (12.↩ 4 Mpix [4112 x 3008])** (p. 223)
IMX255 (Sony)	Color	4096 x 2176	1	88.↩ 7	+	12	40,2	70,9	Global	X / X	3.45	4.5	**BVS CA-↩ BN2-0089↩ ZC (8.9 Mpix [4096 x 2176])** (p. 250)

IMX255 (Sony)	Gray	4112 x 2176	1	88.↩ 7	+	12	40,3	71,0	Global	X / X	3.45	4.5	**BVS CA-↩ BN2-0089↩ ZG (8.9 Mpix [4112 x 2176])** (p. 220)
IMX273 (Sony)	Color	1456 x 1088	1/2.↩ 9	226.↩ 5	+	12	40,2	71,4	Global	X / X	3.45	3.4	**BVS CA-↩ BN2-0016↩ ZC (1.6 Mpix [1456 x 1088])** (p. 243)
IMX273 (Sony)	Gray	1456 x 1088	1/2.↩ 9	226.↩ 5	+	12	40,1	71,1	Global	X / X	3.45	3.4	**BVS CA-↩ BN2-0016↩ ZG (1.6 Mpix [1456 x 1088])** (p. 213)
IMX530 (Sony)	Color	5328 x 4608	1.2	55.↩ 5	+	12	39,8	70,4	Global	X / X	2.74	- - -	**BVS CA-↩ BN4-0246↩ ZC (24.↩ 6 Mpix [5328 x 4608])** (p. 271)

IMX530 (Sony)	Gray	5328 x 4608	1.2	55.↩ 5	+	12	39,8	70,3	Global	X / X	2.74	- - -	**BVS CA-↩ BN4-0246↩ ZG (24.↩ 6 Mpix [5328 x 4608])** (p. 240)
IMX531 (Sony)	Color	4512 x 4512	1.1	66.↩ 0	+	12			Global	X / X	2.74	- - -	**BVS CA-↩ BN4-0204↩ ZC (20.↩ 4 Mpix [4512 x 4512])** (p. 268)
IMX531 (Sony)	Gray	4512 x 4512	1.1	66.↩ 0	+	12	39,7	70,4	Global	X / X	2.74	- - -	**BVS CA-↩ BN4-0204↩ ZG (20.↩ 4 Mpix [4512 x 4512])** (p. 238)
IMX532 (Sony)	Color	5328 x 3040	1.1	82.↩ 8	+	12	39,8	70,5	Global	X / X	2.74	- - -	**BVS CA-↩ BN4-0162↩ ZC (16.↩ 2 Mpix [5328 x 3040])** (p. 266)

IMX532 (Sony)	Gray	5328 x 3040	1.1	82.↩ 8	+	12	39,8	71,7	Global	X / X	2.74	- - -	**BVS CA-↩ BN4-0162↩ ZG (16.↩ 2 Mpix [5328 x 3040])** (p. 235)
IMX535 (Sony)	Color	4128 x 3008	1/1.↩ 1	104.↩ 5	+	12	39,6	69,5	Global	X / X	2.74	- - -	**BVS CA-↩ BN4-0124↩ BC (12.↩ 4 Mpix [4128 x 3008])** (p. 263)
IMX535 (Sony)	Gray	4128 x 3008	1/1.↩ 1	104.↩ 5	+	12	39,6	69,4	Global	X / X	2.74	- - -	**BVS CA-↩ BN4-0124↩ BG (12.↩ 4 Mpix [4128 x 3008])** (p. 233)
IMX536 (Sony)	Color	2848 x 2848	2/3	152.↩ 5	+	12	39,8	69,8	Global	X / X	2.74	- - -	**BVS CA-↩ BN4-0081↩ ZC (8.1 Mpix [2848 x 2848])** (p. 261)

IMX536 (Sony)	Gray	2848 x 2848	2/3	152.↩ 5	+	12	39,7	69,6	Global	X / X	2.74	- - -	**BVS CA-↩ BN4-0081↩ ZG (8.1 Mpix [2848 x 2848])** (p. 230)
IMX537 (Sony)	Color	2464 x 2064	1/1.↩ 8	234.↩ 0	+	12	39,8	69,6	Global	X / X	2.74	- - -	**BVS CA-↩ BN4-0051↩ BC (5.1 Mpix [2464 x 2064])** (p. 259)
IMX537 (Sony)	Gray	2464 x 2064	1/1.↩ 8	234.↩ 0	+	12	39,9	69,9	Global	X / X	2.74	- - -	**BVS CA-↩ BN4-0051↩ BG (5.1 Mpix [2464 x 2064])** (p. 228)
IMX540 (Sony)	Color	5328 x 4608	1.2	24.↩ 1	+	12	39,8	70,5	Global	X / X	2.74	4.0	**BVS CA-↩ BN2-0246↩ AC (24.↩ 6 Mpix [5328 x 4608])** (p. 255)

IMX540 (Sony)	Gray	5328 x 4608	1.2	24.↩ 1	+	12	39,6	70,2	Global	X / X	2.74	4.0	**BVS CA-↩ BN2-0246↩ AG (24.↩ 6 Mpix [5328 x 4608])** (p. 225)

1.11 Filters

MATRIX VISION offers several specific filters for cameras. The **hot mirror filter** (p. 63) is part of the standard delivery condition.

1.11.1 Hot Mirror Filter

The hot mirror filter has great transmission in the visible spectrum and blocks out a significant portion of the IR energy.

The average transmission is specified to be more than 93% from 425 to 675 nm and average reflectance is more than 95% from 750 to 950 nm at normal incidence. Custom angles of incidence can be designed to specification.

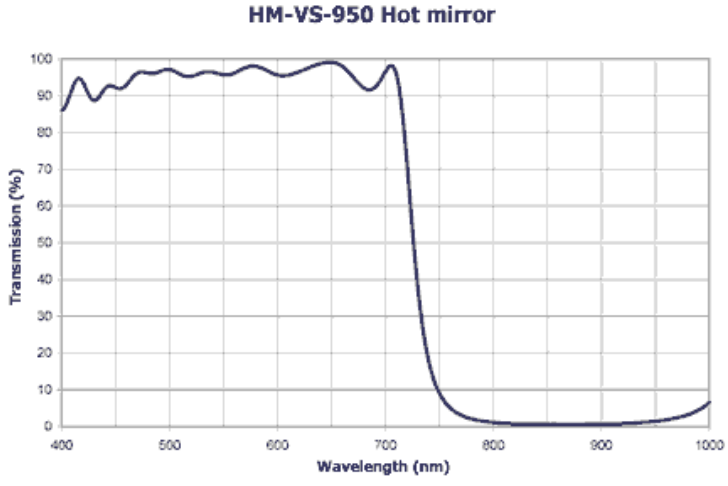


Figure 1: IR-CUT wavelengths and transmission diagram

1.11.2 Cold mirror filter

The high-quality daylight cut filter has optically polished surfaces. The polished surface allows the use of the filter directly in the path of rays in image processing applications. The filter is protected against scratches during the transport by a protection film that has to be removed before the installing the filter.

Half maximum transmission is at 750 nm. This means that the visible light is blocked, infrared light is let through with high transparency. So, the filter is suitable for applications with High-Power-LED-illuminations, as their wavelengths are about 850 or 880 nm.

Technical data
2000nm compared to bright acryl
Tolerances by +/-0.1 mm
Light transmission of 96% from 800 to 2000nm compared to bright acryl
Strengths of 0.5 mm
Steep Cut-on-curve

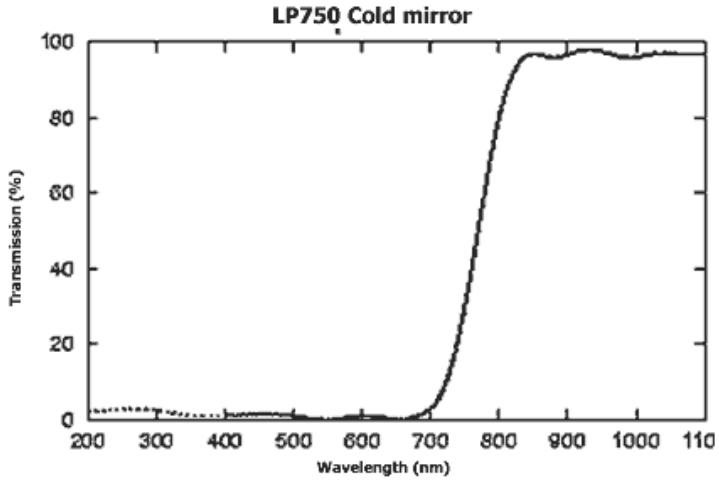


Figure 2: DL-CUT wavelengths and transmission diagram

1.12 GUI tools

1.12.1 Introduction

MATRIX VISION provides several convenient tools with graphical user interface to set up and work with their devices. Please find a short list and description below:

1.12.2 wxPropView

With wxPropView it is possible

- to acquire images
- to configure the device, and
- to display and modify the device properties.

1.12.3 mvDeviceConfigure

mvDeviceConfigure can be used to

- to set the device ID
- to update firmware- to disable CPU sleep states(some version of Windows only)

See also

For further information about the tools, please follow the link to the separate manual describing the GUI tools in great detail on our website: `https://www.matrix-vision.com/manuals/`

1.13 GenICam and advanced features

1.13.1 Introduction

In **wxPropView** (p. 65), you can see them in *"Setting -> Base -> Camera -> GenICam"*:

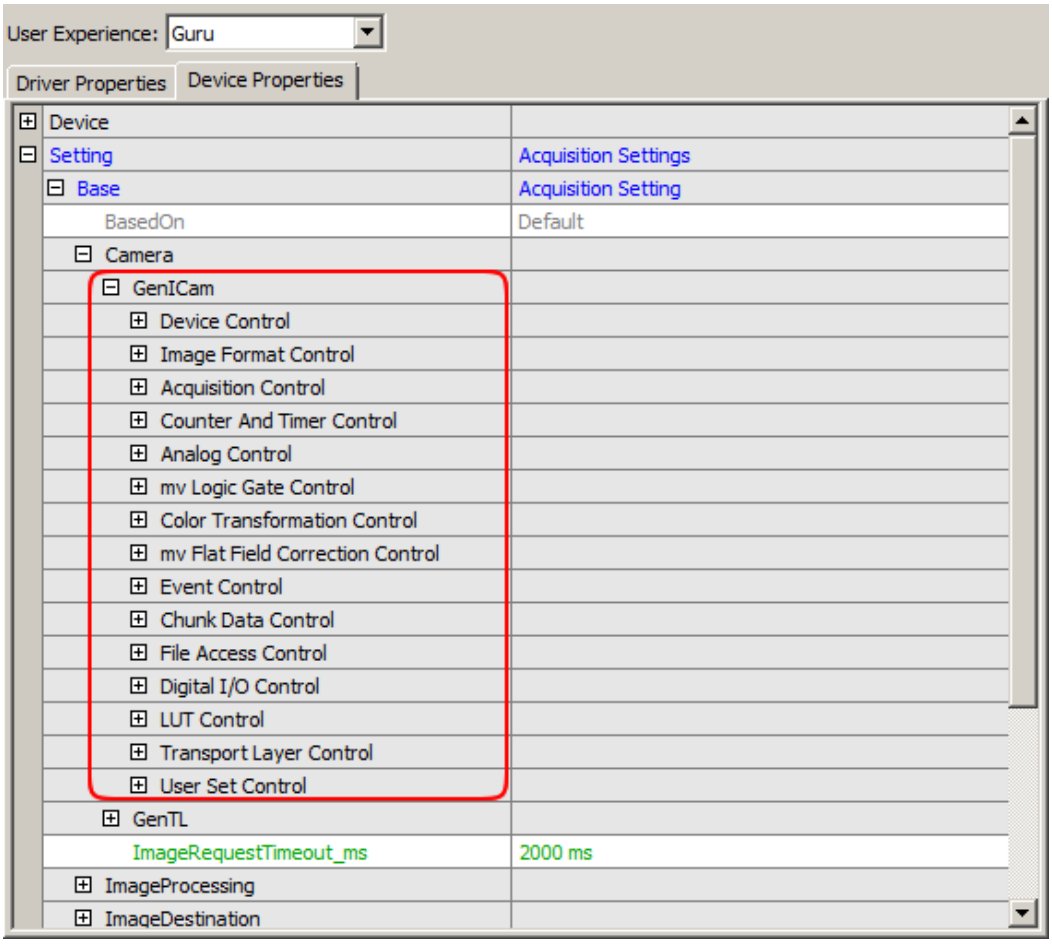


Figure 1: wxPropView - GenICam controls (depends on device and FW version)

As you can see, there are some controls with and without the prefix **"mv"**.

- **"mv"** prefix features are unique non-standard features developed by MATRIX VISION.
- Without **"mv"** are standard features as known from the **Standard Feature Naming Convention of GenICam properties** (p. 121) (SFNC).

All those features are **"camera based** / **device based"** features which can also be accessed using the camera with other **GenICam** (p. 113) compliant third-party software.

Note

Do not mix up the camera based / device based features with the features available in *"Setting -> Base -> Image Processing"*. Theses features are driver based features which are processed by the software and therefore need CPU load.

1.13.2 Device Control

The **"Device Control"** contains the features like

Feature name (acc. to **SFNC** (p. 121))	Property name (acc. to **mvIMPACT Acquire** (p. 119))	Description
DeviceType	*deviceType*	Returns the device type.
DeviceScanType	*deviceScanType*	Scan type of the sensor of the device.
DeviceVendorName	*deviceVendorName*	Name of the manufacturer of the device.
DeviceModelName	*deviceModelName*	Name of the device model.
DeviceManufacturerInfo	*deviceManufacturerInfo*	Manufacturer information about the device.
DeviceVersion	*deviceVersion*	Version of the device.
DeviceFirmwareVersion	*deviceFirmwareVersion*	Firmware version of the device.
DeviceSerialNumber	*deviceSerialNumber*	Serial number of the device.
DeviceUserID	*deviceUserID*	User-programmable device identifier.
DeviceTLVersionMajor	*deviceTLVersionMajor*	Major version of the transport layer of the device.
DeviceTLVersionMinor	*deviceTLVersionMinor*	Minor version of the transport layer of the device.
DeviceLinkSpeed	*deviceLinkSpeed*	Indicates the speed of transmission negotiated on the specified Link.
DeviceTemperature	*deviceTemperature*	Device temperature.
etc.		

related to the device and its sensor.

This control provides a bandwidth control feature . You have to select **DeviceLinkThroughputLimit**. Here you can set the maximum bandwidth in KBps.

MATRIX VISION offers also some information properties about the

- FPGA
  - **mvDeviceFPGAVersion**
- and the image sensor
  - **mvDeviceSensorColorMode**

1.13.3 Image Format Control

The **"Image Format Control"** contains features like

Feature name (acc. to **SFNC** (p. 121))	Property name (acc. to **mvIMPACT Acquire** (p. 119))	Description
SensorWidth	*sensorWidth*	Effective width of the sensor in pixels.
SensorHeight	*sensorHeight*	Effective height of the sensor in pixels.
SensorName	*sensorName*	Name of the sensor.
Width	*width*	Width of the image provided by the device (in pixels).
Height	*height*	Height of the image provided by the device (in pixels).
BinningHorizontal, BinningVertical	*binningHorizontal*, *binningVertical*	Number of horizontal/vertical photo-sensitive cells to combine together.
DecimationHorizontal, DecimationVertical	*decimationHorizontal*, *decimationVertical*	Sub-sampling of the image. This reduces the resolution (width) of the image by the specified decimation factor.
etc.		

related to the format of the transmitted image.

Additionally, MATRIX VISION offers numerous additional features like:

- **mvMultiAreaMode**
  which can be used to define multiple AOIs (Areas of Interests) in one image.

  See also

  The use case **Working with multiple AOIs (mv Multi Area Mode)** (p. 126) shows how this feature works.

1.13.4 Acquisition Control

The **"Acquisition Control"** contains features like

Feature name (acc. to **SFNC** (p. 121))	Property name (acc. to **mvIMPACT Acquire** (p. 119))	Description
AcquisitionMode	*acquisitionMode*	Sets the acquisition mode of the device. The different modes configures a device to send • exactly **one** image (**"SingleFrame"**), • exactly the **set number** of frames (**"MultiFrame"**) or • images constantly until an application explicitly stops the acquisition again (**"Continuous"**). and can be used for asynchronously grabbing and sending image(s). It works with internal and external hardware trigger where the edge is selectable. The external trigger uses ImageRequestTimeout (ms) to time out. • The chapter **"How To See The First Image"** in the **"mvIMPACT Acquire SDK GUI Applications"** manual shows how to acquire images with **wxPropView** (p. 65). • The use case **Acquiring a number of images** (p. 123) shows how to acquire a number of images; also triggered externally.
AcquisitionStart	*acquisitionStart*	Starts the acquisition of the device.
AcquisitionStop	*acquisitionStop*	Stops the acquisition of the device at the end of the current Frame.
AcquisitionAbort	*acquisitionAbort*	Aborts the acquisition immediately.

AcquisitionFrameRate	*acquisitionFrameRate*	Controls the acquisition rate (in Hertz) at which the frames are captured. Some cameras support a special internal trigger mode that allows more exact frame rates. This feature keeps the frame rate constant to an accuracy of +/-0.005 fps at 200 fps. This is achieved using frames with a length difference of up to 1 us. Please check in the **sensor summary** (p. 54) if this feature exists for the requested sensor.
TriggerSelector	*triggerSelector*	Selects the type of trigger to configure. A possible option is **mv↩ TimestampReset**. The use case about **mvTimestampReset is available** (p. 194).
TriggerOverlap[TriggerSelector]	*triggerOverlap*	Specifies the type trigger overlap permitted with the previous frame. *TriggerOverlap* is only intended for external trigger (which is usually non-overlapped: i.e. exposure and readout are sequentially). This leads to minimal latency / jitter between trigger and exposure. However, the camera accepts a new trigger (the exposure time earlier) before the end of the transmission of the current image. `Maximum frame rate in triggered mode = frame rate of continuous mode` This however leads to higher latency / jitter between trigger and exposure. A trigger will be **not** latched if it occurs before this moment (trigger is accurate in time). **See also** **Principles of overlapped and pipelined trigger** (p. 120).
ExposureMode	*exposureMode*	Sets the operation mode of the exposure (or shutter).
ExposureTime	*exposureTime*	Sets the exposure time (in microseconds) when *ExposureMode* is **Timed** and *ExposureAuto* is **Off**.
ExposureAuto	*exposureAuto*	Sets the automatic exposure mode when *ExposureMode* is **Timed**.
etc.		

related to the image acquisition, including the triggering mode.

Additionally, MATRIX VISION offers numerous additional features like:

- **mvShutterMode**
  which selects the shutter mode of the CMOS sensors like rolling shutter or global shutter.
- **mvDefectivePixelEnable**
  which activates the sensor's defective pixel correction.

  See also

  https://www.matrix-vision.com/files/mv11/Glossary/art_image_errors_sensors_en.pdf
- **mvExposureAutoAverageGrey**
  common desired average grey value (in percent) used for Auto Gain Control (AGC) and Auto Exposure Control (AEC).
- **mvExposureAutoAOIMode**
  common AutoControl AOI used for Auto Gain Control (AGC), Auto Exposure Control (AEC) and Auto White Balance (AWB).

Since

FW Revision 2.40.2546.0

Common properties for *AutoExposureControl*, *AutoGainControl* , and *AutoWhiteBalance* are available via **mv Auto Feature Control** (p. 84).

For **"Exposure Auto Mode"**, in **wxPropView** (p. 65) just select in *"Exposure Auto"* **"Continuous"**. Afterwards, you have the possibility to set lower and upper limit, average gray combined with AOI setting:

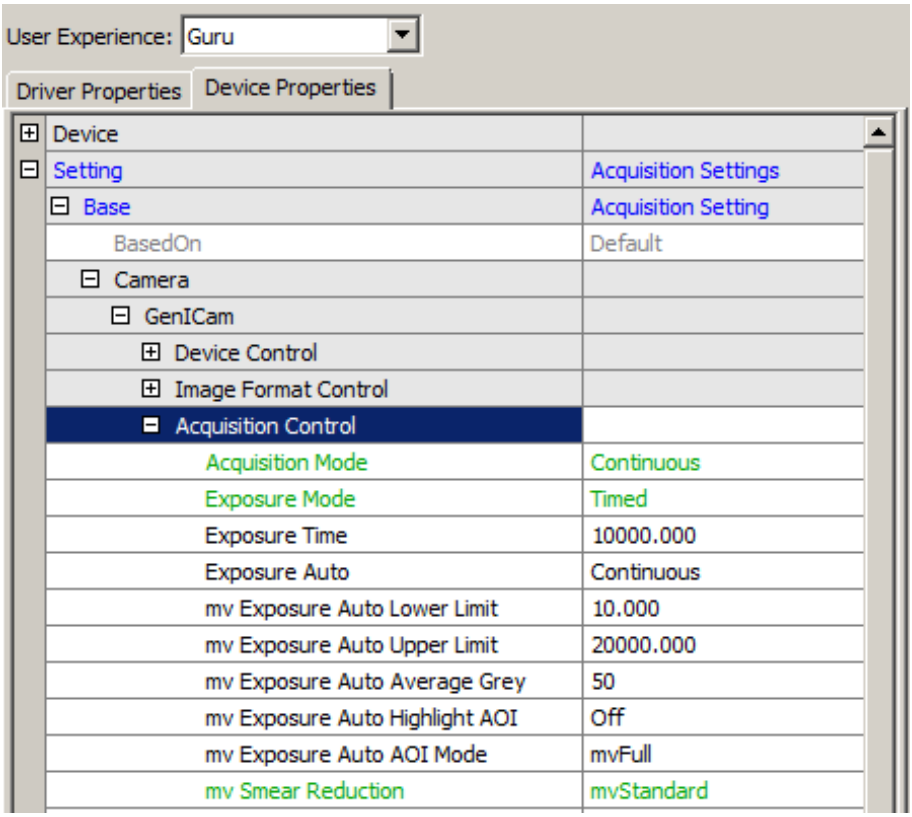


Figure 2: Acquire Control -> Exposure Auto

- **mvSmartFrameRecallFrameSkipRatio**
  When set to a value != 0, the smaller frames get thinned out. AOI requests can still be done for all frames.
- **mvSmartFrameRecallTimestampLookupAccuracy**
  is needed for the SkipRatio feature since you **don't know** the timestamps of the **internal** frames. This value defines the strictness of the timestamp-check for the recalled image (given in us).

1.13.5 Counter And Timer Control

The **"Counter And Timer Control"** is a powerful feature which MATRIX VISION customers already know under the name Hardware Real-Time Controller (HRTC). MATRIX VISION cameras provide:

- 4 counters for counting events or external signals (compare number of triggers vs. number of frames; over-trigger) and
- 2 timers.

Counter and Timers can be used, for example,

- for **pulse width modulation (PWM)** (p. 162) and
- to generate output signals of variable length, depending on conditions in camera.

This achieves complete HRTC functionality which supports following applications:

- frame rate by timer
- exposure time by timer
- pulse width at input

The **"Counter And Timer Control"** contains features like

Feature name (acc. to **SFNC** (p. 121))	Property name (acc. to **mvIMPACT Acquire** (p. 119))	Description
CounterSelector	*counterSelector*	Selects which counter to configure.
CounterEventSource[Counter↩ Selector]	*counterEventSource*	Selects the events that will be the source to increment the counter.
CounterEventActivation[Counter↩ Selector]	*counterEventActivation*	Selects the activation mode event source signal.
etc.		
TimerSelector	*timerSelector*	Selects which timer to configure.
TimerDuration[TimerSelector]	*timerDuration*	Sets the duration (in microseconds) of the timer pulse.
TimerDelay[TimerSelector]	*timerDelay*	Sets the duration (in microseconds) of the delay.
etc.		

related to the usage of programmable counters and timers.

Because there are many ways to use this feature, the list of use cases is long and not finished yet:

- **Processing triggers from an incremental encoder** (p. 160)
- **Creating different exposure times for consecutive images** (p. 166)
- **Creating synchronized acquisitions using timers** (p. 204)
- **Generating a pulse width modulation (PWM)** (p. 162)

- **Outputting a pulse at every other external trigger** (p. 164)
- **Generating very long exposure times** (p. 124)

1.13.6 Analog Control

The **"Analog Control"** contains features like

Feature name (acc. to **SFNC** (p. 121))	Property name (acc. to **mvIMPACT Acquire** (p. 119))	Description
GainSelector	*gainSelector*	Selects which gain is controlled by the various gain features.
Gain[GainSelector]	*gain*	Controls the selected gain as an absolute physical value [in dB].
GainAuto[GainSelector]	*gainAuto*	Sets the automatic gain control (AGC) mode.
GainAutoBalance	*gainAutoBalance*	Sets the mode for automatic gain balancing between the sensor color channels or taps.
BlackLevelSelector	*blackLevelSelector*	Selects which black level is controlled by the various BlackLevel features.
BlackLevel[BlackLevelSelector]	*blackLevel*	Controls the selected BlackLevel as an absolute physical value.
BalanceWhiteAuto	*balanceWhiteAuto*	Controls the mode for automatic white balancing between the color channels.
Gamma	*gamma*	Controls the gamma correction of pixel intensity.
etc.		

related to the video signal conditioning in the analog domain.

Additionally, MATRIX VISION offers:

- **mvBalanceWhiteAuto** functions and
- **mvGainAuto** functions.

In **wxPropView** (p. 65) just select in *"Gain Auto"* (AGC) **"Continuous"**. Afterwards, you have the possibility to set minimum and maximum limit combined with AOI setting:

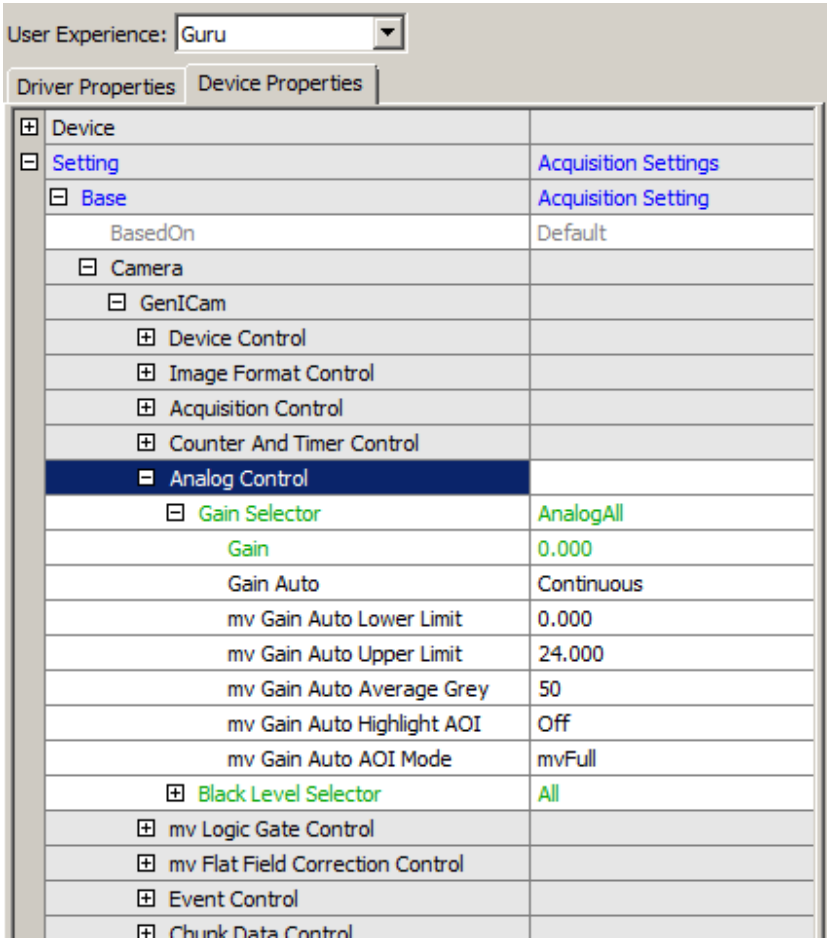


Figure 3: Analog Control -> Gain Auto

See also

Optimizing the color/luminance fidelity of the camera (p. 138)

1.13.7 Color Transformation Control

The **"Color Transformation Control"** contains features like

Feature name (acc. to **SFNC** (p. 121))	Property name (acc. to **mvIMPACT Acquire** (p. 119))	Description
ColorTransformationSelector	*colorTransformationEnable*	Activates the selected color transformation module.
ColorTransformationSelector	*colorTransformationSelector*	Selects which color transformation module is controlled by the various color transformation features.
ColorTransformationValue	*colorTransformationValue*	Represents the value of the selected Gain factor or Offset inside the transformation matrix.
ColorTransformationValueSelector	*colorTransformationValueSelector*	Selects the gain factor or Offset of the Transformation matrix to access in the selected color transformation module.

related to the control of the color transformation.

This control offers an enhanced color processing for optimum color fidelity using a **color correction matrix** (CCM) and enables

- 9 coefficients values ($Gain_{00}$ .. $Gain_{22}$) and
- 3 offset values ($Offset_0$ .. $Offset_2$)

to be entered for RGB_{IN} -> RGB_{OUT} transformation. This can be used to optimize specific colors or specific color temperatures.

$$\begin{pmatrix} R\\ G\\ B \end{pmatrix}_{out} = \begin{pmatrix} Gain00 & Gain01 & Gain02\\ Gain10 & Gain11 & Gain12\\ Gain20 & Gain21 & Gain22 \end{pmatrix} \begin{pmatrix} C0\\ C1\\ C2 \end{pmatrix}_{in} + \begin{pmatrix} Offset0\\ Offset1\\ Offset2 \end{pmatrix}$$

$$\text{Equivalent: } \begin{pmatrix} R\\ G\\ B \end{pmatrix}_{out} = \begin{pmatrix} RR & RG & RB\\ GR & GG & GB\\ BR & BG & BB \end{pmatrix} \begin{pmatrix} R\\ G\\ B \end{pmatrix}_{in} + \begin{pmatrix} R\\ G\\ B \end{pmatrix}_{offset}$$

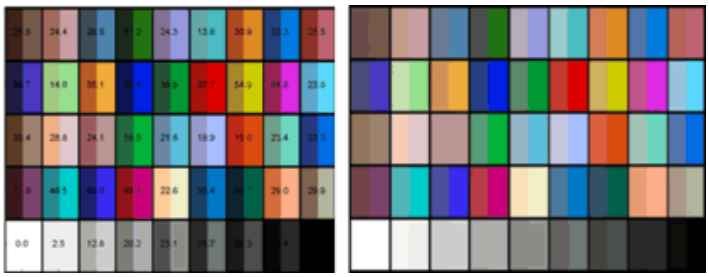


Figure 4: Color correction sample

Coefficients will be made available for sensor models and special requirements on demand.

See also

Optimizing the color/luminance fidelity of the camera (p. 138)

1.13.8 Event Control

The **"Event Control"** contains features like

Feature name (acc. to **SFNC** (p. 121))	Property name (acc. to **mvIMPACT Acquire** (p. 119))	Description
EventSelector	*eventSelector*	Selects which Event to signal to the host application.
EventNotification[EventSelector]	*eventNotification*	Activate or deactivate the notification to the host application of the occurrence of the selected Event.
EventFrameTriggerData	*eventFrameTriggerData*	Category that contains all the data features related to the *FrameTrigger* event.
EventFrameTrigger	*eventFrameTrigger*	Returns the unique identifier of the *FrameTrigger* type of event.
EventFrameTriggerTimestamp	*eventFrameTriggerTimestamp*	Returns the timestamp of the *AcquisitionTrigger* event.
EventFrameTriggerFrameID	*eventFrameTriggerFrameID*	Returns the unique identifier of the frame (or image) that generated the *FrameTrigger* event.
EventExposureEndData	*eventExposureEndData*	Category that contains all the data features related to the *ExposureEnd* event.
EventExposureEnd	*eventExposureEnd*	Returns the unique identifier of the *ExposureEnd* type of event.
EventExposureEndTimestamp	*eventExposureEndTimestamp*	Returns the timestamp of the *ExposureEnd* Event.
EventExposureEndFrameID	*eventExposureEndFrameID*	Returns the unique identifier of the frame (or image) that generated the *ExposureEnd* event.
EventErrorData	*eventErrorData*	Category that contains all the data features related to the error event.
EventError	*eventError*	Returns the unique identifier of the error type of event.
EventErrorTimestamp	*eventErrorTimestamp*	Returns the timestamp of the error event.
EventErrorFrameID	*eventErrorFrameID*	If applicable, returns the unique identifier of the frame (or image) that generated the error event.
EventErrorCode	*eventErrorCode*	Returns an error code for the error(s) that happened.
etc.		

related to the generation of Event notifications by the device.

The use case **Working with Event Control** (p. 130) shows how this control can be used.

1.13.9 Chunk Data Control

The **"Chunk Data Control"** contains features like

Feature name (acc. to **SFNC** (p. 121))	Property name (acc. to **mvIMPACT Acquire** (p. 119))	Description
ChunkModeActive	*chunkModeActive*	Activates the inclusion of chunk data in the payload of the image.
ChunkSelector	*chunkSelector*	Selects which chunk to enable or control.
ChunkEnable[ChunkSelector]	*chunkEnable*	Enables the inclusion of the selected chunk data in the payload of the image.
ChunkImage	*chunkImage*	Returns the entire image data included in the payload.
ChunkOffsetX	*ChunkOffsetX*	Returns the offset x of the image included in the payload.
ChunkOffsetY	*chunkOffsetY*	Returns the offset y of the image included in the payload.
ChunkWidth	*chunkWidth*	Returns the width of the image included in the payload.
ChunkHeight	*chunkHeight*	Returns the height of the image included in the payload.
ChunkPixelFormat	*chunkPixelFormat*	Returns the pixel format of the image included in the payload.
ChunkTimestamp	*chunkTimestamp*	Returns the timestamp of the image included in the payload at the time of the *FrameStart* internal event.
etc.		

related to the Chunk Data Control.

A description can be found in the image acquisition section of the **"mvIMPACT Acquire API"** manuals.

1.13.10 File Access Control

The **"File Access Control"** contains features like

Feature name (acc. to **SFNC** (p. 121))	Property name (acc. to **mvIMPACT Acquire** (p. 119))	Description
FileSelector	*fileSelector*	Selects the target file in the device.
FileOperationSelector[FileSelector]	*fileOperationSelector*	Selects the target operation for the selected file in the device.
FileOperationExecute[FileSelector][FileOperationSelector]	*fileOperationExecute*	Executes the operation selected by *FileOperationSelector* on the selected file.
FileOpenMode[FileSelector]	*fileOpenMode*	Selects the access mode in which a file is opened in the device.
FileAccessBuffer	*fileAccessBuffer*	Defines the intermediate access buffer that allows the exchange.
etc.		

related to the File Access Control that provides all the services necessary for generic file access of a device.

The use case **Working with the UserFile section (Flash memory)** (p. 188) shows how this control can be used.

1.13.11 Digital I/O Control

The **"Digital I/O Control"** contains features like

Feature name (acc. to **SFNC** (p. 121))	Property name (acc. to **mvIMPACT Acquire** (p. 119))	Description
LineSelector	*lineSelector*	Selects the physical line (or pin) of the external device connector to configure.
LineMode[LineSelector]	*lineMode*	Controls if the physical Line is used to Input or Output a signal.
UserOutputSelector	*userOutputSelector*	Selects which bit of the user output register will be set by *UserOutputValue*.
UserOutputValue[UserOutputSelector]	*userOutputValue*	Sets the value of the bit selected by *UserOutputSelector*.
etc.		

related to the control of the general input and output pins of the device.

Additionally, MATRIX VISION offers:

- **mvLineDebounceTimeRisingEdge** and
- **mvLineDebounceTimeFallingEdge** functionality.

A description of these functions can be found in the use case **Creating a debouncing filter at the inputs** (p. 181).

How you can test the digital inputs and outputs is described in **"Testing The Digital Inputs"** in the **"mvIMPACT Acquire SDK GUI Applications"** manual. The use case **Creating synchronized acquisitions using timers** (p. 204) is a further example which shows you how to work with digital inputs and outputs.

1.13.12 Transport Layer Control

The **"Transport Layer Control"** contains features like

Feature name (acc. to **SFNC** (p. 121))	Property name (acc. to **mvIMPACT Acquire** (p. 119))	Description
PayloadSize	*payloadSize*	Provides the number of bytes transferred for each image or chunk on the stream channel.
etc.		

related to the Transport Layer Control.

1.13.13 User Set Control

The **"User Set Control"** contains features like

Feature name (acc. to **SFNC** (p. 121))	Property name (acc. to **mvIMPACT Acquire** (p. 119))	Description
UserSetSelector	*userSetSelector*	Selects the feature user set to load, save or configure.
UserSetLoad[UserSetSelector]	*userSetLoad*	Loads the user set specified by *UserSetSelector* to the device and makes it active.
UserSetSave[UserSetSelector]	*userSetSave*	Endianess of the device registers.
UserSetDefault	*userSetDefault*	Selects the feature user set to load and make active when the device is reset.

related to the User Set Control to save and load the user device settings.

The camera allows the storage of up to four configuration sets in the camera. This feature is similar to storing settings in the registry but this way in the camera. It is possible to store

- exposure,
- gain,
- AOI,
- frame rate,
- etc.

permanently. You can select, which user set comes up after hard reset.

Use case related to the **"User Set Control"** is:

- **Creating user set entries** (p. 186)

Another way to create user data is described here: **Creating user data entries** (p. 184)

1.13.14 mv Logic Gate Control

The **"mv Logic Gate Control"** contains features like

Feature name (acc. to **SFNC** (p. 121))	Property name (acc. to **mvIMPACT Acquire** (p. 119))	Description
	mvLogicGateANDSelector	Selects the AND gate to configure.
	mvLogicGateANDSource1	Selects the first input signal of the AND gate selected by *mvLogicGateANDSelector*.
	mvLogicGateANDSource2	Selects the second input signal of the AND gate selected by *mvLogicGateANDSelector*.
	mvLogicGateORSelector	Selects the OR gate to configure.
	mvLogicGateORSource1	Selects the first input signal of the OR gate selected by *mvLogicGateORSelector*.
	mvLogicGateORSource2	Selects the second input signal of the OR gate selected by *mvLogicGateORSelector*.
	mvLogicGateORSource3	Selects the third input signal of the OR gate selected by *mvLogicGateORSelector*.
	mvLogicGateORSource4	Selects the fourth input signal of the OR gate selected by *mvLogicGateORSelector*.

related to control the devices Logic Gate Control parameters. It performs a logical operation on one or more logic inputs and produces a single logic output.

The use case **Creating different exposure times for consecutive images** (p. 166) shows how you can create different exposure times with timers, counters and the logic gate functionality.

See also

Triggering of an indefinite sequence with precise starting time (p. 173)

1.13.15 mv Serial Interface Control

The **"mv Serial Interface Control"** contains features like

Feature name (acc. to **SFNC** (p. 121))	Property name (acc. to **mvIMPACT Acquire** (p. 119))	Description
	mvSerialInterfaceMode	States the interface mode of the serial interface.
	mvSerialInterfaceEnable	Controls whether the serial interface is enabled or not.
	mvSerialInterfaceBaudRate	Serial interface clock frequency.
	mvSerialInterfaceASCIIBuffer	Buffer for exchanging ASCII data over serial interface.
	mvSerialInterfaceWrite	Command to write data from the serial interface.
	mvSerialInterfaceRead	Command to read data from the serial interface.
etc.		

related to control the devices Serial Interface Control parameters. It enables the camera to be controlled via serial interface.

The use case **Working With The Serial Interface (mv Serial Interface Control)** (p. 200) shows how you can work with the serial interface control.

See also

Working With The Serial Interface (mv Serial Interface Control) (p. 200)

1.13.16 mv Auto Feature Control

The **"mv Auto Feature Control"** contains features like

Feature name (acc. to **SFNC** (p. 121))	Property name (acc. to **mvIMPACT Acquire** (p. 119))	Description
	mvAutoFeatureAOIMode	Common AutoControl AOI used for Auto Gain Control(AGC), Auto Exposure Control(AEC) and Auto White Balancing.
	mvAutoFeatureSensitivity	The controllers sensitivity of brightness deviations. This parameter influences the gain as well as the exposure controller.
	mvAutoFeatureCharacteristic	Selects the prioritization between Auto Exposure Control(AEC) and Auto Gain Control(AGC) controller.
	mvAutoFeatureBrightness↩ Tolerance	The error input hysteresis width of the controller. If the brightness error exceeds the half of the value in positive or negative direction, the controller restarts to control the brightness.

related to the automatic control of exposure, gain, and white balance.

With this control you can influence the characteristic of the controller depending on certain light situations. AEC/AGC can be controlled with the new additional properties **mvAutoFeatureSensitivity**, **mvAutoFeatureCharacteristic** and **mvAutoFeatureBrightnessTolerance**.

1.14 Developing applications using the mvIMPACT Acquire SDK

The mvIMPACT Acquire SDK is a comprehensive software library that can be used to develop applications using the devices described in this manual. A wide variety of programming languages is supported.

For C, C++, .NET, Python or Java developers separate API descriptions can be found on the MATRIX VISION website:

- mvIMPACT Acquire C API
- mvIMPACT Acquire C++ API
- mvIMPACT Acquire Java API
- mvIMPACT Acquire .NET API
- mvIMPACT Acquire Python API

Compiled versions (CHM format) might already be installed on your system. These manuals contain chapters on

- how to link and build applications using mvIMPACT Acquire
- how the log output for **"mvIMPACT Acquire"** devices is configured and how it works in general
- how to create your own installation packages for Windows and Linux
- a detailed API documentation
- etc.

1.15 DirectShow interface

Note

DirectShow can only be used in combination with the Microsoft Windows operating system.

Since Windows Vista, **Movie Maker** does not support capturing from a device registered for DirectShow anymore.

This is the documentation of the MATRIX VISION DirectShow_acquire interface. A MATRIX VISION specific property interface based on the IKsPropertySet has been added. All other features are related to standard DirectShow programming.

- **Supported interfaces** (p. 87)
- **Logging** (p. 87)
- **Setting up devices for DirectShow usage** (p. 87)

1.15.1 Supported interfaces

1.15.1.1 IAMCameraControl

1.15.1.2 IAMDroppedFrames

1.15.1.3 IAMStreamConfig

1.15.1.4 IAMVideoProcAmp

1.15.1.5 IKsPropertySet

The DirectShow_acquire supports the IKsPropertySet Interface. For further information please refer to the Microsoft DirectX 9.0 Programmer's Reference.

Supported property set GUID's:

- AMPROPERTY_PIN_CATEGORY
- DIRECT_SHOW_ACQUIRE_PROPERTYSET

1.15.1.6 ISpecifyPropertyPages

1.15.2 Logging

The DirectShow_acquire logging procedure is equal to the logging of the MATRIX VISION products which uses mvIMPACT Acquire. The log output itself is based on **XML**.

If you want more information about the logging please have a look at the **Logging** chapter of the respective **"mv↩ IMPACT Acquire API"** manual or read on how to configure the log-output using **mvDeviceConfigure** in the **"mv↩ IMPACT Acquire GUI Applications"** manual.

1.15.3 Setting up devices for DirectShow usage

In order to be able to access a device through the mvIMPACT Acquire driver stack from an application through DirectShow a registration procedure is needed. This can either be done using mvDeviceConfigure or by a command line tool that is part of the Windows operating system.

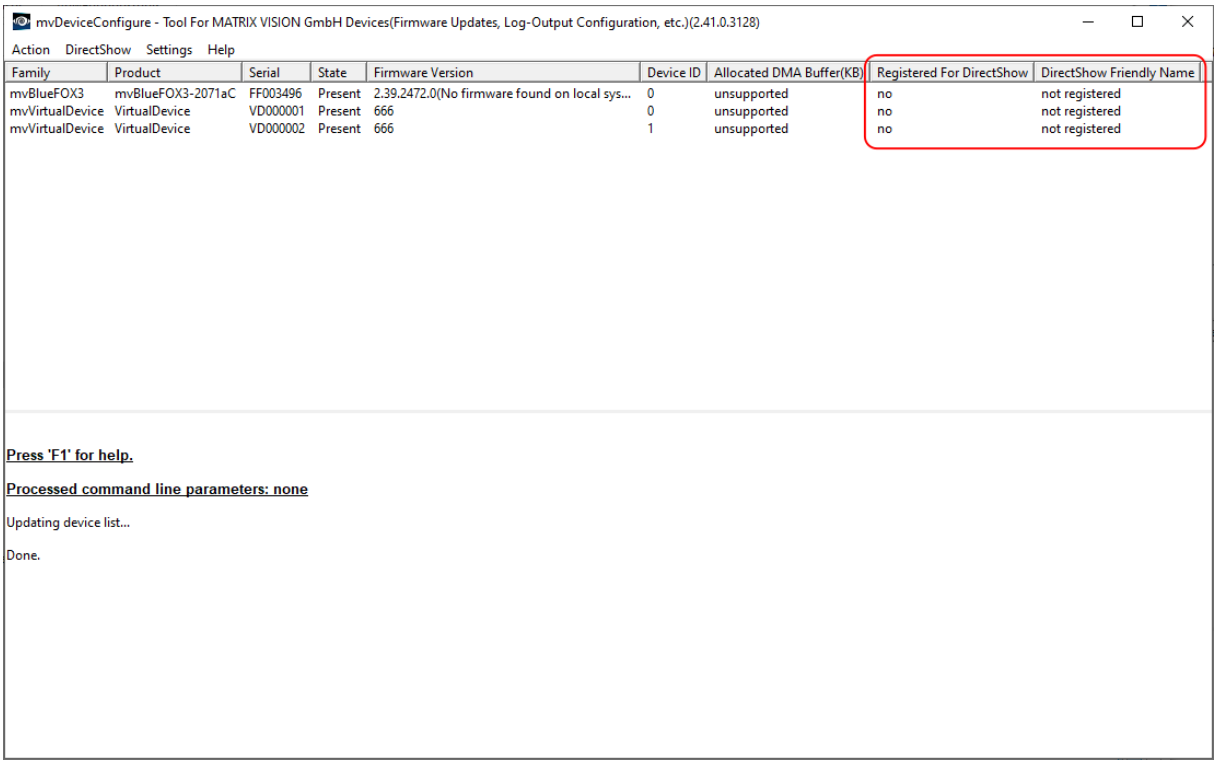
Note

Please be sure to register the MV device for DirectShow with a matching version of **mvDeviceConfigure**. I.e. if you have installed the 32-bit version of the VLC Media Player, Virtual Dub, etc., you have to register devices with the 32-bit version of mvDeviceConfigure (*"C:\Program Files\MATRIX VISION\mvIMPACT Acquire\bin"*, the 64-bit version resides in *"C:\Program Files\MATRIX VISION\mvIMPACT Acquire\bin\x64"*)!

1.15.3.1 Registering devices

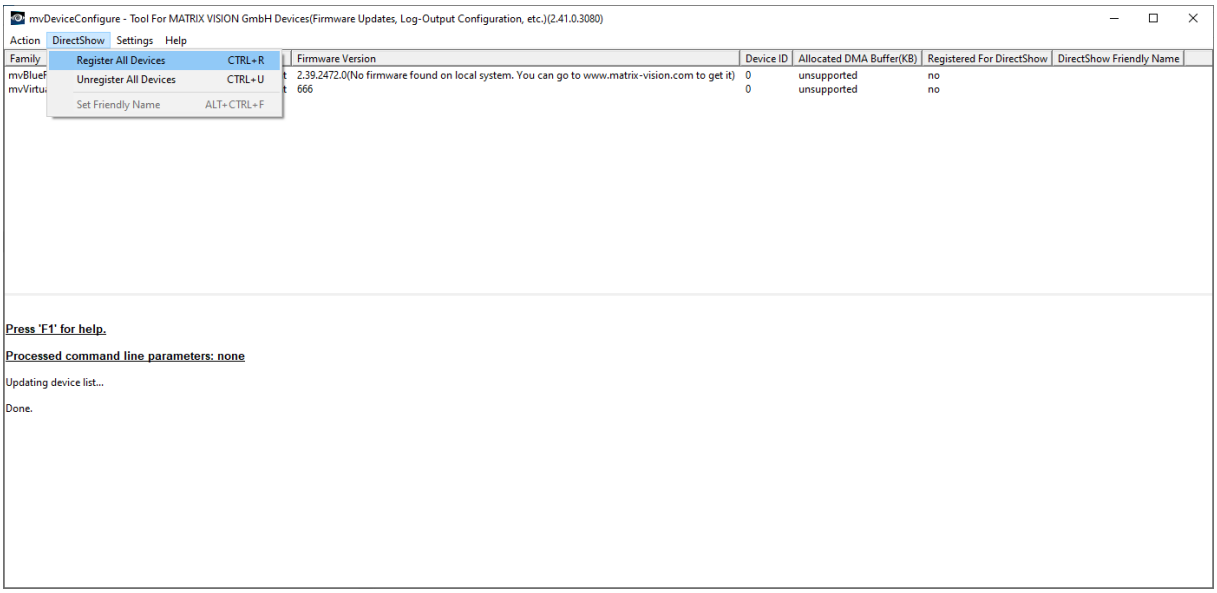
To register all devices currently recognized by the mvIMPACT Acquire driver stack for access with DirectShow the following registration procedure is needed:

1. >mvDeviceConfigure needs to be started(with elevated rights).
   If no device has been registered the application will more or less (depending on the installed devices) look like this.



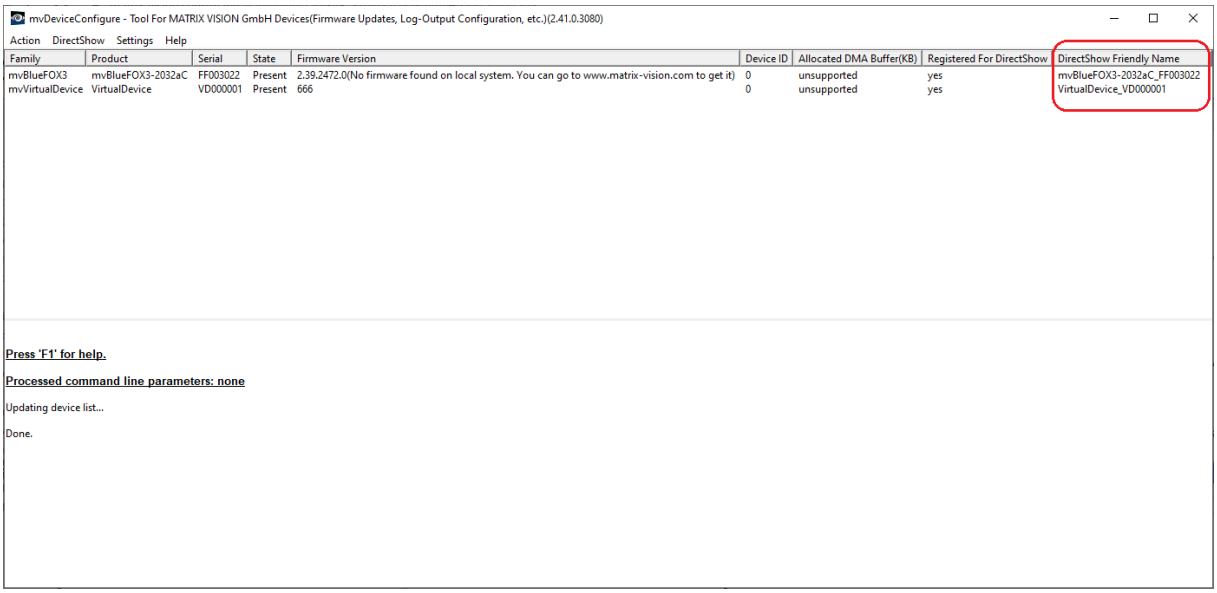
mvDeviceConfigure - After Start

2. To register every installed device for DirectShow access click on the menu item **"DirectShow"** → **"Register All Devices"**.



mvDeviceConfigure - Register All Devices

3. After a successful registration the column "Registered For DirectShow" will display **"yes"** for every device and the devices will be registered with a default DirectShow friendly name which is displayed in the **"DirectShow Friendly Name"** column.

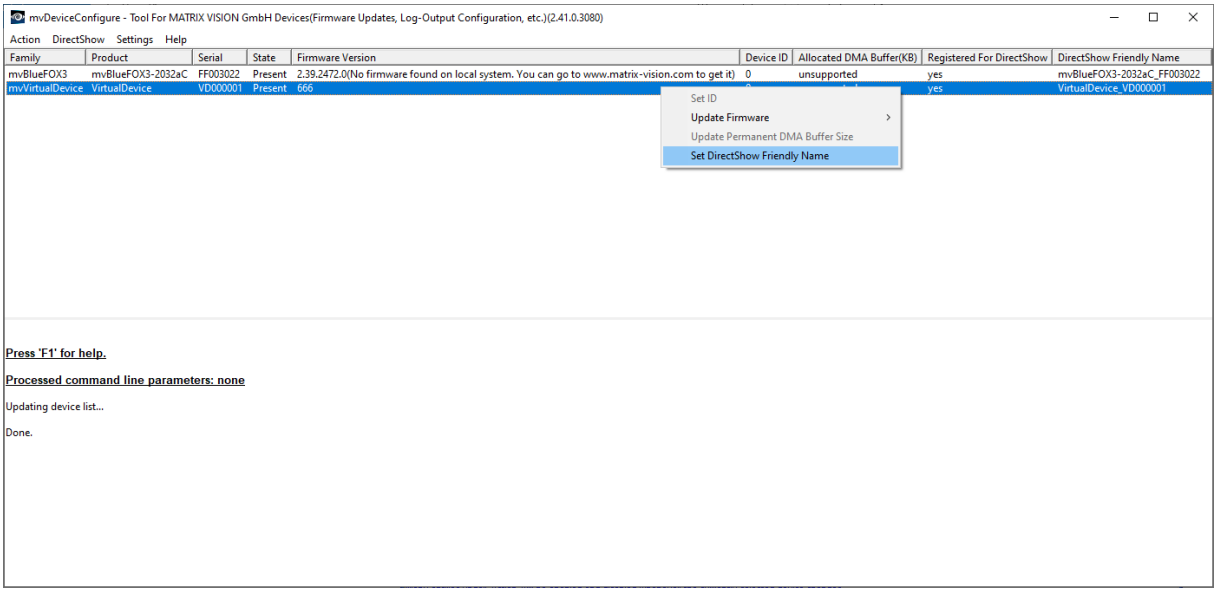


mvDeviceConfigure - All Devices Registered For DirectShow Access

1.15.3.2 Renaming devices

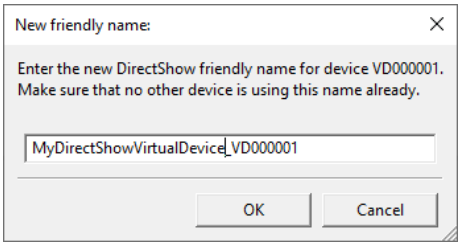
To modify the DirectShow friendly name of a device:

1. mvDeviceConfigure needs to be started(with elevated rights).
2. right-click on the device to rename and select **"Set DirectShow Friendly Name"**:



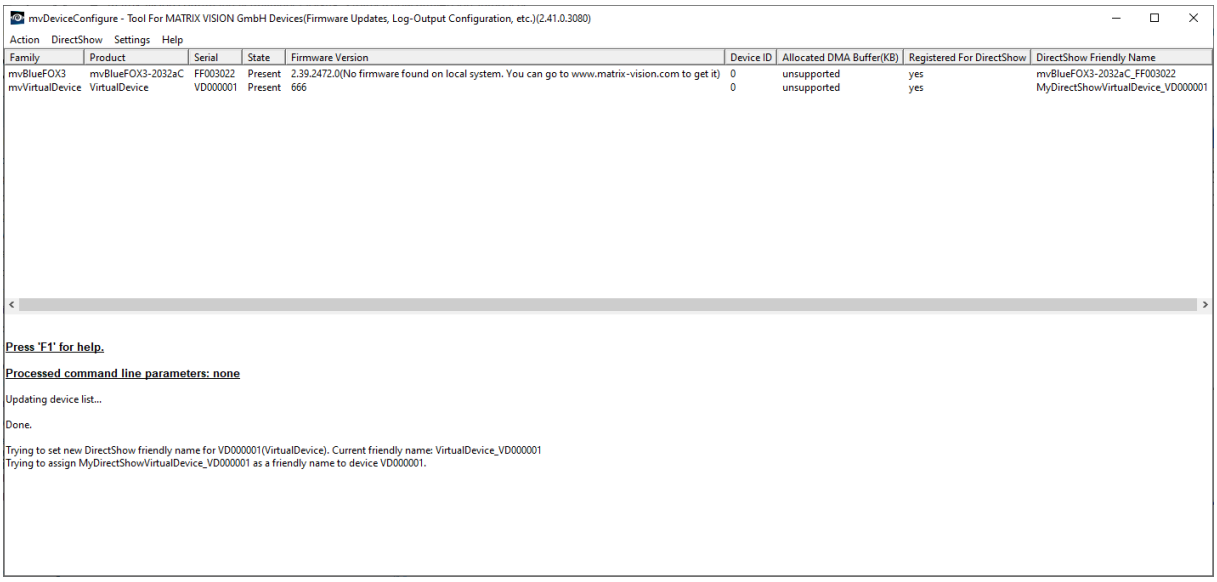
mvDeviceConfigure - Set DirectShow Friendly Name

3. Then, a dialog will appear. Please enter the new name and confirm it with **"OK"**.



mvDeviceConfigure - Dialog For New Name

4. Afterwards the column "DirectShow friendly name" will display the newly assigned friendly name.



mvDeviceConfigure - Renamed Device

Note

Please do not select the same friendly name for two different devices. In theory this is possible, however the mvDeviceConfigure GUI will not allow this to avoid confusion.

1.15.3.3 Using regsvr32

To register all devices currently recognized by the mvIMPACT Acquire driver stack with auto-assigned names, the Windows tool "`regsvr32`" can be used from an elevated command shell.

The following command line options are available and can be passed during the silent registration:

EXAMPLES:

Register ALL devices that are recognized by mvIMPACT Acquire (this will only register devices which have drivers installed) without any user interaction:

```
regsvr32 <path>\DirectShow_acquire.ax /s
```

Unregister ALL devices that have been registered before without any user interaction:

```
regsvr32 <path>\DirectShow_acquire.ax /u /s
```

1.16 Troubleshooting

- **Error code list** (p. 92)
- **Accessing log files** (p. 105)
- **General Issues** (p. 106)
- **Windows** (p. 110) Windows
- **Linux** (p. 111)

1.16.1 Error code list

Numerical Value	String Representation	Brief Description	Detailed Description
-2000	PROPHANDLING_NOT_↩ A_LIST	This component is not a list.	A list operation for this component has been called but this component does not reference a list. [-2000]
-2001	PROPHANDLING_NOT_↩ A_PROPERTY	This component is not a property.	A property operation for this component has been called but this component does not reference a property. [-2001]
-2002	PROPHANDLING_NOT_↩ A_METHOD	This component is not a method.	A method operation for this component has been called but this component does not reference a method. [-2002]
-2003	PROPHANDLING_NO_↩ READ_RIGHTS	The caller has no read rights for this component.	It has been tried to read data from this component, but the caller has no read rights for this component. [-2003]
-2004	PROPHANDLING_NO_↩ WRITE_RIGHTS	The caller has no write rights for this component.	It has been tried to modify data of this component, but the caller has no write rights for this component. [-2004]

-2005	PROPHANDLING_NO_↩ MODIFY_SIZE_RIGHTS	The caller can't modify the size of this component.	It has been tried to modify the size of this list or the number of values stored by a property, but the caller doesn't have the required right to do this. This error will also be reported if the user tried to increase the number of values handled by a property above the maximum number of values it can handle. Therefore before resizing a property check if the new size might exceeds this maximum value by calling the appropriate function. [-2005]
-2006	PROPHANDLING↩ _INCOMPATIBLE_↩ COMPONENTS	The two involved components are not compatible.	An operation requiring two compatible components has been called with two components, which are not compatible. [-2006]
-2008	PROPHANDLING↩ _UNSUPPORTED_↩ PARAMETER	One or more of the specified parameters are not supported by the function.	This error might also be generated if a certain feature is not available on the current platform. [-2008]
-2009	PROPHANDLING_SIZE_↩ MISMATCH	Different sized value buffers have been passed.	While trying to read value pairs the caller passed two different sized value buffers to a function while one is too small to hold all the information. [-2009]
-2010	PROPHANDLING_↩ IMPLEMENTATION_↩ MISSING	A feature that is not implemented so far has been requested.	The caller requested a feature, that hasn't been implemented so far. This error code is only provided for compatibility and will be set in very rare cases only. [-2010]
-2011	PROPHANDLING↩ _ACCESSTOKEN_↩ CREATION_FAILED	An access token object couldn't be created.	This can either happen, because the caller has not the rights required to create an access token or because the system runs very low on memory. DeprecatedThis error code currently is not used anywhere within this framework. It might be removed in a future version. [-2011]

-2012	PROPHANDLING_↩ INVALID_PROP_VALUE	It has been tried to assign an invalid value to a property.	This can either happen if the value lies above or below the min. or max. value for a property or when it has been tried to write a value to a property, which is not in the properties translation dictionary (if it defines one). To find out, which values are allowed for the property in question the user should Check if the property defines a translation dictionary. Check the allowed values within a translation dictionary if one is defined. Check the min and max value for properties, that define limits. [-2012]
-2013	PROPHANDLING_PROP↩ _TRANSLATION_TABLE↩ _CORRUPTED	The properties translation table has been corrupted.	The properties translation table has been corrupted for an unknown reason and can't be used anymore. [-2013]
-2014	PROPHANDLING_PROP↩ _VAL_ID_OUT_OF_↩ BOUNDS	Invalid value index.	The caller tried to read a value from an invalid index from a property. Most properties store one value only, thus the only valid positive value index will be 0 (some negative index values are reserved for special values like e.g. the min/max value of a property). However some properties might store more than one value, thus the max. allowed index might be higher. The highest index allowed will always be the value count of a property minus one for properties with the cfFixedSize flag set. Other properties will automatically adjust the size once the user writes to an index out of bounds. [-2014]
-2015	PROPHANDLING_PROP↩ _TRANSLATION_TABLE↩ _NOT_DEFINED	This property doesn't define a translation table.	The caller tried to modify a translation table, that hasn't been defined for this property. [-2015]

-2016	PROPHANDLING_↩ INVALID_PROP_VALUE_↩ TYPE	An invalid value has been passed to the property.	Although properties are quite tolerant regarding the allowed assignment for them some value types can't be used to write all properties. As an example assigning a float value to an integer property would result in this error. Another reason for this error might be when a user tried to access e.g. a float property with functions meant to be used for int properties. [-2016]
-2017	PROPHANDLING_PROP↩ _VAL_TOO_LARGE	A too large value has been passed.	One or more of the values the caller tried to write to the property are larger than the max. allowed value for this property. [-2017]
-2018	PROPHANDLING_PROP↩ _VAL_TOO_SMALL	A too small value has been passed.	One or more of the values the caller tried to write to the property are smaller than the min. allowed value for this property. [-2018]
-2019	PROPHANDLING_↩ COMPONENT_NOT_↩ FOUND	The specified component could not be found.	[-2019]
-2020	PROPHANDLING_LIST_↩ ID_INVALID	An invalid list has been referenced.	[-2020]
-2021	PROPHANDLING_↩ COMPONENT_ID_INVALID	An invalid component within a list has been referenced.	[-2021]
-2022	PROPHANDLING_LIST_↩ ENTRY_OCCUPIED	The specified list index is occupied.	During the creation of a new component the caller tried the insert the newly created component into a list at a position already used to store another component. [-2022]
-2023	PROPHANDLING_↩ COMPONENT_HAS_↩ OWNER_ALREADY	The specified component already has an owner.	The caller tried to assign an owner to a component that already has an owner. An owner once defined can't be modified anymore. [-2023]
-2024	PROPHANDLING↩ _COMPONENT_↩ ALREADY_REGISTERED	It has been tried to register the same component at twice in the same list.	[-2024]

-2025	PROPHANDLING_LIST_↩ CANT_ACCESS_DATA	The desired data can't be accessed or found.	During loading or saving data this error can occur e.g. if it has been tried to import a setting from a location where the desired setting couldn't be found. Another reason for this error might be that the current user is not allowed to perform a certain operation on the desired data (e.g. a user tries to delete a setting that is stored with global scope but does not have elevated access rights). [-2025]
-2026	PROPHANDLING_↩ METHOD_PTR_INVALID	The function pointer of the referenced method object is invalid.	[-2026]
-2027	PROPHANDLING_↩ METHOD_INVALID_↩ PARAM_LIST	A method object has an invalid parameter list.	[-2027]
-2028	PROPHANDLING_SWIG↩ _ERROR	This indicates an internal error occurred within the SWIG generated wrapper code, when working under Python.	[-2028]
-2029	PROPHANDLING↩ _INVALID_INPUT_↩ PARAMETER	A invalid input parameter has been passed to a function of this module.	In most cases this might be a unassigned pointer, where a valid pointer to a user defined storage location was expected. [-2029]
-2030	PROPHANDLING_↩ COMPONENT_NO_↩ CALLBACK_REGISTERED	The user tried to modify a registered callback, but no callback has been registered for this component.	[-2030]
-2031	PROPHANDLING_INPUT↩ _BUFFER_TOO_SMALL	The user tried to read data into a user supplied storage location, but the buffer was too small to accommodate the result.	[-2031]
-2032	PROPHANDLING_↩ WRONG_PARAM_COUNT	The number of parameters is incorrect.	This error might occur if the user called a function with a variable number of input or output parameters and the number of parameters passed to the function does not match the number of required parameters. [-2032]
-2033	PROPHANDLING↩ _UNSUPPORTED_↩ OPERATION	The user tried to execute an operation, which is not supported by the component he is referring to.	[-2033]

-2034	PROPHANDLING_CANT↩ _SERIALIZE_DATA	The user tried to save(serialize) a property list without having the right to do this.	[-2034]
-2035	PROPHANDLING_↩ INVALID_FILE_CONTENT	The user tried to use a file to update or create a component list, that does not contain valid data for this operation.	This e.g. might happen, if the file does not contain valid XML data or XML data that is not well formed. [-2035]
-2036	PROPHANDLING_CANT↩ _ALLOCATE_LIST	This error will occur when the modules internal representation of the tree structure does not allow the allocation of a new list.	In this case either new list can't be allocated. The only way to solve this problem is to delete another list. [-2036]
-2037	PROPHANDLING_↩ CANT_REGISTER_↩ COMPONENT	The referenced list has no space left to register this component at the desired position.	There might however be an empty space within the list where this element could be registered, but no more components can be registered at the end of this list. [-2037]
-2038	PROPHANDLING_PROP↩ _VALIDATION_FAILED	The user tried to assign a value to a property, that is invalid.	This will result in a detailed error message in the logfile. This error might arise e.g. when a string property doesn't allow the string to contain numbers. In this case trying to set the properties value to 'blabla7bla' would cause this error. [-2038]
-2099	PROPHANDLING_LAST↩ _VALID_ERROR_CODE	Defines the last valid error code value for the property module.	[-2099]
-2100	DMR_DEV_NOT_FOUND	The specified device can't be found.	This error occurs either if an invalid device ID has been passed to the device manager or if the caller tried to close a device which currently isn't initialized. [-2100]
-2101	DMR_INIT_FAILED	The device manager couldn't be initialized.	This is an internal error. [-2101]
-2102	DMR_DRV_ALREADY_IN↩ _USE	The device is already in use.	This error e.g. will occur if this or another process has initialized this device already and an application tries to open the device once more or if a certain resource is available only once but shall be used twice. [-2102]

-2103	DMR_DEV_CANNOT_↩ OPEN	The specified device couldn't be initialized.	[-2103]
-2104	DMR_NOT_INITIALIZED	The device manager or another module hasn't been initialized properly.	This error occurs if the user tries e.g. to close the device manager without having initialized it before or if a library used internally or a module or device associated with that library has not been initialized properly or if [-2104]
-2105	DMR_DRV_CANNOT_↩ OPEN	A device could not be initialized.	In this case the log-file will contain detailed information about the source of the problem. [-2105]
-2106	DMR_DEV_REQUEST_↩ QUEUE_EMPTY	The devices request queue is empty.	This error e.g. occurs if the user waits for an image request to become available at a result queue without having send an image request to the device before. It might also arise when trying to trigger an image with a software trigger mechanism before the acquisition engine has been completely started. In this case a small delay and then again calling the software trigger function will succeed. [-2106]
-2107	DMR_DEV_REQUEST_↩ CREATION_FAILED	A request object couldn't be created.	The creation of a request object failed. This might e.g. happen, if the system runs extremely low on memory. [-2107]
-2108	DMR_INVALID_↩ PARAMETER	An invalid parameter has been passed to a function.	This might e.g. happen if a function requiring a pointer to a structure has been passed an unassigned pointer or if a value has been passed, that is either too large or too small in that context. [-2108]

-2109	DMR_EXPORTED_↩ SYMBOL_NOT_FOUND	One or more symbols needed in a detected driver library couldn't be resolved.	In most cases this is an error handled internally. So the user will not receive this error code as a result of a call to an API function. However when the user tries to get access to an IMPACT buffer type while the needed IMPACT Base libraries are not installed on the target system this error code also might be returned to the user. [-2109]
-2110	DEV_UNKNOWN_ERROR	An unknown error occurred while processing a user called driver function.	[-2110]
-2111	DEV_HANDLE_INVALID	A driver function has been called with an invalid device handle.	[-2111]
-2112	DEV_INPUT_PARAM_↩ INVALID	A driver function has been called but one or more of the input parameters are invalid.	There are several possible reasons for this error↩ : an unassigned pointer has been passed to a function, that requires a valid pointer one or more of the passed parameters are of an incorrect type one or more parameters contain an invalid value (e.g. a filename that points to a file that can't be found, a value, that is larger or smaller than the allowed values. within the current setup one or more parameters impose restrictions on the requested operation that don't allow its execution. [-2112]
-2113	DEV_WRONG_INPUT_↩ PARAM_COUNT	A function has been called with an invalid number of input parameters.	[-2113]
-2114	DEV_CREATE_SETTING↩ _FAILED	The creation of a setting failed.	This can either happen, when a setting with the same name as the one the user tried to create already exists or if the system can't allocate memory for the new setting. [-2114]

-2115	DEV_REQUEST_CANT_↩ BE_UNLOCKED	The unlock for a Request object failed.	This might happen, if the Request is not locked at the time of calling the unlock function. It either has been unlocked by the user already or this request has never been locked as the request so far has not been used to capture image data into its buffer. Another reason for this error might be that the user tries to unlock a request that is currently processed by the device driver. [-2115]
-2116	DEV_INVALID_↩ REQUEST_NUMBER	The number for the Request object is invalid.	The max. number for a Request object is the value of the property RequestCount in the SystemSettings list - 1. [-2116]
-2117	DEV_LOCKED_↩ REQUEST_IN_QUEUE	A Request that hasn't been unlocked has been passed back to the driver.	This error might occur if the user requested an image from the driver but hasn't unlocked the Request that will be used for this new image. [-2117]
-2118	DEV_NO_FREE_↩ REQUEST_AVAILABLE	The user requested a new image, but no free Request object is available to process this request.	[-2118]
-2119	DEV_WAIT_FOR_↩ REQUEST_FAILED	The wait for a request failed.	This might have several reasons: The user waited for an image, but no image has been requested before. The user waited for a requested image, but the image is still not ready(e.g. because of a short timeout and a long exposure time). A triggered image has been requested but no trigger signal has been detected within the wait period. A plug and play device(e.↩ g. an USB device) has been unplugged and therefore can't deliver images anymore. In this case the 'state' property should be checked to find out if the device is still present or not. [-2119]
-2120	DEV_UNSUPPORTED_↩ PARAMETER	The user tried to get/set a parameter, which is not supported by this device.	[-2120]

-2121	DEV_INVALID_RTC_↩ NUMBER	The requested real time controller is not available for this device.	[-2121]
-2122	DMR_INTERNAL_ERROR	Some kind of internal error occurred.	More information can be found in the ∗.log-file or the debug output. [-2122]
-2123	DMR_INPUT_BUFFER_↩ TOO_SMALL	The user allocated input buffer is too small to accommodate the result.	[-2123]
-2124	DEV_INTERNAL_ERROR	Some kind of internal error occurred in the device driver.	More information can be found in the ∗.log-file or the debug output. [-2124]
-2125	DMR_LIBRARY_NOT_↩ FOUND	One or more needed libraries are not installed on the system.	[-2125]
-2126	DMR_FUNCTION_NOT_↩ IMPLEMENTED	A called function or accessed feature is not available for this device.	[-2126]
-2127	DMR_FEATURE_NOT_↩ AVAILABLE	The feature in question is (currently) not available for this device or driver.	This might be because another feature currently blocks the one in question from being accessible. More information can be found in the ∗.log-file or the debug output. [-2127]
-2128	DMR_EXECUTION_↩ PROHIBITED	The user is not permitted to perform the requested operation.	This e.g. might happen if the user tried to delete user data without specifying the required password. [-2128]
-2129	DMR_FILE_NOT_FOUND	The specified file can't be found.	This might e.g. happen if the current working directory doesn't contain the file specified. [-2129]
-2130	DMR_INVALID_LICENCE	The licence doesn't match the device it has been assigned to.	When e.g. upgrading a device feature each licence file is bound to a certain device. If the device this file has been assigned to has a different serial number then the one used to create the licence this error will occur. [-2130]
-2131	DEV_SENSOR_TYPE_↩ ERROR	There is no sensor found or the found sensor type is wrong or not supported.	[-2131]

-2132	DMR_CAMERA_↩ DESCRIPTION_INVALID	A function call was associated with a camera description, that is invalid.	One possible reason might be, that the camera description has been deleted(driver closed?). 1.5.0 [-2132]
-2133	DMR_NEWER_LIBRARY↩ _REQUIRED	A suitable driver library to work with the device manager has been detected, but it is too old to work with this version of the mvDevice↩ Manager library.	This might happen if two different drivers have been installed on the target system and one introduces a newer version of the device manager that is not compatible with the older driver installed on the system. In this case this error message will be written into the logfile together with the name of the library that is considered to be too old. The latest drivers will always be available online under www.↩ matrix-vision.de. There will always be an updated version of the library considered to be too old for download from here. 1.6.6 [-2133]
-2134	DMR_TIMEOUT	A general timeout occurred.	This is the typical result of functions that wait for some condition to be met with a timeout among their parameters. More information can be found in the ∗.log-file or the debug output. 1.7.2 [-2134]
-2135	DMR_WAIT_ABANDONED	A wait operation has been aborted.	This e.g. might occur if the user waited for some message to be returned by the driver and the device driver has been closed within another thread. In order to inform the user that this waiting operation terminated in an unusual wait, DMR_↩ WAIT_ABANDONED will be returned then. 1.7.2 [-2135]
-2136	DMR_EXECUTION_↩ FAILED	The execution of a method object or reading/writing to a feature failed.	More information can be found in the log-file. 1.9.0 [-2136]

-2137	DEV_REQUEST_↩ ALREADY_IN_USE	This request is currently used by the driver.	This error may occur if the user tries to send a certain request object to the driver by a call to the corresponding image request function. 1.10.31 [-2137]
-2138	DEV_REQUEST_↩ BUFFER_INVALID	A request has been configured to use a user supplied buffer, but the buffer pointer associated with the request is invalid.	1.10.31 [-2138]
-2139	DEV_REQUEST_↩ BUFFER_MISALIGNED	A request has been configured to use a user supplied buffer, but the buffer pointer associated with the request has an incorrect alignment.	Certain devices need aligned memory to perform efficiently thus when a user supplied buffer shall be used to capture data into this buffer must follow these alignment constraints 1.10.31 [-2139]
-2140	DEV_ACCESS_DENIED	The requested access to a device could not be granted.	There are multiple reasons for this error code. Detailed information can be found in the ∗.log-file. POSSIBLE CAUSES: an application tries to access a device exclusively that is already open in another process a network device has already been opened with control access from another system and the current system also tries to establish control access to the device an application tried to execute a function that is currently not available an application tries to write to a read-only location. 1.↩ 10.39 [-2140]

-2141	DMR_PRELOAD_↩ CHECK_FAILED	A pre-load condition for loading a device driver failed.	Certain device drivers may depend on certain changes applied to the system in order to operate correctly. E.g. a device driver might need a certain environment variable to exist. When the device manager tries to load a device driver it performs some basic checks to detect problems like this. When one of these checks fails the device manager will not try to load the device driver and an error message will be written to the selected log outputs. 1.10.52 [-2141]
-2142	DMR_CAMERA_↩ DESCRIPTION_INVALID↩ _PARAMETER	One or more of the camera descriptions parameters are invalid for the grabber it is used with.	There are multiple reasons for this error code. Detailed information can be found in the ∗.log-file. POSSIBLE CAUSES: The TapsXGeometry or Taps↩ YGeometry parameter of the selected camera description cannot be used with a user defined AOI. A scan standard has been selected, that is not supported by this device. An invalid scan rate has been selected. ... This error code will be returned by frame grabbers only. 1.10.57 [-2142]
-2143	DMR_FILE_ACCESS_↩ ERROR	A general error returned whenever there has been a problem with accessing a file.	There can be multiple reasons for this error and a detailed error message will be sent to the log-output whenever this error code is returned. POSSIBLE CAUSES: The driver tried to modify a file, for which it has no write access. The driver tried to read from a file, for which it has no read access. ... 1.10.87 [-2143]
-2144	DMR_INVALID_QUEUE_↩ SELECTION	An error returned when the user application attempts to operate on an invalid queue.	1.11.0 [-2144]

-2145	DMR_ACQUISITION_↩ ENGINE_BUSY	An error returned when the user application attempts to start the acquisition engine at a time, where it is already running.	2.5.3 [-2145]
-2146	DMR_BUSY	An error returned when the user application attempts to perform any operation that currently for any reason cannot be started because something else already running.	The log-output will provide additional information. 2.↩ 32.0 [-2146]
-2147	DMR_OUT_OF_MEMORY	An error returned when for any reason internal resources (memory, handles, ...) cannot be allocated.	The log-output will provide additional information. 2.↩ 32.0 [-2147]
-2199	DMR_LAST_VALID_↩ ERROR_CODE	Defines the last valid error code value for device and device manager related errors.	[-2199]

1.16.2 Accessing log files

If you need support using our products, you can shorten response times by sending us your log files. Accessing the log files is different in Windows and Linux:

1.16.2.1 Windows

Since

mvIMPACT Acquire 2.11.9

You can access the log files in Windows using **wxPropView** (p. 65). The way to do this is described in **"Accessing Log Files"** in the **"mvIMPACT Acquire SDK GUI Applications"** manual.

1.16.2.2 Linux

Since

mvIMPACT Acquire 2.24.0

You can access the log files in Linux via `/opt/mvIMPACT_Acquire/data/logs`.

You can also extract the directory using the following command

```
env | grep MVIMPACT_ACQUIRE_DATA_DIR
```

or change the directory directly via

```
cd $MVIMPACT_ACQUIRE_DATA_DIR/logs
```

For older versions:

Like on Windows, log files will be generated, if the activation flag for logging called **mvDebugFlags.mvd** is available in the same folder as the application (however, using Windows log files will be generated automatically, because the applications are started from the same folder). By default, on Linux the **mvDebugFlags.mvd** will be installed in the installation's destination folder in the sub-folder "apps". For example, if the destination folder was "/home/workspace", you can locate the **mvDebugFlags.mvd** like the following way:

```
user@linux-desktop:~$
// <- Starting the console window, you will be in the home directory: /home/
user@linux-desktop:~$ cd workspace/apps/
// <- Change the directory
user@linux-desktop:/home/workspace/apps$ ls -l
// <- List the directory
insgesamt 144
drwxr-xr-x  9 user user  4096 Mai 21 15:08 Callback
drwxr-xr-x  8 user user  4096 Mai 21 15:08 Callback_C
drwxr-xr-x  9 user user  4096 Mai 21 15:08 CaptureToUserMemory_C
drwxr-xr-x  3 user user  4096 Mai 21 15:03 Common
drwxr-xr-x 11 user user  4096 Mai 21 15:09 ContinuousCapture
drwxr-xr-x  9 user user  4096 Mai 21 15:09 ContinuousCaptureAllDevices
drwxr-xr-x  6 user user  4096 Mai 21 15:09 ContinuousCaptureFLTK
drwxr-xr-x  9 user user  4096 Mai 21 15:09 ContinuousCapture_C
drwxr-xr-x 11 user user  4096 Mai 21 15:09 DigitalIOs
drwxr-xr-x  9 user user  4096 Mai 21 15:09 FirmwareUpgrade
drwxr-xr-x 11 user user  4096 Mai 21 15:09 GenericInterfaceLayout
drwxr-xr-x 11 user user  4096 Mai 21 15:09 GenICamInterfaceLayout
-rw-r--r--  1 user user   854 Mai 21 15:03 Makefile
-rw-r--r--  1 user user  7365 Mai 21 15:03 Makefile.samp.inc
-rw-r--r--  1 user user 20713 Mai 21 15:03 mvDebugFlags.mvd
// <- Log activation flag
drwxr-xr-x  7 user user  4096 Mai 21 15:09 mvDeviceConfigure
drwxr-xr-x  6 user user  4096 Mai 21 15:10 mvIPConfigure
drwxr-xr-x  6 user user  4096 Mai 21 15:11 mvPropView
drwxr-xr-x  9 user user  4096 Mai 21 15:11 SingleCapture
drwxr-xr-x  9 user user  4096 Mai 21 15:11 SingleCaptureStorage
```

For log file generation you have to execute your app from the folder where **mvDebugFlags.mvd** is located. E.g. if you want to start wxPropView:

```
user@linux-desktop:/home/workspace/apps$ ./mvPropView/x86/wxPropView
// <- Start the executable from the folder, where \b mvDebugFlags.mvd is located.
```

Another possibility would be, to copy the **mvDebugFlags.mvd** file to the folder of the executable:

```
user@linux-desktop:/home/workspace/apps$ cp mvDebugFlags.mvd ./mvPropView/x86/wxPropView
// <- Copy the log activation flag
user@linux-desktop:/home/workspace/apps$ cd ./mvPropView/x86/
// <- Change the directory
user@linux-desktop:/home/workspace/apps/mvPropView/x86/$ ./wxPropView
// <- Start the executable
```

Afterwards, several log files are generated which are listed in files.mvloglist. The log files have the file extension .mvlog. Please send these files to our support team.

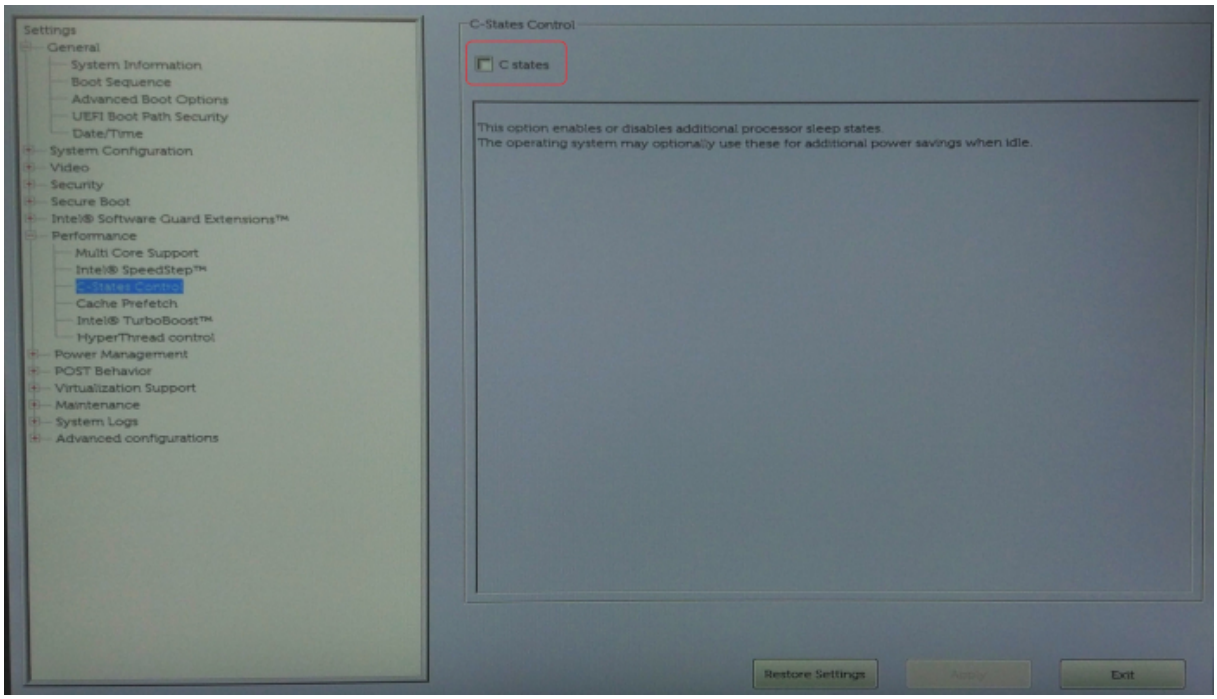
1.16.3 General Issues

- **The error counter increases** (p. 107)
- **Which PC power settings are useful and possibly necessary?** (p. 108)
- **Why does updating the device list take so long** (p. 109)

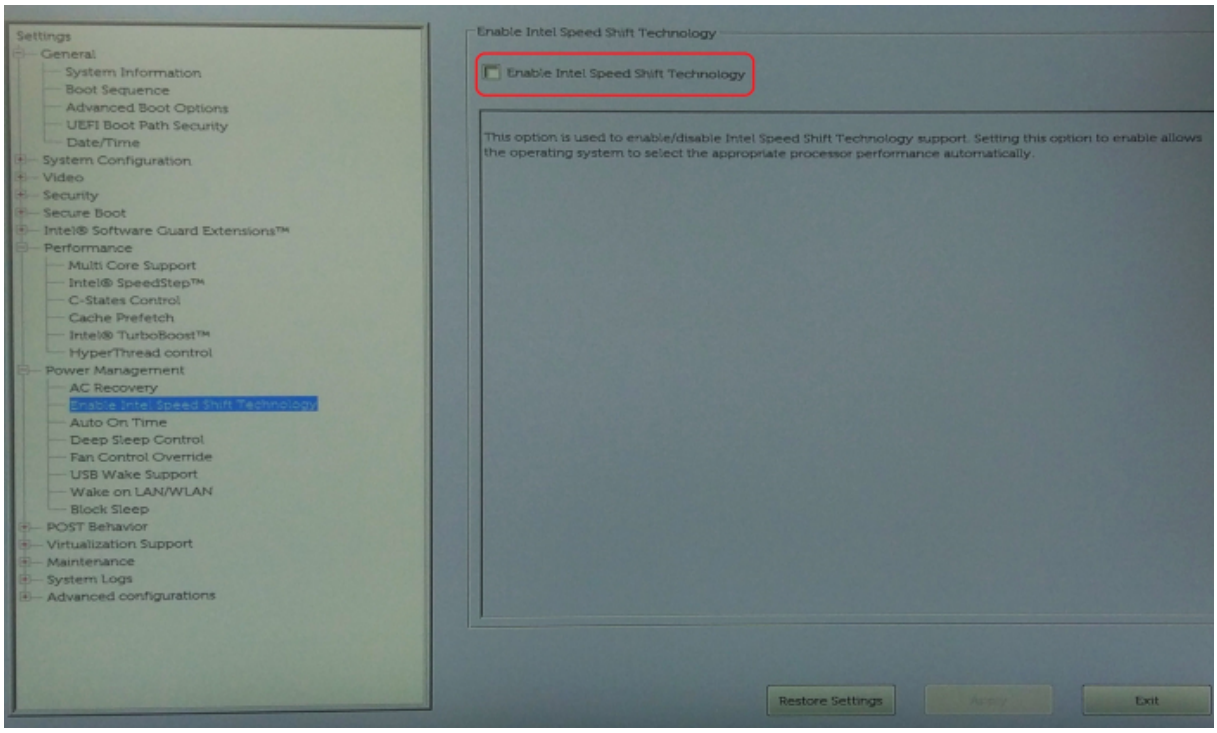
1.16.3.1 The error counter increases

Modern PC's, notebook's, etc. try to save energy by using a smart power management. If processor performance is not needed the processor will change to a power saving (sleep) state automatically and vice versa. Every state change will stop the processor for microseconds. This time is enough to cause image error counts!

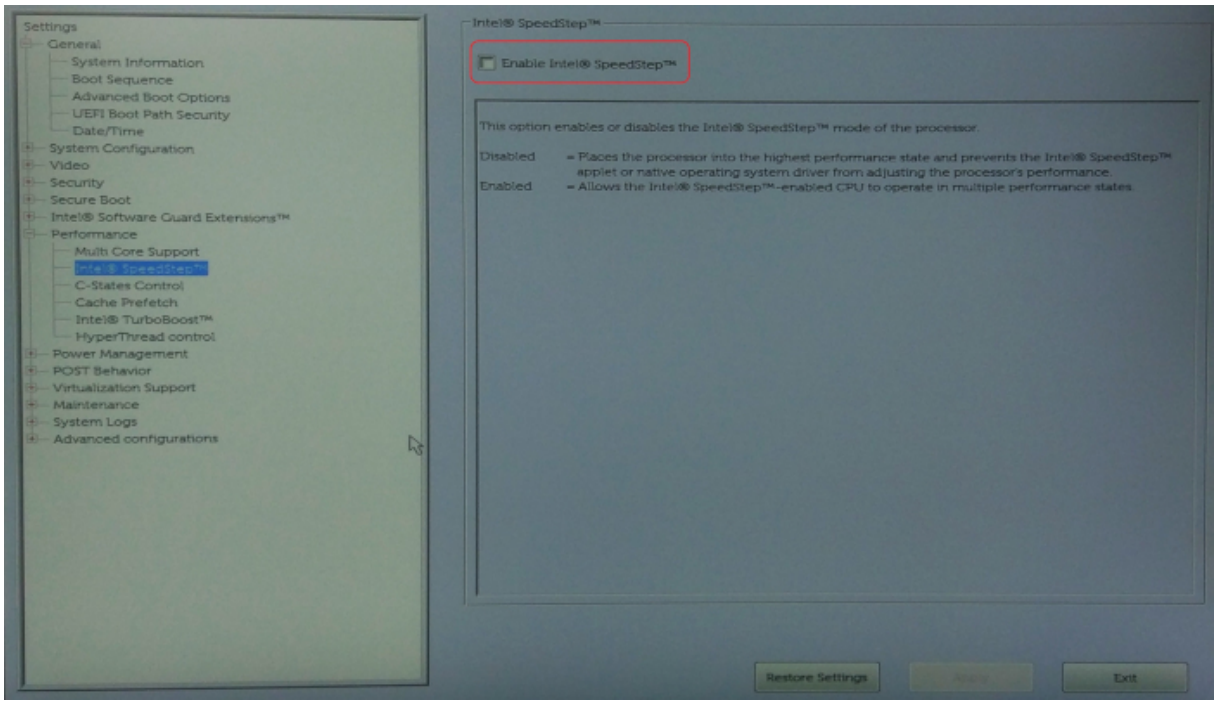
With mvDeviceConfigure running on Windows 7 or below you can disable the CPU sleep states on Windows systems. As the API needed for this has been removed in Windows 8 and above doing the same on Windows 8 or higher or when working with Linux system must be done by using BIOS options. Below are some exemplary screen-shots from BIOS settings. Your system might use different names or look different of course:



Swiching off the C states might improve the situation



Swiching off Intel SpeedShift options might improve the situation



Swiching off Intel SpeedStep options might improve the situation

Note

Another that is always worth considering is to look for an updated version of your systems BIOS! Whenever you encounter problems like described above and you are sure that you have set up the system as perfect as possible (e.g. by configuring network cards of USB interfaces like described in the corresponding sections of the mvBlueFOX3 or mvBlueCOUGAR series) a BIOS update might solve all remaining issues!

1.16.3.2 Which PC power settings are useful and possibly necessary?

Symptoms

Wrong power settings can lead to different problems like:

- unstable link speed,
- camera is available in **wxPropView** (p. 65) but streaming is not possible, or
- differences seen between cold start and warm start of the PC.

Cause

Embedded vision components are optimized products. Alternating power states can affect the connection between devices.

Resolution

Please check the following settings, some or all of which may be needed to cure the problem.

BIOS

Component	Recommendation
C-States	OFF
EIST	Disabled
ASPM	Disabled

Windows

Component	Recommendation
High Perfomance power plan	Selected
PCI Express Link State Power Management (Advanced power settings)	OFF

Alternatively:

Component	Recommendation
Ultimate Perfomance power plan	Selected

See also

`https://www.howtogeek.com/368781/how-to-enable-ultimate-performance-power-plan-in-w`

Linux

Boot the Linux kernel with the parameter:

Component	Recommendation
pcie_aspm	OFF

1.16.3.3 Why does updating the device list take so long

Since

mvIMPACT Acquire 2.24.0

The GenTL driver scans both the GigE Vision and the USB3 Vision interfaces for compliant devices. Given that the GigE Vision standard allows detection timeouts of 1000 ms, it could take long, especially for USB3 Vision users which even do not have GigE Vision in their application, until the device update is finished.

For this reason, we upgraded the **MATRIX VISION application tools** (p. 65) to excluded interfaces or single devices to be scanned or detected. The easiest way to do so is as follows:

1. Open **wxPropView** (p. 65) .
2. Open the device list drop down and
3. select **"Missing A Device? Click here..."**.

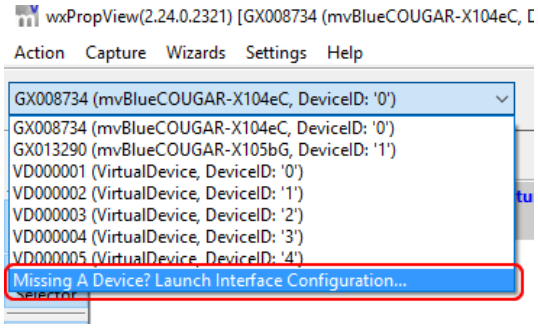


Figure 1: "Missing A Device? Click here..."

Now, you will see the interfaces and the detected device.

4. Finally, you can deselect the interfaces or devices which you do not want to scan for updating the device list. The following figure shows that the GigE Vision interface should be skipped:

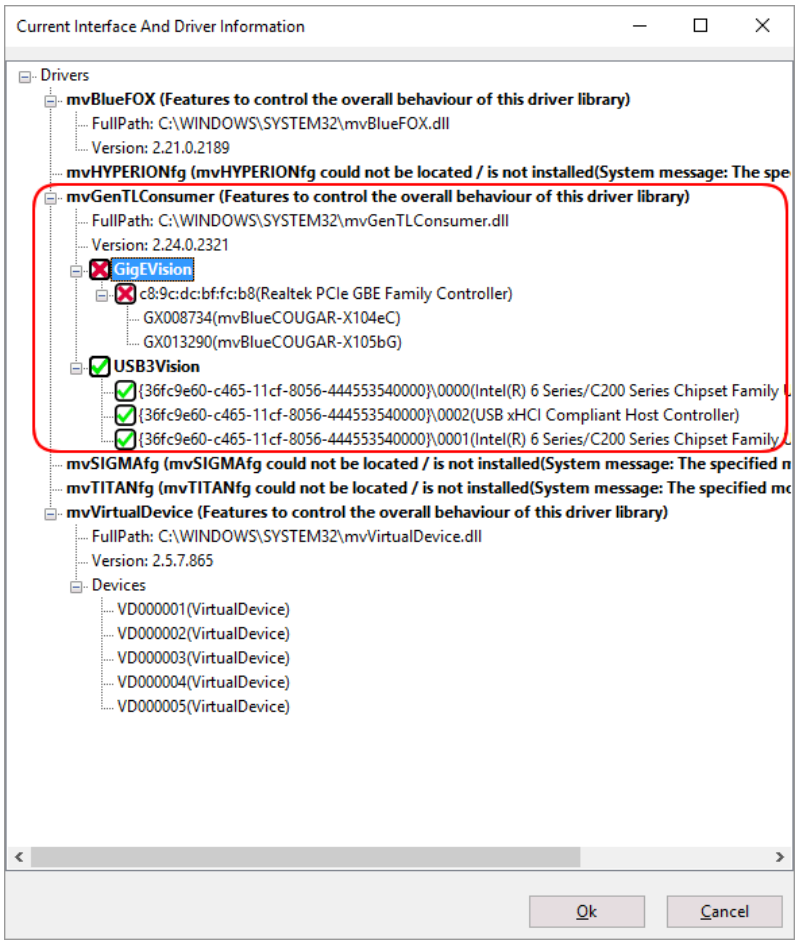


Figure 2: Skip the GigE Vision interface

1.16.4 Windows

- **Bluescreen On Machines Running Windows 10 Or Higher When Streaming Is Started** (p. 111)

1.16.4.1 Bluescreen On Machines Running Windows 10 Or Higher When Streaming Is Started

Symptoms

On systems running Windows 10 or higher a Bluescreen "DRIVER-VERIFIER-DMA-VIOLATION" shows up when the streaming is started.

Cause

The root cause is that on newer systems a protection mechanism is used which prevents PCIe devices accessing DMA memory per default.

It is possible to check the status of the DMA access protection settings within msinfo32. To investigate the settings please:

- Hit the keys Windows + R
- Type msinfo32 and hit enter
- Search for "Kernel DMA Protection" (it is usually located at the "System Summary")

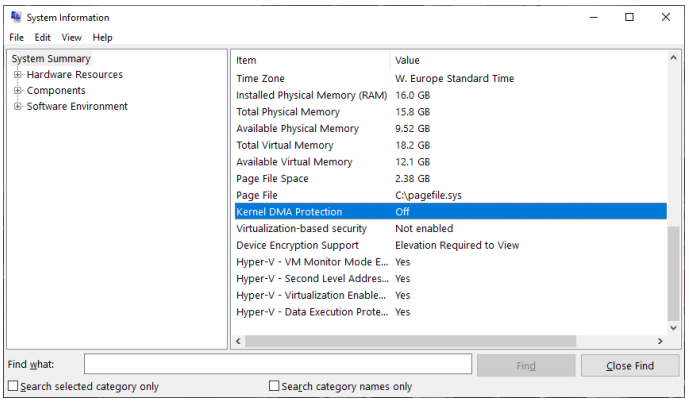
Resolution

Update to mvIMPACT Acquire 2.47.0 or higher!

If this does not solve the problem, several other things can be done to solve this problem manually:

- Disable the IOMMU the system's BIOS. This option is typically called "VT-d", "Intel VT for Directed I/O" or "AMD Vi".
- Disable "DMA Protection" or "DMA Remapping Protection" in the system's BIOS.

Afterwards msinfo32 should report the "Kernel DMA Protection" as "Off"



msinfo32 - Kernel DMA Protection

1.16.5 Linux

- **No GenICam devices are detected on a Linux system** (p. 112)

1.16.5.1 No GenICam devices are detected on a Linux system

Symptoms

After starting any mvIMPACT Acquire based application no GenICam compliant devices are accessible or detected

Cause

The environment variables which are necessary to us the mvGenTL Acquire package are defined by two shell scripts which are set by profile.d. This causes the variables to be available within every login-shell run on the system. Once an application is started from a non-login shell which is using mvIMPACT Acquire libraries the variables from profile.d are not used. This results in a situation that the necessary runtime libraries can not be found which leads to missing interfaces and devices.

Resolution

To avoid this situation there are two possible solutions:

- Use a login-shell to run the application using the MATRIX VISION driver package.
- Insert the variables from: /etc/profile.d/genicam.sh and /etc/profile.d/acquire.sh to a shell configuration file which is read to configure environment variables.

1.17 Glossary

ADC	Analog-to-digital converter (A/D converter).
API	Application programming interface (API). The standard API for MATRIX VISION products is called **mvIMPACT_Acquire** (p. 119).
BusyBox	Configurable monolithic application including shell and other useful command line tools - often called the "swiss army knife" for embedded systems. Even desktop distributions are sometimes relying on BusyBox due to its robustness. Please see `http://www.busybox.net` for details.
CIFS	Common Internet file system (CIFS) replaced Samba in 2006. It gets rid of NetBIOS packets an introduced Unix features like soft/hard links and allows larger files.
CPU	Central processing unit aka processor.
DAC	Digital to analog converter (D/A converter).
Defaults	Standard system settings.
DHCP	Dynamic Host Configuration Protocol (DHCP). DHCP is a protocol used by networked devices (clients) to obtain various parameters necessary for the clients to operate in an Internet Protocol (IP) network.
Digital I/O	Digital inputs and outputs.
GDB	GDB, the GNU Project debugger.

GenICam	GenICam stands for **GEN**eric programming **I**nterface for **CAM**eras. It's a generic way to access and modify device parameters with a unified interface. A GenICam compliant device either directly provides a GenICam compliant description file (in internal memory) or the description file can be obtained from a local (hard disk etc.) or web location. A GenICam description file is something like a machine readable device manual. It provides a user readable name and value range for **parameters** (p. 121) that are offered by the device for reading and/or writing and instructions on what command must be sent to a device when the user modifies or reads a certain parameter from the device. These description files are written in XML. An excerpt from such a file can be seen in the figure below:
	
	`<Float Name="GainAbs" NameSpace="Standard">`
	`  <ToolTip>Sets the camera's gain in dB</ToolTip>`
	`  <Description>This float value sets the camera's gain in dB.</Description>`
	`  <DisplayName>Gain (Abs)</DisplayName>`
	`  <pIsLocked>TLParamsLocked</pIsLocked>`
	`  <pValue>GainAbs_CtrlValueFao</pValue>`
	`  <Unit>dB</Unit>`
	`  <Representation>Linear</Representation>`
	`</Float>`
	`<FloatReg Name="GainAbs_CtrlValueFao" NameSpace="Custom">`
	`  <Address>0x00000004</Address>`
	`  <pAddress>CamCtrl_AnalogCtrlBaseAddress</pAddress>`
	`  <Length>8</Length>`
	`  <AccessMode>RW</AccessMode>`
	`  <pPort>Device</pPort>`
	`  <Endianess>BigEndian</Endianess>`
	`</FloatReg>`
	
	Excerpt of a GenICam description file (in XML)
	For further information on this topic please have a look at `https://en.wikipedia.org/wiki/GenICam`.

GenTL	GenTL is the transport layer interface for cameras, acquiring images from the camera, and moving them to the user application.

Gigabit Ethernet (GigE)	The term Gigabit Ethernet (defined by the IEEE 802.3-2008 standard) represents various technologies for transmitting Ethernet frames at a rate of a gigabit per second (1,000,000,000 bits per second).

GigE Vision

GigE Vision is a network protocol designed for the communication between an imaging device and an application. This protocol completely describes:

- device discovery
- data transmission
  - image data
  - additional data
- read/write of parameters.

GigE Vision uses UDP for data transmission to reduce overhead introduced by TCP.

Note

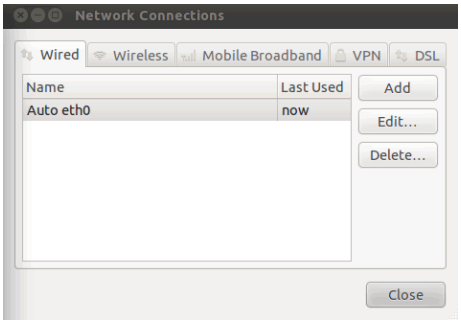
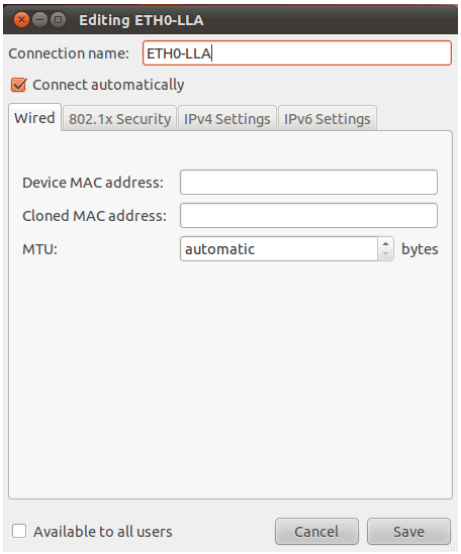
UDP does not guarantee the order in which packets reach the client nor does it guarantee that packets arrive at the client at all. However, GigE Vision defines mechanisms that can detect lost packets. This allows capture driver manufacturers to implement algorithms that can reconstruct images and other data by requesting the device to resend lost data packets until the complete buffer has been assembled. For further information please have a look at `https://en.wikipedia.org/wiki/GigE_Vision` The MATRIX VISION GigE Vision capture filter driver as well as the socket based acquisition driver and all MATRIX VISION GigE Vision compliant devices support resending thus lost data can be detected and in most cases reconstructed. This of course can not enhance the max. bandwidth of the transmission line thus if e.g. parts of the transmission line are overloaded for a longer period of time data will be lost anyway.

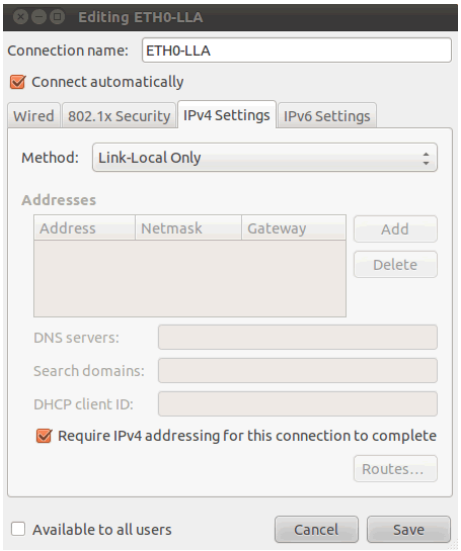
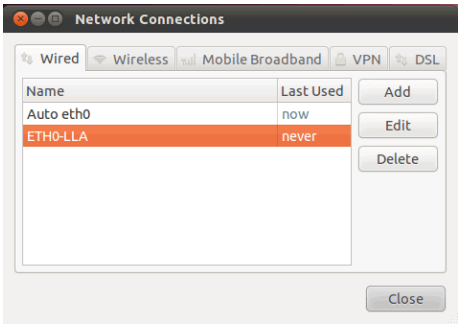
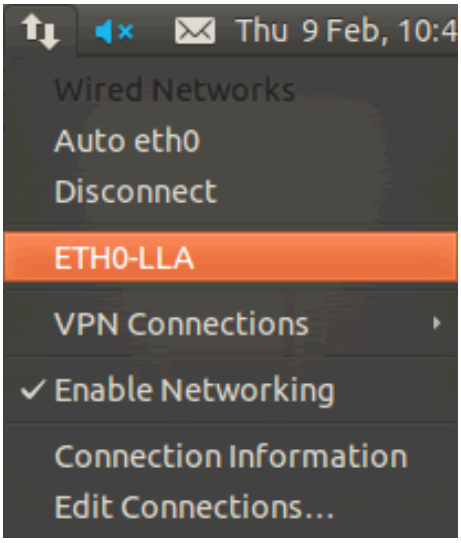
Both capture drivers will allow to fine tune the resend algorithm used internally and both drivers will also provide information about the amount of data lost and the amount of data that was re-requested. This information/configuration will be part of the drivers SDK. More information about it can be found in the corresponding interface description.

Note

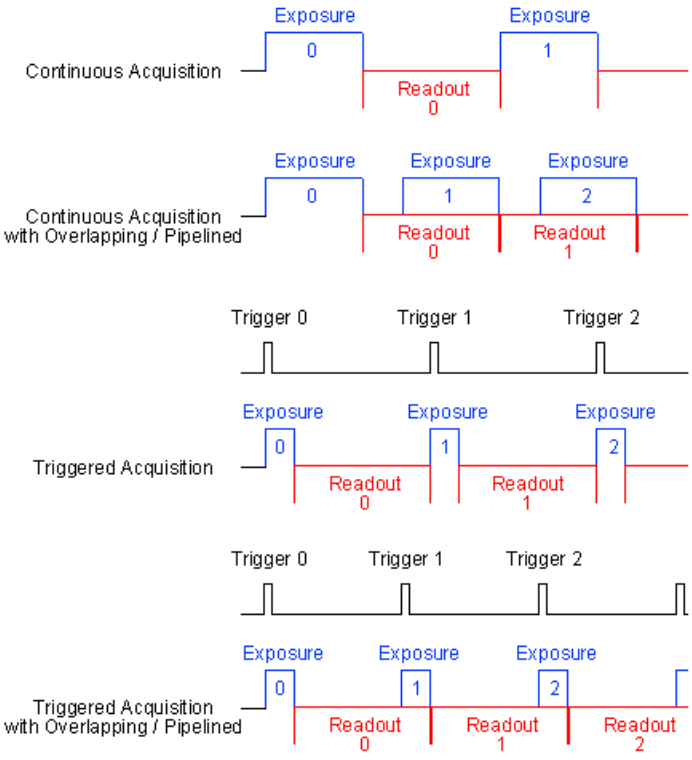
On Windows 2000 the filter driver does not support the **"Resend"** mechanism.

High Dynamic Range (HDR)	The HDR (High Dynamic Range) mode increases the usable contrast range. This is achieved by dividing the integration time in two or three phases. The exposure time proportion of the three phases can be set independently. Furthermore, it can be set, how many signal of each phase is charged.
HRTC	With a *Hardware Real-Time Controller* (HRTC) built inside the FPGA users can define a PLC like sequence of operating steps to control basic time critical functions like exposure time, image trigger and I/O ports. Timing is hard real-time with sub microsecond high resolution.
IDE	a software application that provides comprehensive facilities to computer programmers for software development. An IDE normally consists of a source code editor, a compiler and/or interpreter, build automation tools, and (usually) a debugger.
IPKG	Itsy package management system originally designed for embedded systems. Please have a look at `https://en.wikipedia.org/wiki/Ipkg` or a more sophisticated documentation at `http://buffalo.nas-central.org/index.php` "Overview_of_the_ipkg_package_management_system" MATRIX VISION distributes all non-firmware, i.e. optional software as ipk packages.
JFFS2	JFFS2 is a file system which supports wear leveling. **See also** Sources about the JFFS file system: • `http://sources.redhat.com/jffs2/` • `http://www.linux-mtd.infradead.org/faq/jffs2.html`
Link Aggregation (LAG)	Dual-GigE cameras need a network interface card with two network interfaces. However, both network interfaces have to work as a unit. Link aggregation (LAG) or bonding is the name of the game and has to be supported by the network interface card's driver. With it you can bond the two network interfaces so that the work as one interface.

LLA	Logical link address (LLA) is a type of mechanism to obtain a valid IP address without a DHCP server being present. Whether an IP address is available or not is resolved using address resolution protocol (ARP) packets. If no ARP response is received on a given address it is considered unused and will be assigned to the interface. LLA space is 169.254.x.y, i.e. 16bit netmask yielding 64K possible device addresses. With Linux you have to add LLA as an additional interface. By default, you can find one interface in *Connections:* (This description uses *"Gnome Network Manager"*, however using KDE should be similar)  In *Wired*, you can add interfaces via **Add:** 

	In the tab *"IPv4 Setting"* you have to set **"Link-Local Only"**:  After saving, you will find both connections in the summary:  Now, you can select the wished connection using the left mouse button in the *"Network Manager"* menu. In the LLA case it is just the new created connection: 
MAC address	Media Access Control address (MAC address) is a quasi-unique identifier attached to most network adapters (NICs) in computer networking.

MTU	Maximum transmission unit (MTU) refers to the size (in bytes) of the largest packet that a given layer of a communications protocol can pass onwards. The default MTU for Ethernet is 1500. The optimum for Gigabit Ethernet is 8000 - 12000. Different MTU settings in the same subnet can cause package losses, i.e. **never ever** change the MTU unless you know what you're doing. Changing the MTU to other values than 1500 when using file or web services from other computers are almost always a bug. It's save to increase MTU when working in peer-to-peer mode with both devices sharing the same MTU. Please not that few network cards support 16K, most Gigabit Ethernet cards are limited to 9k, some don't support Jumbo Frames (MTU $>$ 1500) at all.
mvIMPACT Acquire	This driver supplied with MATRIX VISION products represents the port between the programmer and the hardware. The driver concept of MATRIX VISION provides a standardized programming interface to all image processing products made by MATRIX VISION GmbH. The advantage of this concept for the programmer is that a developed application runs without the need for any major modifications to the various image processing products made by MATRIX VISION GmbH. You can also incorporate new driver versions, which are available for download free of charge on our website: `https://www.matrix-vision.com`. The developer interface description of the mvIMPACT Acquire is called • mvIMPACT_Acquire_API_CPP_manual.chm • mvIMPACT_Acquire_API_C_manual.chm • mvIMPACT_Acquire_API_NET_manual.chm and can be downloaded from our website.
Netboot	With netboot you can boot mvBlueCOUGAR over network. This is especially useful when several devices share the same pieces of software, i.e. same root file system, which might be subject to change frequently.
NFS	Network File System (NFS) is a network file system protocol, allowing clients to access files over LAN. Given that you need a NFS server are uncommon on Windows, this protocol best fits for Linux-Linux connections.
NIC	Network interface card - synonym for network controller.

Overlapped / pipelined transfer	By default, the steps exposure and readout out of an image sensor are done one after the other. • By design, CCD sensors support overlap capabilities also combined with trigger (see figure). • In contrast, so-called pipelined CMOS sensors only support the overlapped mode. Even less CMOS sensors support the overlapped mode combined with trigger. Please check the **sensor summary** (p. 54). In overlapping mode, the exposure starts the exposure time earlier during readout.  **Note** In overlapped trigger mode, you have to keep in mind the following formula `interval between two trigger events >= (readout time - exposure time)`
PSE	Power sourcing equipment. The network PoE element that inserts power onto an Ethernet cable.
ROI/AOI	Region/Area of interest.
SDK	Sofware development kit (SDK). The standard image processing software library from MARTIX VISION is **mvIMPACT**.

SFNC	**S**tandard **F**eature **N**aming **C**onvention of **GenICam** (p. 113). **See also** The latest GenICam properties list can be found here: **http://www.emva.org/standards-technology/genicam/genicam-downloads/** The file is called "GenICam Standard Features Naming Convention (PDF)"
Shell	In computing, a shell is a piece of software that provides an interface for users. Command-line shells provide a command-line interface (CLI) to the operating system. The primary purpose of the shell is to invoke or "launch" another program; however, shells frequently have additional capabilities such as viewing the contents of directories.
USB3 Vision	A closed source framework, defined and administered by the Automated Imaging Association (AIA), for transmitting video and related control data over USB 3. Sometimes U3V is used as an acronym.
UDP	The User Datagram Protocol (UDP) is an Internet protocol. It is used by applications to send messages to other hosts on an Internet Protocol (IP) network.
Virtual Network Computing (VNC)	Virtual Network Computing (VNC) is a graphical desktop sharing system that uses the RFB protocol to remotely control another computer. Over a network, it transmits the mouse and keyboard events from one computer to another, relaying the graphical screen updates back in the other direction. To access the camera's desktop from a PC via VNC, • you have to know the IP address of the remote system. • Start a VNC viewer and • point it to the remote system. You won't need a password. Of course, you won't get a very fast live image display via the network with VNC but you should be able to start **wxPropView** (p. 65) and capture images.
wxWidgets	wxWidgets is a cross-platform GUI library. It can be used from languages such as C++, Python, Perl, and C#/.NET. **See also** http://www.wxwidgets.org

1.18 Use Cases

- **Introducing acquisition / recording possibilities** (p. 123)
- **Improving the acquisition / image quality** (p. 133)
- **Working with triggers** (p. 160)
- **Working with I/Os** (p. 177)
- **Saving data on the device** (p. 184)
- **Working with device features** (p. 192)
- **Working with several cameras simultaneously** (p. 204)
- **Working with 3rd party tools** (p. 210)

1.18.1 Introducing acquisition / recording possibilities

There are several use cases concerning the acquisition / recording possibilities of the camera:

- **Acquiring a number of images** (p. 123)
- **Generating very long exposure times** (p. 124)
- **Working with multiple AOIs (mv Multi Area Mode)** (p. 126)
- **Working with Event Control** (p. 130)

1.18.1.1 Acquiring a number of images

As described in chapter **Acquisition Control** (p. 69), if you want to acquire a number of images, you can use as *"Setting -> Base -> Camera -> Acquisition Control -> Acquisition Mode"* **"MultiFrame"** and you have to set the **"Acquisition Frame Count"**.

Afterwards, if you start the acquisition via the button *"Acquire"*, the camera will deliver the number of images.

The **"MultiFrame"** functionality can be combined with an external signal to start the acquisition.

There are several ways and combinations possible, e.g.:

- A trigger starts the acquisition (Figure 1).
- A trigger starts the acquisition start event and a second trigger starts the grabbing itself (Figure 2).

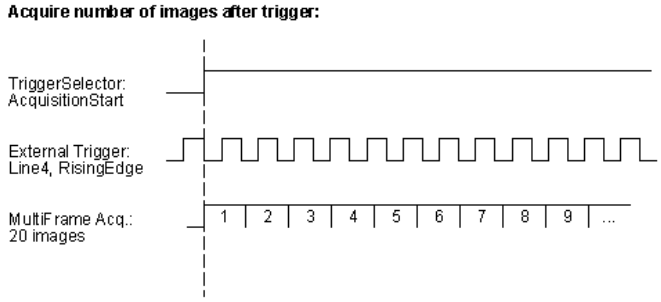


Figure 1: Starting an acquisition after one trigger event

For this scenario, you have to use the *"Setting -> Base -> Camera -> Acquisition Control -> Trigger Selector"* **"AcquisitionStart"**.

The following figure shows, how to set the scenario shown in Figure 1 with **wxPropView** (p. 65)

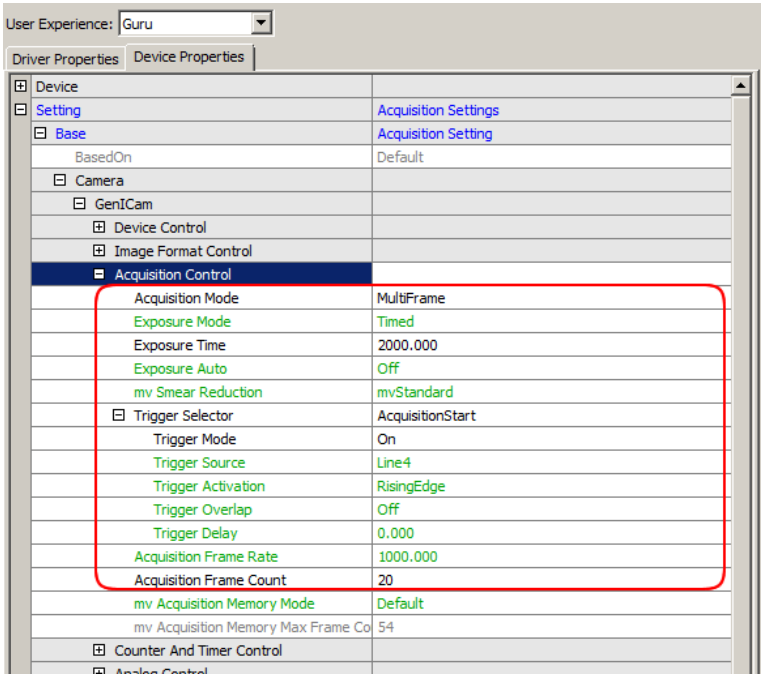


Figure 2: wxPropView - Setting acquisition of a number of images started by an external signal

A rising edge at line 4 will start the acquisition of 20 images.

1.18.1.2 Generating very long exposure times

1.18.1.2.1 Basics

At the moment the exposure time is limited to a maximum of 1 up to 20 seconds depending on certain internal sensor register restrictions. So each device might report a different maximum exposure time.

Since

Firmware version 2.28.0

Firmware version 2.28 did contain a major overhaul here so updating to at least this version can result in a much higher maximum exposure time. However, current sensor controllers can be configured to use even longer exposure times if needed using one of the devices timers to create an external exposure signal that can be fed back into the sensor. This use case will explain how this can be done.

This approach of setting up long exposure times requires the sensor of the camera to allow the configuration of an external signal to define the length of the exposure time, so only devices offering the `ExposureMode` `Trigger↩Width` can be used for this setup.

Note

The maximum exposure time in microseconds that can be achieved in this configuration is the maximum value offered by the timer used.

With **GenICam** (p. 113) compliant devices that support all the needed features the setup is roughly like this:

1. Select *"Setting -> Base -> Camera -> GenICam -> Counter And Timer Control -> Timer Selector -> Timer 1"* and .

2. set *"Timer Trigger Source"* = *"UserOutput0"*.

3. set *"Timer Trigger Activation"* = *"RisingEdge"*.
   I.e. a rising edge on **UserOutput0** will start **Timer1**.

4. Then set the *"Timer Duration"* property to the desired exposure time in *us*.

5. In *"Setting -> Base -> Camera -> GenICam -> Acquisition Control"* set the *Trigger* Selector = *"FrameStart"*.
   I.e. the acquisition of one frame will start when

6. **Timer1** is active: *"Trigger Source"* = *"Timer1Active"*.

7. **Exposure** time will be the trigger width: *"Exposure Mode"* = *"TriggerWidth"*.

The following diagram illustrates all the signals involved in this configuration:

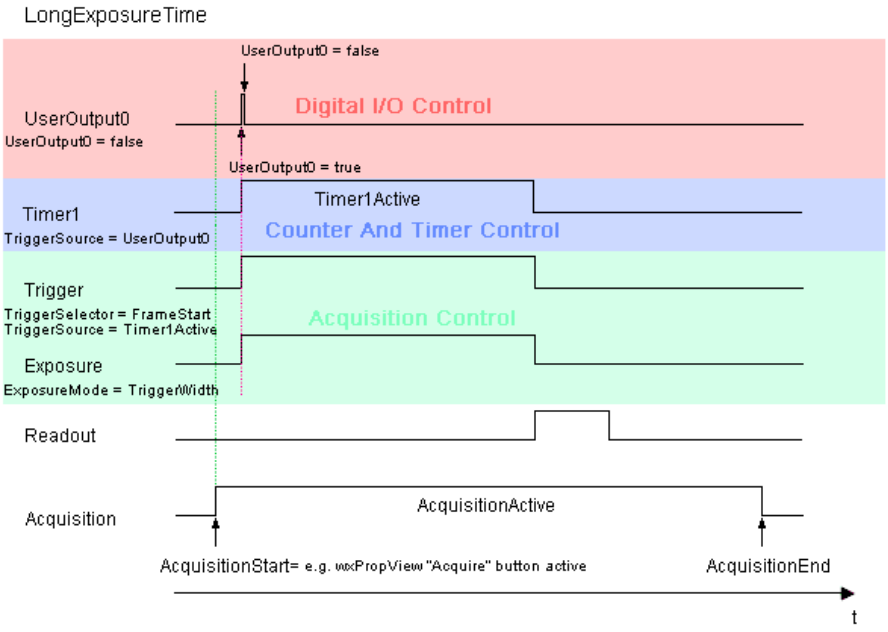


Figure 1: Long exposure times using GenICam

To start the acquisition of one frame a rising edge must be detected on **UserOutput0** in this example but other configurations are possible as well.

1.18.1.2.2 Setting up the device

The easiest way to define a long exposure time would be by using a single timer. The length of the timer active signal is then used as trigger signal and the sensor is configured to expose while the trigger signal is active. This allows to define exposure time with micro-second precision up the the maximum value of the timer register. With a 32 bit timer register this results in a maximum exposure time of roughly 4295 seconds (so roughly 71.5 minutes). When writing code e.g. in C# this could look like this:

```
private static void configureDevice(Device pDev, int exposureSec, GenICam.DigitalIOControl ioc)
{
  try
  {
    // establish access to the CounterAndTimerControl interface
    GenICam.CounterAndTimerControl ctc = new mv.impact.acquire.GenICam.CounterAndTimerControl(pDev);
    // set TimerSelector to Timer1 and TimerTriggerSource to UserOutput0
    ctc.timerSelector.writeS("Timer1");
```

```
    ctc.timerTriggerSource.writeS("UserOutput0");
    ctc.timerTriggerActivation.writeS("RisingEdge");

    // Set timer duration for Timer1 to value from user input
    ctc.timerDuration.write(exposureSec * 1000000);

    // set userOutputSelector to UserOutput0 and set UserOutput0 to inactive.
    // We will later generate a pulse here to initiate the exposure
    ioc.userOutputSelector.writeS("UserOutput0");
    ioc.userOutputValue.write(TBoolean.bFalse);

    // establish access to the AcquisitionCotrol interface
    GenICam.AcquisitionControl ac = new mv.impact.acquire.GenICam.AcquisitionControl(pDev);
    // set TriggerSelector to FrameStart and try to set ExposureMode to TriggerWidth
    ac.triggerSelector.writeS("FrameStart");
    // set TriggerSource for FrameStart to Timer1Active and activate TriggerMode
    ac.triggerSource.writeS("Timer1Active");
    ac.triggerMode.writeS("On");

    // expose as long as we have a high level from Timer1
    ac.exposureMode.writeS("TriggerWidth");
  }
  catch (Exception e)
  {
    Console.WriteLine("ERROR: Selected device does not support all features needed for
    this long time exposure approach: {0}, terminating...", e.Message);
    System.Environment.Exit(1);
  }
}
```

Note

Make sure that you adjust the `ImageRequestTimeout_ms` either to 0 (infinite)(this is the default value) or to a reasonable value that is larger than the actual exposure time in order not to end up with timeouts resulting from the buffer timeout being smaller than the actual time needed for exposing, transferring and capturing the image:

```
ImageRequestTimeout_ms = 0 # or reasonable value
```

See also

Counter And Timer Control (p. 73)

Digital I/O Control (p. 80)

Acquisition Control (p. 69)

1.18.1.3 Working with multiple AOIs (mv Multi Area Mode)

Since

mvIMPACT Acquire 2.18.3

1.18.1.3.1 Introduction

A special feature of Pregius sensors (a.k.a. IMX) from Sony is the possibility to define multiple AOIs (Areas of Interests - a.k.a. ROI - Regions of Interests) and to transfer them at the same time. Because many applications just need one or several specific parts in an image to be checked, this functionality can increase the frame rate.

Once activated, the *"mv Multi Area Mode"* allows you, depending on the sensor, to define up to eight AOIs (*mvArea0* to *mvArea7*) in one image. There are several parameters in combination with the AOIs which are illustrated in the following figure:

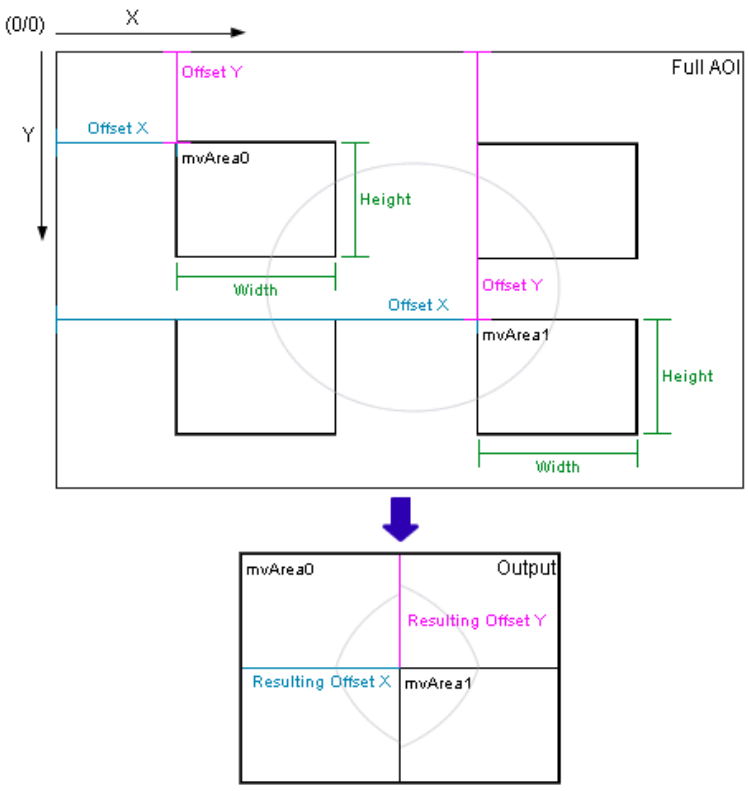


Figure 1: Multiple AOIs principle

The *"Resulting Offset X"* and *"Resulting Offset Y"* indicates the starting point of the specific AOI in the output image. To complete the rectangular output image, the "missing" areas are filled up with the image data horizontally and vertically. We recommend to use the wizard as a starting point - the wizard provides a live preview of the final merged output image.

1.18.1.3.2 Using wxPropView

To create multiple AOIs with **wxPropView** (p. 65), you have to do the following step:

1. Start **wxPropView** (p. 65) and
2. connect to the camera.
3. Then change in *"Setting -> Base -> Camera -> GenICam -> Image Format Control"* the **"mv Multi Area Mode"** to **"mvMultiAreasCombined"**. Afterwards, **"mv Area Selector"** is available.
4. Now, select the area a.k.a. AOI you want to create via *"mv Area Selector"*, e.g. **"mvArea3"** and
5. set the parameters **"mv Area Width"**, **"mv Area Height"**, **"mv Area Offset X"**, and **"mv Area Offset Y"** to your needs.
6. Activate the area a.k.a. AOI by checking the box of **"mv Area Enable"**.

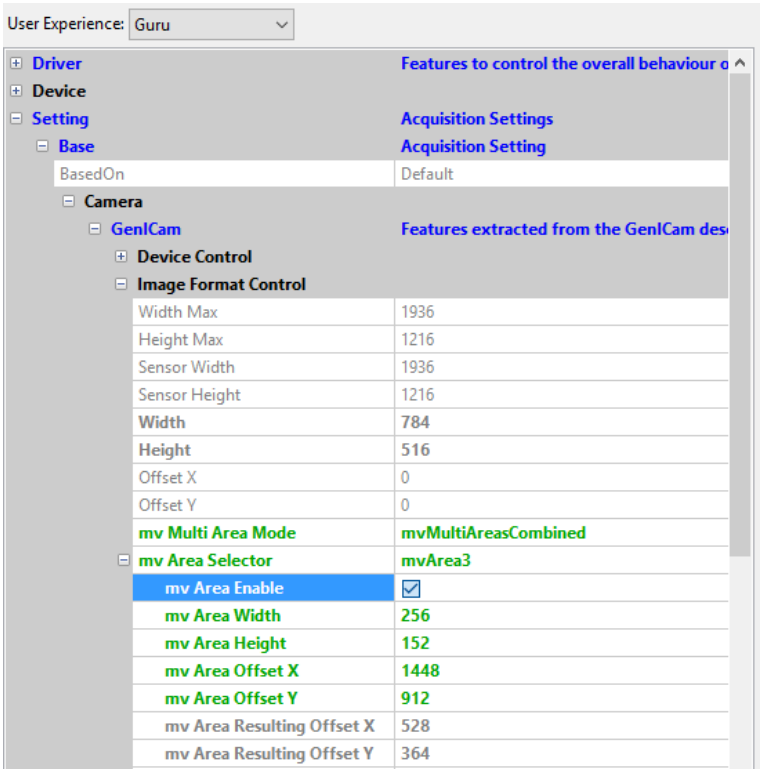


Figure 2: wxPropView - Multiple AOIs

7. Finally, start the acquisition by clicking the button *"Acquire"*.

1.18.1.3.2.1 Using the Multi AOI wizard

Since

mvIMPACT Acquire 2.19.0

wxPropView (p. 65) offers a wizard for the Multi AOI usage:

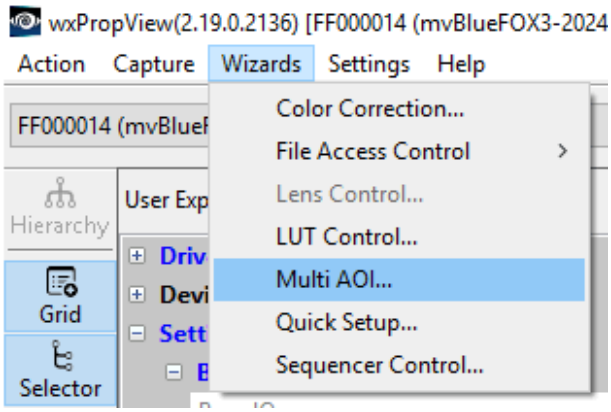


Figure 3: wxPropView - Wizard menu

The wizard can be used to get a comfortable overview about the settings of the AOIs and to create and set the AOIs in a much easier way:

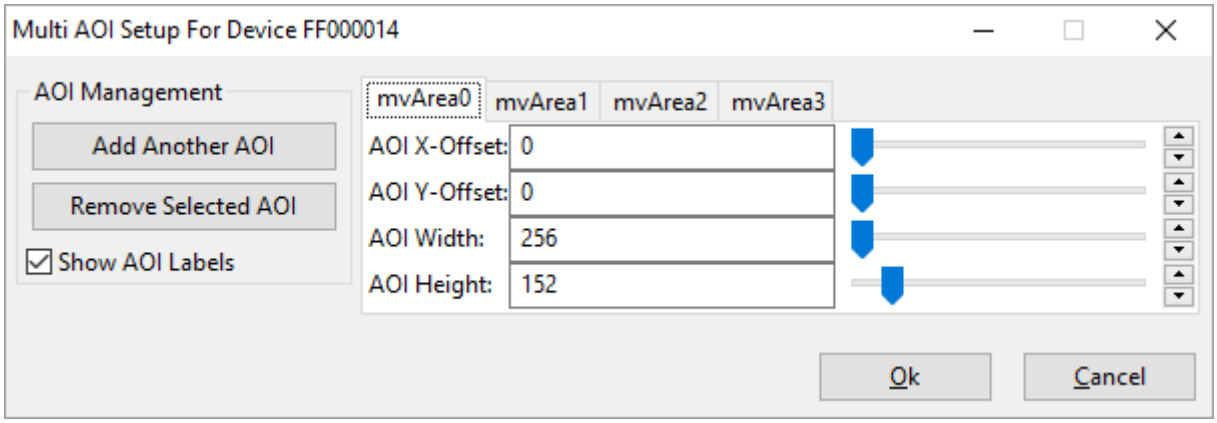


Figure 4: wxPropView - Multi AOI wizard

Just

- select the desired **mvArea** tabs,
- set the properties like offset, width, and height in the table directly, and
- confirm the changes at the end using the **Ok** button.

The live image shows the created AOIs and the merged or "missing" areas which are used to get the final rectangular output image.

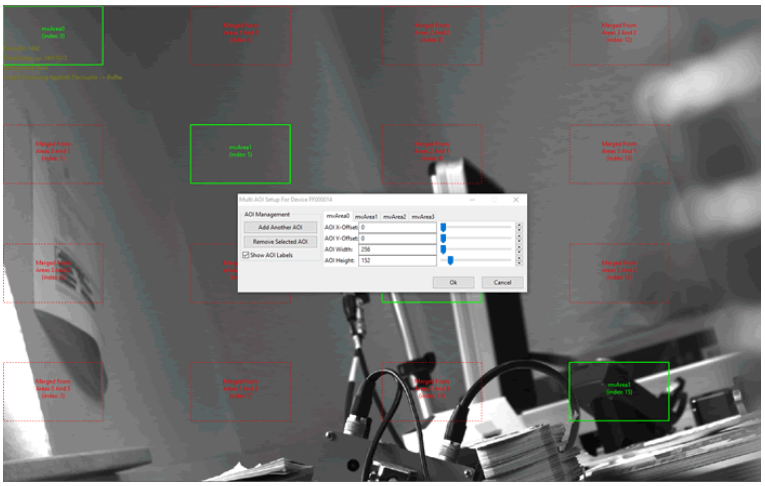


Figure 5: wxPropView - Multi AOI wizard - Live image

1.18.1.3.3 Programming multiple AOIs

```
#include <mvIMPACT_CPP/mvIMPACT_acquire.h>
#include <mvIMPACT_CPP/mvIMPACT_acquire_GenICam.h>

...
    GenICam::ImageFormatControl ifc( pDev );
    ifc.mvMultiAreaMode.writeS( "mvMultiAreasCombined" );
    ifc.mvAreaSelector.writeS( "mvArea0" );
    ifc.mvAreaEnable.write( bTrue );
    ifc.mvAreaOffsetX.write( 0 );
    ifc.mvAreaOffsetY.write( 0 );
    ifc.mvAreaWidth.write( 256 );
    ifc.mvAreaHeight.write( 152 );
    ifc.mvAreaSelector.writeS( "mvArea1" );
    ifc.mvAreaEnable.write( bFalse );
    ifc.mvAreaSelector.writeS( "mvArea2" );
    ifc.mvAreaEnable.write( bFalse );
    ifc.mvAreaSelector.writeS( "mvArea3" );
    ifc.mvAreaEnable.write( bTrue );
    ifc.mvAreaOffsetX.write( 0 );
    ifc.mvAreaOffsetY.write( 0 );
    ifc.mvAreaWidth.write( 256 );
    ifc.mvAreaHeight.write( 152 );
    ifc.mvAreaOffsetX.write( 1448 );
    ifc.mvAreaOffsetY.write( 912 );
...
```

1.18.1.4 Working with Event Control

MATRIX VISION devices can generate Event notifications. An Event is a message that is sent to the host application to notify it of the occurrence of an internal event. With *"Setting -> Base -> Camera -> GenICam -> Event Control"* you can handle these notifications.

At the moment, it is possible to handle

- Exposure End (= sensor's exposure end)
- Line 4 (= DigIn0) Rising Edge
- Line 5 (= DigIn1) Rising Edge
- Frame End (= the camera is ready for a new trigger)

1.18.1.4.1 Setting Event notifications using wxPropView

To activate the notifications, just

1. Select the Event via *"Setting -> Base -> Camera -> GenICam -> Event Control -> Event Selector"*, e.g. **ExposureEnd** .
2. Set the *"Event Notification"* to **"On"** .

Afterwards, it is possible to attach a custom callback that gets called whenever the property is modified. E.g. if you want to attach a callback to the Frame ID after the exposure was finished, you have to

1. select *"Setting -> Base -> Camera -> GenICam -> Event Control -> Event Exposure End Data -> Event Exposure End Frame ID"*,
2. right-click on the property, and

3. click on **"Attach Callback"**.

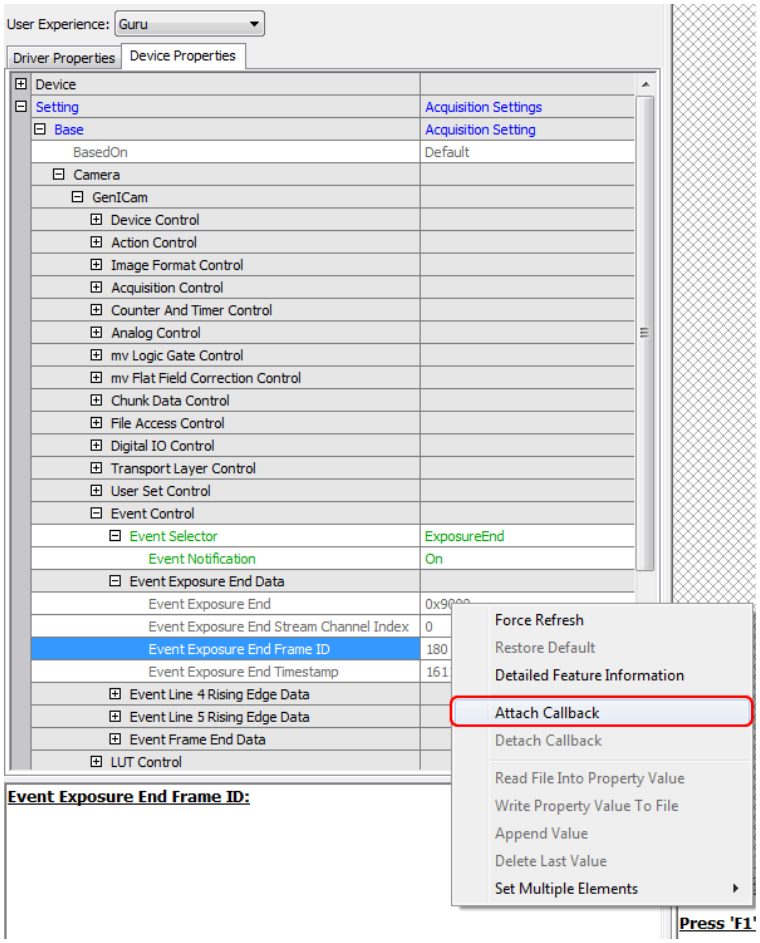


Figure 1: wxPropView - "Attach Callback" to Event Exposure End Frame ID

Now, you can track the property modifications in the output window:

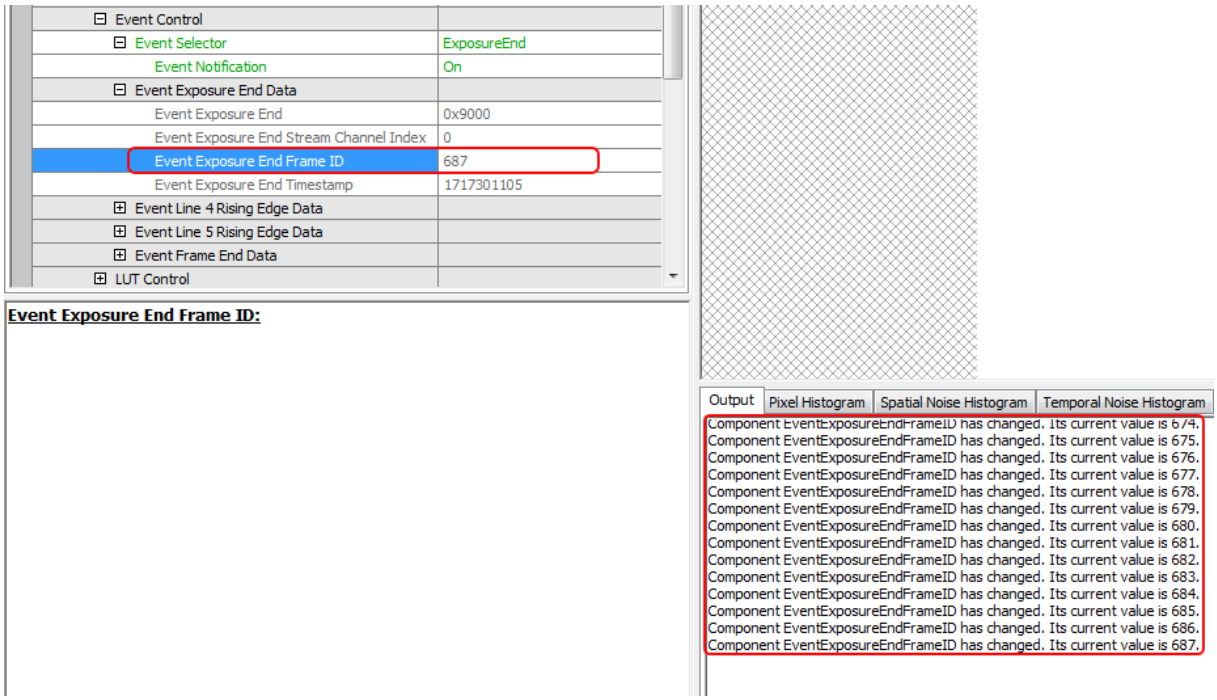


Figure 2: wxPropView - Output window with the Event notifications

You can find a detailed `Callback code example in the C++ API manual`.

1.18.2 Improving the acquisition / image quality

There are several use cases concerning the acquisition / image quality of the camera:

- **Correcting image errors of a sensor** (p. 133)
- **Optimizing the color/luminance fidelity of the camera** (p. 138)
- **Setting a flicker-free auto expose and auto gain** (p. 153)
- **Working with binning / decimation** (p. 157)

1.18.2.1 Correcting image errors of a sensor

1.18.2.1.1 Introduction Due to random process deviations, technical limitations of the sensors, etc. there are different reasons that image sensors have image errors. MATRIX VISION provides several procedures to correct these errors, by default these are host-based calculations, however some camera families support camera-based corrections, which saves dozens of % CPU load and lowers latency.

Camera Family	Algorithm-↩ Based detection and correction	List-Based correction	Storing facility for defective-pixel list	Flat-Field Correction (Host)	Flat-Field Correction (Camera)
mvBlue↩ COUGAR-X	-	-	X	X	X
mvBlue↩ COUGAR-XD	If binning/decimation is on -> no list is stored	X	X	X	
mvBlue↩ COUGAR-XT	X	-	-	X	X
mvBlueFOX3	If binning/decimation is on -> no list is stored	X	X	X	-

Generally, removing defect pixels requires two sub-tasks:

- Detection of defective pixels
- Correction of defective pixels

Both tasks can performed in different "locations":

- Detection and correction on the host using mvIMPACT Acquire
- Detection on the host using mvIMPACT Acquire, correction on the camera using the camera's `mv↩ DefectivePixelCorrectionControl` in the list-based mode
- Detection and correction on the camera using `mvDefectivePixelCorrectionControl` in the algorithm-based mode.

If detection is not happening in real-time, meaning during the image acquisition itself, it is necessary to store the detected defects somewhere. This can be either on the camera or the host or both.

1.18.2.1.2 Defect pixel detection using mvIMPACT Acquire As mentioned, the defect pixel list can be generated using mvIMPACT Acquire. Since there are three types of defects, mvIMPACT Acquire offers three calibration methods for detection:

1. **leaky pixel** (in the dark)
   which indicates pixels that produce a higher read out code than the average
2. **hot pixel** (in standard light conditions)
   which indicates pixels that produce a higher non-proportional read out code when temperatures are rising
3. **cold pixel** (in standard light conditions)
   which indicates pixels that produce a lower read out code than average when the sensor is exposed (e.g. caused by dust particles on the sensor)

Note

Please use either an Mono or RAW Bayer image format when detecting defective pixel data in the image.

1.18.2.1.2.1 Detecting leaky pixels To detect leaky pixels the following steps are necessary:

1. Set gain ("`Setting -> Base -> Camera -> GenICam -> Analog Control -> Gain = 0 dB`") and exposure time "`Setting -> Base -> Camera -> GenICam -> Acquisition Control -> ExposureTime = 360 msec`" to the given operating conditions
   The total number of defective pixels found in the array depend on the gain and the exposure time.
2. Black out the lens completely
3. Set the (Filter-) "`Mode = Calibrate leaky pixel`"
4. Acquire an image (e.g. by pressing **Acquire** in wxPropView with "`Acquisition Mode = Single↩ Frame`")

The filter checks:

```
Pixel > LeakyPixelDeviation_ADCLimit // (default value: 50)
```

All pixels above this value are considered as leaky pixel.

1.18.2.1.2.2 Detecting hot or cold pixels

Note

With "Mode = Calibrate Hot And Cold Pixel" you can execute both detections at the same time.

To detect hot or cold pixels the following steps are necessary:

1. You will need a uniform sensor illumination approx. 50 - 70 % saturation (which means an average gray value between 128 and 180)
2. Set the (Filter-) "`Mode = Calibrate Hot Pixel`" or "`Mode = Calibrate Cold Pixel`" or "`Mode = Calibrate Hot And Cold Pixel`"
3. Acquire an image (e.g. by pressing **Acquire** in wxPropView with "`Acquisition Mode = Single↩ Frame`")

The filter checks:

```
Pixel > T[hot] // (default value: 15 %)

// T[hot] = deviation of the average gray value


Pixel < T[cold] // (default value: 15 %)

// T[cold] = deviation of the average gray value
```

Note

Repeating the defective pixel corrections will accumulate the correction data which leads to a higher value in "`DefectivePixelsFound`". If you want to reset the correction data or repeat the correction process you have to set the filter mode to "`Reset Calibration Data`". In oder to limit the amount of defective pixels detected the "`DefectivePixelsMaxDetectionCount`" property can be used.

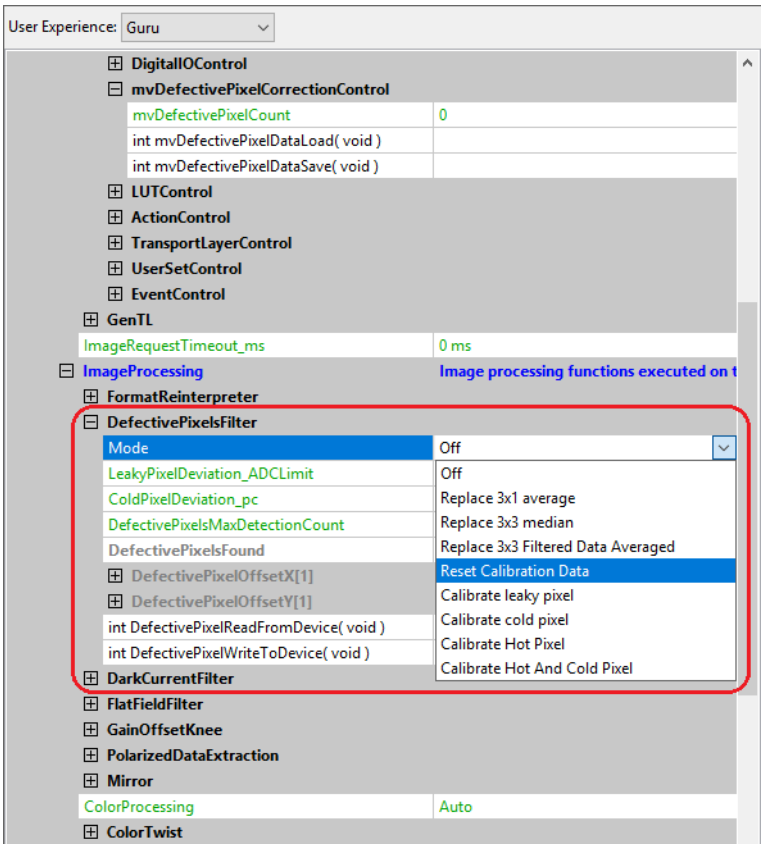


Figure 1: Image corrections: DefectivePixelsFilter

1.18.2.1.3 Storing defective pixel data on the device To save and load the defective pixel data, appropriate functions are available:

- **"int mvDefectivePixelDataLoad(void)"**
- **"int mvDefectivePixelDataSave(void)"**

The section *"Setting -> Base -> ImageProcessing -> DefectivePixelsFilter"* was also extended (see Figure 2). First, the "DefectivePixelsFound" indicates the number of found defective pixels. The coordinates are available through the properties "DefectivePixelOffsetX" and "DefectivePixelOffsetY" now. In addition to that it is possible to edit, add and delete these values manually (via right-click on the "DefectivePixel↩Offset" and select **"Append Value"** or **"Delete Last Value"**). Second, with the functions

- **"int mvDefectivePixelReadFromDevice(void)"**
- **"int mvDefectivePixelWriteToDevice(void)"**

you can exchange the data from the filter with the camera and vice versa.

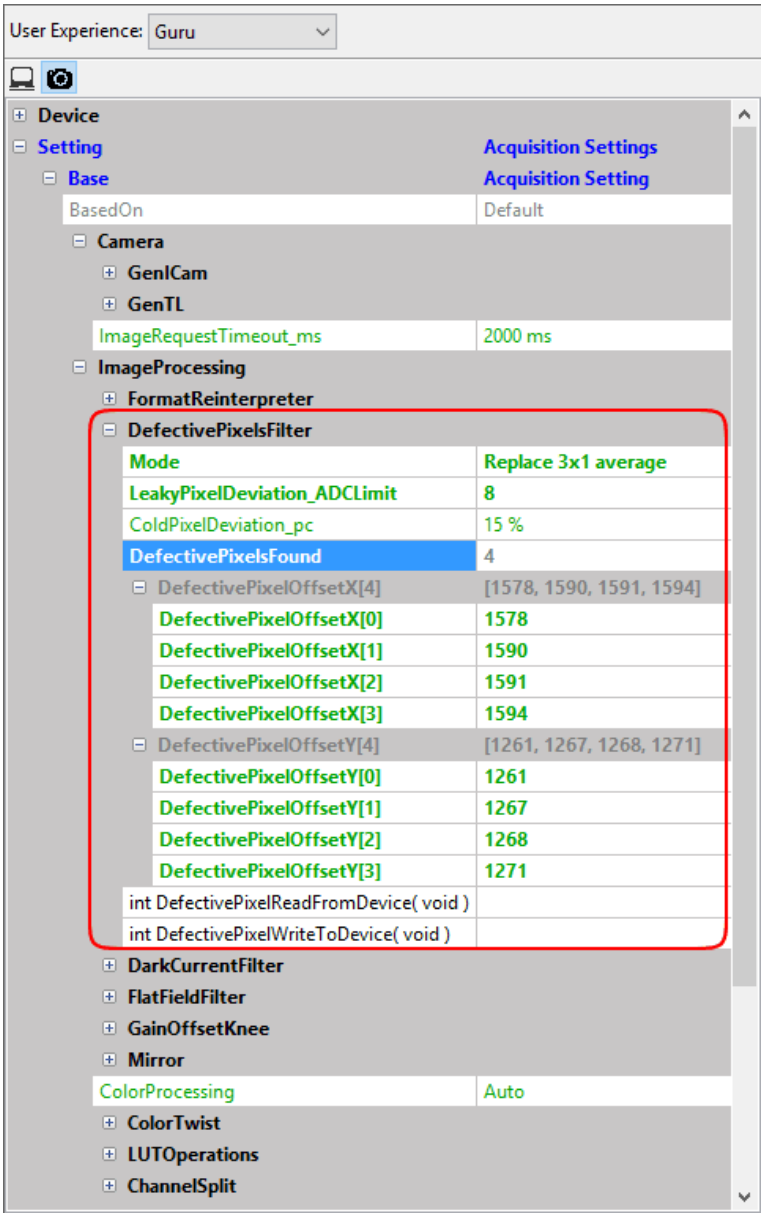


Figure 2: Image corrections: DefectivePixelsFilter (since driver version 2.17.1 and firmware version 2.12.406)

Just right-click on "mvDefectivePixelWriteToDevice" and click on **"Execute"** to write the data to the camera (and hand over the data to the Storing pixel data on the device). To permanently store the data inside the cameras non-volatile memory afterwards "mvDefectivePixelDataSave" must be called as well!

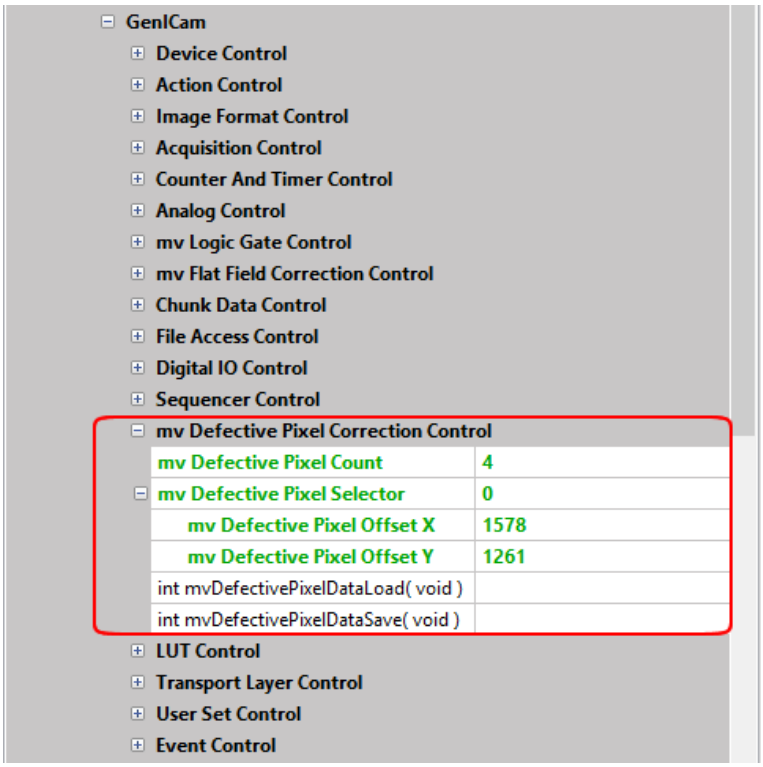


Figure 3: Defective pixel data are written to the camera (since driver version 2.17.1 and firmware version 2.12.406)

While opening the camera, the camera will load the defective pixel data from the camera. If there are pixels in the filter available (via calibration), nevertheless you can load the values from the camera. In this case the values will be merged with the existing ones. I.e., new ones are added and duplicates are removed.

1.18.2.1.4 Defect Pixel correction using mvImpactAcquire After a defect-list is generated, a correction can be performed using mvIMPACT Acquire.

To correct the defective pixels various substitution methods exist:

1. **"Replace 3x1 average"**
   which substitutes the detected defective pixels with the average value from the left and right neighboring pixel (3x1)
2. **"Replace 3x3 median"**
   which substitutes the detected defective pixels with the median value calculated from the nearest neighboring in a 3 by 3 region
3. **"Replace 3x3 Filtered Data Averaged"**
   which substitutes and treats the detected defective pixels as if they have been processed with a 3 by 3 filter algorithm before reaching this filter
   Only recommended for devices which do not offer a defective pixel compensation; packed RGB or packed YUV444 data is needed. See enumeration value `dpfmReplaceDefectivePixelAfter3x3Filter` in the corresponding API manual for additional details about this algorithm and when and why it is needed

1.18.2.1.5 List-based defect Pixel correction on the camera As described before, it is possible to upload lists of defect pixel onto the camera. Different algorithms can be used to determine whether a pixel is defective or not, which is dependent of how much it is allowed a pixel to deviate, temperature, gain, and exposure time. As described before, the list-based correction is deterministic, meaning it is exactly known which pixels will be corrected.

Anyhow, the list-based correction has some disadvantages:

- A default list is stored in the camera during production, but this might to fit to the target application because of much different temperature / exposure time setting
  → It is necessary to create the list using a detection algorithm (or mvIMPACT Acquire support)
- During time and sensor aging, new defects could/will appear
- It doesn't work in binning/decimation modes
- The memory for storing defective pixels is limited

1.18.2.1.6 Adaptive / algorithm-based correction on the camera In this case, the camera performs detection and correction on-the-fly without using any defect-list.

The adaptive correction addresses the above-mentioned disadvantages of the list-based method. While the correction itself (this is which pixels are used to correct an identified defect) is the same, no static information from a list is used, instead they are detected "on the fly".

To use reasonable thresholds, knowledge of the noise statistics of the sensor is used to detect the outliers. These will be corrected also on the fly. Because this is a dynamic approach, it also works in binning/decimation modes and would also detect new appearing defects.

Nevertheless, there are some disadvantages:

- It is non-deterministic
- Wrong positives can be detected, meaning non-defect pixels could be treated as defect
- If pixels are at the edge of the used thresholds, it could be corrected in one frame, but not in the next

On mvBlueFOX3 cameras, the adaptive correction is always used if:

- There is no list stored on the camera
- Binning or decimation is used

1.18.2.2 Optimizing the color/luminance fidelity of the camera

Purpose of this chapter is to optimize the color image of a camera, so that it looks as natural as possible on different displays and for human vision.

This implies some linear and nonlinear operations (e.g. display color space or Gamma viewing LUT) which are normally not necessary or recommended for machine vision algorithms. A standard monitor offers, for example, several display modes like *sRGB*, *"Adobe RGB"*, etc., which reproduce the very same color of a camera color differently.

It should also be noted that users can choose for either

- camera based settings and adjustments or
- host based settings and adjustments or
- a combination of both.

Camera based settings are advantageous to achieve highest calculating precision, independent of the transmission bit depth, lowest latency, because all calculations are performed in FPGA on the fly and low CPU load, because the host is not invoked with these tasks. These camera based settings are

- **gamma correction** (p. 141)
- **negative gain / gain** (p. 141)
- **look-up table (LUT)** (p. 141)
- **white balance** (p. 146)
- **offset** (p. 148)
- **saturation and color correction** (p. 150)

Host based settings save transmission bandwidth at the expense of accuracy or latency and CPU load. Especially performing gain, offset, and white balance in the camera while outputting RAW data to the host can be recommended.

Of course host based settings can be used with all families of cameras (e.g. also mvBlueFOX).

Host based settings are:

- look-up table (*LUTOperations*)
- color correction (*ColorTwist*)

To show the different color behaviors, we take a color chart as a starting point:

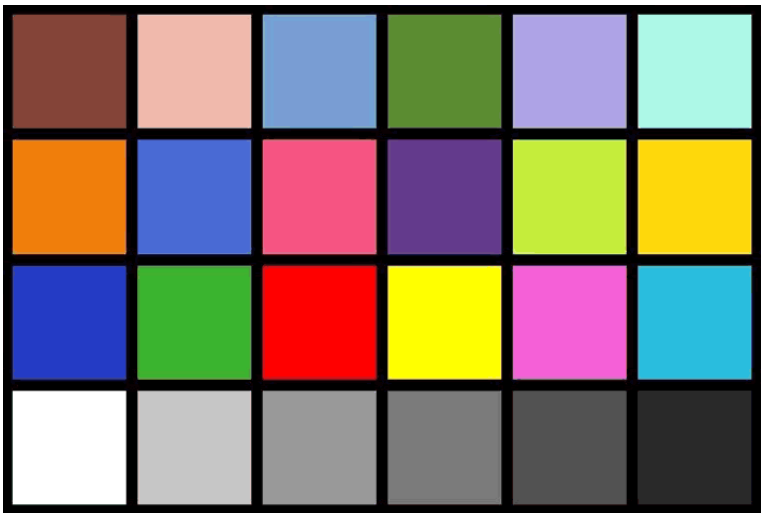


Figure 1: Color chart as a starting point

If we take a *SingleFrame* image without any color optimizations, an image can be like this:



Figure 2: SingleFrame snap without color optimization



Figure 3: Corresponding histogram of the horizontal white to black profile

As you can see,

- saturation is missing,
- white is more light gray,
- black is more dark gray,
- etc.

Note

You have to keep in mind that there are two types of images: the one generated in the camera and the other one displayed on the computer monitor. Up-to-date monitors offer different display modes with different color spaces (e.g. sRGB). According to the chosen color space, the display of the colors is different.

The following figure shows the way to a perfect colored image

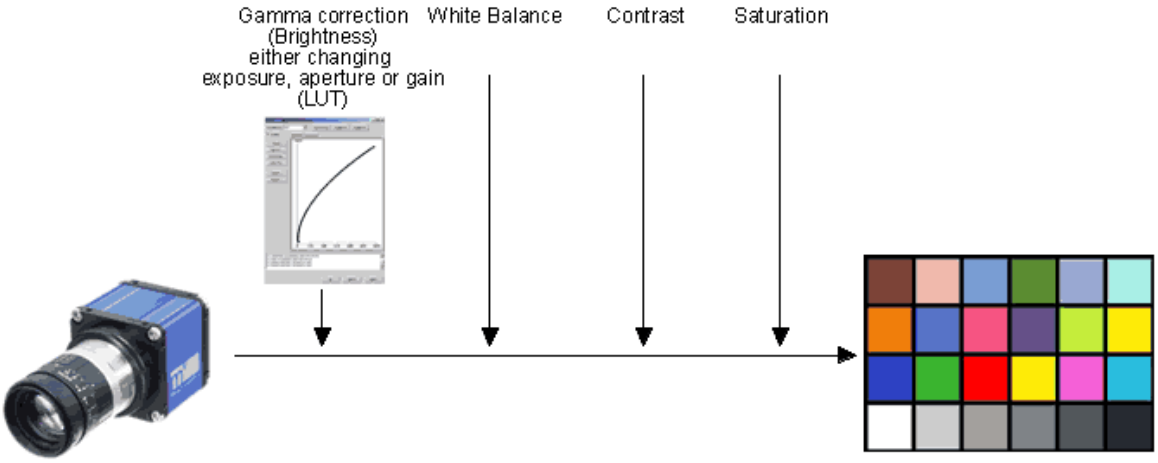


Figure 4: The way to a perfect colored image

including these process steps:

1. Do a **Gamma correction (Luminance)** (p. 141),
2. make a **White balance** (p. 146) and
3. Improve the **Contrast** (p. 148).
4. Improve **Saturation** (p. 150), and use a **"color correction matrix"** for both
   (a) the sensor and / or
   (b) the monitor.

The following sections will describe the single steps in detail.

1.18.2.2.1 Step 1: Gamma correction (Luminance)

First of all, a **Gamma correction** can be performed to change the image in a way how humans perceive light and color.

To do this, you have the following 3 possibilities:

- set Gamma using Gamma feature in AnalogControl,
- set Gamma using a LUT in LUTControl or
- set Gamma using a LUT in ImageProcessing.

Note

The first two possibilities are running on the camera, whereas the third is running on host system

The first and easiest way to set a Gamma is using the gamma feature in AnalogControl (*"Setting -> Base -> Camera -> AnalogControl"*)

mvGammaSelector is used to switch between a fixed mvSRGB gamma curve and mvuser which enables the Gamma property to set gamma between a value of 0.1 and 10.0 The Gamma feature is working on the camera and enabled by setting mvGammaEnable

Note

mvGammaEnable can only be set if no LUT in LUTControl is active

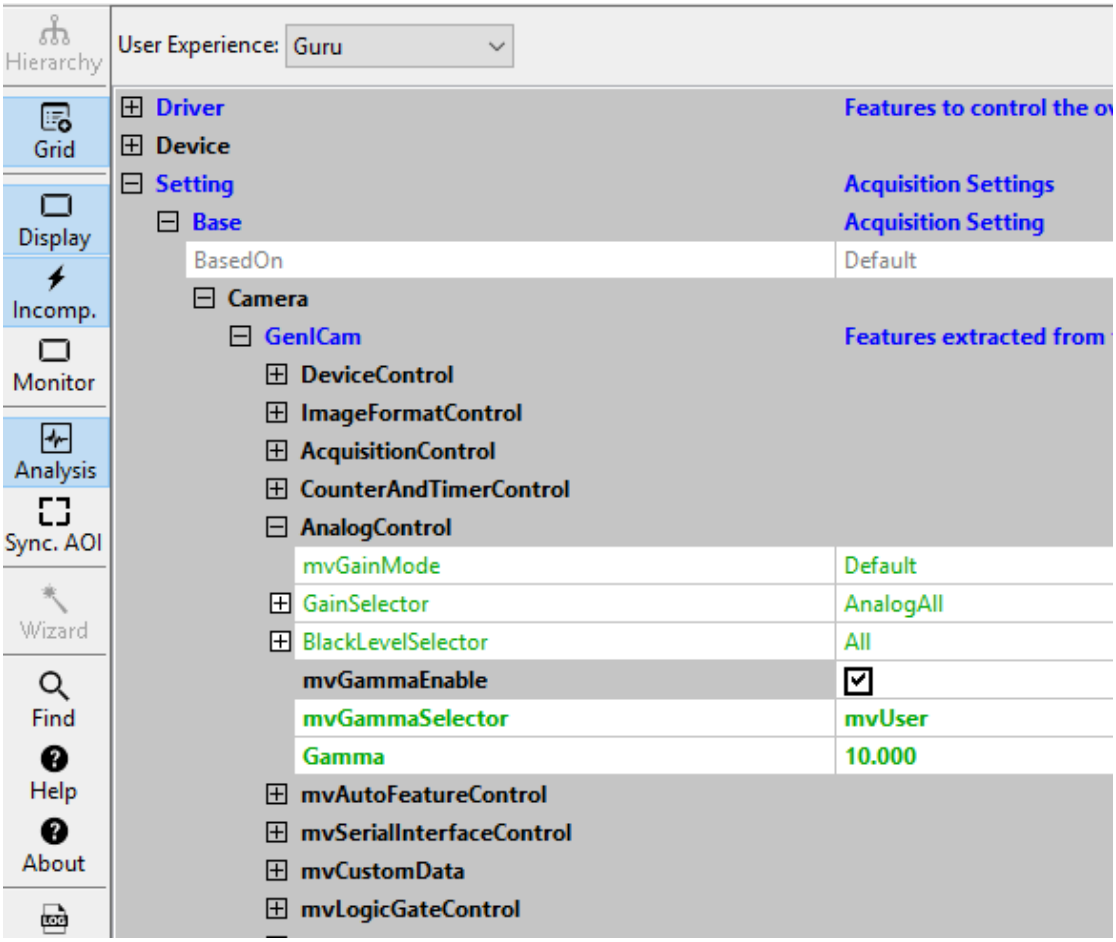


Figure 5: Gamma using AnalogControl

Note

To avoid artifacts while changing the Gamma value, stop Acquisition before changing values and afterwards, start Acquisition

The second way is setting Gamma via LUTControl using **wxPropView** (p. 65), here, the following steps have to be done:

1. Click on *"Setting -> Base -> Camera -> GenICam -> LUT Control -> LUT Selector"*.
2. Afterwards, click on **"Wizard"** to start the LUT Control wizard tool.
   The wizard will load the data from the camera.

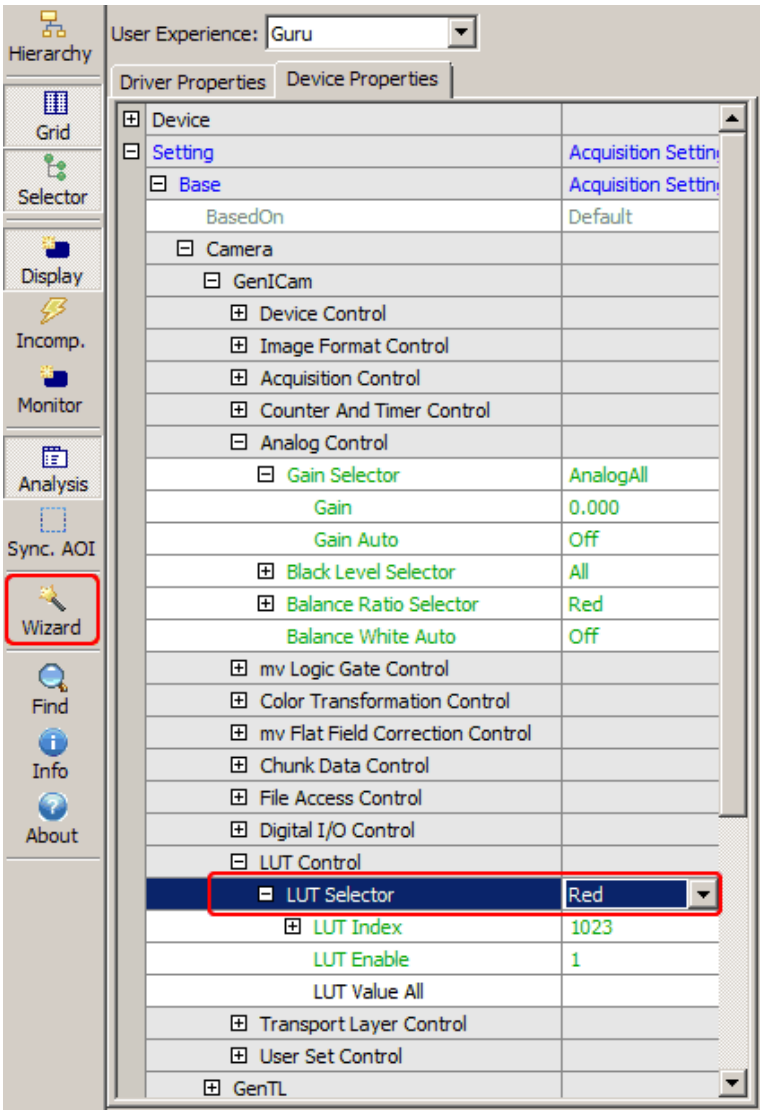


Figure 6: Selected LUT Selector and click on wizard will start wizard tool

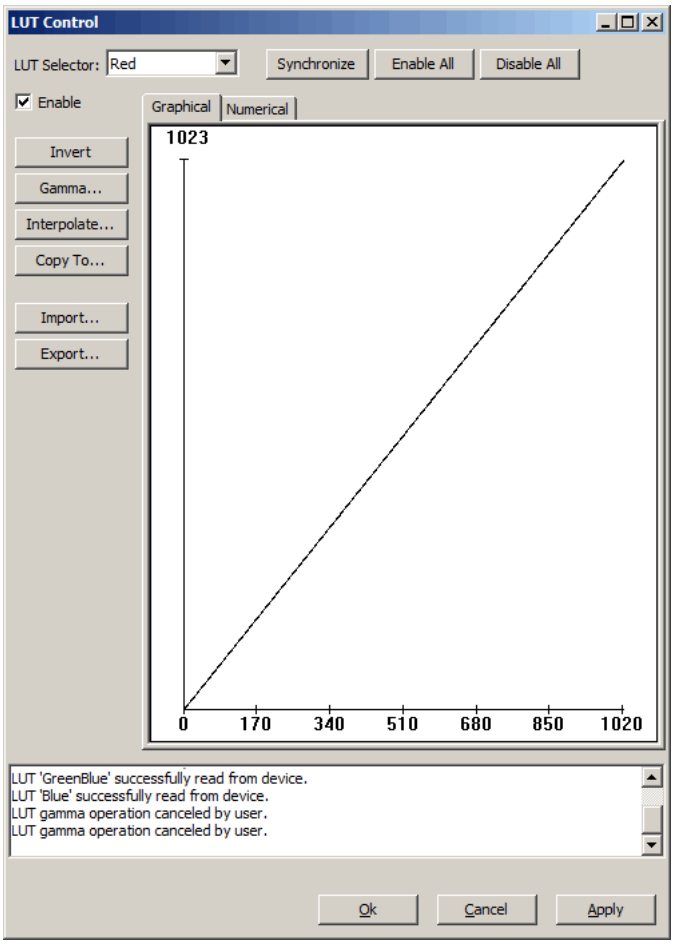


Figure 7: LUT Control

3. Now, click on the **"Gamma..."** button
4. and enter e.g. *"2.2"* as the Gamma value:

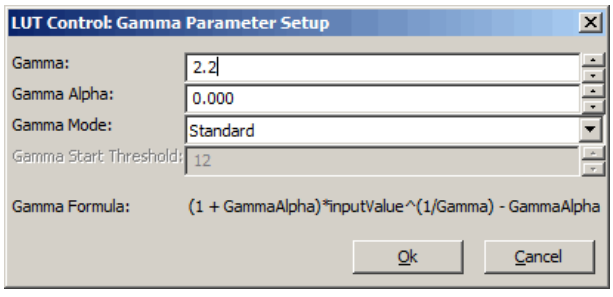


Figure 8: Gamma Parameter Setup

5. Then, click on **"Copy to..."** and select *"All"* and
6. and click on **"Enable All"**.
7. Finally, click on **Synchronize** and play the settings back to the device (via *"Cache -> Device"*).

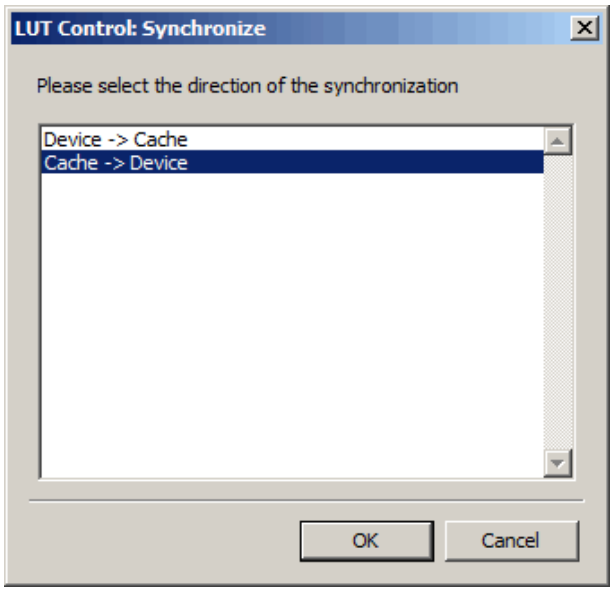


Figure 9: Synchronize

After gamma correction, the image will look like this:

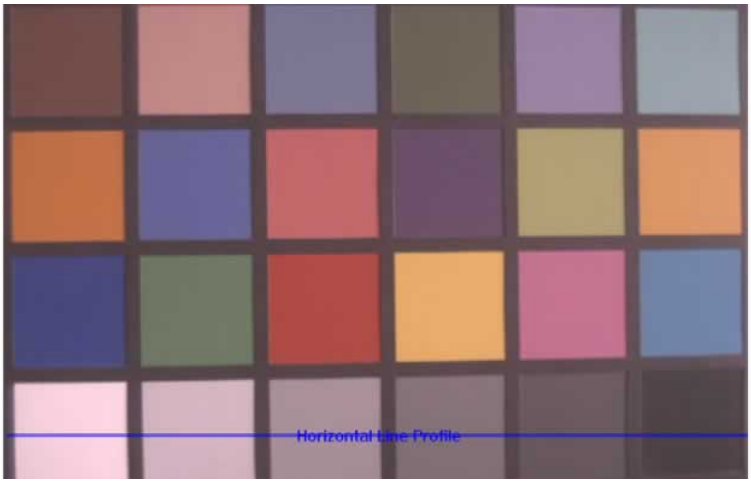


Figure 10: After gamma correction



Figure 11: Corresponding histogram after gamma correction

The third way to set Gamma, is gamma correction via (*"Setting -> Base -> ImageProcessing -> LUTControl"*). Here, the changes will affect the 8 bit image data and the processing is running on the CPU of the host system:

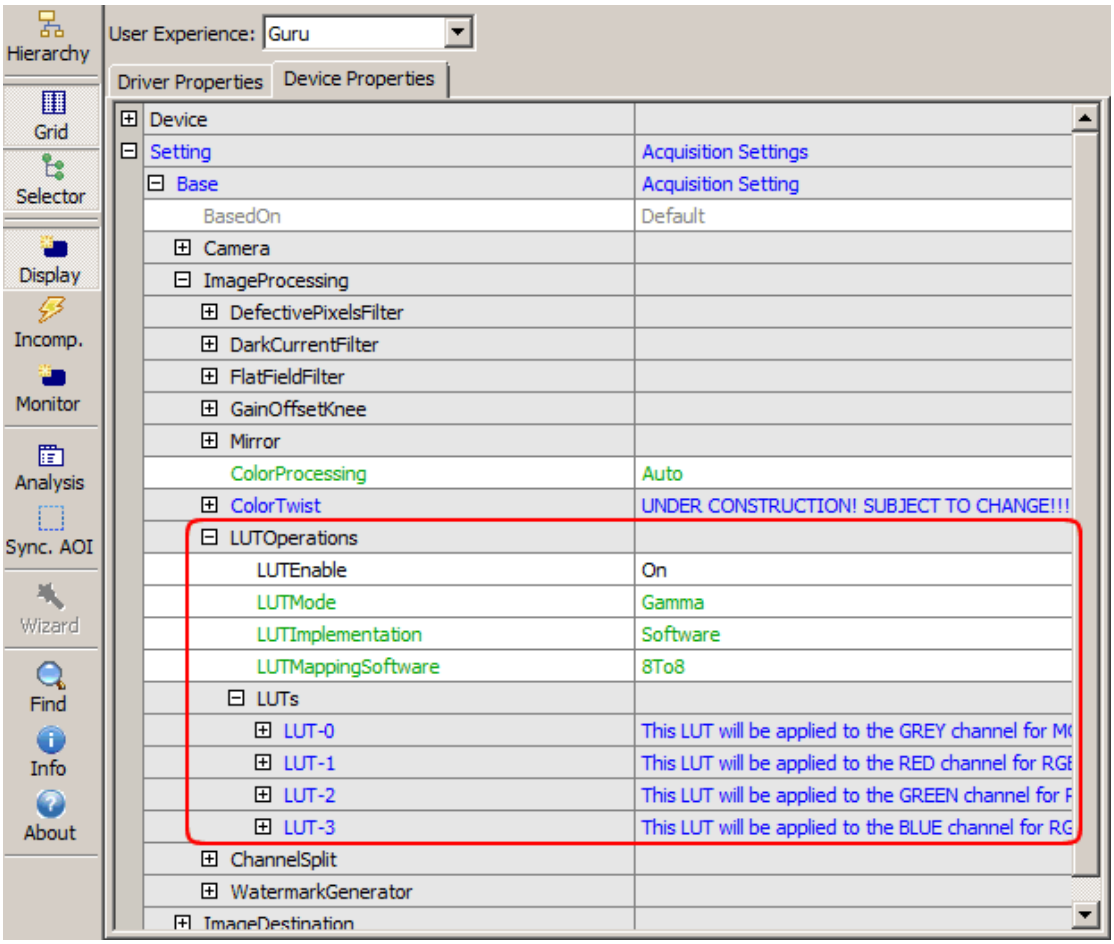


Figure 12: LUTControl dialog

Just set *"LUTEnable"* to **"On"** and adapt the single LUTs like (LUT-0, LUT-1, etc.).

1.18.2.2.2 Step 2: White Balance

As you can see in the histogram, the colors red and blue are above green. Using green as a reference, we can optimize the white balance via *"Setting -> Base -> Camera -> GenICam -> Analog Control -> Balance Ratio Selector"* (*"Balance White Auto"* has to be *"Off"*):

1. Just select *"Blue"* and
2. adjust the *"Balance Ratio"* value until the blue line reaches the green one.

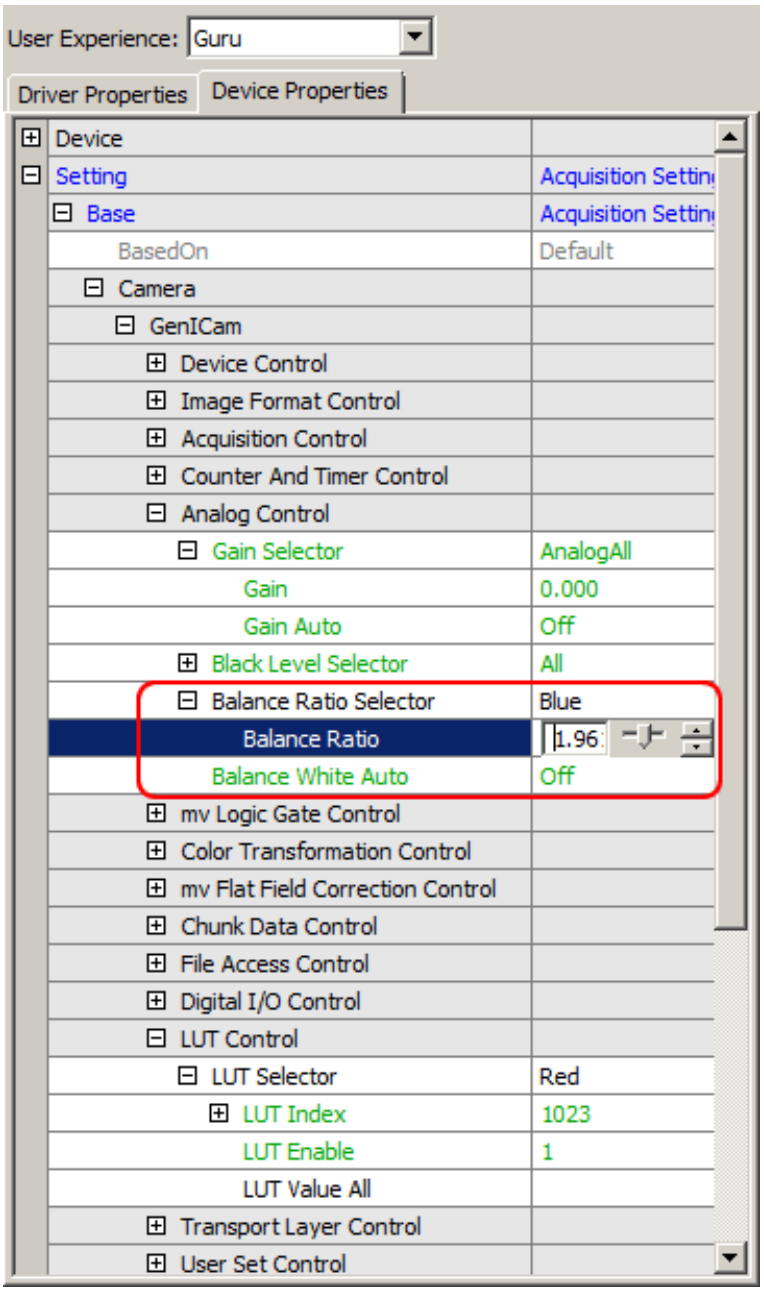


Figure 13: Optimizing white balance

3. Repeat this for *"Red"*.

After optimizing white balance, the image will look like this:

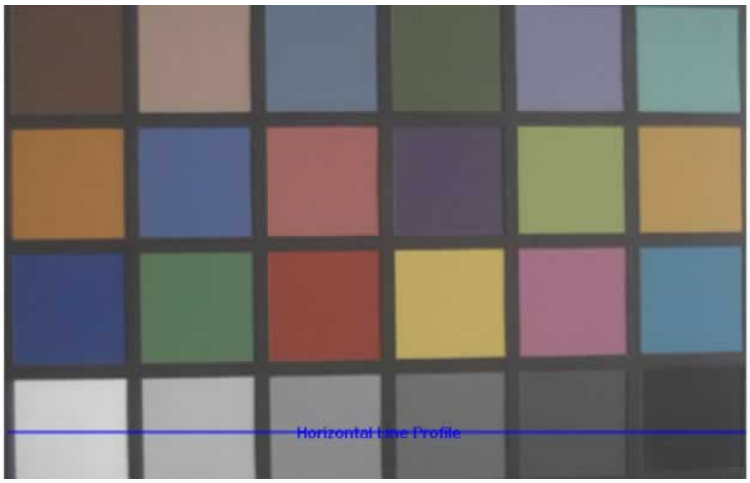


Figure 14: After white balance

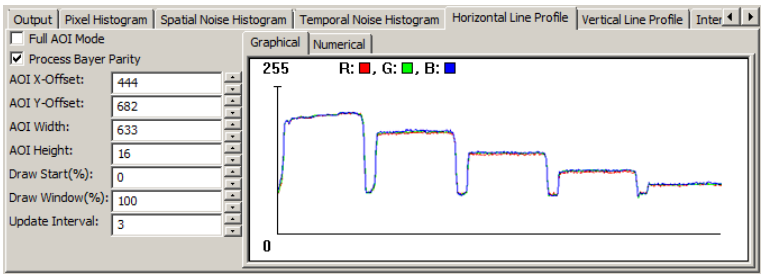


Figure 15: Corresponding histogram after white balance

1.18.2.2.3 Step 3: Contrast

Still, black is more a darker gray. To optimize the contrast you can use *"Setting -> Base -> Camera -> GenICam -> Analog Control -> Black Level Selector"*:

1. Select *"DigitalAll"* and
2. adjust the *"Black Level"* value until black seems to be black.

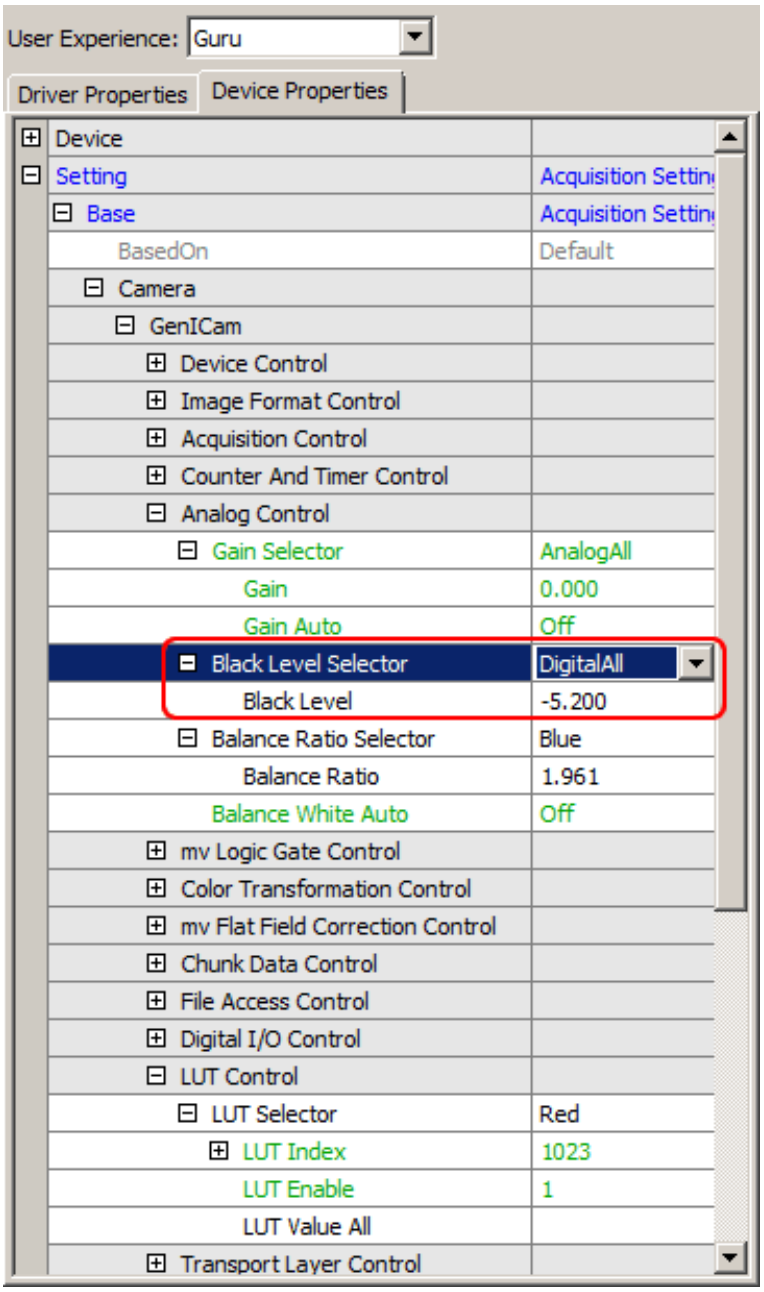


Figure 16: Back level adjustment

The image will look like this now:

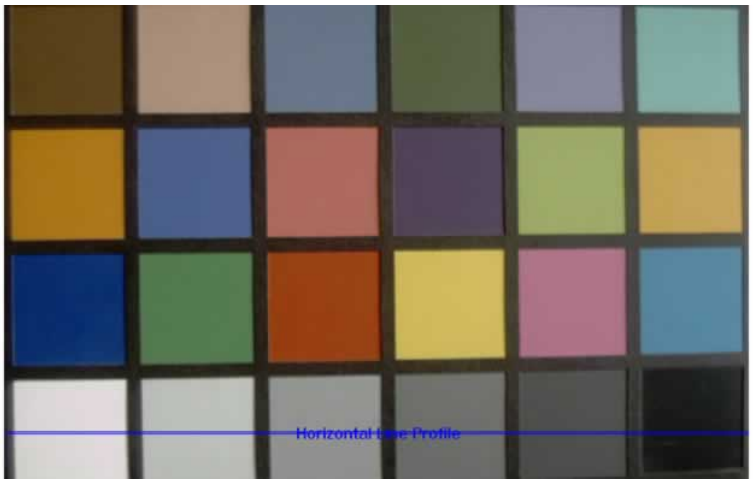


Figure 17: After adapting contrast



Figure 18: Corresponding histogram after adapting contrast

1.18.2.2.4 Step 4: Saturation and Color Correction Matrix (CCM)

Still saturation is missing. To change this, the **"Color Transformation Control"** can be used (*"Setting -> Base -> Camera -> GenICam -> Color Transformation Control"*).

Note

In our optical laboratory we determine the best correction data for the sRGB color space for each sensor type and these parameters are shipped with our cameras. If you want to use these parameters, you have to activate them using the wizard of the Color Transformation Control.

1. Click on **"Color Transformation Enable"** and
2. click on **"Wizard"** to start the saturation via *"Color Transformation Control"* wizard tool (since firmware version 1.4.57).

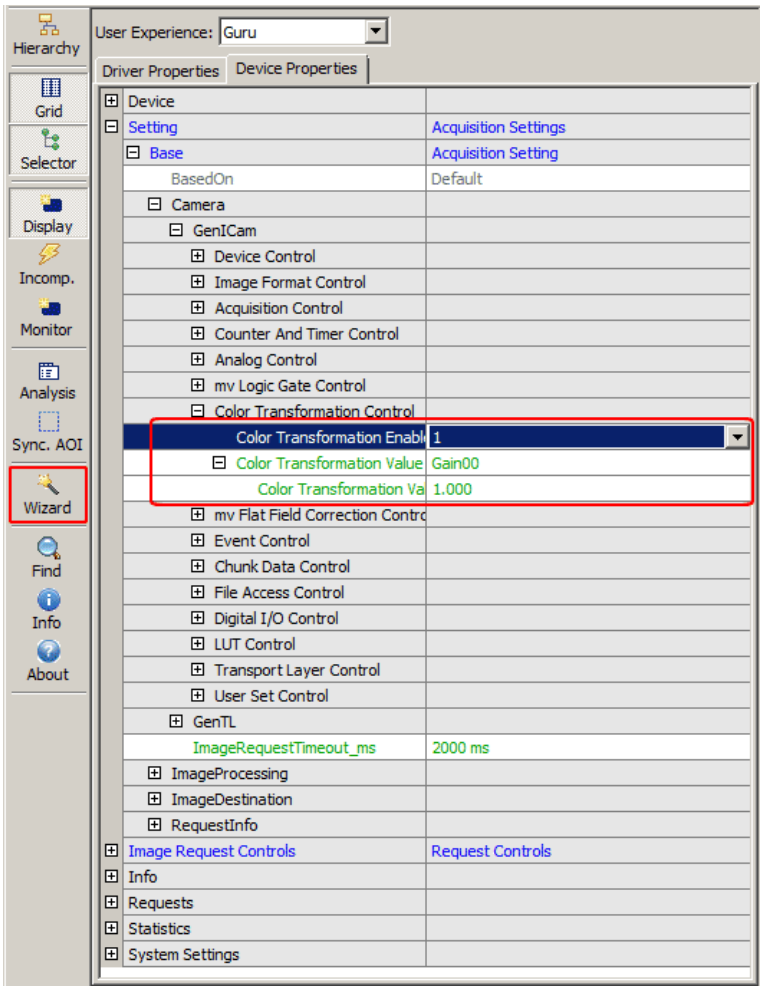


Figure 19: Selected Color Transformation Enable and click on wizard will start wizard tool

3. Now, you can adjust the saturation e.g. *"1.1"*.

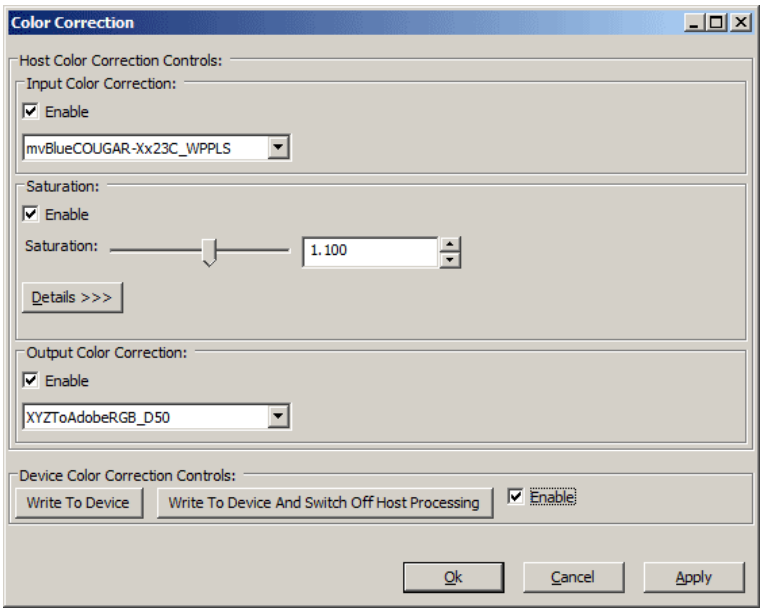


Figure 20: Saturation Via Color Transformation Control dialog

4. Afterwards, click on **"Enable"**.

5. Since driver version 2.2.2, it is possible to set the special color correction matrices at
   (a) the input (sensor),
   (b) the output side (monitor) and
   (c) the saturation itself using this wizard.
6. Select the specific input and output matrix and
7. click on **"Enable"**.
8. As you can see, the correction is done by the host by default (*"Host Color Correction Controls"*). However, you can decide, if the color correction is done by the device by clicking on **"Write To Device And Switch Off Host Processing"**. The wizard will take the settings of the *"Host Color Correction Controls"* and will save it in the device.
9. Finally, click on **"Apply"**.

After the saturation, the image will look like this:

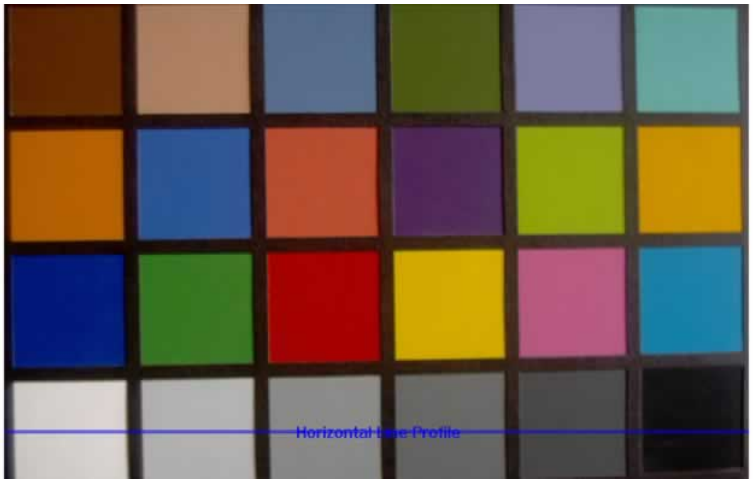


Figure 21: After adapting saturation

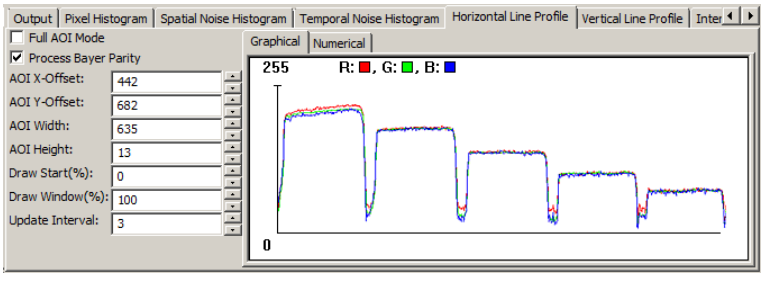


Figure 22: Corresponding histogram after adapting saturation

Note

As mentioned above, you can change the saturation and the color correction matrices via (*"Setting -> Base -> ImageProcessing -> ColorTwist"*). Here, the changes will affect the 8 bit image data and the processing needs the CPU of the host system:

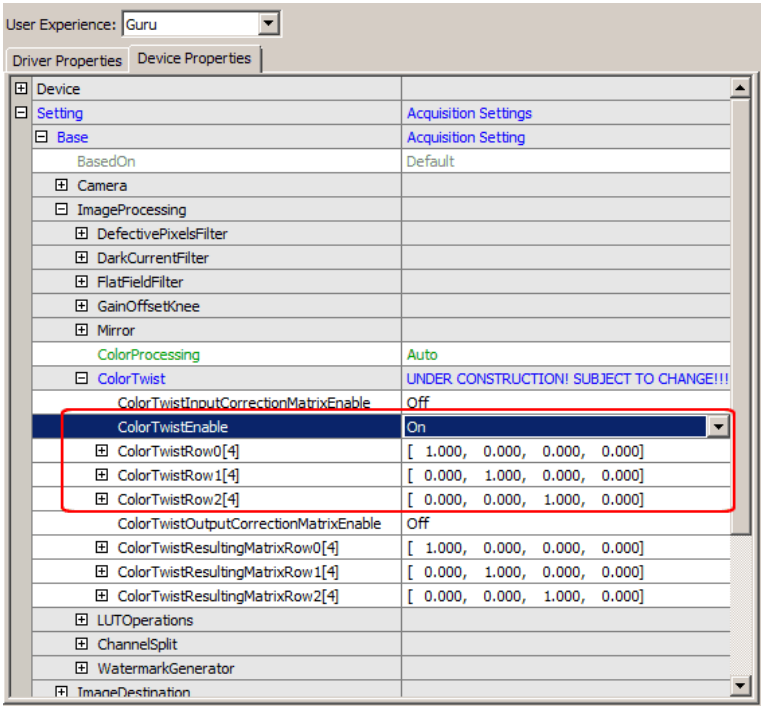


Figure 23: ColorTwist dialog

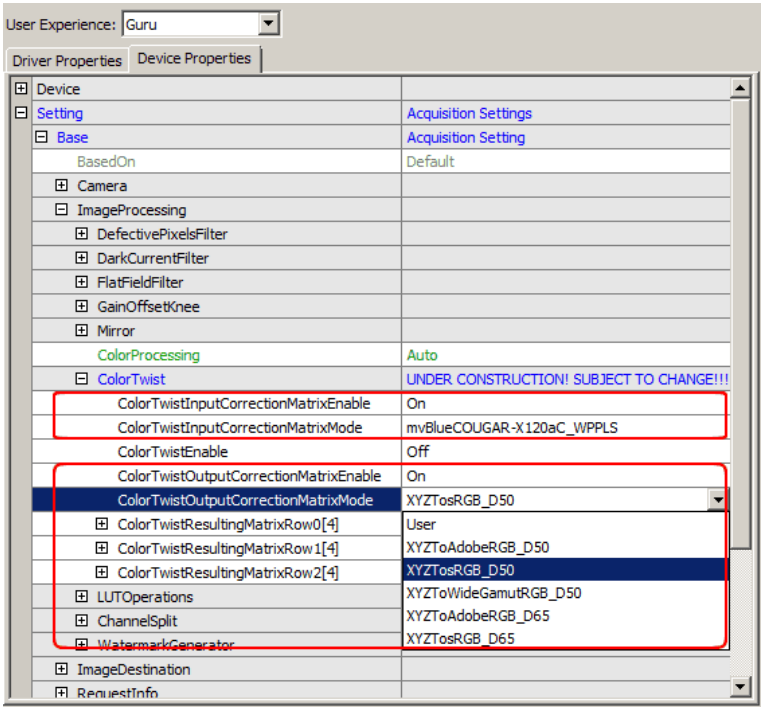


Figure 24: Input and output color correction matrix

1.18.2.3 Setting a flicker-free auto expose and auto gain

1.18.2.3.1 Introduction

In order to prevent oscillations it is important to adapt the camera frequency to the frequency of AC light.

This is, for example, in

- Europe 50 cycles (100 fluctuations/s) whereas in
- USA, Japan and other countries it is 60 Hz.

This means the camera must strictly be coupled to this frequency. In conjunction with auto exposure this can only be maintained by using a timer based generation of external trigger pulses. This is a behavior of both sensor types: CCD and CMOS.

Note

It is not enough to use *"Setting -> Base -> Camera -> GenICam -> Acquisition Control -> Acquisition Frame Rate"* for this, as there are small fluctuations in this frame rate if the exposure time changes. These fluctuations lead to oscillations (see settings marked with red boxes in Figure 1). The *"Acquisition Frame Rate"* will only provide the exact frame rate if auto exposure is turned off.

As shown in Figure 1, it is possible to set (*"mv Exposure Auto Mode"*) which part of the camera handles the auto expose (device or the sensor itself; pink boxes). Using *"mvSensor"* as *"mv Exposure Auto Mode"*, it is possible to avoid oscillations in some cases. The reason for this behavior is that you can set more parameters like *"mv Exposure Auto Delay Images"* in contrast to *"mvDevice"*. However, as mentioned above it is recommended to use a timer based trigger when using auto expose together with continuous acquisition.

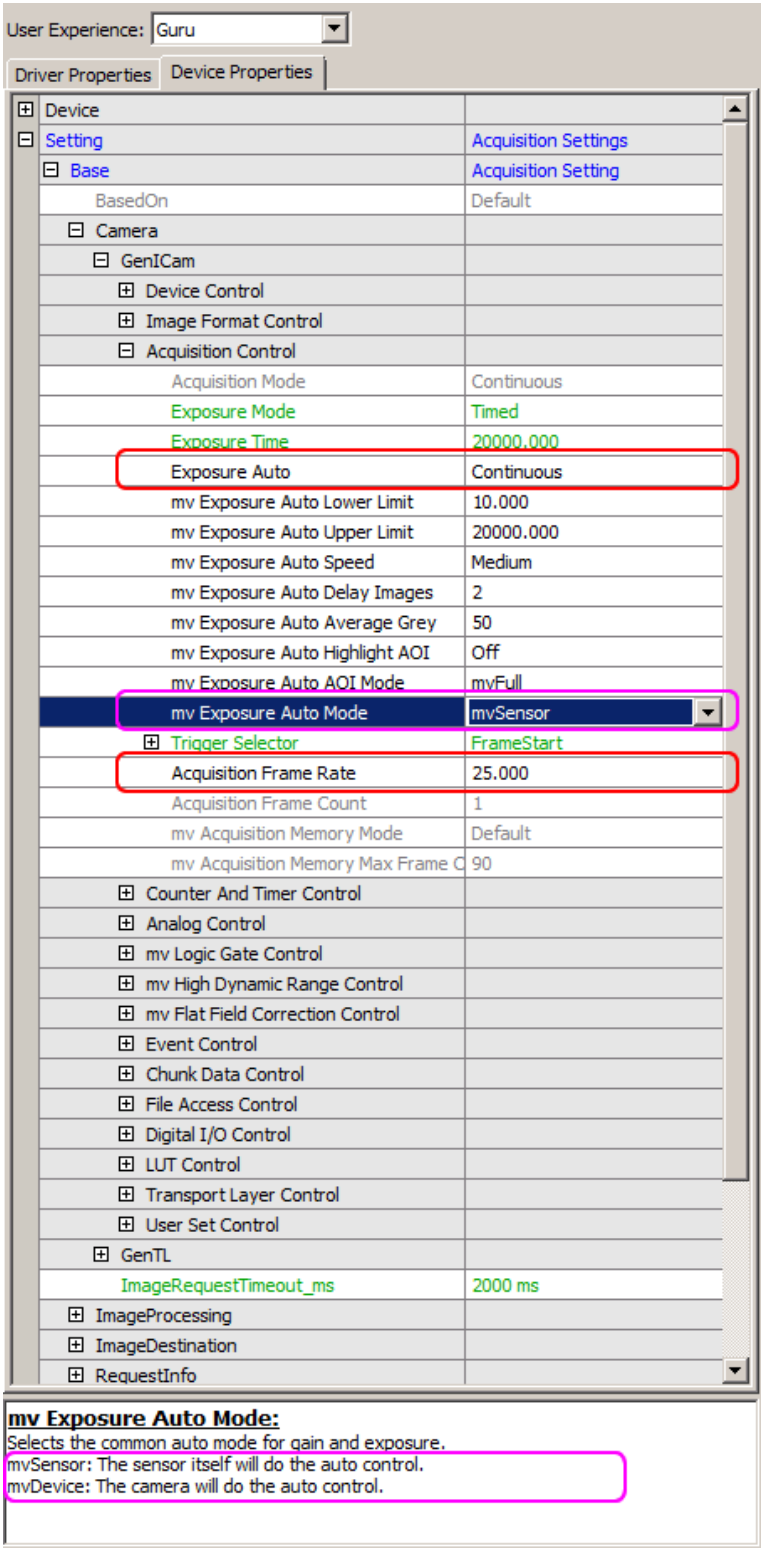


Figure 1: wxPropView - Auto expose is turned on and the frame rate is set to 25 fps

1.18.2.3.2 Example of using a timer for external trigger

Figure 2 shows how to generate a 25 Hz signal, which triggers the camera:

- *"Setting -> Base -> Camera -> GenICam -> Counter & Timer Control -> Timer Selector -> Timer 1"*:

  - *"Timer Trigger Source" = "Timer1End"*
  - *"Timer Duration" = "40000"*

$$FPS_{max} = \frac{1}{40000\ us} = 25$$

- *"Setting -> Base -> Camera -> GenICam -> Acquisition Control -> Trigger Selector -> FrameStart"*:
  - *"Trigger Mode" = "On"*
  - *"Trigger Source" = "Timer1End"*

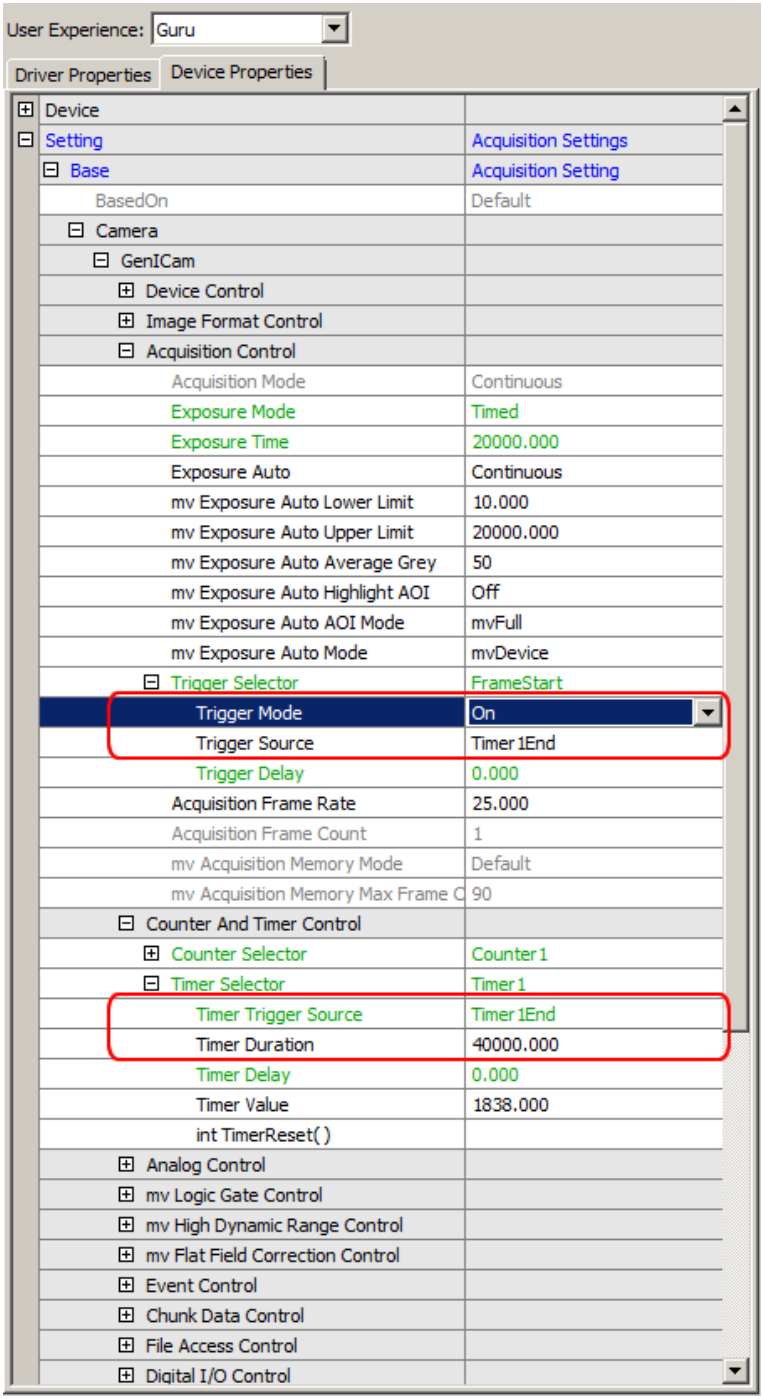


Figure 2: wxPropView - 25 Hz timer for external trigger

No oscillation occurs, regardless of DC ambient vs. AC indoor light.

This operation mode is known as flicker-free or flicker-less operation.

What it mainly does is to adjust the frame frequency to precisely the frequency of the power line. Usually the line frequency is very stable and therefore is the harmonic frequency difference of the two signals are very slow; probably in the range of < 0.1 Hz.

The fact that we do not know the phase relation between the two frequencies means that we scan the alternating ambient light source with our camera. The shorter the exposure time, the more we see a slow change in brightness.

Using AutoExposure/AutoGain can completely eliminate this change because the frequency of change is very low. That means it will be legal if we calculate a brightness difference in one picture and apply it for the next one, because the change is also valid in the next one; as we fulfill the Nyquist theorem.

If we use an arbitrary scanning frequency like 20 fps or whatever your algorithm and data flow is accepting, is wrong in this aspect and leads to oscillations and undesired flicker.

Pointing to a 60 Hz display with flashing backlight an oscillation of 10 Hz can be seen of course.

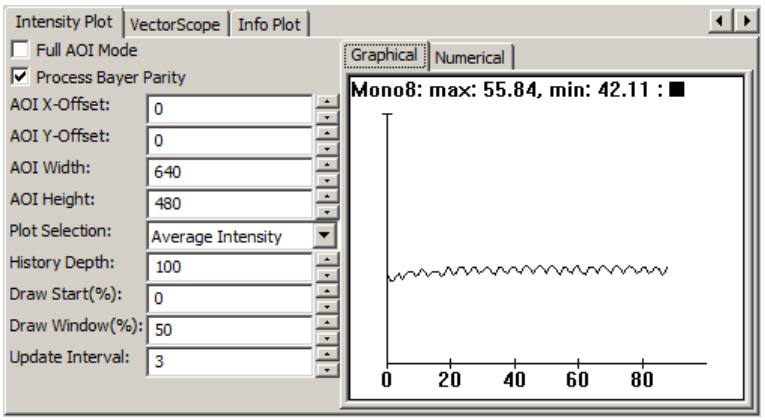


Figure 3: wxPropView - Intensity plot while pointing the camera to a 60 Hz display

1.18.2.3.3 Conclusion

To avoid oscillations, it is necessary to adapt the camera frequency to the frequency of AC light. When using auto expose a flicker-free mode (timer based external trigger) is needed. If the camera is used throughout the world it is necessary that the frequency of AC light can be set in the software and the software adapts the camera to this specific environment.

1.18.2.4 Working with binning / decimation

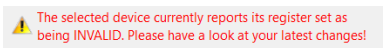
With binning / decimation it is possible to combine / reduce adjacent pixels vertically and/or horizontally. Depending on the sensor, up to 16 adjacent pixels can be combined / reduced.

See also

`https://www.matrix-vision.com/manuals/SDK_CPP/Binning_modes.png`

To optimize the sensor speed, the firmware will check if the binning / decimation functionality is provided by the sensor. If it is available it will use the sensor for binning / decimation, if not, binning / decimation will be processed in the camera's FPGA.

If a binning / decimation combination is not possible, the following note will be displayed:



In this case no acquired images will be displayed until you have changed the settings.

1.18.2.4.1 Binning

Possible binning modes are:

- **Sum:** The response from the combined pixels will be added, resulting in increased sensitivity.
- **Average:** The response from the combined pixels will be averaged, resulting in increased signal/noise ratio.

Binning can be used to lighten the image at the expense of the resolution. This is a neat solution for applications with low light and low noise.

The following results were achieved with the mvBlueFOX3-2124G:

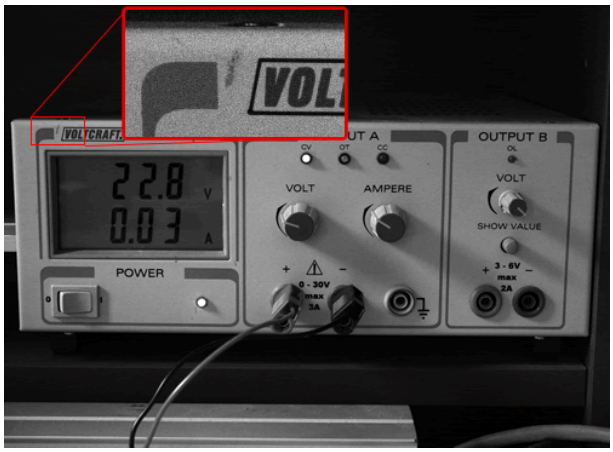
Exposure [in us]	Binning	Gain [in dB]	Averager
2500	-	0	-



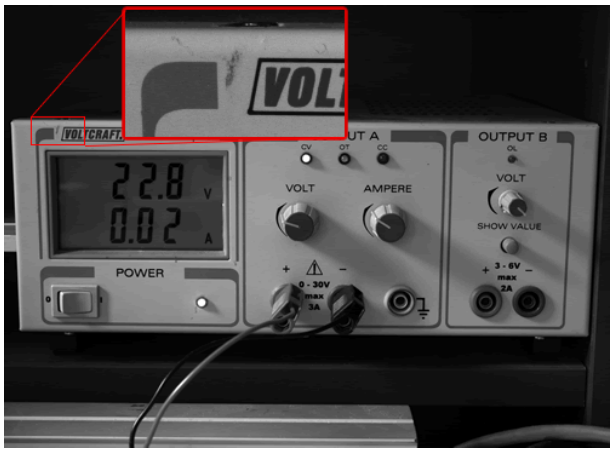
Exposure [in us]	Binning	Gain [in dB]	Averager
2500	-	30	-



Exposure [in us]	Binning	Gain [in dB]	Averager
2500	2H 2V	30	-



Exposure [in us]	Binning	Gain [in dB]	Averager
2500	2H 2V	30	Averaging using 24 frames



1.18.2.4.2 Decimation

Possible decimation modes are:

- **Discard:** The value of every Nth pixel is kept, others are discarded.
- **Average:** The value of a group of N adjacents pixels are averaged.

1.18.3 Working with triggers

There are several use cases concerning trigger:

- **Processing triggers from an incremental encoder** (p. 160)
- **Generating a pulse width modulation (PWM)** (p. 162)
- **Outputting a pulse at every other external trigger** (p. 164)
- **Creating different exposure times for consecutive images** (p. 166)
- **Detecting overtriggering** (p. 169)
- **Triggering of an indefinite sequence with precise starting time** (p. 173)
- **Low latency triggering** (p. 176)

1.18.3.1 Processing triggers from an incremental encoder

1.18.3.1.1 Basics

The following figure shows the principle of an incremental encoder:

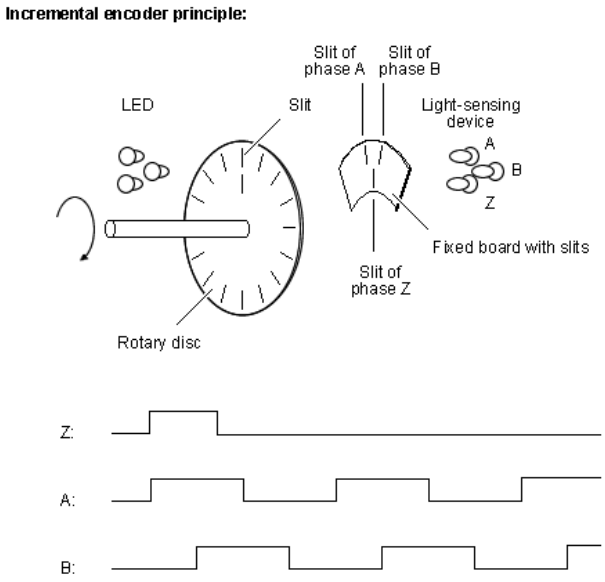


Figure 1: Principle of an incremental encoder

This incremental encoder will send a A, B, and Z pulse. With these pulses there are several ways to synchronize an image with an incremental encoder.

1.18.3.1.2 Using Counter

It is also possible to use **Counter** and **CounterEnd** as the trigger event for synchronizing images with an incremental encoder

To create an external trigger event by an incremental encoder, please follow these steps:

1. Connect the incremental encoder output signal A, for example, to the digital input 0 (**"Line4"**) of the mv↩ BlueCOUGAR-X .
   This line counts the forward pulses of the incremental encoder.
2. Set *"Setting -> Base -> Camera -> GenICam -> Counter and Timer Control -> Counter Selector"* to **"↩ Counter1"** and
3. **"Counter Event Source"** to **"Line4"** to count the number of pulses e.g. as per revolution (e.g. **"Counter Duration"** to **3600**).
4. Then set the trigger *"Setting -> Base -> Camera -> GenICam -> Acquisition Control -> Trigger Selector"* **"FrameStart"** to the **"Counter1End"** (*"Trigger Source"*) signal.

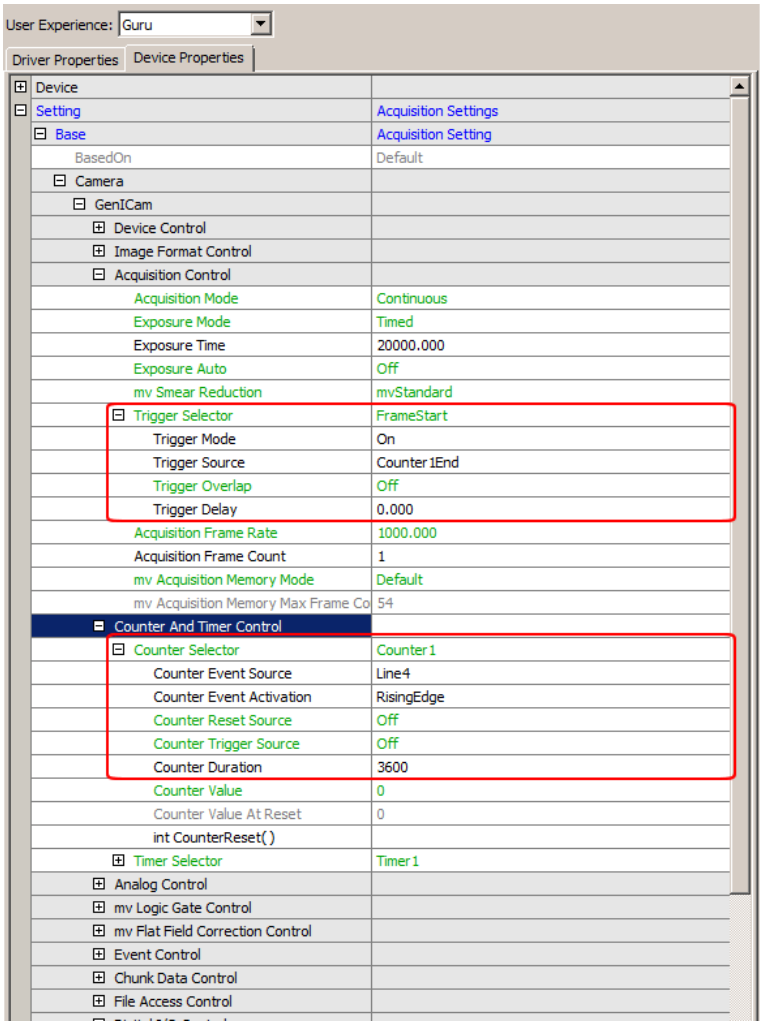


Figure 3: wxPropView setting

To reset **"Counter1"** at Zero degrees, you can connect the digital input 1 (**"Line5"**) to the encoder Z signal.

1.18.3.2 Generating a pulse width modulation (PWM)

1.18.3.2.1 Basics

To dim a laser line generator, for example, you have to generate a pulse width modulation (PWM).

For this, you will need

- 2 timers and
- the active signal of the second timer at an output line

1.18.3.2.2 Programming the pulse width modulation

You will need two timers and you have to set a trigger.

- *Timer1* defines the interval between two triggers.
- *Timer2* generates the trigger pulse at the end of *Timer1*.

The following sample shows a trigger

- which is generated every second and
- the pulse width is 10 ms:

```
#include <mvIMPACT_CPP/mvIMPACT_acquire.h>
#include <mvIMPACT_CPP/mvIMPACT_acquire_GenICam.h>

...

// Master: Set timers to trig image: Start after queue is filled
    GenICam::CounterAndTimerControl catcMaster(pDev);
    catcMaster.timerSelector.writeS( "Timer1" );
    catcMaster.timerDelay.write( 0. );
    catcMaster.timerDuration.write( 1000000. );
    catcMaster.timerTriggerSource.writeS( "Timer1End" );

    catcMaster.timerSelector.writeS( "Timer2" );
    catcMaster.timerDelay.write( 0. );
    catcMaster.timerDuration.write( 10000. );
    catcMaster.timerTriggerSource.writeS( "Timer1End" );
```

See also

Counter And Timer Control (p. 73)

Note

Make sure the *Timer1* interval must be larger than the processing time. Otherwise, the images are lost.

Now, the two timers will work like the following figure illustrates, which means

- *Timer1* is the trigger event and
- *Timer2* the trigger pulse width:

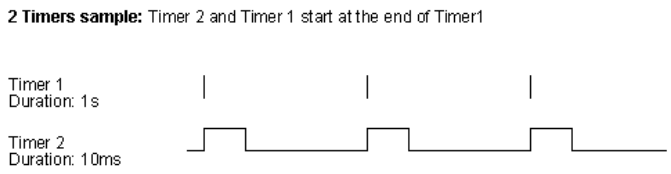


Figure 1: Timers

The timers are defined, now you have to set the digital output, e.g. **"Line 0"**:

```
// Set Digital I/O
    GenICam::DigitalIOControl io(pDev);
    io.lineSelector.writeS( "Line0" );
    io.lineSource.writeS( "Timer2Active" );
```

See also

Digital I/O Control (p. 80)

This signal has to be connected with the digital inputs of the application.

1.18.3.2.3 Programming the pulse width modulation with wxPropView

The following figures show, how you can set the timers using the GUI tool **wxPropView** (p. 65)

1. Setting of *Timer1* (blue box) on the master camera:

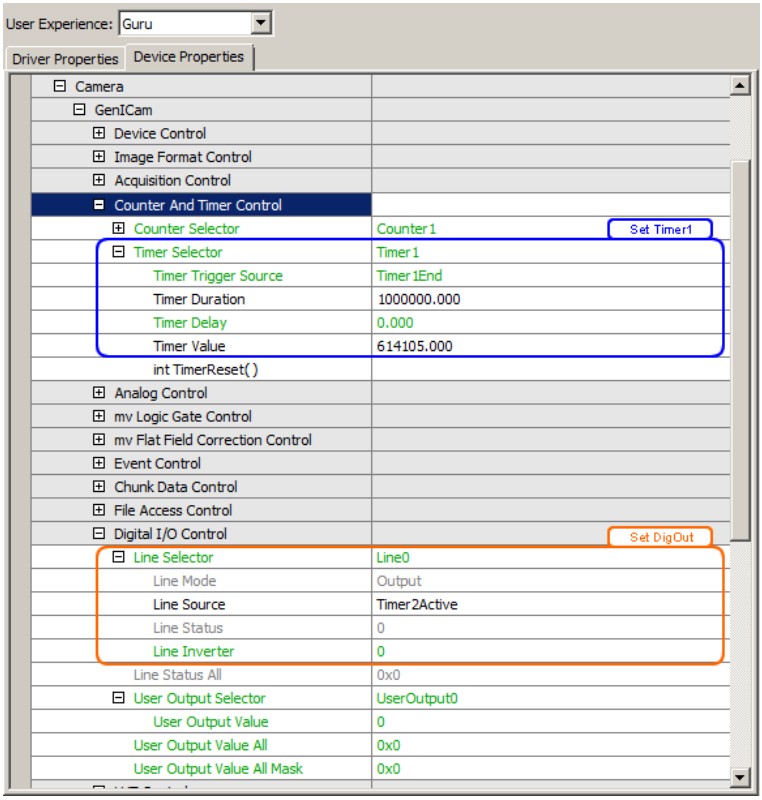


Figure 2: wxPropView - Setting of *Timer1*

2. Setting of *Timer2* (purple on the master camera):

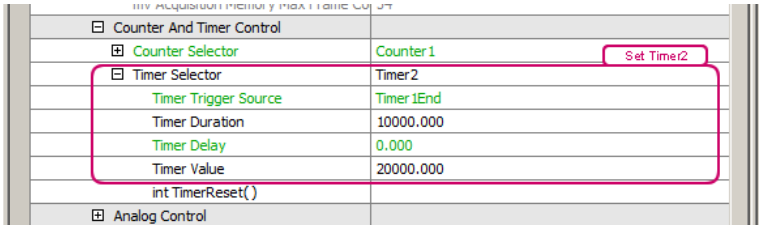


Figure 3: wxPropView - Setting of *Timer2*

3. Assigning timer to DigOut (orange box in Figure 2).

1.18.3.3 Outputting a pulse at every other external trigger

To do this, please follow these steps:

1. Switch *"Trigger Mode"* to **"On"** and
2. Select the *"Trigger Source"*, e.g. **"Line5"**.
3. Use *"Counter1"* and count the number of input trigger by setting the *"Counter Duration"* to **"2"**.
4. Afterwards, start *"Timer1"* at the end of *"Counter1"*:

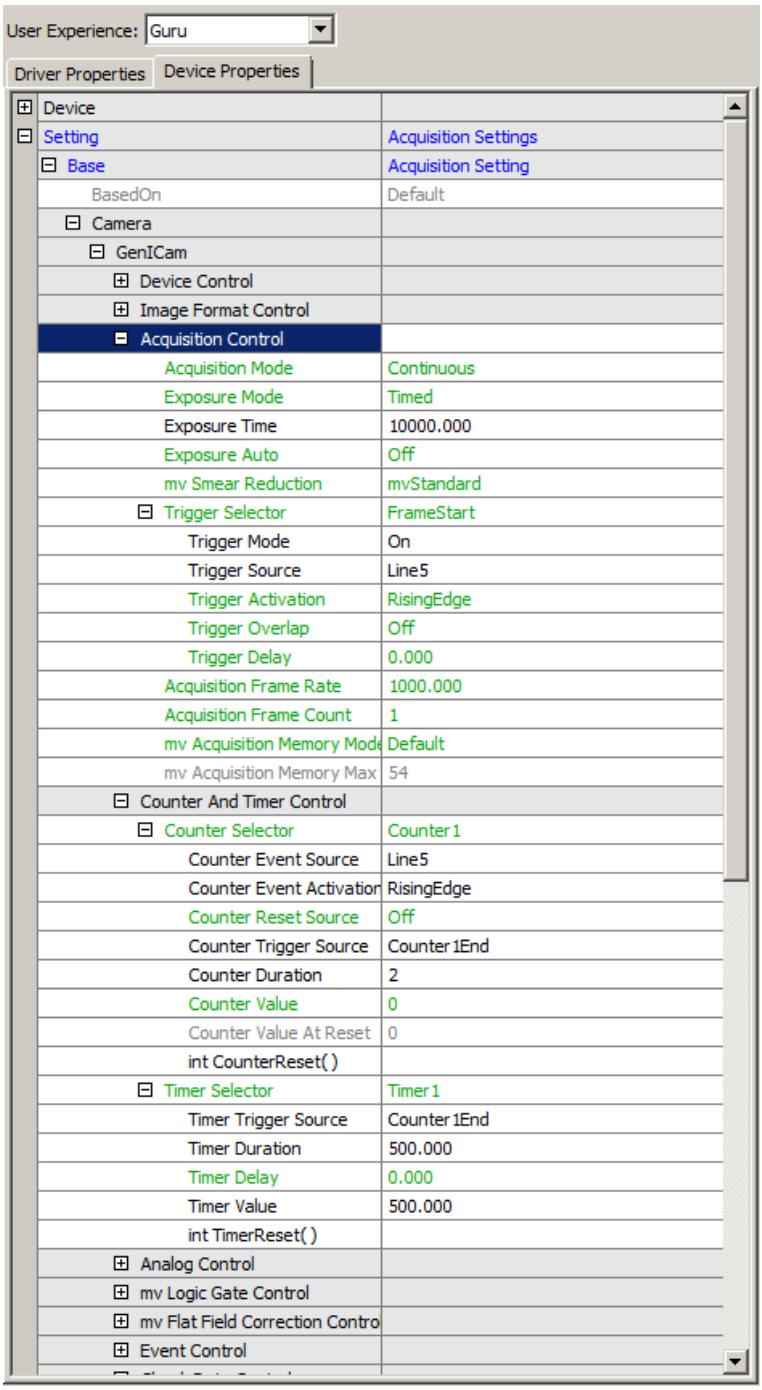


Figure 1: wxPropView - Setting the sample

The *"Timer1"* appears every second image.

Now, you can assign *"Timer1Active"* to a digital output e.g. **"Line3"**:

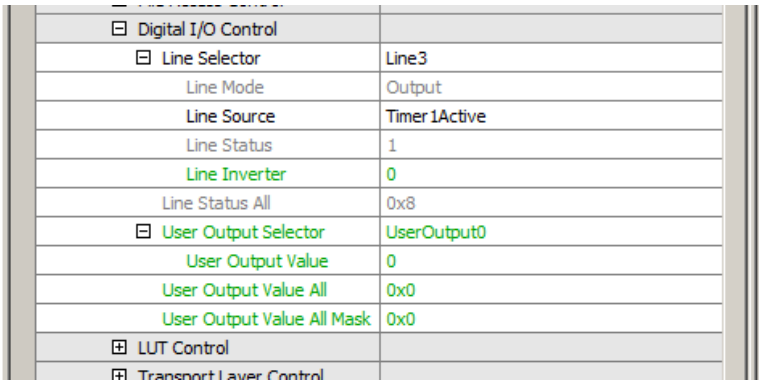


Figure 2: Assigning the digital output

Note

You can delay the pulse if needed.

1.18.3.4 Creating different exposure times for consecutive images

If you want to create a sequence of exposure times, you have to trigger the camera "externally" via pulse width:

1. Use **Timer** and **Counter** to build a sequence of different pulse widths.
2. Use the **Counter** for the time between the exposures (with respect to the readout times).
3. Afterwards, use an **AND** gate followed by **OR** gate to combine different exposure times.

Note

Please be sure that the sensor can output the complete image during **Counter1** or **Counter2**. Otherwise, only one integration time will be used.

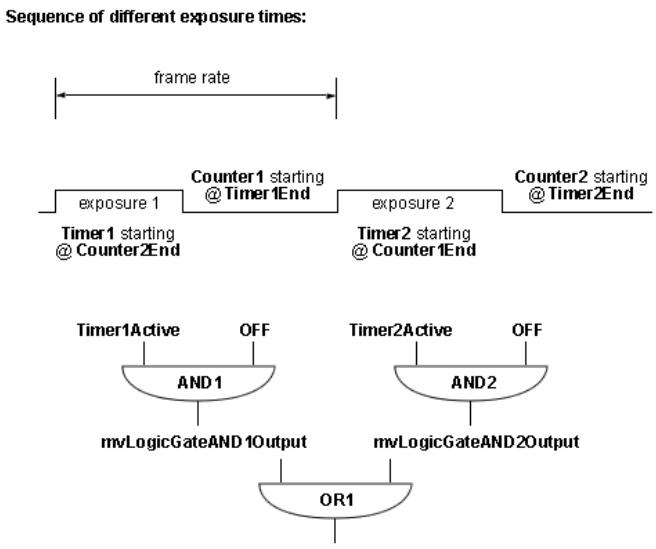


Figure 1: wxPropView - Logic gate principle

You can set this sample in **wxPropView** (p. 65). E.g. the sensor makes 22.7 frames per second in **Continuous Mode** (p. 69). This means that the sensor needs 44 ms to output the complete image.

```
    1
--------- = approx. 44 ms
  22.7
```

We take 55 ms to be sure. Now, as different exposure times we take 1 ms (**Timer1**) and 5 ms (**Timer2**). To get the 55 ms, we have to add 54000 us (**Counter1**) and 50000 us (**Counter2**).

Finally, you have to set the logic gate as shown in the figure:

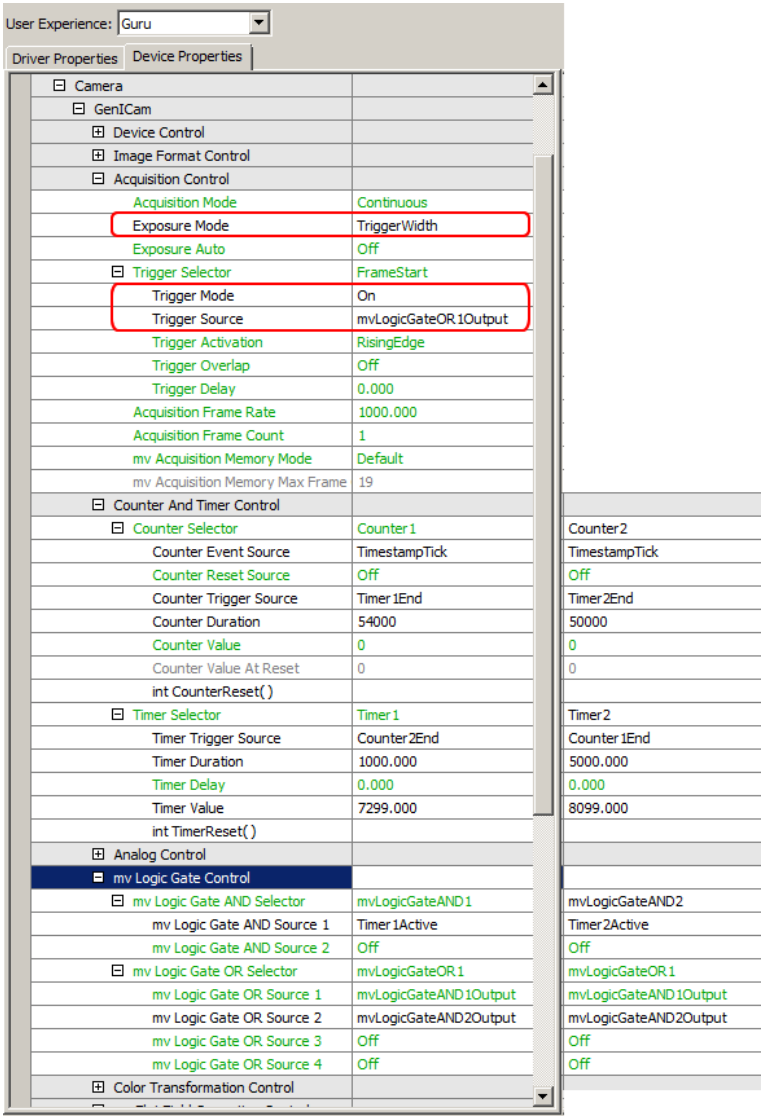


Figure 2: wxPropView - Logic gate setting

Note

Because there are 4 counters and 2 timers you can only add one further exposure time using one counter as a timer.

So if you want other sequences, you have to use the counters and timers in a flexible way as show in the next sample:

1.18.3.4.1 Sequence with 4 times exposure A followed by 1 time exposure B

If you have an external trigger, you can use the counter and timer to create longer exposure sequences.

For example, if you want a sequence with 4 times exposure A followed by 1 time exposure B you can count the trigger events. That means practically:

1. Use **Counter1** to count 5 trigger signals then
2. issuing **Timer2** for the long exposure time (re-triggered by **Counter1End**).
3. Every trigger issues **Timer1** for the short exposure.
4. Afterwards, an **AND** gate followed by **OR** gate combines the different exposure times.

In **wxPropView** (p. 65) it will look like this:

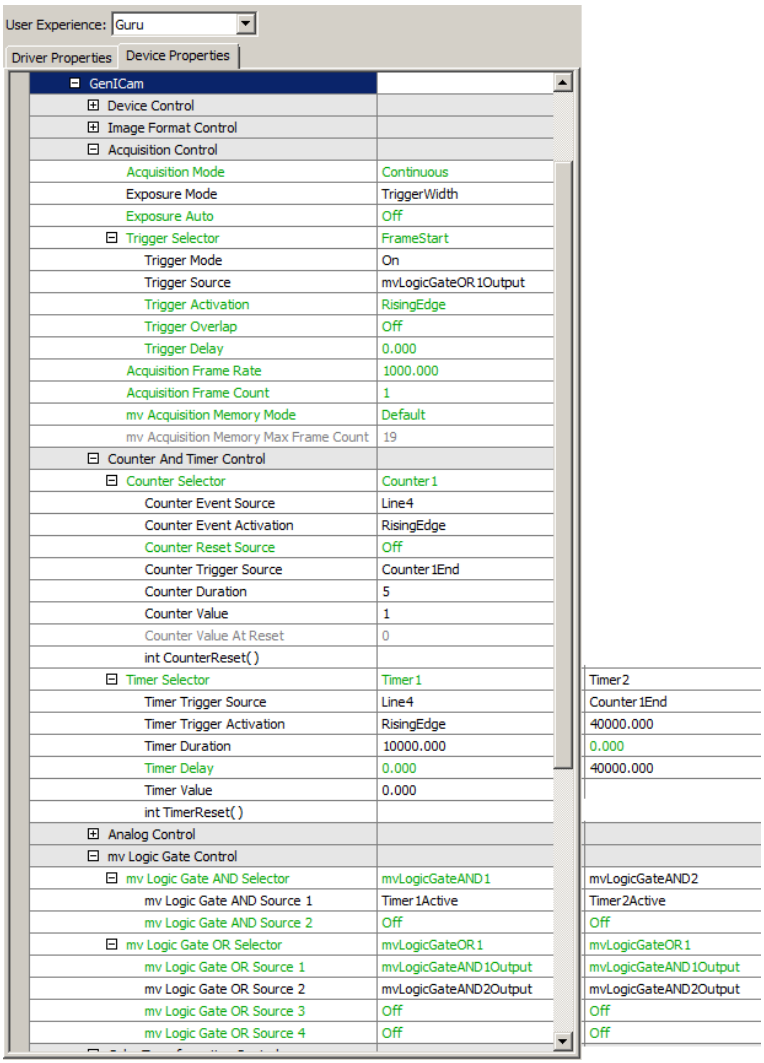


Figure 3: wxPropView - Logic gate setting 2

1.18.3.5 Detecting overtriggering

1.18.3.5.1 Scenario

The image acquisition of a camera consists of two steps:

- exposure of the sensor and
- readout of the sensor data

During these steps, a trigger signal will be skipped:

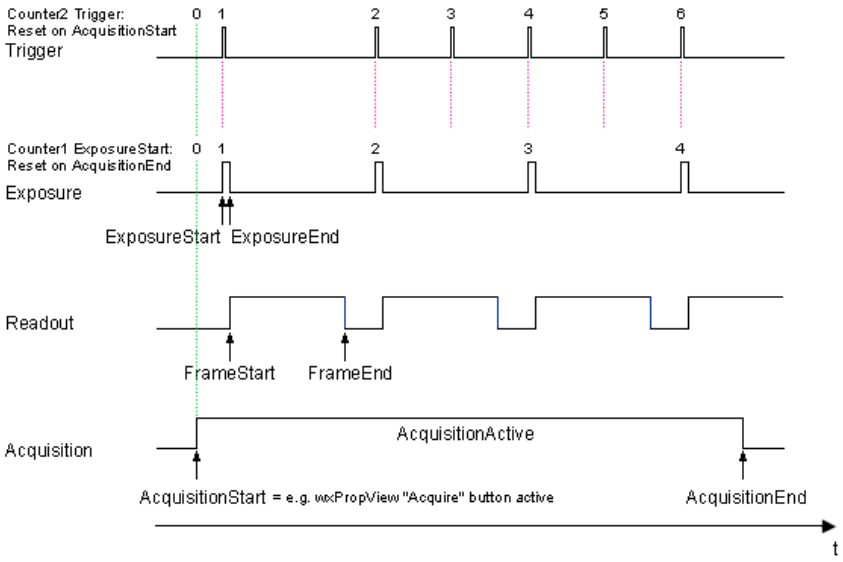


Figure 1: Trigger counter increases but the start exposure counter not

To notice overtriggering, you can use **counters** (p. 73):

- One counter counts the incoming trigger signals, the
- second counter counts the `ExposureStart` signals.

Using the **chunk data** (p. 79) you can overlay the counters in the live image.

1.18.3.5.2 Setting the overtrigger detector using wxPropView

First of all, we have to set the **trigger** in *"Setting -> Base -> Camera -> GenICam -> Acquisition Control"* with following settings:

Property name wxPropView	Setting
Trigger Selector	`FrameStart`
Trigger Mode	`On`
Trigger Source	`Line4`
Trigger Activation	`RisingEdge`
Exposure Mode	`Timed`

This trigger will start an acquisition after a rising edge signal on line 4 (= DigIn0).

Now, set the two counters. Both counters (`Counter1` and `Counter2`) will be reset and start after the acquisition (`AcquisitionStart`) has started.

While `Counter1` increases with every `ExposureStart` event (see figure above for the event and acquisition details) ...

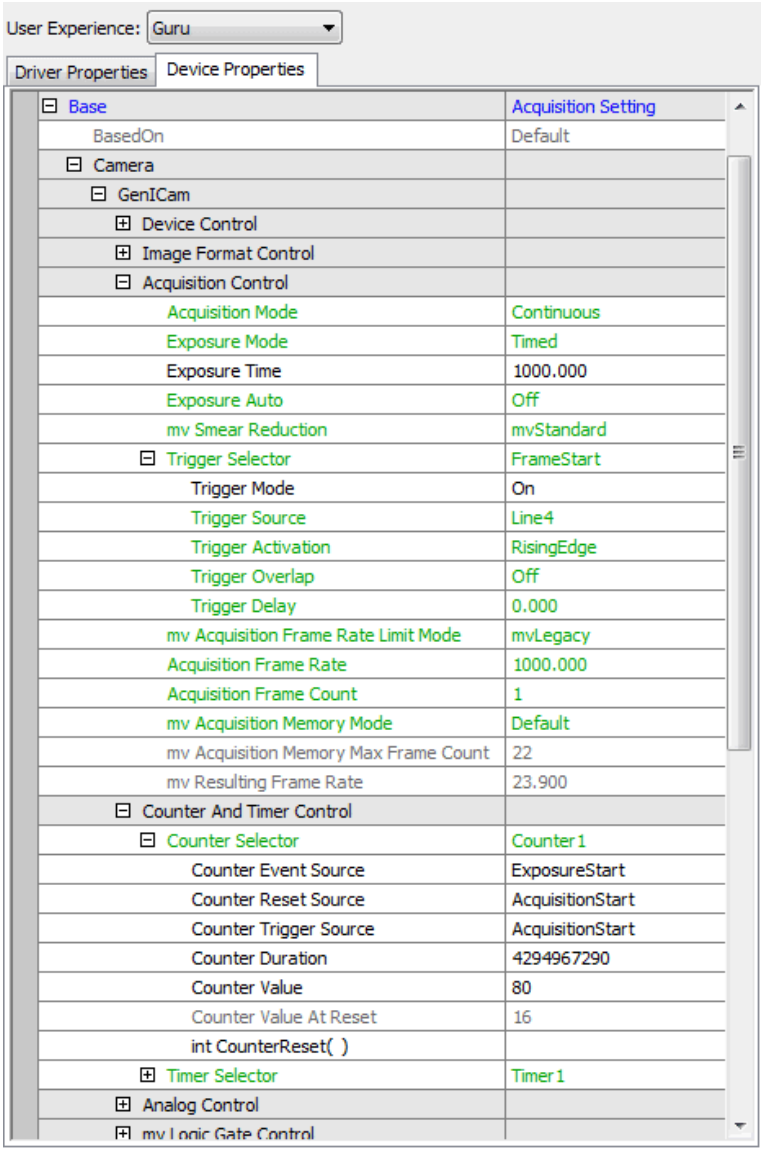


Figure 2: Setting Counter1

... `Counter2` increases with every `RisingEdge` of the trigger signal:

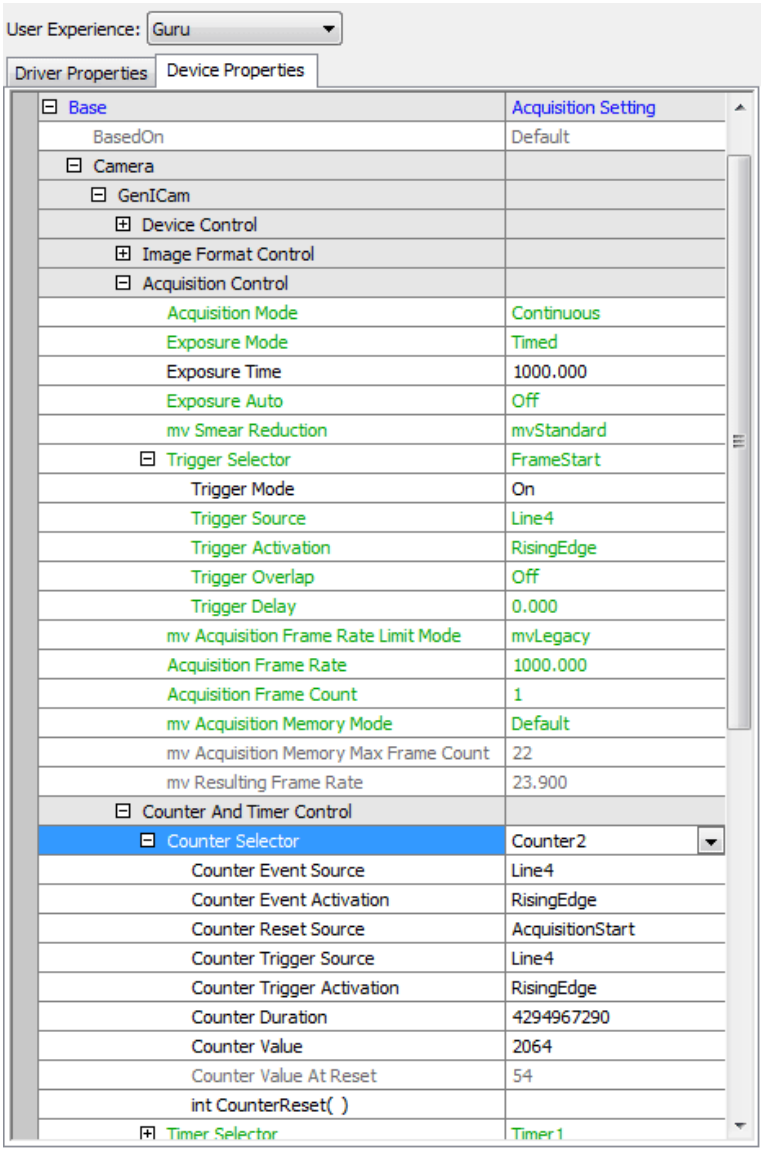


Figure 3: Setting Counter2

Now, you can check if the trigger signal is skipped (when a rising edge signal is active during readout) or not by comparing the two counters.

Enable the inclusion of the selected chunk data ("`Chunk Mode Active = 1`") in the payload of the image in *"Setting -> Base -> Camera -> GenICam -> Chunk Data Control"*:

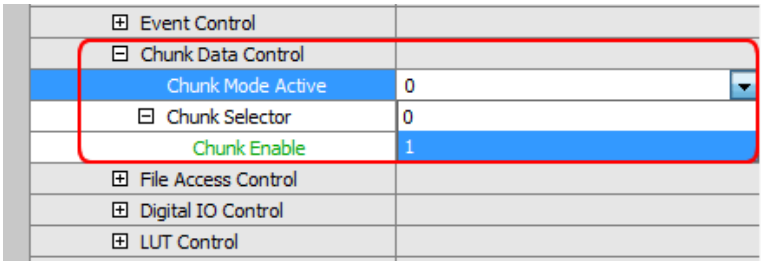


Figure 4: Enable chunk data

Activate the info overlay in the display area. Right-click on the live display and select: *"Request Info Overlay"*

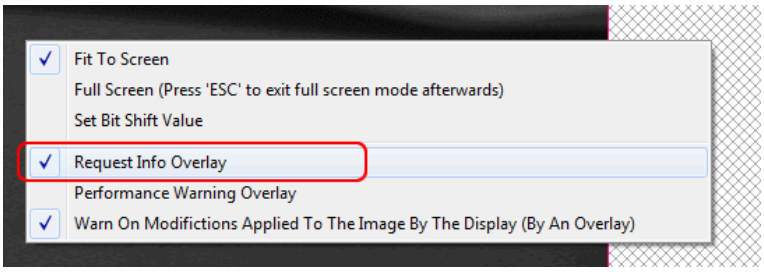


Figure 5: Show chunk data

The following figure shows that no trigger signal is skipped:



Figure 6: Trigger Signal counter equals ExposureStart counter

The following figure shows that the acquisition is overtriggered:



Figure 7: Trigger Signal counter is higher than ExposureStart counter

1.18.3.6 Triggering of an indefinite sequence with precise starting time

1.18.3.6.1 Scenario

Especially in the medical area, there are applications where a triggered acquisition is started, for example, with a foot switch. Following challenges have to be solved in combination with these applications:

- The user wants the acquired image immediately (precise starting time).
- It is not known, when the user stops the acquisition (indefinite sequence).

Using *AcquisitionStart* as the trigger source, it could take between 10 and 40 ms until the camera acquires the first frame. That's not really an immediately acquisition start. It is recommended to use *FrameStart* as the trigger source instead. However, according to the time the trigger event occurs, there will be a timeout during the first frame in nearly all cases.

You can avoid this using a timer which generates a "high" every 100 us and which is connected to the trigger input *Line4* using a logical *AND* gate. I.e. if the timer is "high" and there is a trigger signal at *Line4* then the logical conjunction is true. The AND gate result is then connected as *TriggerSource* of the *FrameStart* trigger using a logical *OR* gate. I.e. as soon as the logical AND conjunction is true, the trigger source is true and the image acquisition will start.

The following figure illustrates the settings:

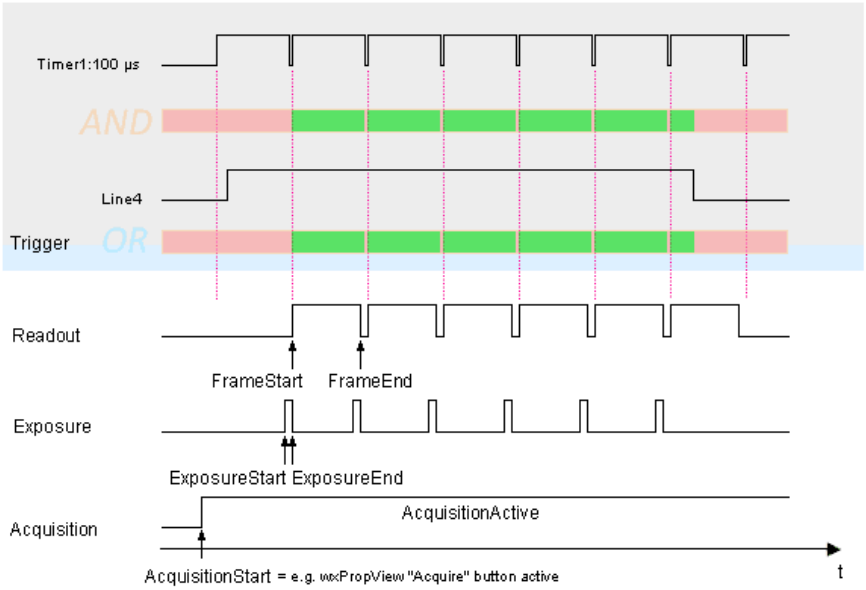


Figure 1: Schematic illustration of the settings

With this setting, there is still an acceptable time delay of approx. 100 to 130 us possible.

1.18.3.6.2 Creating the use case using wxPropView

First of all, we have to set the **timer** in *"Setting -> Base -> Camera -> GenICam -> Counter And Timer Control"* with following settings:

Property name wxPropView	Setting
Timer Selector	`Timer1`
Timer Trigger Source	`Timer1End`
Timer Duration	`100.000`

Afterwards, we have to set the **logical gates** in *"Setting -> Base -> Camera -> GenICam -> mv Logic Gate Control"* with following settings:

Property name wxPropView	Setting
mv Logic Gate AND Selector	`mvLogicGateAND1`
mv Logic Gate AND Source 1	`Line4`
mv Logic Gate AND Source 2	`Time1Active`
mv Logic Gate OR Selector	`mvLogicGateOR1`
mv Logic Gate OR Source 1	`mvLogicGateAND1Output`

Finally, we have to set the **trigger** in *"Setting -> Base -> Camera -> GenICam -> Acquisition Control"* with following settings:

Property name wxPropView	Setting
Trigger Selector	`FrameStart`
Trigger Mode	`On`
Trigger Source	`mvLogicGateAND1Output`
Trigger Activation	`RisingEdge`
Exposure Mode	`Timed`

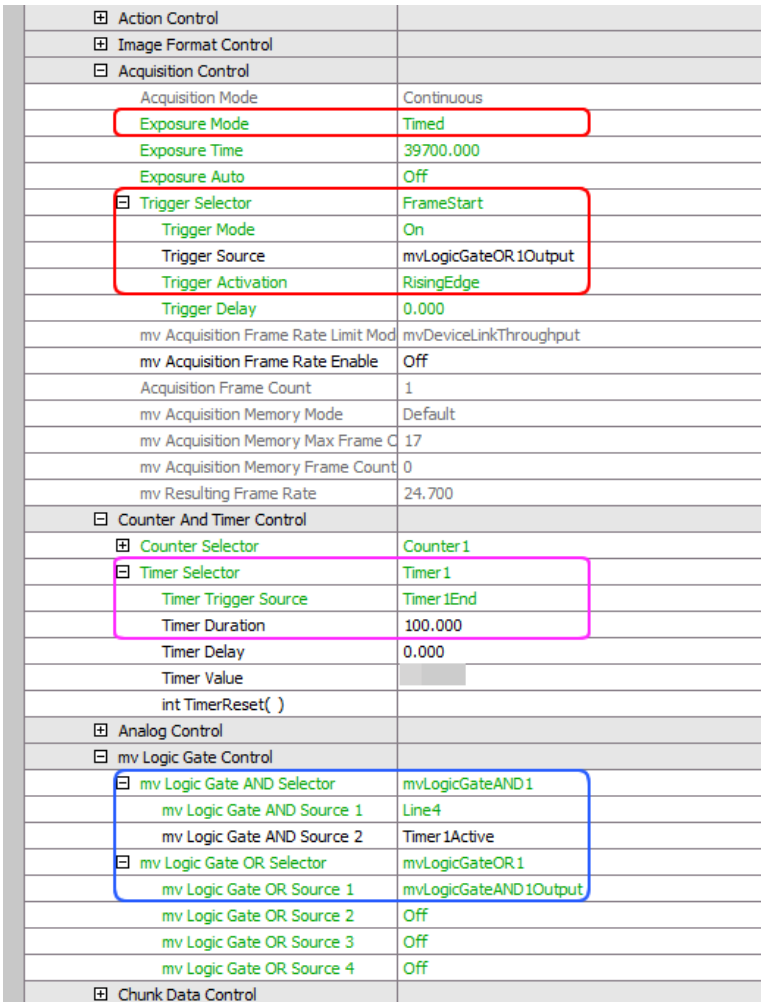


Figure 2: Sample settings

1.18.3.7 Low latency triggering

Since

Firmware revision 2.45

1.18.3.7.1 Introduction

Because the exposure start is synced with the line-period, there is a jitter of the start of up to one line. Additionally, there is a latency of 2 or 4 line-periods (depending on the sensor) until the exposure actually starts. The following image shows this visually:

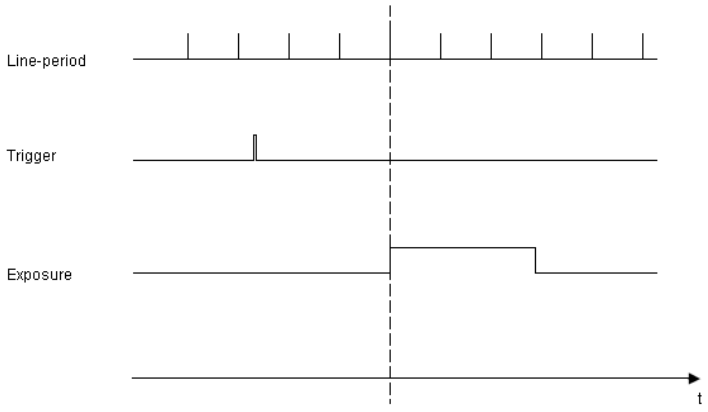


Figure 1: Normal trigger mode

The exposure time increment is a multiple of the line-period. This trigger behavior works also with overlapped exposure and readout.

With the Pregius global shutter sensors of the 2nd, 3rd, and 4th generation from Sony, it is possible to start the exposure on the trigger. There is no latency and no jitter. The length of exposure is also not based on the line-period, because the sensor does not depend on it in this mode. The minimum exposure time is limited by several line periods though. Because during readout the sensor obviously requires the line-periods this mode is only supported in non-overlapped exposure and readout:

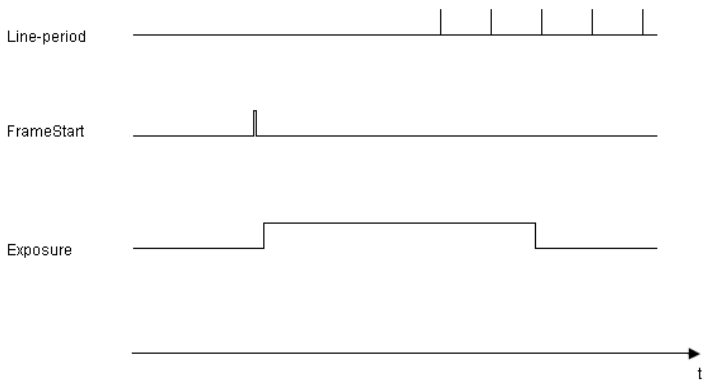


Figure 2: LowLatency trigger mode

1.18.3.7.2 Using wxPropView

To activate the LowLatency trigger mode, please follow these steps:

1. Set the **TriggerSelector** in *"Setting → Base → Camera → GenICam → AcquisitionControl"* to **FrameStart** .
2. Afterwards, change to **TriggerMode** and
3. Select **mvLowLatency**.

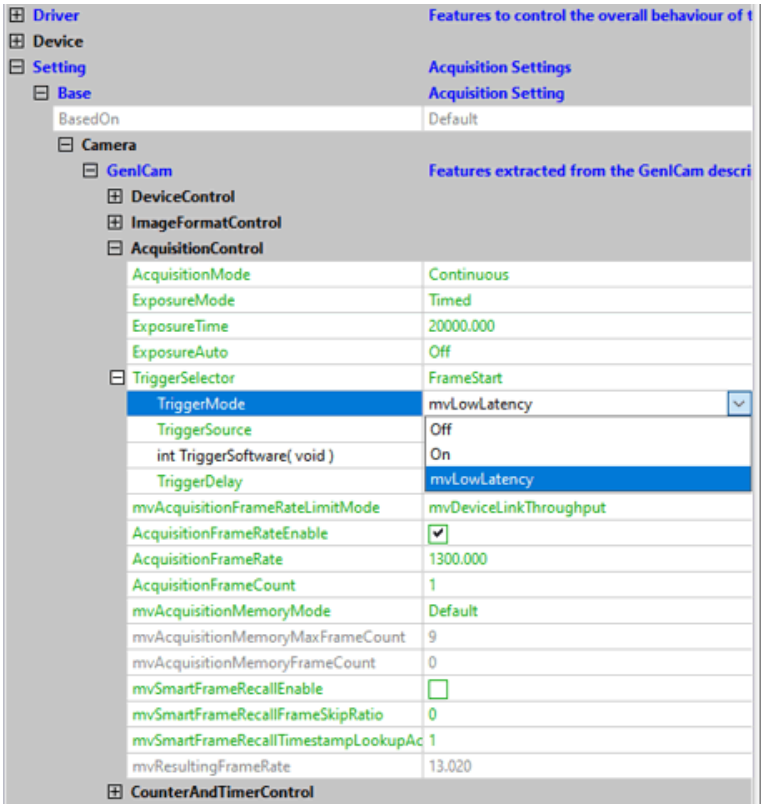


Figure 3: mvLowLatency

1.18.4 Working with I/Os

There are several use cases concerning I/Os:

- **Controlling strobe or flash at the outputs** (p. 178)
- **Creating a debouncing filter at the inputs** (p. 181)

1.18.4.1 Controlling strobe or flash at the outputs

Of course, MATRIX VISION devices support strobe or flash lights. However, there are several things you have to keep in mind when using strobes or flash:

1. Be sure that the illumination fits with the movement of the device under test.
2. Bright illumination and careful control of exposure time are usually required.
3. To compensate blur in the image, short exposure times are needed.

Alternatively, you can use flash with short burn times. For this, you can control the flash using the camera. The following figures show, how you can do this using **wxPropView** (p. 65)

1. Select in *"Setting -> Base -> Camera -> Digital I/O Control"* the output line with the **"Line Selector"** to which the strobe or flash is connected.
2. Now, set the **"Line Source"** to **"mvExposureAndAcquisitionActive"**.
   This means that the signal will be high for the exposure time and only while acquisition of the camera is active.

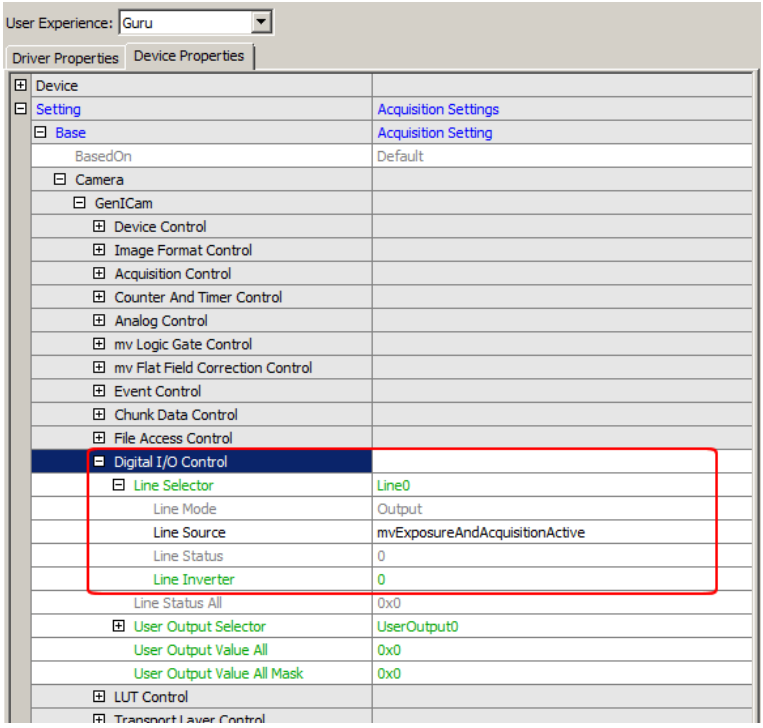


Figure 1: Setting the **"Line Source"** to **"mvExposureAndAcquisitionActive"**

Additionally, you can shift the signal controlling the flash or strobe using the camera. The camera will send the signal at the selected output line either sooner or later. The following figure demonstrates the procedure using **wxPropView** (p. 65)

1. Enable in *"Setting -> Base -> Camera -> AcquisitionControl"* the property **"mvExposureActiveShift↩ Enable"**.
   The property **"mvExposureActiveShift"** will appear.

2. Now you can select a shift value and **"mvExposureAndAcquisitionActive"** will be available either later (with positive values) or sooner (with negative values).

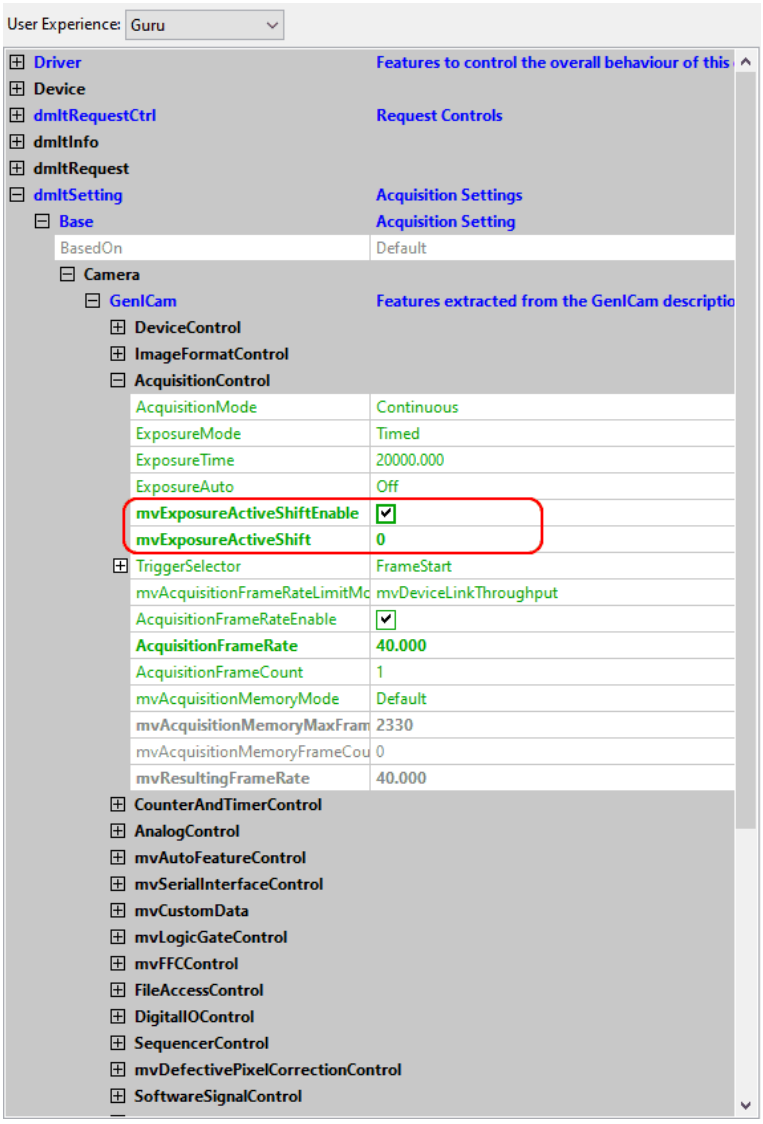


Figure 2: Controlling the shift of **"mvExposureAndAcquisitionActive"** at the output line

Note

This can be combined using an external trigger.

1.18.4.1.1 Special case: Rolling shutter Starvis sensors

Since

Firmware version 2.24.975

With the rolling shutter flash mode, the exposure time which is set corresponds to the time where all lines are exposed simultaneously. The exposure signal corresponds to the time where the last line starts until the first line ends. The rolling shutter flash mode avoids motion effects and is suitable for exposure times $>$ 300 usec.

Additional to the digital output settings in **Controlling strobe or flash at the outputs** (p. 178) you have to set

1. the **"mvShutterMode"** to **"mvRollingShutterFlash"** and
2. the desired **"ExposureTime"**.

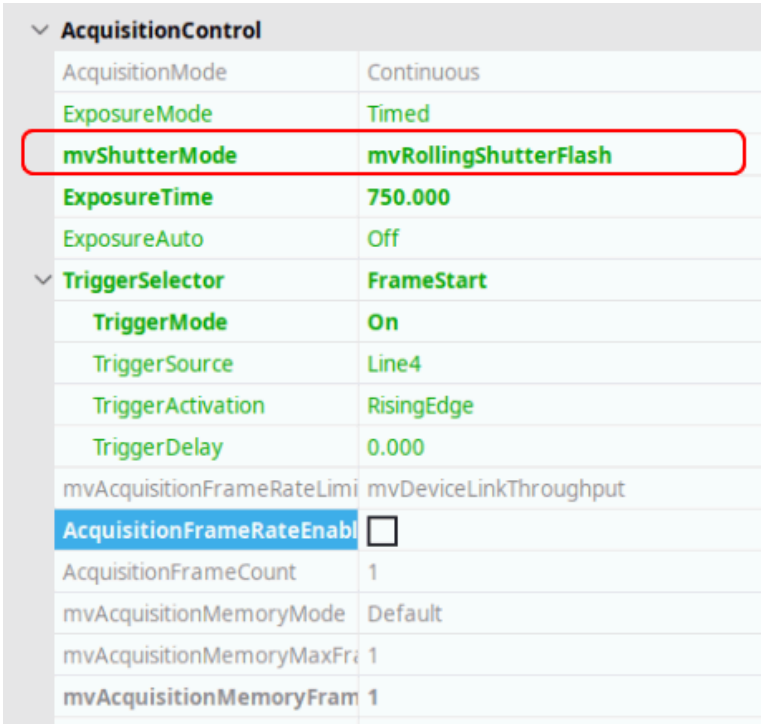


Figure 3: Setting the **"mvShutterMode"** to **"mvRollingShutterFlash"**

Note

If you have an external trigger, please adjust the settings according to **this trigger sample** (p. 169).

1.18.4.1.2 Compensating delay of strobe or flash

Normally, the input circuitry of flash has a delay (e.g. low pass filtering). Using **"ExposureActive"** to fire strobe would actually illuminate delayed with respect to exposure of the sensor. Figure 4 shows the problem:

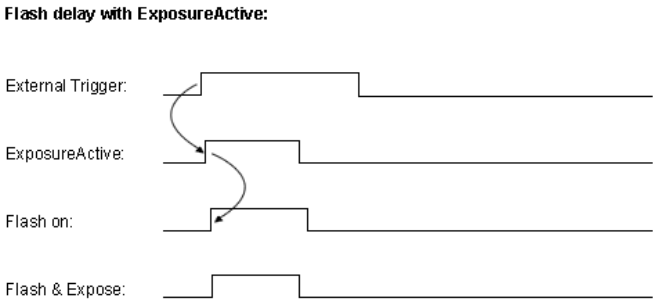


Figure 4: Flash delay with **"ExposureActive"**

To solve this issue, you can use following procedure:

1. Do not use **"ExposureActive"** for triggering strobe.

2. Build flash signal with **Timer**,
3. trigger **Timer** with external trigger (e.g. **"Line5"**).
4. Use **"Trigger Delay"** to delay exposure of the sensor accordingly.

In **wxPropView** (p. 65) it will look like this:

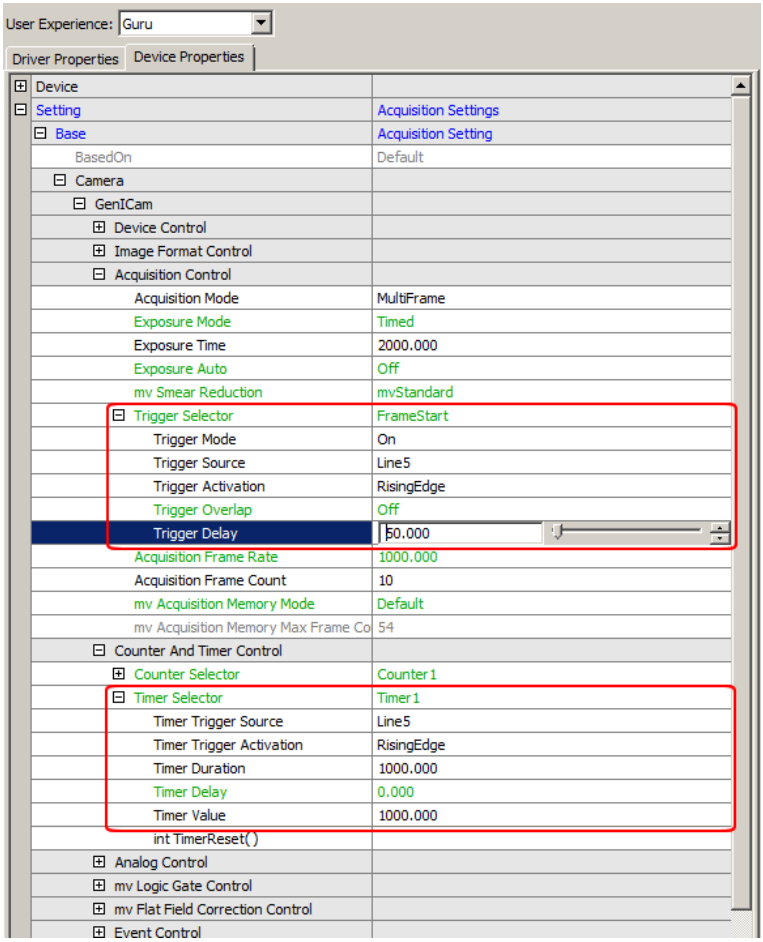


Figure 5: Working with **Timer** and **"Trigger Delay"**

1.18.4.2 Creating a debouncing filter at the inputs

In some cases, it is necessary to eliminate noise on trigger lines. This can become necessary when either

- the edges of a trigger signal are not perfect in terms of slope or
- if, because of the nature of the trigger signal source, multiple trigger edges are generated within a very short period of time even if there has just been a single trigger event.

The latter one is also called *bouncing*.

Bouncing is the tendency of any two metal contacts in an electronic device to generate multiple signals as the contacts close or open; now debouncing is any kind of hardware device or software that ensures that only a single signal will be acted upon for a single opening or closing of a contact.

To address problems that can arise from these kinds of trigger signals MATRIX VISION offers debouncing filters at the digital inputs of a device.

The debouncing filters can be found under *"Setting -> Base -> Camera -> GenICam -> Digital I/O Control -> LineSelector"* (red box in Figure 1) for each digital input:

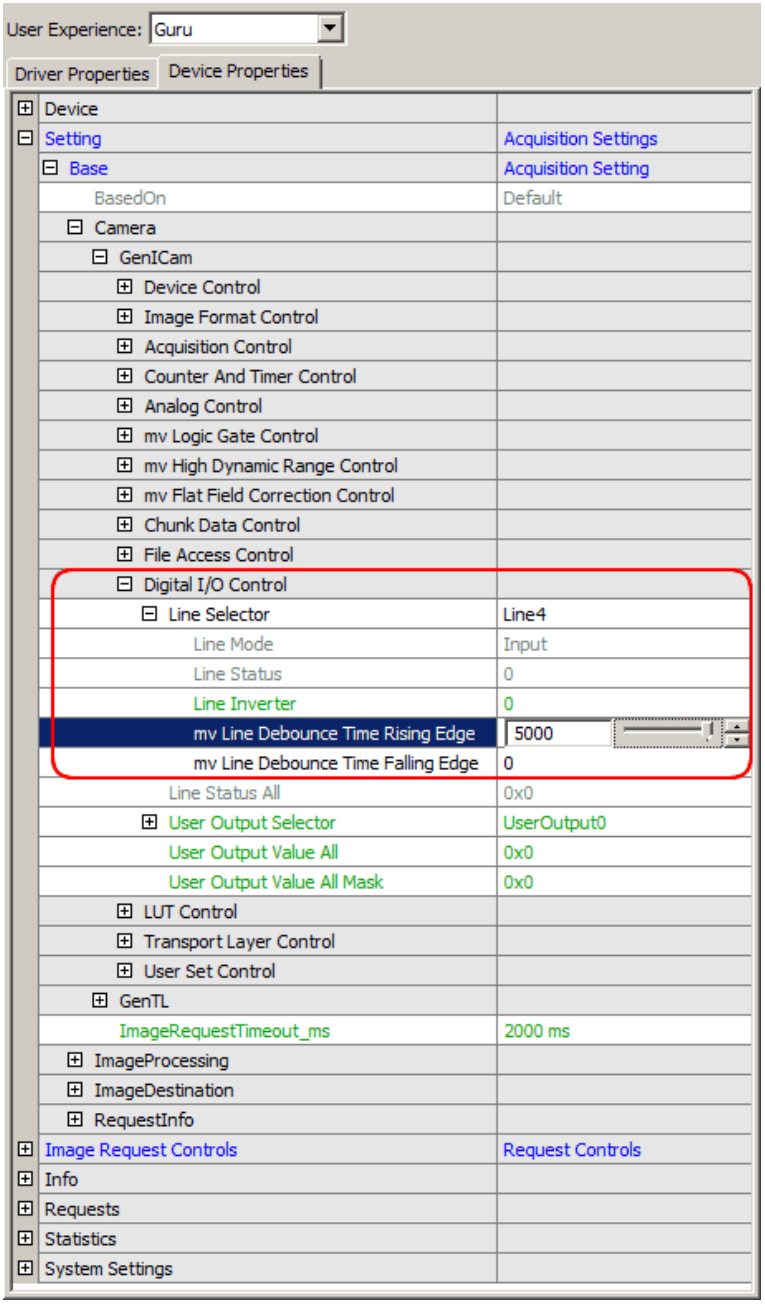


Figure 1: wxPropView - Configuring Digital Input Debounce Times

Each digital input (*LineMode* equals *Input*) that can be selected via the *LineSelector* property will offer its own property to configure the debouncing time for falling edge trigger signals (*"mv Line Debounce Time Falling Edge"*) and rising edge (*"mv Line Debounce Time Rising Edge"*) trigger signals.

The line debounce time can be configured in micro-seconds with a maximum value of up to 5000 micro-seconds. Internally each time an edge is detected at the corresponding digital input a timer will start (orange $*$ in figure 2 and 3), that is reset whenever the signal applied to the input falls below the threshold again. Only if the signal stays at a constant level for a full period of the defined *mvLineDebounceTime* the input signal will be considered as a valid trigger signal.

Note

Of course this mechanism delays the image acquisition by the debounce time.

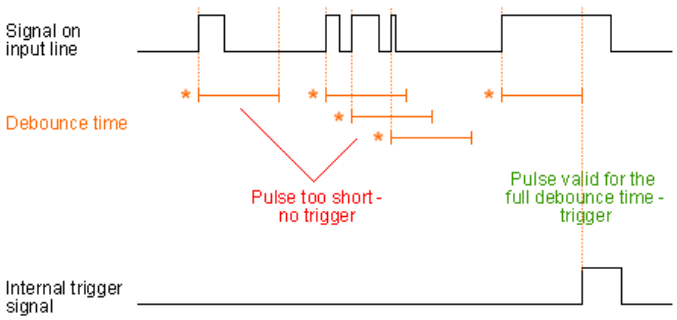


Figure 2: mvLineDebounceTimeRisingEdge Behaviour

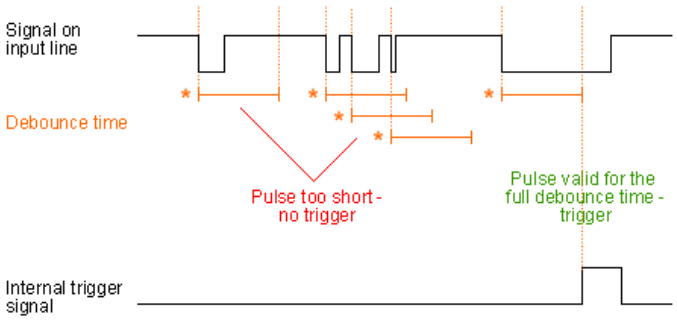


Figure 3: mvLineDebounceTimeFallingEdge Behaviour

1.18.5 Saving data on the device

Note

As described in **"Storing, Restoring And Managing Settings"** in the **"mvIMPACT Acquire SDK GUI Applications"** manual, it is also possible to save the settings as an XML file on the host system. You can find further information about for example the XML compatibilities of the different driver versions in the mvIMPACT Acquire SDK manuals and the according setting classes: `https://www.matrix-vision.com/manuals/SDK_CPP/classmvIMPACT_1_1acquire_1_1FunctionInterface.html` (C++)

- **Creating user data entries** (p. 184)
- **Creating user set entries** (p. 186)
- **Working with the UserFile section (Flash memory)** (p. 188)

1.18.5.1 Creating user data entries

1.18.5.1.1 Basics about user data

It is possible to save arbitrary user specific data on the hardware's non-volatile memory. The amount of possible entries depends on the length of the individual entries as well as the size of the devices non-volatile memory reserved for storing:

- mvBlueFOX,
- mvBlueFOX-M,
- mvBlueFOX-MLC,
- mvBlueFOX3,
- mvBlueCOUGAR-X,
- mvBlueCOUGAR-XD,
- mvBlueCOUGAR-XT and
- mvBlueNAOS

currently offer 512 bytes of user accessible non-volatile memory of which 12 bytes are needed to store header information leaving 500 bytes for user specific data.

```
One entry will currently consume:
1 + <length_of_name (up to 255 chars)> + 2 + <length_of_data (up to 65535 bytes)> + 1 (access mode) bytes

as well as an optional:
1 + <length_of_password> bytes per entry if a password has been defined for this particular entry
```

It is possible to save either **String** or **Binary** data in the *data* property of each entry. When storing binary data please note, that this data internally will be stored in `Base64` format thus the amount of memory required is 4/3 time the binary data size.

The UserData can be accessed and created using **wxPropView** (p. 65) (the device has to be closed). In the section `"UserData"` you will find the entries and following methods:

- `"CreateUserDataEntry"`

- "DeleteUserDataEntry"
- "WriteDataToHardware"

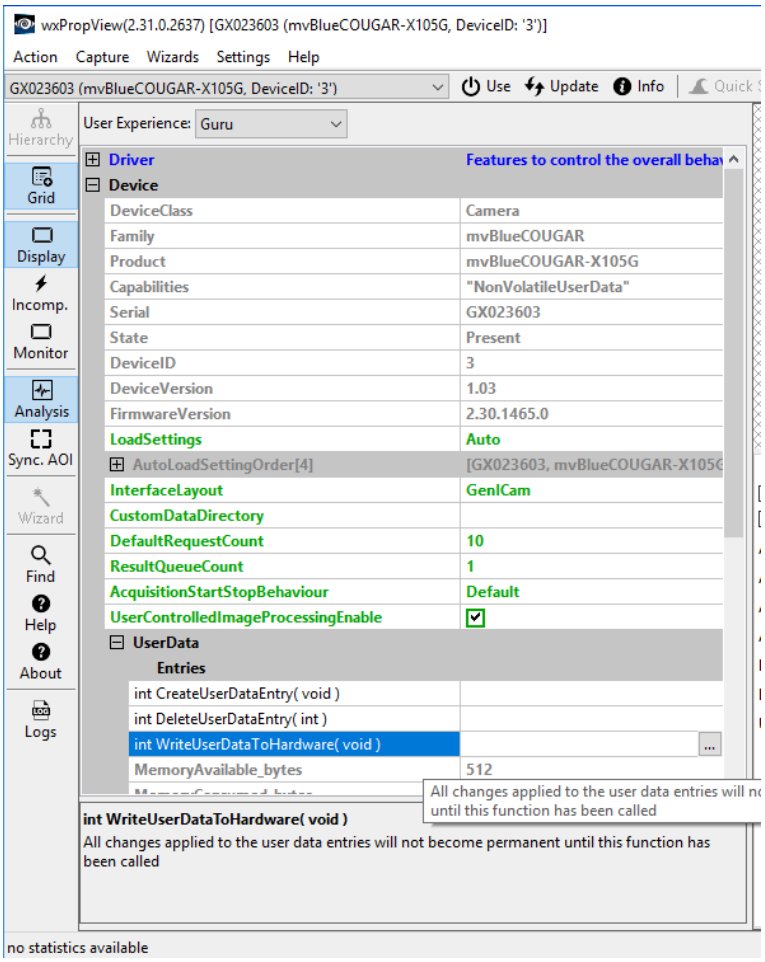


Figure 1: wxPropView - section "UserData -> Entries"

To create a user data entry, you have to

- Right click on "CreateUserDataEntry"
- Select **"Execute"** from the popup menu.
  An entry will be created.
- In "Entries" click on the entry you want to adjust and modify the data fields.
  To permanently commit a modification made with the keyboard the **ENTER** key **must be pressed**.
- To save the data on the device, you have to execute "WriteDataToHardware". Please have a look at the "Output" tab in the lower right section of the screen as shown in Figure 2, to see if the write process returned with no errors. If an error occurs a message box will pop up.

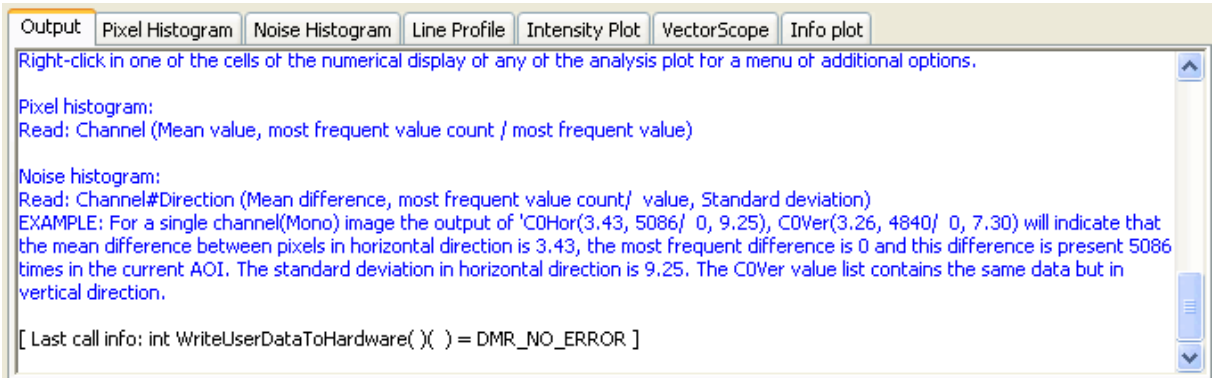


Figure 2: wxPropView - analysis tool "Output"

1.18.5.1.2 Coding sample

If you e.g. want to use the UserData as dongle mechanism (with binary data), it is not suggested to use **wxPropView** (p. 65). In this case you have to program the handling of the user data.

See also

mvIMPACT::acquire::UserDataEntry in mvIMPACT_Acquire_API_CPP_manual.chm.

1.18.5.2 Creating user set entries

With MATRIX VISION GenICam compliant devices it is possible to store up to five configuration sets (4 user plus one factory default) in the camera.

This feature is similar to the **storing settings functionality** (p. 184), which saves the settings in the registry. However, as mentioned before the user sets are stored in the camera.

The user set stores

- exposure,
- gain,
- AOI,
- frame rate,
- etc.

permanently and is independent of the computer which in used.

Additionally, you can select, which user set comes up after hard reset.

Note

The storage of user data in the registry can still override user set data!
User sets are cleared after firmware change.

1.18.5.2.1 List of ignored properties

1.18.5.2.2 Working with the user sets

You can find the user set control in *"Setting -> Base -> Camera -> GenICam -> User Set Control"*:

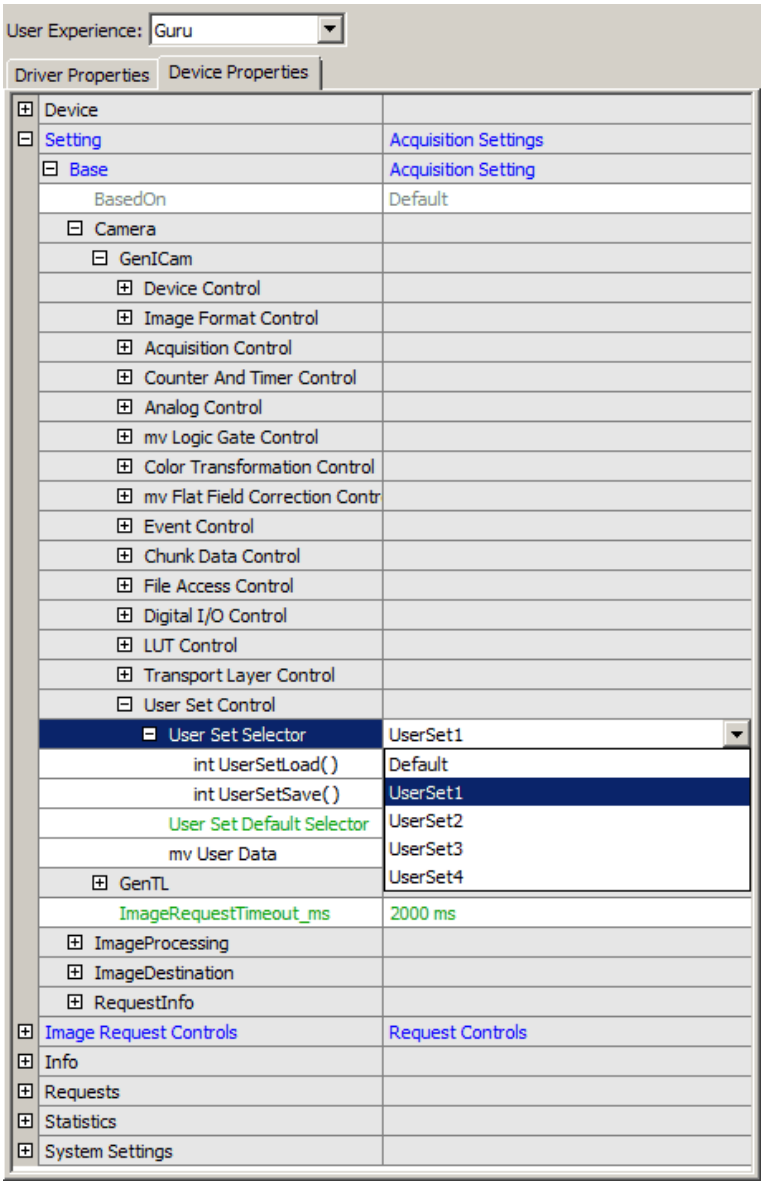


Figure 1: User Set Control

With *"User Set Selector"* you can select the user set (*"Default"*, *"UserSet1 - UserSet4"*). To save or load the specific user set, you have two functions:

- *"int UserSetLoad()"* and
- *"int UserSetSave()"*.

"User Set Default" is the property, where you can select the user set, which comes up after hard reset.

Finally, with *"mv User Data"* you have the possibility to store arbitrary user data.

1.18.5.3 Working with the UserFile section (Flash memory)

The mvBlueNAOS offers a 64 KByte section in the Flash memory that can be used to upload a custom file to (**UserFile**).

To read or write this file you can use the following GenICam **File Access Control** (p. 80) and its interfaces:

- **IDevFileStream** (read)
- **ODevFileStream** (write)

Note

The **UserFile** is lost each time a firmware update is applied to the device.

1.18.5.3.1 Using wxPropView

wxPropView (p. 65) offers a wizard for the **File Access Control** (p. 80) usage:

1. Click on *"Setting -> Base -> Camera -> GenICam -> File Access Control -> File Selector -> File Operator Selector"*.
   Now, the *"Wizard"* button becomes active.

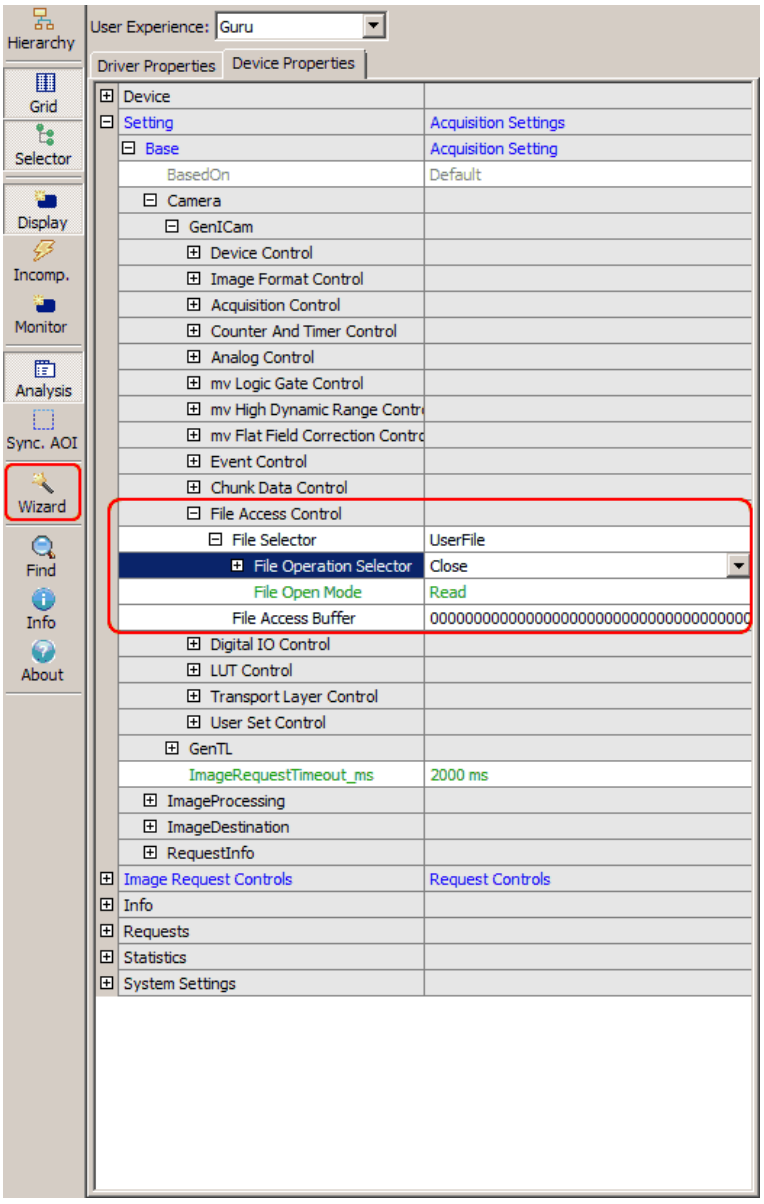


Figure 1: wxPropView - UserFile wizard

2. Click on the *"Wizard"* button.
   Now, a dialog appears where you can choose either to upload or download a file.

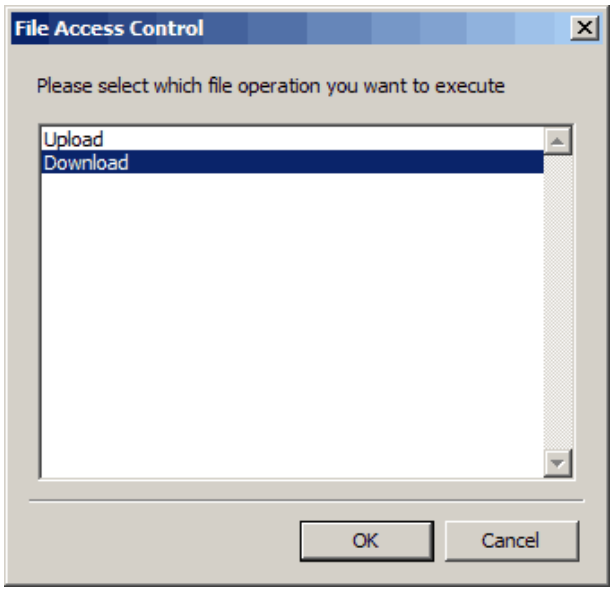


Figure 2: wxPropView - Download / Upload dialog

3. Make your choice and click on *"OK"*.
   Now, a dialog appears where you can select the File.

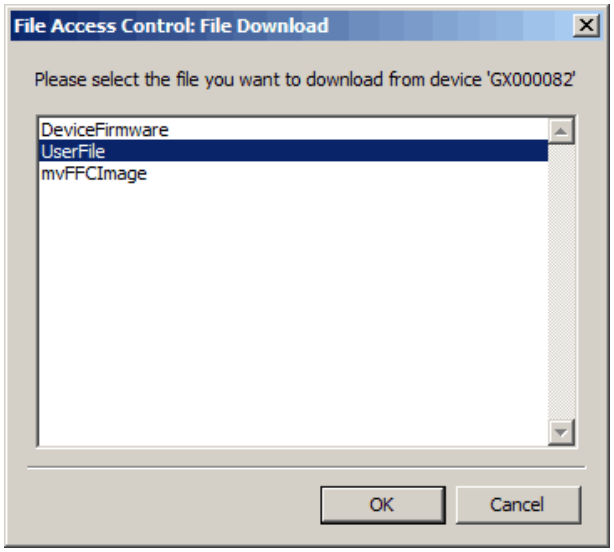


Figure 3: wxPropView - Download / Upload dialog

4. Select *"UserFile"* follow the instructions.

1.18.5.3.2 Manually control the file access from an application (C++)

The header providing the file access related classes must be included into the application:

```
#include <mvIMPACT_CPP/mvIMPACT_acquire_GenICam_FileStream.h>
```

A write access then will look like:

```
const string fileNameDevice("UserFile");

// uploading a file
mvIMPACT::acquire::GenICam::ODevFileStream file;
file.open( pDev, fileNameDevice.c_str() );

if( !file.fail() )
{
  // Handle the successful upload.
}
else
{
  // Handle the error.
}
```

A read access will look like:

```
const string fileNameDevice("UserFile");

// downloading a file works in a similar way
mvIMPACT::acquire::GenICam::IDevFileStream file;
file.open( pDev, fileNameDevice.c_str() );

if( !file.fail() )
{
  // Handle the successful upload.
}
else
{
  // Handle the error.
}
```

You can find a detailed code example in the C++ API manual in the documentation of the classes **mvIMPACT::acquire::GenICam::IDevFileStream** and **mvIMPACT::acquire::GenICam::ODevFileStream**

1.18.6 Working with device features

- **Working with the temperature sensors** (p. 192)
- **Reset timestamp by hardware** (p. 194)
- **Synchronizing camera timestamps** (p. 195)
- **Working With The Serial Interface (mv Serial Interface Control)** (p. 200)

1.18.6.1 Working with the temperature sensors

The MATRIX VISION devices offer two different temperature sensors:

- on the sensor board (typically lower)
- on FGPA board (typically higher)

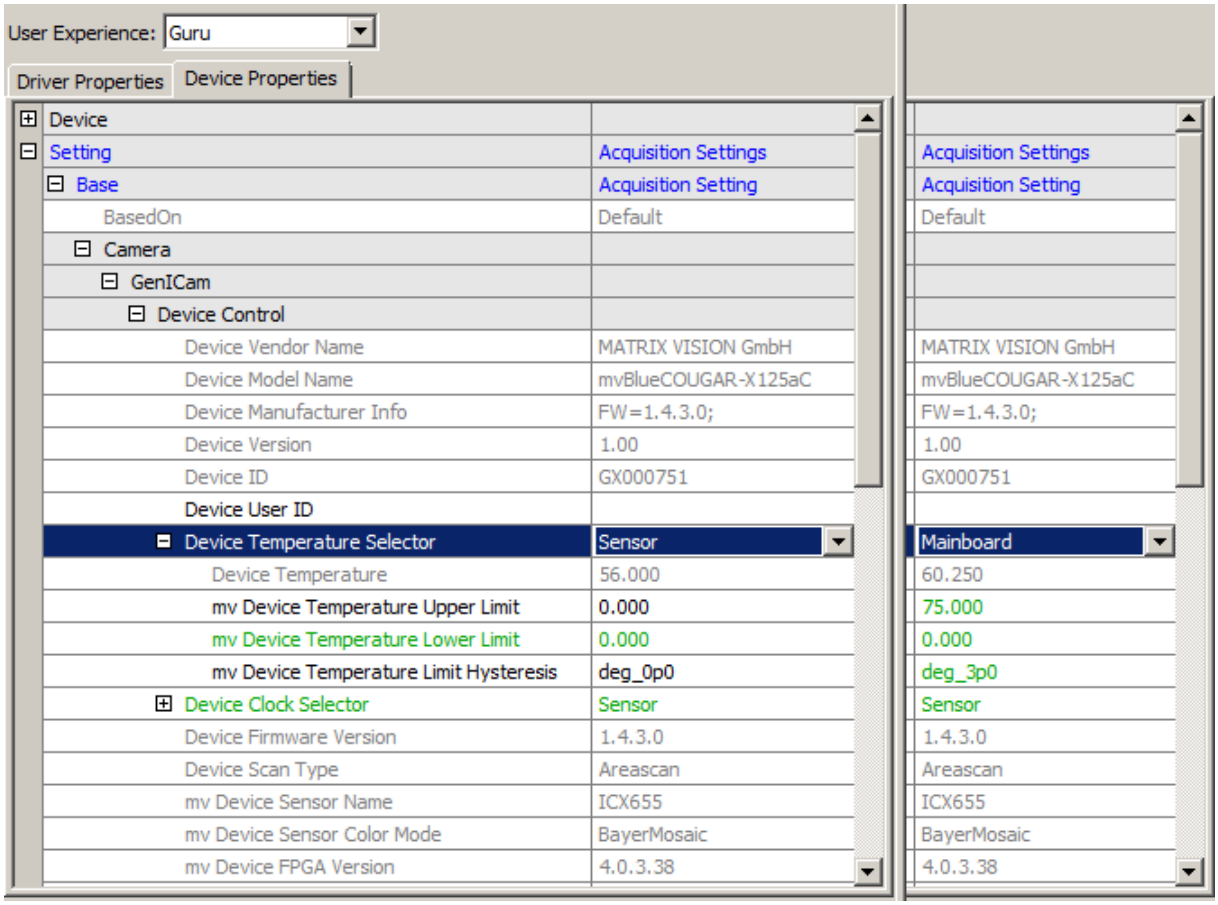


Figure 1: wxPropView - Device Temperature Selector

Note

Avoid temperatures higher than 80 C by lowering thermal resistance of camera housing to connecting structure active cooling means.

It is possible to regulate the temperature of the camera. The limits of this feature are

- upper limit (<= 255 C) and
- lower limit (>= 0 C).

Furthermore, the hysteresis (0,1.5,3,6 C) difference between on/off prevents oscillation. The temperature out of range will set selected output high (see Figure 2). The output can directly drive a fan or a (modest) heating system (but not both due to limit of IC).

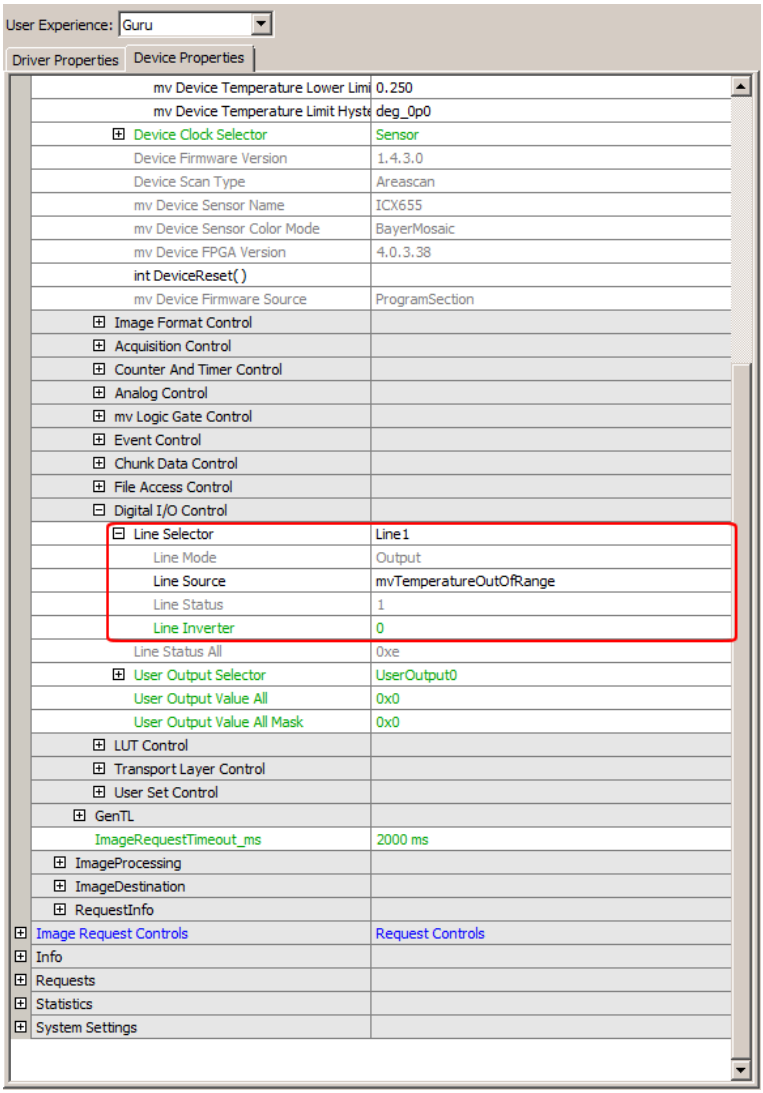


Figure 2: wxPropView - Temperature I/O setting

1.18.6.2 Reset timestamp by hardware

This feature can be used

- for precise control of timestamp
  - for one camera or
  - to synchronize timestamp of multitude of cameras.

The latter sample, can be achieved by following steps:

1. Define the input line (*"TriggerSource"*) to reset the timestamp, e.g. **"Line5"** and
2. set the *"Trigger Selector"* to **"mvTimestampReset"**.
3. Connect all input lines of all cameras together.
4. Finally, use one output of one camera to generate reset edge:

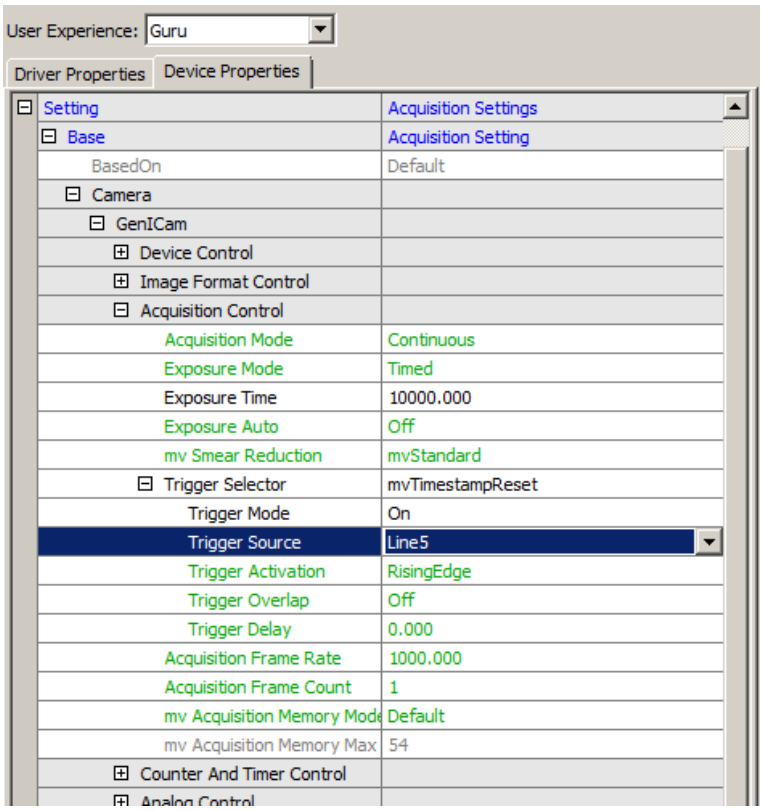


Figure 1: wxPropView - Setting the sample

Note

Be aware of the drift of the individual timestamps.

The timestamp is generated via FPGA in the camera which itself is clocked by a crystal oscillator. This is done independently in each camera and by default not synchronized among cameras or the host system.

Typical stability of crystal oscillators is in the range 100ppm (parts per million).

I.e. for longer operation times (say in excess of hours) there is a tendency that timestamps of individual cameras drift against each other and against the time in the operating system of the host.

Customers wishing to use the individual camera timestamps for synchronization and identification of images via timestamps for multi-camera systems will have in the meantime
- to reset all timestamps either by hardware signal or by command and regularly resynchronize or check the drift algorithmically
- in order to make sure that the drift is less half an image frame time.

1.18.6.3 Synchronizing camera timestamps

1.18.6.3.1 Introduction

Camera timestamps are a recommended **Genicam** (p. 113) / **SFNC** (p. 121) feature to add the information when an image was taken (exactly: when the exposure of the image started).

Without additional synchronization it is merely a camera individual timer with a vendor specific increment and implementation dependent accuracy. Each camera starts its own timestamp beginning with zero and there are no means to adjust or synchronize them among cameras or host PCs. MATRIX VISION cameras offer several ways of synchronizing:

- **mvTimestampReset** (p. 195)
- **Pulse-per-second signal (PPS)** (p. 197)
- **mvDeviceTimeSync** (p. 199)

The usage is described below.

1.18.6.3.2 Without Precision Time Protocol (PTP) IEEE1588

There are many applications which do not or cannot profit from **"IEEE1588"** but have certain synchronization needs. Solutions for these scenarios are describes as follows.

1.18.6.3.3 Synchronizing using mvTimestampReset

First of all the standard does not provide hardware means to reset the timestamp in a camera other than plug off and on again. Therefore MATRIX VISION has created its own mechanism *mvTimestampReset* to reset the timestamp by a hardware input.

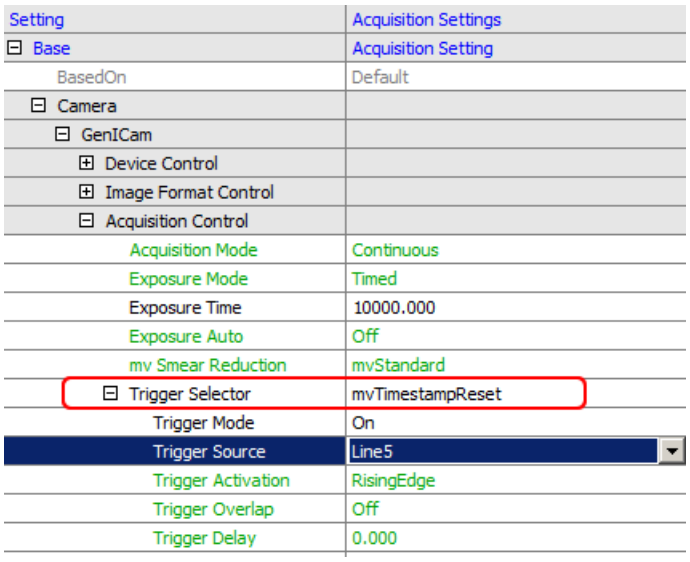


Figure 1: mvTimestampReset

This can be elegantly used for synchronization purposes by means of wiring an input of all cameras together and reset all camera timestamps at the beginning by a defined signal edge from the process. From this reset on all cameras start at zero local time and will increment independently their timestamp so that we achieve a basic accuracy only limited by drift of the clock main frequency (e.g. a 1 MHz oscillator in the FPGA) over time.

In order to compensate for this drift we can in addition reset the timestamp every second or minute or so and count the reset pulse itself by a counter in each camera. Assuming this reset pulse is generated by the master camera itself by means of a timer and output as the hardware reset signal for all cameras, we now can count the reset pulse with all cameras and put both the count and the reset timestamp as so called chunk data in the images.

We thus have achieved a synchronized timestamp with the precision of the master camera among all connected cameras.

Settings required are shown using MATRIX VISION's **wxPropView** (p. 65) tool:

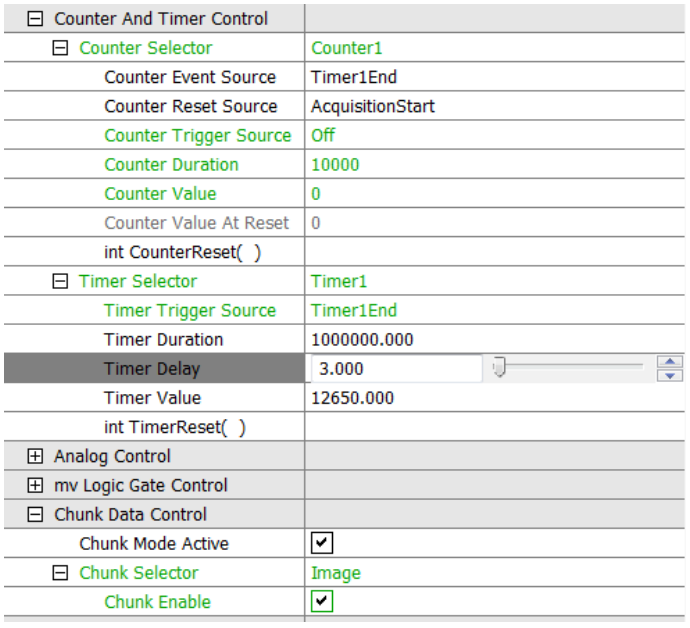


Figure 2: Reset the timestamp every second

An example of the chunk data attached to the image can be seen below. The timestamp is in µs and *Counter1* counts the reset pulses, in this case itself generated by the camera via *Timer1*.

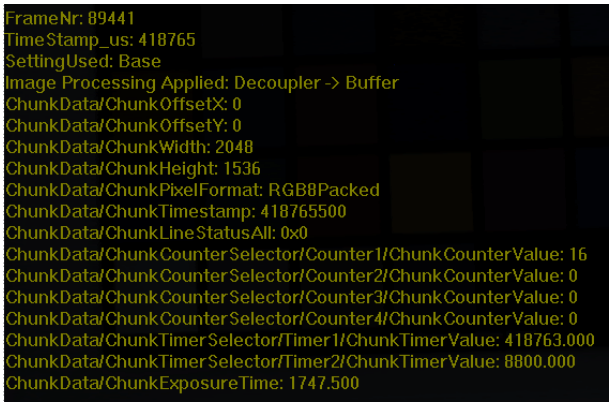


Figure 3: ChunkData

The task of resetting the counter at the beginning of the acquisition can be done by setting the reset property accordingly. Of course is this all independent whether the camera is acquiring images in triggered or continuous mode.

1.18.6.3.4 Synchronizing using a pulse-per-second signal

In order to eliminate the unknown drifts of different devices, a PPS (pulse per second) signal can be fed into each camera using a PC with NTP (network time protocol software), GPS devices or even a looped-back camera timer.

From these pulses the device can extract *how long one second is*. When a device detects that it is no longer running precisely it will adapt its internal clock leading to a "stabilized oscillator".

The device would then maintain their timestamp-differences over long times and stay synchronized. The initial difference between the timers - before the PPS was used - remains however. If you aim to eliminate that as well, you can use the `mvTimestampReset` up front with the same PPS input signal. In an application this can be configured like this (C# syntax):

```
// -----------------------------------------------------------------------
bool waitForNextPulse(Device pDev, String triggerLine)
// -----------------------------------------------------------------------
{
    GenICam.CounterAndTimerControl ctc = new GenICam.CounterAndTimerControl(pDev);
    ctc.counterEventSource.writeS(triggerLine);
    long momentaryValue = ctc.counterValue.read();
    for (int i=0; i<12; i++)
    {
        System.Threading.Thread.Sleep(100);
        if (momentaryValue != ctc.counterValue.read())
        {
            return true;
        }
    }
    return false;
}

// -----------------------------------------------------------------------
void SetupPPS(Device[] pDevs)
// -----------------------------------------------------------------------
{
    string TriggerLine = "Line4";

    if (!waitForNextPulse(pDevs[0],TriggerLine))
    {
```

```
        Console.WriteLine("No pulse seems to be present");
        return;
    }

    // No configure all the devices to reset their timestamp with each pulse coming
    // on the trigger line that the PPS signal is connected to.
    foreach(Device aDevice in pDevs)
    {
        GenICam.AcquisitionControl ac = new GenICam.AcquisitionControl(aDevice);
        ac.triggerSelector.writeS("mvTimestampReset");
        ac.triggerSource.writeS(TriggerLine);
        ac.triggerMode.writeS("On");
    }

    // wait for the next pulse that will then reset the timestamps of all the devices
    if (!waitForNextPulse(pDevs[0],TriggerLine))
    {
        Console.WriteLine("the pulses aren't coming fast enough ...");
        return;
    }

    // Now switch off the reset of the timestamp again. All devices did restart their
    // timestamp counters and will stay in sync using the PPS signal now
    foreach(Device aDevice in pDevs)
    {
        GenICam.AcquisitionControl ac = new GenICam.AcquisitionControl(aDevice);
        ac.triggerMode.writeS("Off");
        GenICam.DeviceControl dc = new GenICam.DeviceControl(aDevice);
        dc.mvTimestampPPSSync.writeS("Line4");
    }
}
```

1.18.6.3.4.1 Using a looped-back camera timer for the PPS signal

To reduce the amount of hardware needed you might want to sacrifice some timestamp validity and use one of the cameras as a master clock. This can be done like this:

- setting *Timer1* to duration 1s
- starting *Timer2* with every *Timer1End* and generating a short pulse (duration = 1000 us)
- placing *Timer2Active* on one of the digital I/O lines and using that as the source for PPS signal

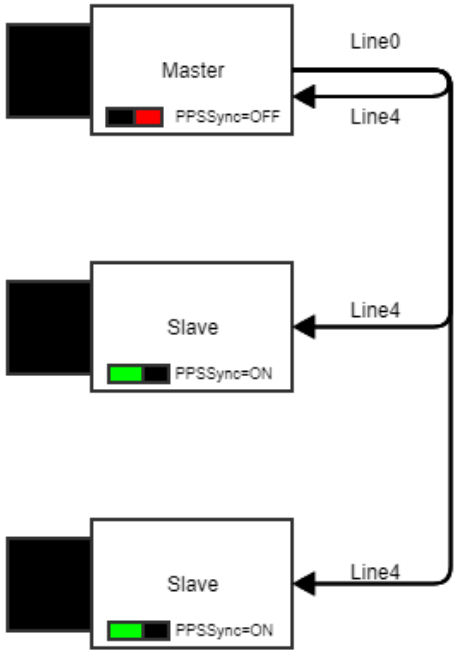


Figure 4: Setup for looped-back Timer

Setting the *Timer1* to 1 s seems like an easy task but due to some internal dependencies you should be carefully here.

- Please use 999997 us duration since the self-triggering will consume the other 3 us
- Please refrain from switching on PPSSync inside the Master camera since (at least in Type 1 cameras) since this will lead to an instable feedback loop.

1.18.6.3.5 Synchronizing using mvDeviceTimeSync

The **Device Control** (p. 66) property **"mvDeviceTimeSync"** is another way to synchronize the timestamp. The **"mvDeviceTimeSync"** provides three possibilites:

- **mvSystemTime** (Default)
  The camera uses the timestamp from the host PC and synchronizes approx. every second. The timestamp reference used is the time in microseconds since January 1st 1970 (i.e. UNIX time).
- **mvNoSynchronization**
  There will be no synchronization, i.e. the timestamps drift apart.
- **mvPushSynchronization**
  If this possibility is selected you may write another property called **"mvDeviceTimeSyncPushValue"** to transfer a timestamp to the camera from another source, for example from a GPS receiver connected to the host PC. You can also use this property to write the normal system time manually to the camera. This is similar to using **"mvSystemTime"** but there will be no automatic resynchronization and you can choose exactly when to synchronize the times of the camera and PC yourself.

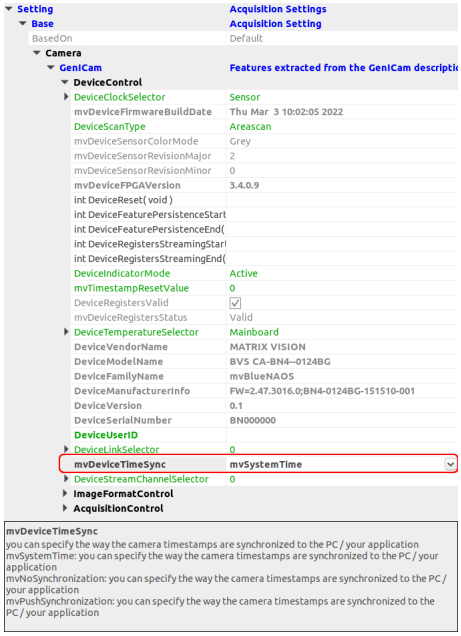


Figure 6: mvDeviceTimeSync

1.18.6.4 Working With The Serial Interface (mv Serial Interface Control)

1.18.6.4.1 Introduction

As mentioned in **GenICam And Advanced Features** section of this manual, the **mv Serial Interface Control** (p. 84) is a feature which allows an easy integration of motor lenses or other peripherals based on RS232.

- Available message buffer size: 128 Bytes

Note

Use the Power GND for the RS232 signal.

1.18.6.4.2 Setting up the device

Follow these steps to prepare the camera for the communication via RS232:

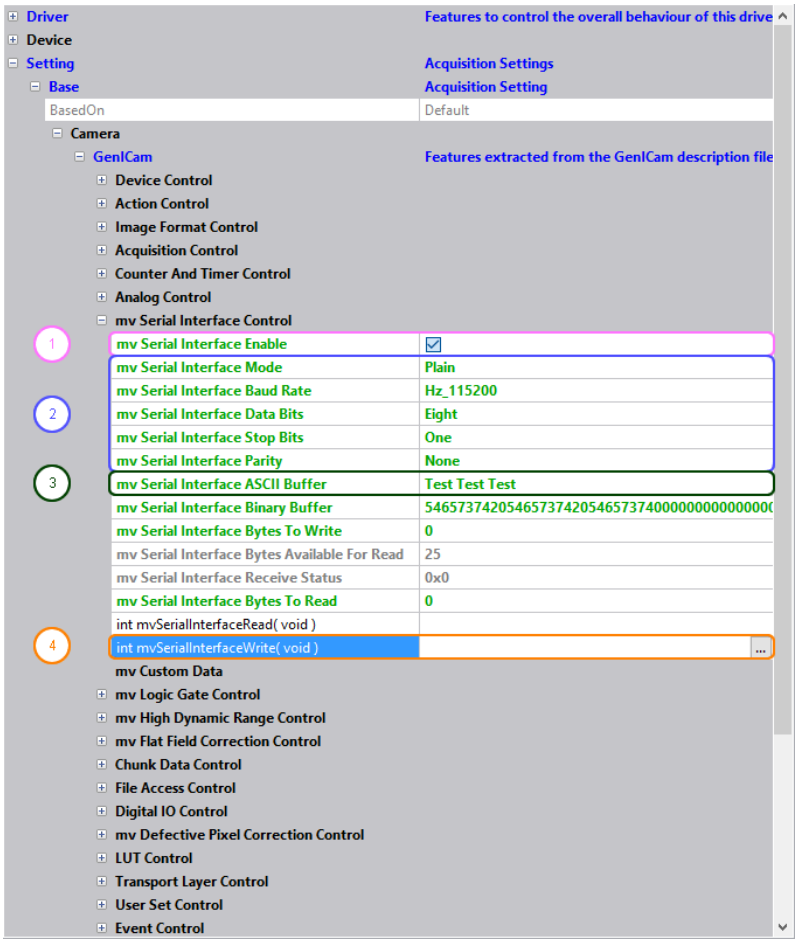


Figure 1: wxPropView - mv Serial Interface Control

1. Start **wxPropView** (p. 65)
2. Connect to the camera
3. Under *"Setting -> Base -> Camera -> GenICam -> mv Serial Interface Control"* activate the serial interface by enabling
   "mv Serial Interface Enable" (1).
   Afterwards **"mv Serial Interface Control"** is available.
4. Set up the connection settings to your needs (2).
5. To test the settings you can send a test message (3).
6. Send messages by executing the function *"int mvSerialInterfaceWrite(void)"* by either clicking on the 3 dots next to the function name or by right-clicking on the command and then selecting *Execute* from the context menu.(4)

Note

Please enter a hexadecimal value in **mvSerialInterfaceBinaryBuffer**.

If you are listening to the RS232 serial line using a tool like *PuTTY* with matching settings...

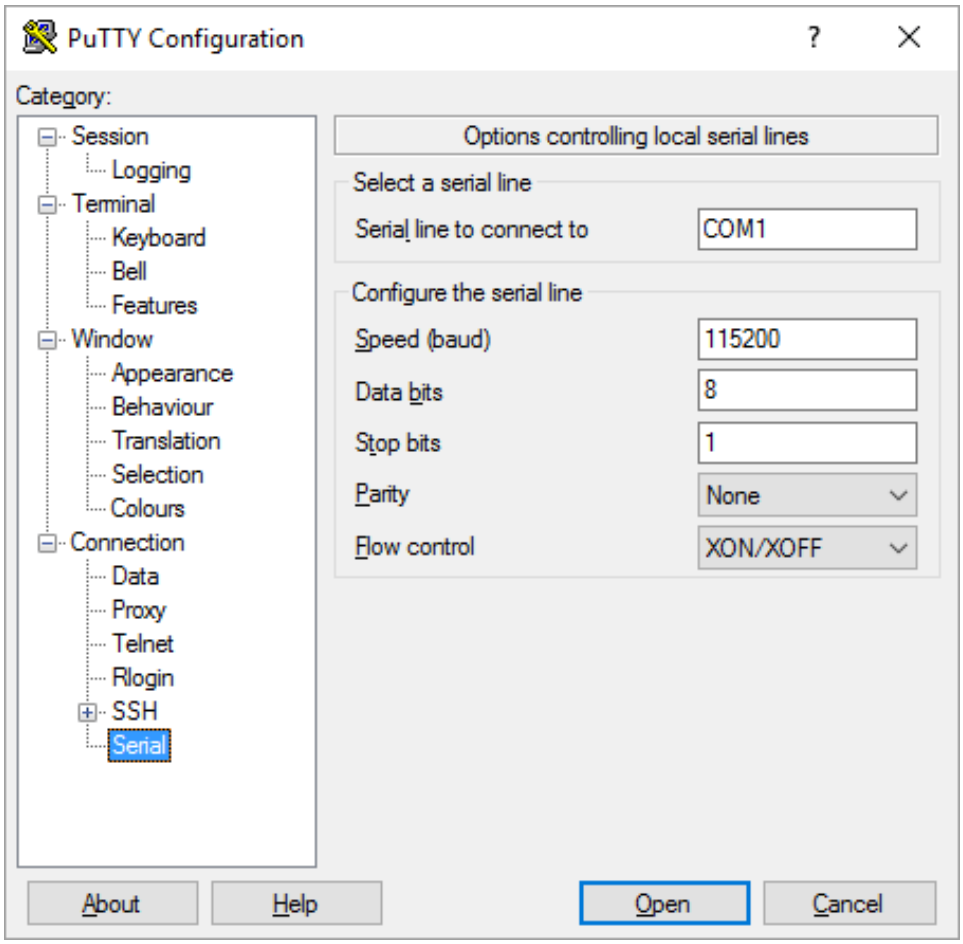


Figure 2: PuTTY - Setting up the serial interface

you will see the test message:



Figure 3: PuTTY - Receiving the test message

1.18.6.4.3 Programming the serial interface

```
#include <mvIMPACT_CPP/mvIMPACT_acquire_GenICam.h>

  // more code
  GenICam::mvSerialInterfaceControl sic( pDev );
  sic.mvSerialInterfaceBaudRate.writeS( "Hz_115200" );
  sic.mvSerialInterfaceASCIIBuffer.writeS( "Test Test Test" );
  sic.mvSerialInterfaceWrite();
  // more code
```

1.18.7 Working with several cameras simultaneously

- **Creating synchronized acquisitions using timers** (p. 204)

1.18.7.1 Creating synchronized acquisitions using timers

1.18.7.1.1 Basics

Getting images from several cameras exactly at the same time is a major task in

- 3D image acquisitions
  (the images must be acquired at the same time using two cameras) or
- acquisitions of larger objects
  (if more than one camera is required to span over the complete image, like in the textile and printing industry).

To solve this task MATRIX VISION devices offer timers that can be used to generate pulse at regular intervals. This pulse can be connected to a digital output. The digital output can be connected digital to the digital input of one or more cameras to use it as a trigger.

1.18.7.1.2 Connecting the hardware

1.18.7.1.2.1 On the master camera

1. Connect power supply GND to **"pin 1"** and **"pin 7"**.
2. Connect power supply +12 V to **"pin 2"** and **"pin 10 (power supply for the outputs)"**.
3. Connect DigOut0 of master (**"pin 6"**) to DigIn0 of master (**"pin 4"**).

1.18.7.1.2.2 On each slave camera

1. Connect power supply GND to **"pin 1"** and **"pin 7"**.
2. Connect power supply +12 V to **"pin 2"** and **"pin 10 (power supply for the outputs)"**.

1.18.7.1.2.3 Between the cameras

1. Connect DigOut0 of master (**"pin 6"**) to DigIn0 of slave (**"pin 4"**).

Note

If more than one slave is used, connect the same **"pin 6"** of master to all **"pin 4"** of the slaves.

If each camera has its own power supply, then connect all grounds (GND) together.

For the master camera, there are 2 possibilities how it is triggered:

1. The master camera triggers itself logically (so called **"Master - Slave"**, see Figure 1), or
2. the master camera uses the external trigger signal, which was created by itself, via digital input (so called **"Slave - Slave"**, see Figure 2).

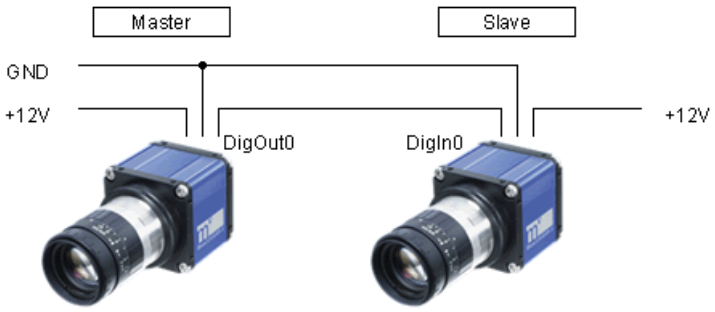


Figure 1: Master - Slave connecting

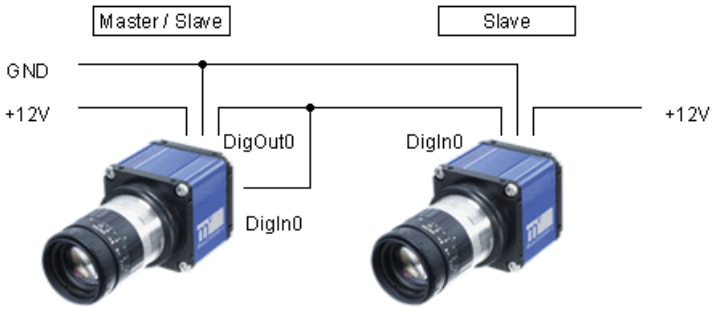


Figure 2: Slave - Slave connecting

Note

With **"Master - Slave"** and according to the delay of the opto-isolated inputs of the slave cameras, you have to adapted the property *"Trigger delay"* of the master camera to synchronize the cameras exactly.

1.18.7.1.3 Programming the acquisition

You will need two timers and you have to set a trigger.

1.18.7.1.3.1 Start timer

Two timers are used for the **"start timer"**. *Timer1* defines the interval between two triggers. *Timer2* generates the trigger pulse at the end of *Timer1*.

The following sample shows a trigger

- which is generated every second and
- the pulse width is 10 ms:

```
#include <mvIMPACT_CPP/mvIMPACT_acquire.h>
#include <mvIMPACT_CPP/mvIMPACT_acquire_GenICam.h>

...

// Master: Set timers to trig image: Start after queue is filled
    GenICam::CounterAndTimerControl catcMaster(pDev);
    catcMaster.timerSelector.writeS( "Timer1" );
    catcMaster.timerDelay.write( 0. );
    catcMaster.timerDuration.write( 1000000. );
    catcMaster.timerTriggerSource.writeS( "Timer1End" );

    catcMaster.timerSelector.writeS( "Timer2" );
    catcMaster.timerDelay.write( 0. );
    catcMaster.timerDuration.write( 10000. );
    catcMaster.timerTriggerSource.writeS( "Timer1End" );
```

See also

Counter And Timer Control (p. 73)

Note

Make sure the *Timer1* interval must be larger than the processing time. Otherwise, the images are lost.

The timers are defined, now you have to do following steps:

1. Set the digital output, e.g. **"Line 0"**,
2. connect the digital output with the inputs of the slave cameras, and finally
3. set the trigger source to the digital input, e.g. **"Line4"**.

1.18.7.1.3.2 Set digital I/O

In this step, the signal has to be connected to the digital output, e.g. **"Line0"**:

```
// Set Digital I/O
    GenICam::DigitalIOControl io(pDev);
    io.lineSelector.writeS( "Line0" );
    io.lineSource.writeS( "Timer2Active" );
```

See also

Digital I/O Control (p. 80)

This signal has to be connected with the digital inputs of the slave cameras as shown in Figure 1 and 2.

1.18.7.1.3.3 Set trigger

"If you want to use Master - Slave":

```
// Set Trigger of Master camera
    GenICam::AcquisitionControl ac(pDev);
    ac.triggerSelector.writeS( "FrameStart" );
    ac.triggerMode.writeS( "On" );
    ac.triggerSource.writeS( "Timer1Start" );
// or ac.triggerSource.writeS( "Timer1End" );


// Set Trigger of Slave camera
    GenICam::AcquisitionControl ac(pDev);
    ac.triggerSelector.writeS( "FrameStart" );
    ac.triggerMode.writeS( "On" );
    ac.triggerSource.writeS( "Line4" );
    ac.triggerActivation.writeS( "RisingEdge" );
```

See also

Acquisition Control (p. 69)

Now, the two timers will work like the following figure illustrates, which means

- *Timer1* is the trigger event and
- *Timer2* the trigger pulse width:

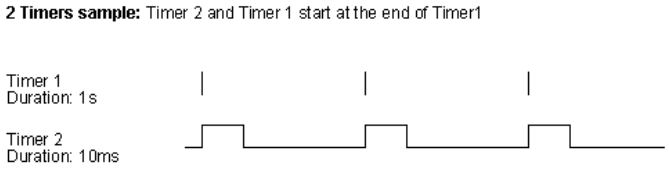


Figure 2: Timers

By the way, this is a simple **"pulse width modulation (PWM)"** example.

1.18.7.1.4 Setting the synchronized acquisition using wxPropView

The following figures show, how you can set the timers and trigger using the GUI tool **wxPropView** (p. 65)

1. Setting of *Timer1* (blue box) on the master camera:

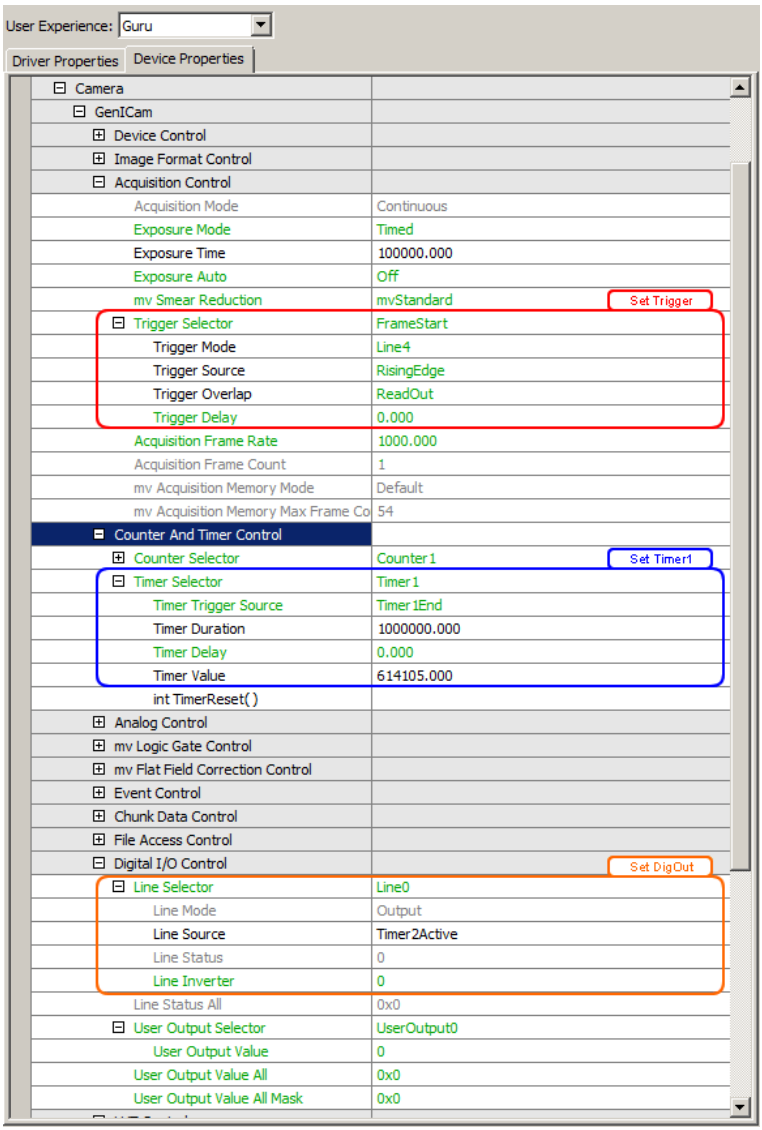


Figure 3: wxPropView - Setting of *Timer1* on the master camera

2. Setting of *Timer2* (purple box) on the master camera:

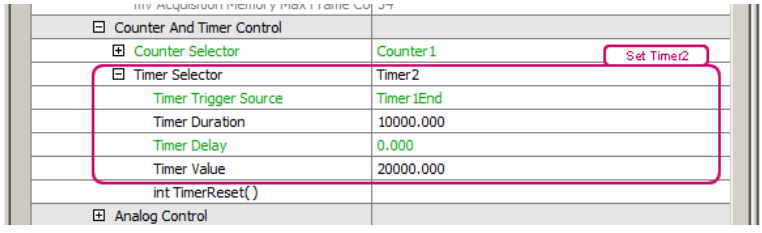


Figure 4: wxPropView - Setting of *Timer2* on the master camera

3. Setting the trigger slave camera(s)
   - The red box in Figure 5 is showing **"Master - Slave"**), which means that the master is triggered internally and the slave camera is set as shown in Figure 4.

4. Assigning timer to DigOut (orange box in Figure 3).

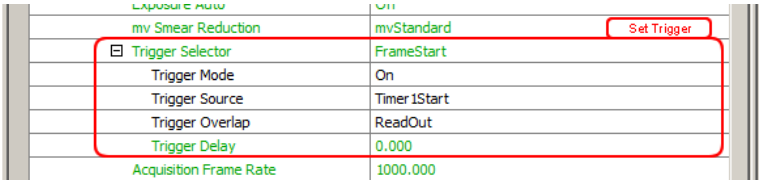


Figure 5: Trigger setting of the master camera using **"Master - Slave"**

1.18.8 Working with 3rd party tools

- **Using VLC Media Player** (p. 210)

1.18.8.1 Using VLC Media Player

With the **DirectShow interface** (p. 87) MATRIX VISION devices become a (acquisition) video device for the VLC Media Player.

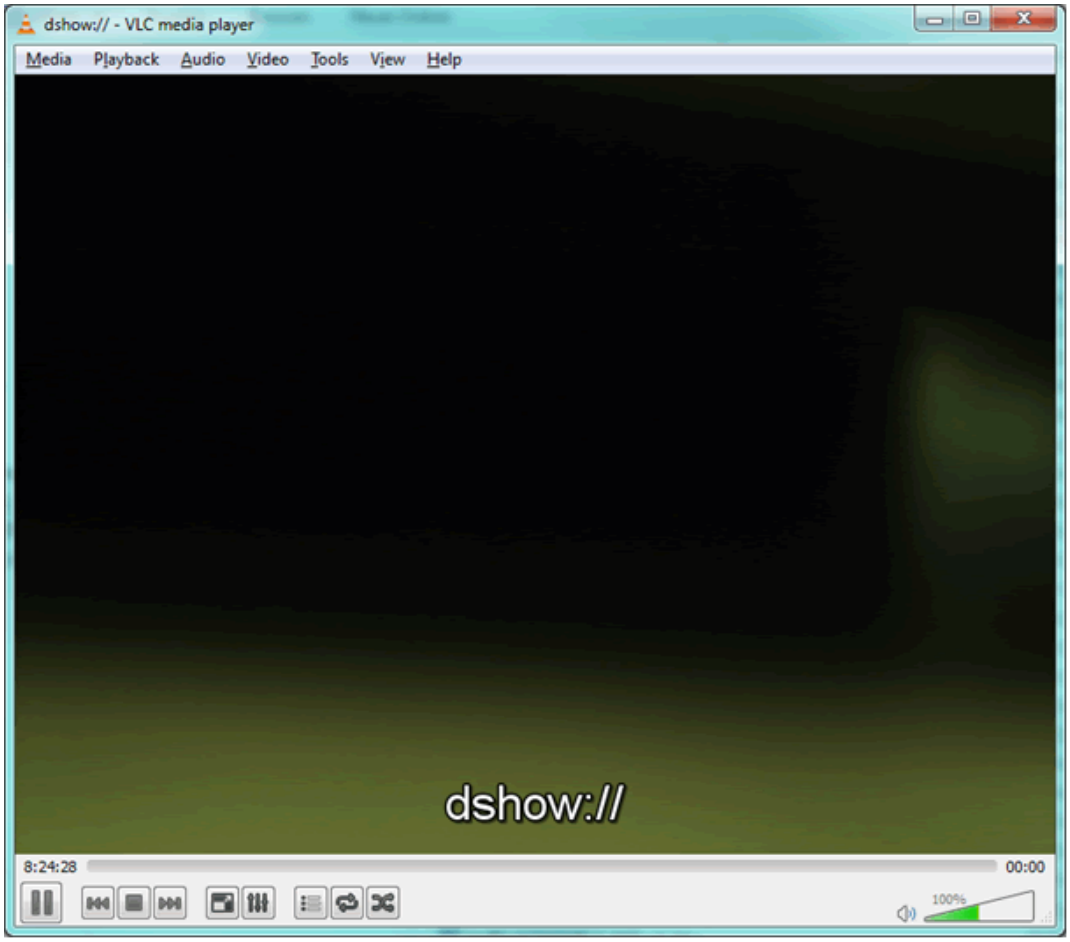


Figure 1: VLC Media Player with a connected device via DirectShow

1.18.8.1.1 System requirements

It is necessary that following drivers and programs are installed on the host device (laptop or PC):

- Windows 7 or higher, 32-bit or 64-bit
- up-do-date VLC Media Player, 32-bit or 64-bit (here: version 2.0.6)
- up-do-date MATRIX VISION driver, 32-bit or 64-bit (here: version 2.5.6)

Note

Using **Windows 10** or **Windows 7**: VLC Media Player with versions 2.2.0 have been tested successfully with older versions of mvIMPACT Acquire. Since version 3.0.0 of VLC at least mvIMPACT Acquire 2.34.0 will be needed to work with devices through the DirectShow interface!

1.18.8.1.2 Installing VLC Media Player

1. Download a suitable version of the VLC Media Player from the VLC Media Player website mentioned below.
2. Run the setup.
3. Follow the installation process and use the default settings.

A restart of the system is not required.

See also

`http://www.videolan.org/`

1.18.8.1.3 Setting up MV device for DirectShow

Note

Please be sure to register the MV device for DirectShow with the right version of **mvDeviceConfigure** (p. 65) . I.e. if you have installed the 32 bit version of the VLC Media Player, you have to register the MV device with the 32 bit version of **mvDeviceConfigure** (p. 65) (*"C:/Program Files/MATRIX VISION/mv↩ IMPACT Acquire/bin"*) !

1. Connect the MV device to the host device directly or via GigE switch using an Ethernet cable.
2. Power the camera using a power supply at the power connector.
3. Wait until the status LED turns blue.
4. Open the tool **mvDeviceConfigure** (p. 65) ,
5. **set a friendly name** (p. 90) ,
6. and **register the MV device for DirectShow** (p. 87) .

Note

In some cases it could be necessary to repeat step 5.

1.18.8.1.4 Working with VLC Media Player

1. Start VLC Media Player.
2. Click on **"Media ->** **Open Capture Device..."** .

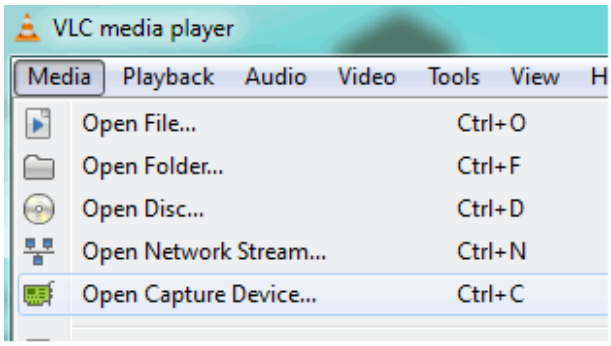


Figure 2: Open Capture Device...

3. Select the tab **"Device Selection"** .
4. In the section **"Video device name"** , select the friendly name of the MV device:

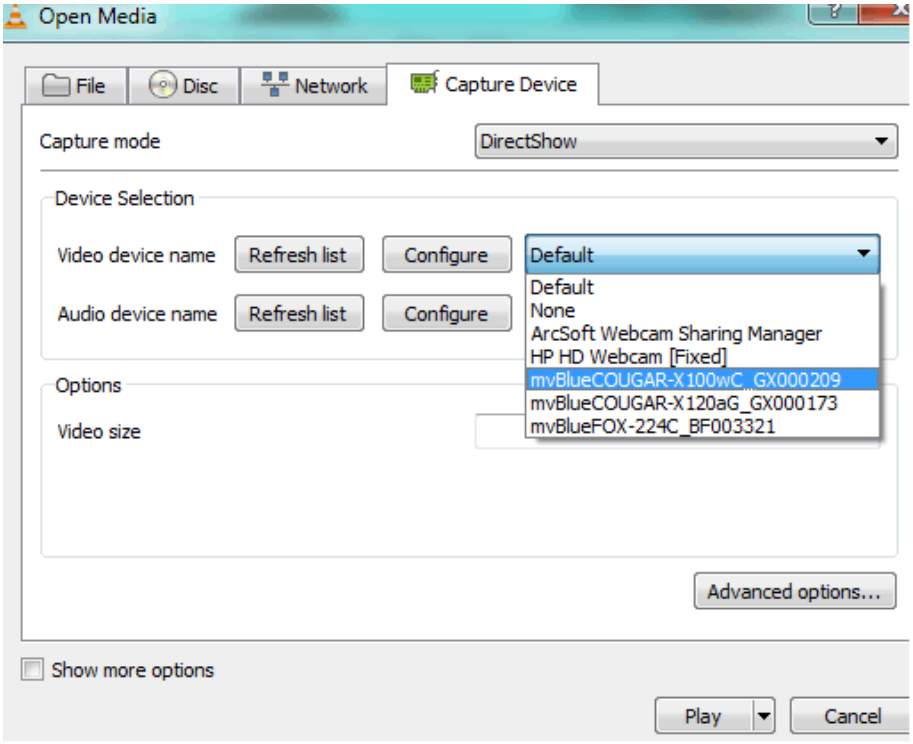


Figure 3: Video device name

5. Finally, click on **"Play"** .
   After a short delay you will see the live image of the camera.

1.19 Appendix A. Specific Camera / Sensor Data

- **A.1 Pregius CMOS (Mono)** (p. 213)
- **A.2 Pregius CMOS (Color)** (p. 243)

1.19.1 A.1 Pregius CMOS (Mono)

- **mvBlueNAOS2** (p. 213)
- **mvBlueNAOS4** (p. 228)

1.19.1.1 mvBlueNAOS2

- **BVS CA-BN2-0016ZG (1.6 Mpix [1456 x 1088])** (p. 213)
- **BVS CA-BN2-0032ZG (3.2 Mpix [2064 x 1544])** (p. 215)
- **BVS CA-BN2-0051ZG (5.1 Mpix [2464 x 2056])** (p. 218)
- **BVS CA-BN2-0089ZG (8.9 Mpix [4112 x 2176])** (p. 220)
- **BVS CA-BN2-0124ZG (12.4 Mpix [4112 x 3008])** (p. 223)
- **BVS CA-BN2-0246AG (24.6 Mpix [5328 x 4608])** (p. 225)

1.19.1.1.1 BVS CA-BN2-0016ZG (1.6 Mpix [1456 x 1088])

1.19.1.1.1.1 Introduction The sensor uses a global shutter, i.e. light exposure takes place on all pixels in parallel, although subsequent readout is sequential.

Feature	Description
Manufacturer	Sony
Sensor name	IMX273
Max. frames per second	226.5
Device Structure	CMOS image sensor
SNR_{max} [dB][1/]	40,1
DNR (normal / HDR) [dB][1/]	71,1 / -
Image size	1/2.9
Number of effective pixels	1456 (H) x 1088 (V)
Unit cell size	3.45µm (H) x 3.45µm (V)
ADC resolution / output	12 bit → 8/10/(12)

[1] Measured accord. to EMVA1288 with gray scale version of the camera

1.19.1.1.1.2 Spectral Sensitivity

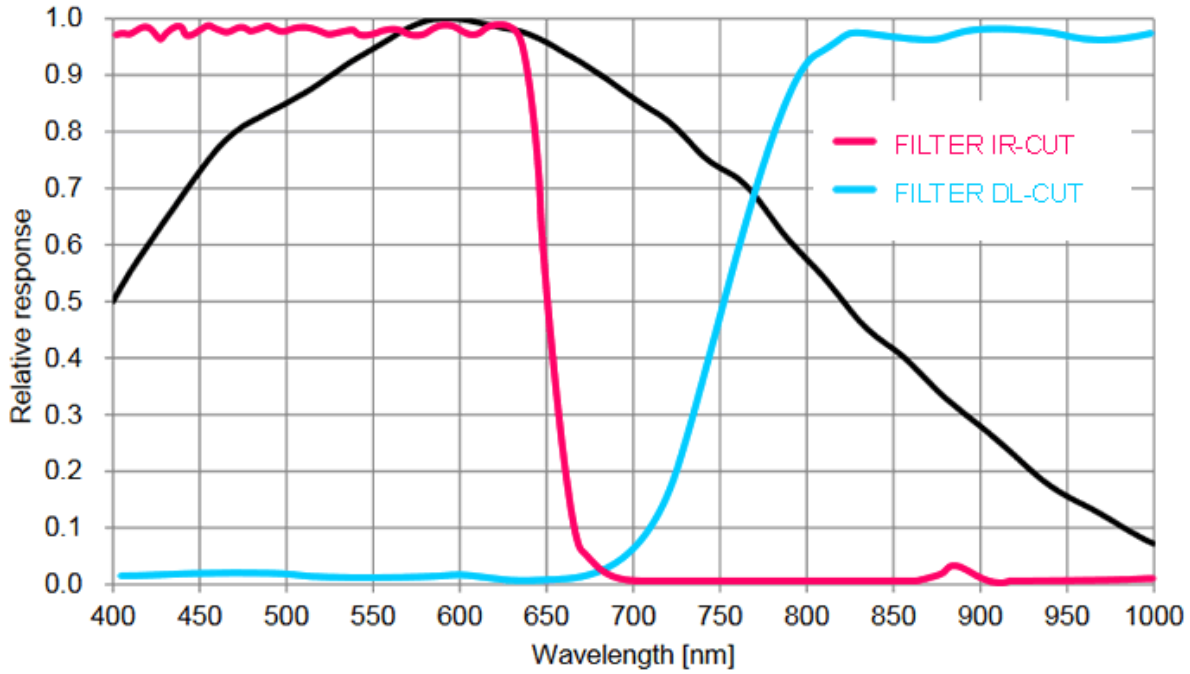


Figure 1: Spectral sensitivity BVS CA-BN2-0016ZG

1.19.1.1.1.3 Timings For calculating the frame rates, please use the frame rate calculator of the online manual:

See also

`https://www.matrix-vision.com/manuals/mvBlueNAOS/mvBlue_models_BVS_CA-BN2-0016ZG_.html#mvBC_subsubsection_sensors_CMOS_models_BVS_CA-BN2-0016ZG_Calc`

The following trigger modes are available:

Setting (GenICam)	Mode / Setting (obsolete "Device Specific")	Description
`"TriggerSelector = FrameStart"` `"TriggerMode = Off"`	Continuous	Free running, no external trigger signal needed.

`"TriggerSelector = FrameStart"`
`"TriggerMode = On"`
`"TriggerSource = Software"`
`"ExposureMode = Timed"`
 	
To trigger one frame execute the *TriggerSoftware@i* command then.
	OnDemand	Image acquisition triggered by command (software trigger).

"TriggerSelector = AcquisitionActive" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = LevelLow" "ExposureMode = Timed"	OnLowLevel	Start an exposure of a frame as long as the trigger input is below the trigger threshold. (No Frame↩ Trigger!)
"TriggerSelector = AcquisitionActive" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = LevelHigh" "ExposureMode = Timed"	OnHighLevel	Start an exposure of a frame as long as the trigger input is above the trigger threshold. (No Frame↩ Trigger!)
"TriggerSelector = FrameStart" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = FallingEdge" "ExposureMode = Timed"	OnFallingEdge	Each falling edge of trigger signal acquires one image.
"TriggerSelector = FrameStart" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = RisingEdge" "ExposureMode = Timed"	OnRisingEdge	Each rising edge of trigger signal acquires one image.
"TriggerSelector = FrameStart" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = AnyEdge" "ExposureMode = Timed"	OnAnyEdge	Start the exposure of a frame when the trigger input level changes from high to low or from low to high.

Device Feature And Property List (p. 215)

1.19.1.1.1.4 Device Feature And Property List

1.19.1.1.2 BVS CA-BN2-0032ZG (3.2 Mpix [2064 x 1544])

1.19.1.1.2.1 Introduction The sensor uses a global shutter, i.e. light exposure takes place on all pixels in parallel, although subsequent readout is sequential.

Feature	Description
Manufacturer	Sony
Sensor name	IMX252
Max. frames per second	191.5
Device Structure	CMOS image sensor
SNR_{max} [dB][1/]	40,2
DNR (normal / HDR) [dB][1/]	71,3 / -
Image size	1/1.8
Number of effective pixels	2064 (H) x 1544 (V)
Unit cell size	3.45µm (H) x 3.45µm (V)
ADC resolution / output	12 bit → 8/10/(12)

[1] Measured accord. to EMVA1288 with gray scale version of the camera

1.19.1.1.2.2 Spectral Sensitivity

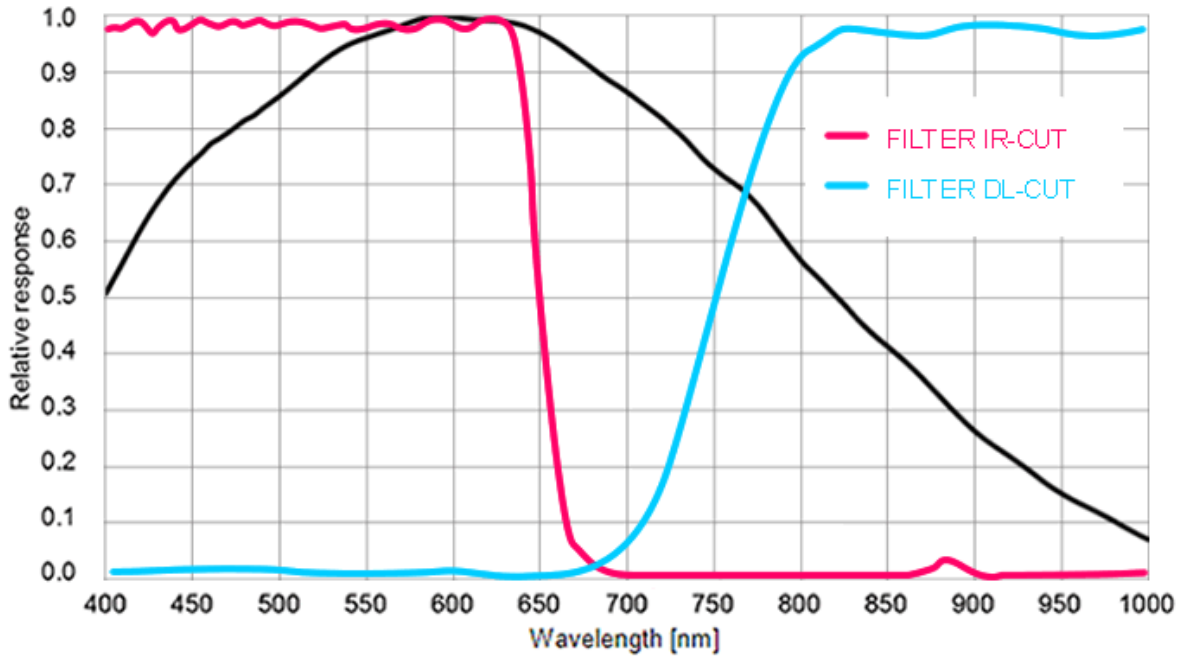


Figure 1: Spectral sensitivity BVS CA-BN2-0032ZG

1.19.1.1.2.3 Timings For calculating the frame rates, please use the frame rate calculator of the online manual:

See also

https://www.matrix-vision.com/manuals/mvBlueNAOS/mvBlue_models_BVS_CA-BN2-0032ZG_.html#mvBC_subsubsection_sensors_CMOS_models_BVS_CA-BN2-0032ZG_Calc

The following trigger modes are available:

Setting (GenICam)	Mode / Setting (obsolete "Device Specific")	Description
"TriggerSelector = FrameStart" "TriggerMode = Off"	Continuous	Free running, no external trigger signal needed.
"TriggerSelector = FrameStart" "TriggerMode = On" "TriggerSource = Software" "ExposureMode = Timed" To trigger one frame execute the *TriggerSoftware@i* command then.	OnDemand	Image acquisition triggered by command (software trigger).
"TriggerSelector = AcquisitionActive" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = LevelLow" "ExposureMode = Timed"	OnLowLevel	Start an exposure of a frame as long as the trigger input is below the trigger threshold. (No Frame↩ Trigger!)
"TriggerSelector = AcquisitionActive" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = LevelHigh" "ExposureMode = Timed"	OnHighLevel	Start an exposure of a frame as long as the trigger input is above the trigger threshold. (No Frame↩ Trigger!)
"TriggerSelector = FrameStart" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = FallingEdge" "ExposureMode = Timed"	OnFallingEdge	Each falling edge of trigger signal acquires one image.
"TriggerSelector = FrameStart" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = RisingEdge" "ExposureMode = Timed"	OnRisingEdge	Each rising edge of trigger signal acquires one image.

"TriggerSelector = FrameStart" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = AnyEdge" "ExposureMode = Timed"	OnAnyEdge	Start the exposure of a frame when the trigger input level changes from high to low or from low to high.

Device Feature And Property List (p. 218)

1.19.1.1.2.4 Device Feature And Property List

1.19.1.1.3 BVS CA-BN2-0051ZG (5.1 Mpix [2464 x 2056])

1.19.1.1.3.1 Introduction The sensor uses a global shutter, i.e. light exposure takes place on all pixels in parallel, although subsequent readout is sequential.

Feature	Description
Manufacturer	Sony
Sensor name	IMX250
Max. frames per second	140
Device Structure	CMOS image sensor
SNR_{max} [dB][1/]	40,3
DNR (normal / HDR) [dB][1/]	71,3 / -
Image size	2/3
Number of effective pixels	2464 (H) x 2056 (V)
Unit cell size	3.45µm (H) x 3.45µm (V)
ADC resolution / output	12 bit → 8/10/(12)

[1] Measured accord. to EMVA1288 with gray scale version of the camera

1.19.1.1.3.2 Spectral Sensitivity

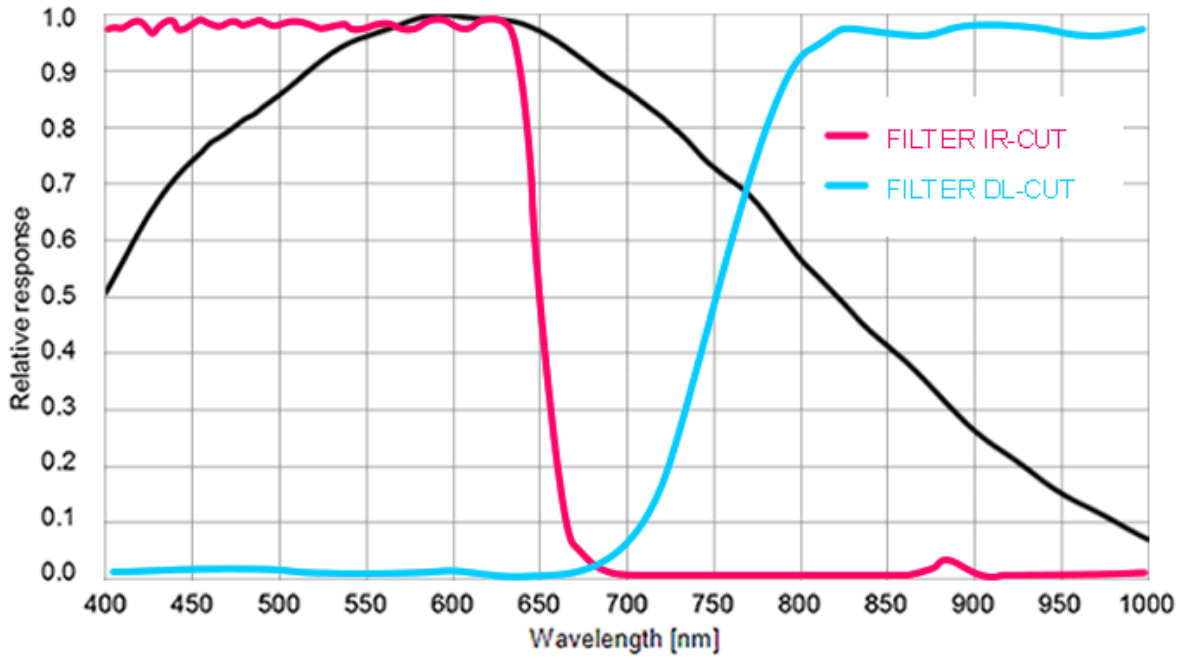


Figure 1: Spectral sensitivity BVS CA-BN2-0051ZG

1.19.1.1.3.3 Timings For calculating the frame rates, please use the frame rate calculator of the online manual:

See also

`https://www.matrix-vision.com/manuals/mvBlueNAOS/mvBlue_models_BVS_CA-BN2-0051ZG_.html#mvBC_subsubsection_sensors_CMOS_models_BVS_CA-BN2-0051ZG_Calc`

The following trigger modes are available:

Setting (GenICam)	Mode / Setting (obsolete "Device Specific")	Description
`"TriggerSelector = FrameStart"` `"TriggerMode = Off"`	Continuous	Free running, no external trigger signal needed.

`"TriggerSelector = FrameStart"`
`"TriggerMode = On"`
`"TriggerSource = Software"`
`"ExposureMode = Timed"`
 	
To trigger one frame execute the *TriggerSoftware@i* command then.
	OnDemand	Image acquisition triggered by command (software trigger).

"TriggerSelector = AcquisitionActive" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = LevelLow" "ExposureMode = Timed"	OnLowLevel	Start an exposure of a frame as long as the trigger input is below the trigger threshold. (No Frame↩ Trigger!)
"TriggerSelector = AcquisitionActive" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = LevelHigh" "ExposureMode = Timed"	OnHighLevel	Start an exposure of a frame as long as the trigger input is above the trigger threshold. (No Frame↩ Trigger!)
"TriggerSelector = FrameStart" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = FallingEdge" "ExposureMode = Timed"	OnFallingEdge	Each falling edge of trigger signal acquires one image.
"TriggerSelector = FrameStart" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = RisingEdge" "ExposureMode = Timed"	OnRisingEdge	Each rising edge of trigger signal acquires one image.
"TriggerSelector = FrameStart" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = AnyEdge" "ExposureMode = Timed"	OnAnyEdge	Start the exposure of a frame when the trigger input level changes from high to low or from low to high.

Device Feature And Property List (p. 220)

1.19.1.1.3.4 Device Feature And Property List

1.19.1.1.4 BVS CA-BN2-0089ZG (8.9 Mpix [4112 x 2176])

1.19.1.1.4.1 Introduction The sensor uses a global shutter, i.e. light exposure takes place on all pixels in parallel, although subsequent readout is sequential.

Feature	Description
Manufacturer	Sony
Sensor name	IMX255
Max. frames per second	88.7
Device Structure	CMOS image sensor
SNR_{max} [dB][1/]	40,3
DNR (normal / HDR) [dB][1/]	71,0 / -
Image size	1
Number of effective pixels	4112 (H) x 2176 (V)
Unit cell size	3.45µm (H) x 3.45µm (V)
ADC resolution / output	12 bit → 8/10/(12)

[1] Measured accord. to EMVA1288 with gray scale version of the camera

1.19.1.1.4.2 Spectral Sensitivity

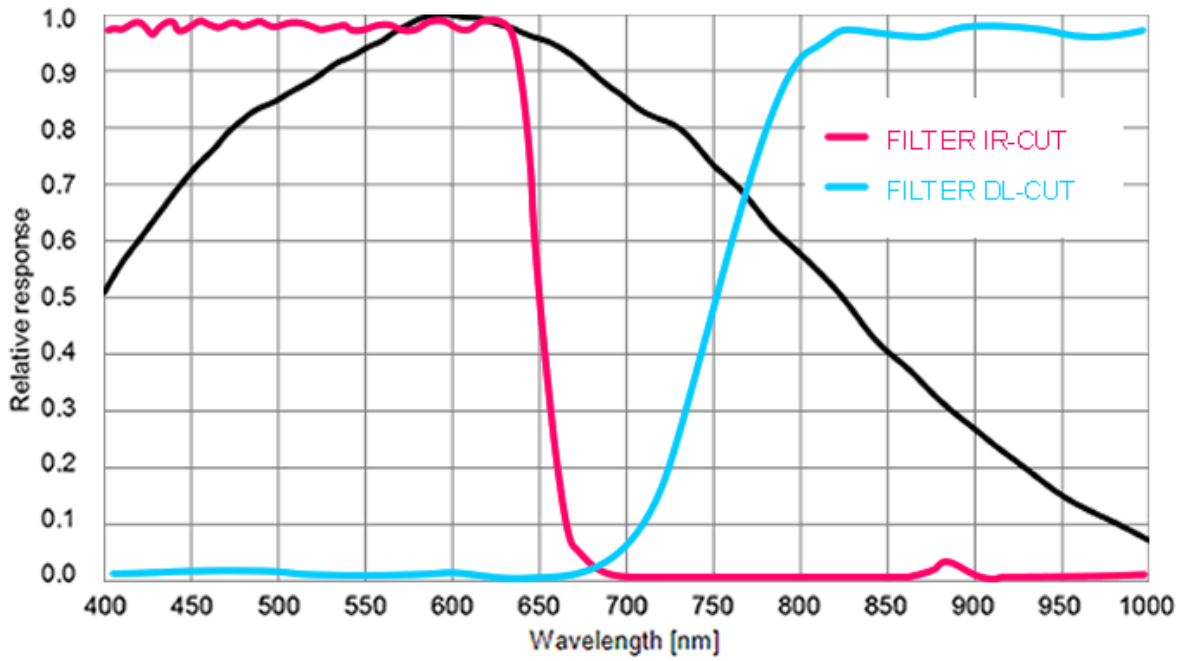


Figure 1: Spectral sensitivity BVS CA-BN2-0089ZG

1.19.1.1.4.3 Timings For calculating the frame rates, please use the frame rate calculator of the online manual:

See also

https://www.matrix-vision.com/manuals/mvBlueNAOS/mvBlue_models_BVS_CA-BN2-0089ZG_.html#mvBC_subsubsection_sensors_CMOS_models_BVS_CA-BN2-0089ZG_Calc

The following trigger modes are available:

Setting (GenICam)	Mode / Setting (obsolete "Device Specific")	Description
"TriggerSelector = FrameStart" "TriggerMode = Off"	Continuous	Free running, no external trigger signal needed.
"TriggerSelector = FrameStart" "TriggerMode = On" "TriggerSource = Software" "ExposureMode = Timed" To trigger one frame execute the *TriggerSoftware@i* command then.	OnDemand	Image acquisition triggered by command (software trigger).
"TriggerSelector = AcquisitionActive" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = LevelLow" "ExposureMode = Timed"	OnLowLevel	Start an exposure of a frame as long as the trigger input is below the trigger threshold. (No Frame↩ Trigger!)
"TriggerSelector = AcquisitionActive" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = LevelHigh" "ExposureMode = Timed"	OnHighLevel	Start an exposure of a frame as long as the trigger input is above the trigger threshold. (No Frame↩ Trigger!)
"TriggerSelector = FrameStart" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = FallingEdge" "ExposureMode = Timed"	OnFallingEdge	Each falling edge of trigger signal acquires one image.
"TriggerSelector = FrameStart" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = RisingEdge" "ExposureMode = Timed"	OnRisingEdge	Each rising edge of trigger signal acquires one image.

"TriggerSelector = FrameStart" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = AnyEdge" "ExposureMode = Timed"	OnAnyEdge	Start the exposure of a frame when the trigger input level changes from high to low or from low to high.

Device Feature And Property List (p. 223)

1.19.1.1.4.4 Device Feature And Property List

1.19.1.1.5 BVS CA-BN2-0124ZG (12.4 Mpix [4112 x 3008])

1.19.1.1.5.1 Introduction The sensor uses a global shutter, i.e. light exposure takes place on all pixels in parallel, although subsequent readout is sequential.

Feature	Description
Manufacturer	Sony
Sensor name	IMX253
Max. frames per second	64.6
Device Structure	CMOS image sensor
SNR_{max} [dB][1/]	40,1
DNR (normal / HDR) [dB][1/]	71,1 / -
Image size	1.1
Number of effective pixels	4112 (H) x 3008 (V)
Unit cell size	3.45µm (H) x 3.45µm (V)
ADC resolution / output	12 bit → 8/10/(12)

[1] Measured accord. to EMVA1288 with gray scale version of the camera

1.19.1.1.5.2 Spectral Sensitivity

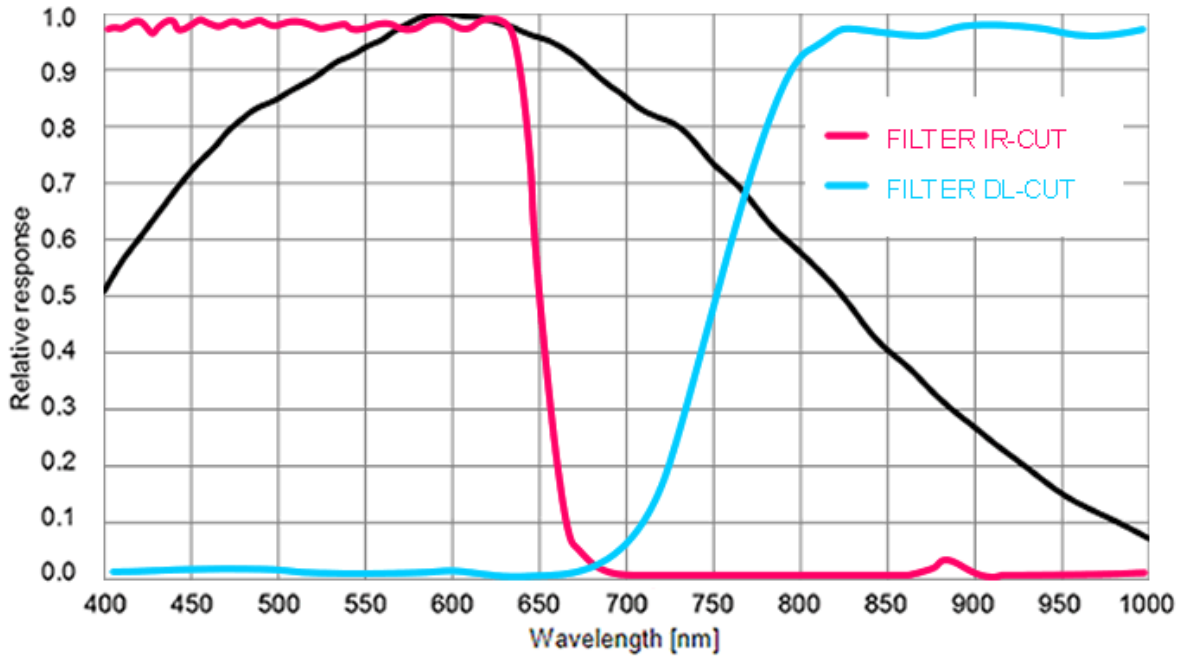


Figure 1: Spectral sensitivity BVS CA-BN2-0124ZG

1.19.1.1.5.3 Timings For calculating the frame rates, please use the frame rate calculator of the online manual:

See also

`https://www.matrix-vision.com/manuals/mvBlueNAOS/mvBlue_models_BVS_CA-BN2-0124ZG_.html#mvBC_subsubsection_sensors_CMOS_models_BVS_CA-BN2-0124ZG_Calc`

The following trigger modes are available:

Setting (GenICam)	Mode / Setting (obsolete "Device Specific")	Description
`"TriggerSelector = FrameStart"` `"TriggerMode = Off"`	Continuous	Free running, no external trigger signal needed.

`"TriggerSelector = FrameStart"`
`"TriggerMode = On"`
`"TriggerSource = Software"`
`"ExposureMode = Timed"`
 	
To trigger one frame execute the *TriggerSoftware@i* command then.
	OnDemand	Image acquisition triggered by command (software trigger).

"TriggerSelector = AcquisitionActive" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = LevelLow" "ExposureMode = Timed"	OnLowLevel	Start an exposure of a frame as long as the trigger input is below the trigger threshold. (No Frame↩ Trigger!)
"TriggerSelector = AcquisitionActive" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = LevelHigh" "ExposureMode = Timed"	OnHighLevel	Start an exposure of a frame as long as the trigger input is above the trigger threshold. (No Frame↩ Trigger!)
"TriggerSelector = FrameStart" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = FallingEdge" "ExposureMode = Timed"	OnFallingEdge	Each falling edge of trigger signal acquires one image.
"TriggerSelector = FrameStart" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = RisingEdge" "ExposureMode = Timed"	OnRisingEdge	Each rising edge of trigger signal acquires one image.
"TriggerSelector = FrameStart" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = AnyEdge" "ExposureMode = Timed"	OnAnyEdge	Start the exposure of a frame when the trigger input level changes from high to low or from low to high.

Device Feature And Property List (p. 225)

1.19.1.1.5.4 Device Feature And Property List

1.19.1.1.6 BVS CA-BN2-0246AG (24.6 Mpix [5328 x 4608])

1.19.1.1.6.1 Introduction The sensor uses a global shutter, i.e. light exposure takes place on all pixels in parallel, although subsequent readout is sequential.

Feature	Description
Manufacturer	Sony
Sensor name	IMX540
Max. frames per second	24.1
Device Structure	CMOS image sensor
SNR_{max} [dB][1/]	39,6
DNR (normal / HDR) [dB][1/]	70,2 / -
Image size	1.2
Number of effective pixels	5328 (H) x 4608 (V)
Unit cell size	2.74µm (H) x 2.74µm (V)
ADC resolution / output	12 bit → 8/10/(12)

[1] Measured accord. to EMVA1288 with gray scale version of the camera

1.19.1.1.6.2 Spectral Sensitivity

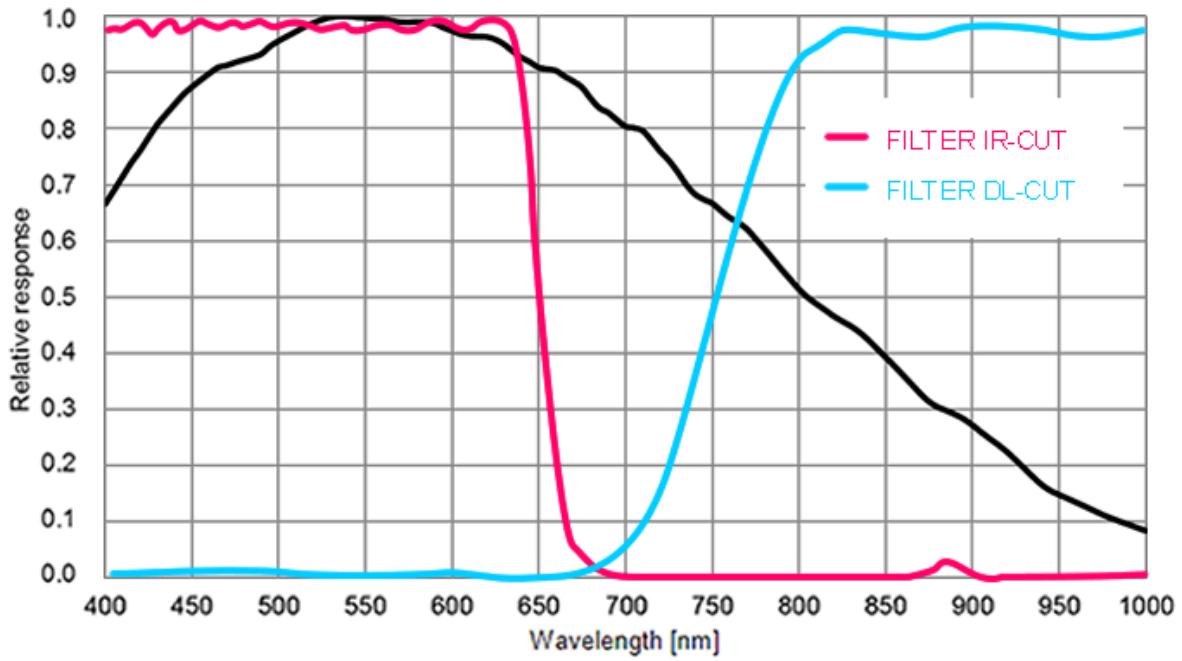


Figure 1: Spectral sensitivity BVS CA-BN2-0246AG

1.19.1.1.6.3 Timings For calculating the frame rates, please use the frame rate calculator of the online manual:

See also

https://www.matrix-vision.com/manuals/mvBlueNAOS/mvBlue_models_BVS_CA-BN2-0246AG_.html#mvBC_subsubsection_sensors_CMOS_models_BVS_CA-BN2-0246AG_Calc

The following trigger modes are available:

Setting (GenICam)	Mode / Setting (obsolete "Device Specific")	Description
"TriggerSelector = FrameStart" "TriggerMode = Off"	Continuous	Free running, no external trigger signal needed.
"TriggerSelector = FrameStart" "TriggerMode = On" "TriggerSource = Software" "ExposureMode = Timed" To trigger one frame execute the *TriggerSoftware@i* command then.	OnDemand	Image acquisition triggered by command (software trigger).
"TriggerSelector = AcquisitionActive" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = LevelLow" "ExposureMode = Timed"	OnLowLevel	Start an exposure of a frame as long as the trigger input is below the trigger threshold. (No Frame↩ Trigger!)
"TriggerSelector = AcquisitionActive" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = LevelHigh" "ExposureMode = Timed"	OnHighLevel	Start an exposure of a frame as long as the trigger input is above the trigger threshold. (No Frame↩ Trigger!)
"TriggerSelector = FrameStart" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = FallingEdge" "ExposureMode = Timed"	OnFallingEdge	Each falling edge of trigger signal acquires one image.
"TriggerSelector = FrameStart" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = RisingEdge" "ExposureMode = Timed"	OnRisingEdge	Each rising edge of trigger signal acquires one image.

"TriggerSelector = FrameStart" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = AnyEdge" "ExposureMode = Timed"	OnAnyEdge	Start the exposure of a frame when the trigger input level changes from high to low or from low to high.

Device Feature And Property List (p. 228)

1.19.1.1.6.4 Device Feature And Property List

1.19.1.2 mvBlueNAOS4

- **BVS CA-BN4-0051BG (5.1 Mpix [2464 x 2064])** (p. 228)
- **BVS CA-BN4-0081ZG (8.1 Mpix [2848 x 2848])** (p. 230)
- **BVS CA-BN4-0124BG (12.4 Mpix [4128 x 3008])** (p. 233)
- **BVS CA-BN4-0162ZG (16.2 Mpix [5328 x 3040])** (p. 235)
- **BVS CA-BN4-0204ZG (20.4 Mpix [4512 x 4512])** (p. 238)
- **BVS CA-BN4-0246ZG (24.6 Mpix [5328 x 4608])** (p. 240)

1.19.1.2.1 BVS CA-BN4-0051BG (5.1 Mpix [2464 x 2064])

1.19.1.2.1.1 Introduction The sensor uses a global shutter, i.e. light exposure takes place on all pixels in parallel, although subsequent readout is sequential.

Feature	Description
Manufacturer	Sony
Sensor name	IMX537
Max. frames per second	234.0
Device Structure	CMOS image sensor
SNR_{max} [dB][1/]	39,9
DNR (normal / HDR) [dB][1/]	69,9 / -
Image size	1/1.8
Number of effective pixels	2464 (H) x 2064 (V)
Unit cell size	2.74µm (H) x 2.74µm (V)
ADC resolution / output	12 bit → 8/10/(12)

[1] Measured accord. to EMVA1288 with gray scale version of the camera

1.19.1.2.1.2 Spectral Sensitivity

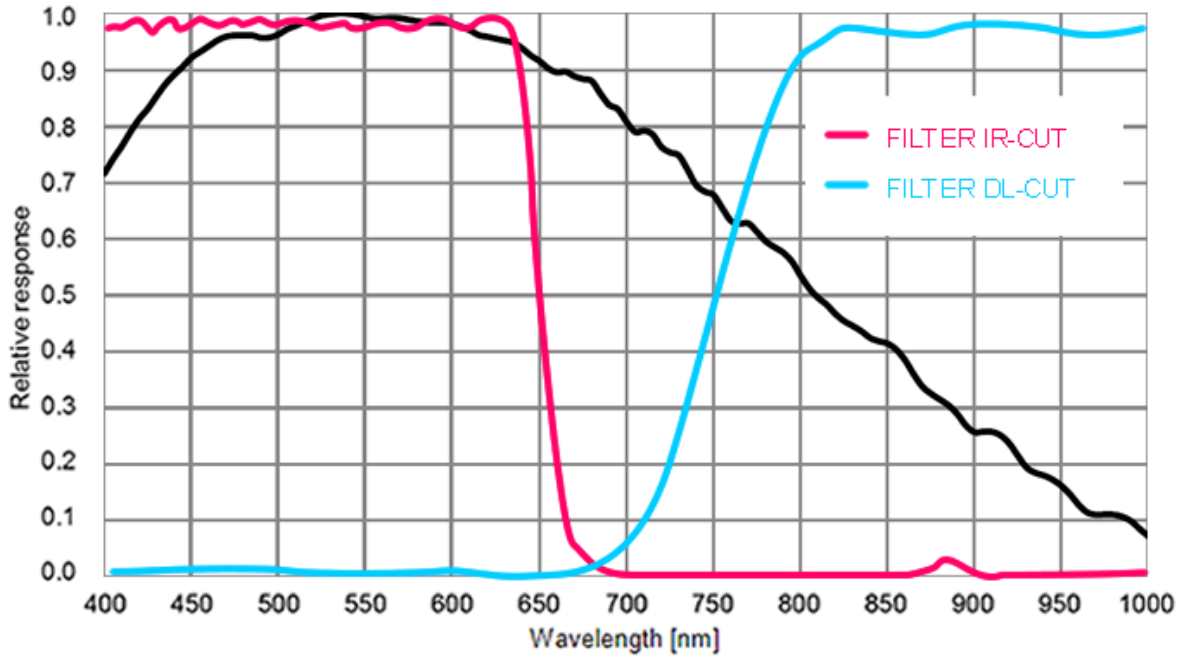


Figure 1: Spectral sensitivity BVS CA-BN4-0051BG

1.19.1.2.1.3 Timings For calculating the frame rates, please use the frame rate calculator of the online manual:

See also

`https://www.matrix-vision.com/manuals/mvBlueNAOS/mvBlue_models_BVS_CA-BN4-0051BG_.html#mvBC_subsubsection_sensors_CMOS_models_BVS_CA-BN4-0051BG_Calc`

The following trigger modes are available:

Setting (GenICam)	Mode / Setting (obsolete "Device Specific")	Description
`"TriggerSelector = FrameStart"` `"TriggerMode = Off"`	Continuous	Free running, no external trigger signal needed.
`"TriggerSelector = FrameStart"` `"TriggerMode = On"` `"TriggerSource = Software"` `"ExposureMode = Timed"` To trigger one frame execute the *TriggerSoftware@i* command then.	OnDemand	Image acquisition triggered by command (software trigger).

"TriggerSelector = AcquisitionActive" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = LevelLow" "ExposureMode = Timed"	OnLowLevel	Start an exposure of a frame as long as the trigger input is below the trigger threshold. (No Frame↩ Trigger!)
"TriggerSelector = AcquisitionActive" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = LevelHigh" "ExposureMode = Timed"	OnHighLevel	Start an exposure of a frame as long as the trigger input is above the trigger threshold. (No Frame↩ Trigger!)
"TriggerSelector = FrameStart" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = FallingEdge" "ExposureMode = Timed"	OnFallingEdge	Each falling edge of trigger signal acquires one image.
"TriggerSelector = FrameStart" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = RisingEdge" "ExposureMode = Timed"	OnRisingEdge	Each rising edge of trigger signal acquires one image.
"TriggerSelector = FrameStart" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = AnyEdge" "ExposureMode = Timed"	OnAnyEdge	Start the exposure of a frame when the trigger input level changes from high to low or from low to high.

Device Feature And Property List (p. 230)

1.19.1.2.1.4 Device Feature And Property List

1.19.1.2.2 BVS CA-BN4-0081ZG (8.1 Mpix [2848 x 2848])

1.19.1.2.2.1 Introduction The sensor uses a global shutter, i.e. light exposure takes place on all pixels in parallel, although subsequent readout is sequential.

Feature	Description
Manufacturer	Sony
Sensor name	IMX536
Max. frames per second	152.5
Device Structure	CMOS image sensor
SNR_{max} [dB][1/]	39,7
DNR (normal / HDR) [dB][1/]	69,6 / -
Image size	2/3
Number of effective pixels	2848 (H) x 2848 (V)
Unit cell size	2.74µm (H) x 2.74µm (V)
ADC resolution / output	12 bit → 8/10/(12)

[1] Measured accord. to EMVA1288 with gray scale version of the camera

1.19.1.2.2.2 Spectral Sensitivity

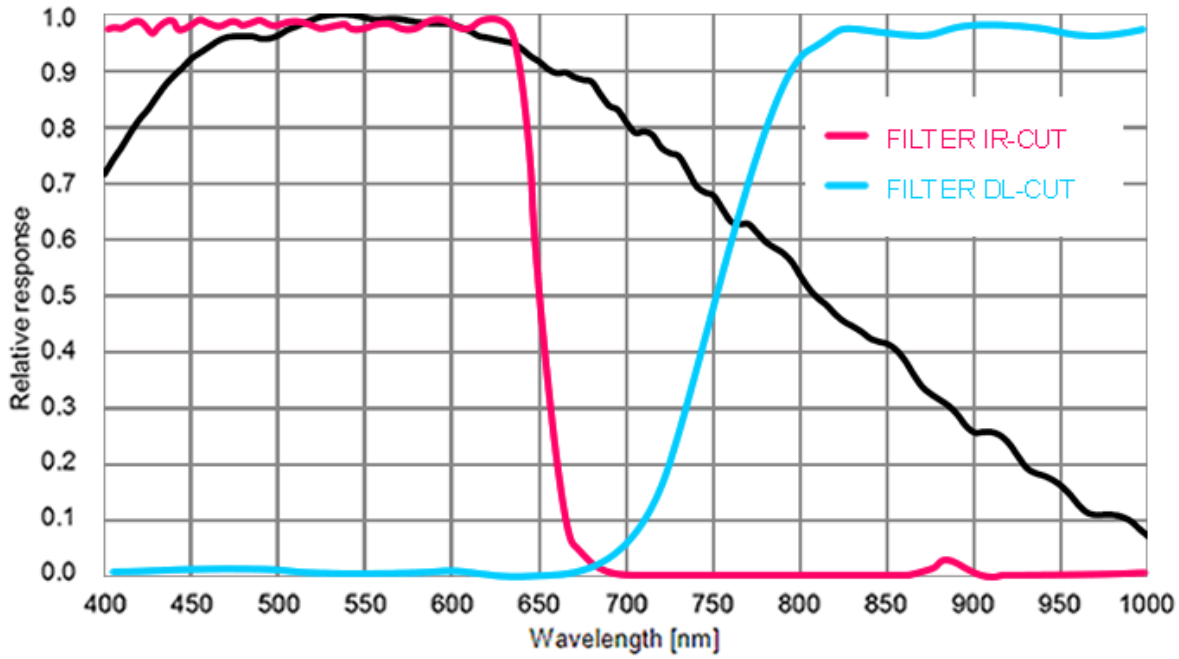


Figure 1: Spectral sensitivity BVS CA-BN4-0081ZG

1.19.1.2.2.3 Timings For calculating the frame rates, please use the frame rate calculator of the online manual:

See also

https://www.matrix-vision.com/manuals/mvBlueNAOS/mvBlue_models_BVS_CA-BN4-0081ZG_.html#mvBC_subsubsection_sensors_CMOS_models_BVS_CA-BN4-0081ZG_Calc

The following trigger modes are available:

Setting (GenICam)	Mode / Setting (obsolete "Device Specific")	Description
"TriggerSelector = FrameStart" "TriggerMode = Off"	Continuous	Free running, no external trigger signal needed.
"TriggerSelector = FrameStart" "TriggerMode = On" "TriggerSource = Software" "ExposureMode = Timed" To trigger one frame execute the *TriggerSoftware@i* command then.	OnDemand	Image acquisition triggered by command (software trigger).
"TriggerSelector = AcquisitionActive" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = LevelLow" "ExposureMode = Timed"	OnLowLevel	Start an exposure of a frame as long as the trigger input is below the trigger threshold. (No Frame↩ Trigger!)
"TriggerSelector = AcquisitionActive" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = LevelHigh" "ExposureMode = Timed"	OnHighLevel	Start an exposure of a frame as long as the trigger input is above the trigger threshold. (No Frame↩ Trigger!)
"TriggerSelector = FrameStart" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = FallingEdge" "ExposureMode = Timed"	OnFallingEdge	Each falling edge of trigger signal acquires one image.
"TriggerSelector = FrameStart" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = RisingEdge" "ExposureMode = Timed"	OnRisingEdge	Each rising edge of trigger signal acquires one image.

"TriggerSelector = FrameStart" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = AnyEdge" "ExposureMode = Timed"	OnAnyEdge	Start the exposure of a frame when the trigger input level changes from high to low or from low to high.

Device Feature And Property List (p. 233)

1.19.1.2.2.4 Device Feature And Property List

1.19.1.2.3 BVS CA-BN4-0124BG (12.4 Mpix [4128 x 3008])

1.19.1.2.3.1 Introduction The sensor uses a global shutter, i.e. light exposure takes place on all pixels in parallel, although subsequent readout is sequential.

Feature	Description
Manufacturer	Sony
Sensor name	IMX535
Max. frames per second	104.5
Device Structure	CMOS image sensor
SNR_{max} [dB][1/]	39,6
DNR (normal / HDR) [dB][1/]	69,4 / -
Image size	1/1.1
Number of effective pixels	4128 (H) x 3008 (V)
Unit cell size	2.74µm (H) x 2.74µm (V)
ADC resolution / output	12 bit → 8/10/(12)

[1] Measured accord. to EMVA1288 with gray scale version of the camera

1.19.1.2.3.2 Spectral Sensitivity

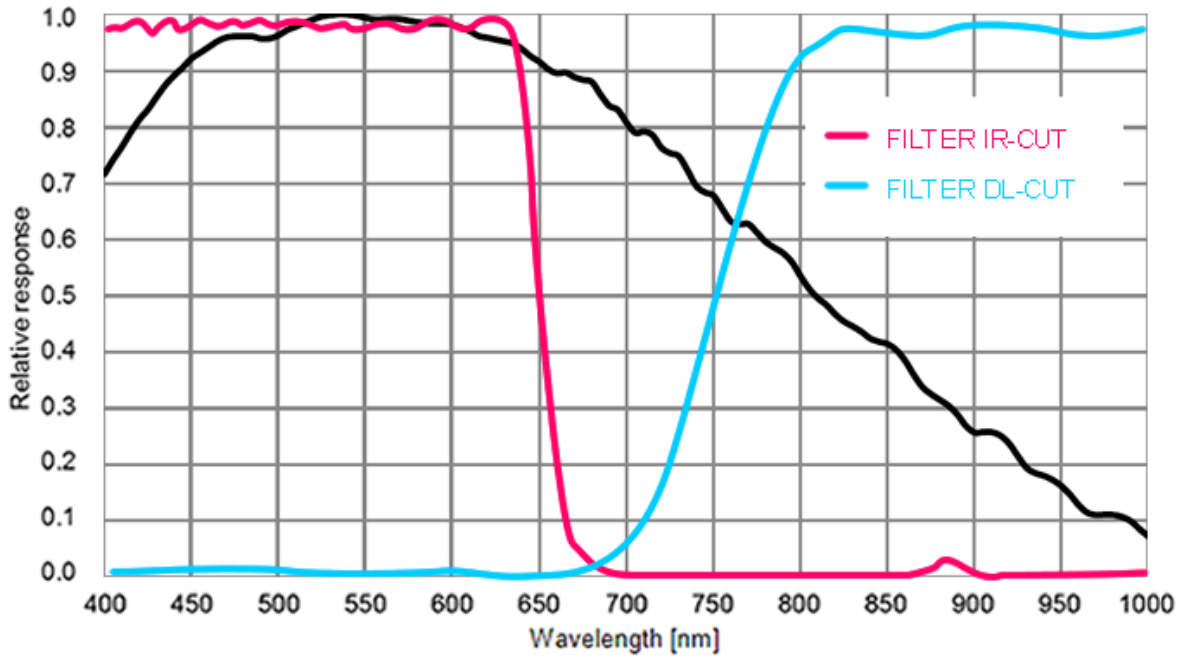


Figure 1: Spectral sensitivity BVS CA-BN4-0124BG

1.19.1.2.3.3 Timings For calculating the frame rates, please use the frame rate calculator of the online manual:

See also

`https://www.matrix-vision.com/manuals/mvBlueNAOS/mvBlue_models_BVS↩_CA-BN4-0124BG_.html#mvBC_subsubsection_sensors_CMOS_models_BVS_CA-↩BN4-0124BG_Calc`

The following trigger modes are available:

Setting (GenICam)	Mode / Setting (obsolete "Device Specific")	Description
`"TriggerSelector = FrameStart"` `"TriggerMode = Off"`	Continuous	Free running, no external trigger signal needed.

`"TriggerSelector = FrameStart"`
`"TriggerMode = On"`
`"TriggerSource = Software"`
`"ExposureMode = Timed"`
 	
To trigger one frame execute the *TriggerSoftware@i* command then.
	OnDemand	Image acquisition triggered by command (software trigger).

"TriggerSelector = AcquisitionActive" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = LevelLow" "ExposureMode = Timed"	OnLowLevel	Start an exposure of a frame as long as the trigger input is below the trigger threshold. (No Frame↩ Trigger!)
"TriggerSelector = AcquisitionActive" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = LevelHigh" "ExposureMode = Timed"	OnHighLevel	Start an exposure of a frame as long as the trigger input is above the trigger threshold. (No Frame↩ Trigger!)
"TriggerSelector = FrameStart" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = FallingEdge" "ExposureMode = Timed"	OnFallingEdge	Each falling edge of trigger signal acquires one image.
"TriggerSelector = FrameStart" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = RisingEdge" "ExposureMode = Timed"	OnRisingEdge	Each rising edge of trigger signal acquires one image.
"TriggerSelector = FrameStart" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = AnyEdge" "ExposureMode = Timed"	OnAnyEdge	Start the exposure of a frame when the trigger input level changes from high to low or from low to high.

Device Feature And Property List (p. 235)

1.19.1.2.3.4 Device Feature And Property List

1.19.1.2.4 BVS CA-BN4-0162ZG (16.2 Mpix [5328 x 3040])

1.19.1.2.4.1 Introduction The sensor uses a global shutter, i.e. light exposure takes place on all pixels in parallel, although subsequent readout is sequential.

Feature	Description
Manufacturer	Sony
Sensor name	IMX532
Max. frames per second	82.8
Device Structure	CMOS image sensor
SNR_{max} [dB][1/]	39,8
DNR (normal / HDR) [dB][1/]	71,7 / -
Image size	1.1
Number of effective pixels	5328 (H) x 3040 (V)
Unit cell size	2.74µm (H) x 2.74µm (V)
ADC resolution / output	12 bit → 8/10/(12)

[1] Measured accord. to EMVA1288 with gray scale version of the camera

1.19.1.2.4.2 Spectral Sensitivity

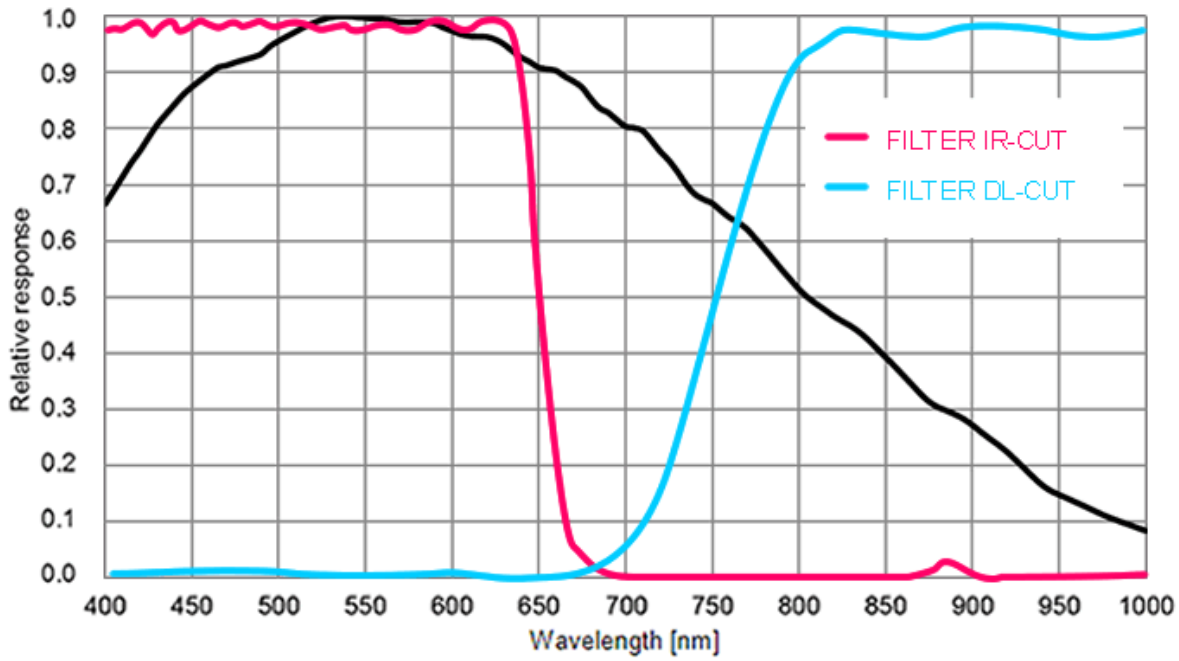


Figure 1: Spectral sensitivity BVS CA-BN4-0162ZG

1.19.1.2.4.3 Timings For calculating the frame rates, please use the frame rate calculator of the online manual:

See also

https://www.matrix-vision.com/manuals/mvBlueNAOS/mvBlue_models_BVS_CA-BN4-0162ZG_.html#mvBC_subsubsection_sensors_CMOS_models_BVS_CA-BN4-0162ZG_Calc

The following trigger modes are available:

Setting (GenICam)	Mode / Setting (obsolete "Device Specific")	Description
"TriggerSelector = FrameStart" "TriggerMode = Off"	Continuous	Free running, no external trigger signal needed.
"TriggerSelector = FrameStart" "TriggerMode = On" "TriggerSource = Software" "ExposureMode = Timed" To trigger one frame execute the *TriggerSoftware@i* command then.	OnDemand	Image acquisition triggered by command (software trigger).
"TriggerSelector = AcquisitionActive" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = LevelLow" "ExposureMode = Timed"	OnLowLevel	Start an exposure of a frame as long as the trigger input is below the trigger threshold. (No Frame↩ Trigger!)
"TriggerSelector = AcquisitionActive" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = LevelHigh" "ExposureMode = Timed"	OnHighLevel	Start an exposure of a frame as long as the trigger input is above the trigger threshold. (No Frame↩ Trigger!)
"TriggerSelector = FrameStart" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = FallingEdge" "ExposureMode = Timed"	OnFallingEdge	Each falling edge of trigger signal acquires one image.
"TriggerSelector = FrameStart" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = RisingEdge" "ExposureMode = Timed"	OnRisingEdge	Each rising edge of trigger signal acquires one image.

"TriggerSelector = FrameStart" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = AnyEdge" "ExposureMode = Timed"	OnAnyEdge	Start the exposure of a frame when the trigger input level changes from high to low or from low to high.

Device Feature And Property List (p. 238)

1.19.1.2.4.4 Device Feature And Property List

1.19.1.2.5 BVS CA-BN4-0204ZG (20.4 Mpix [4512 x 4512])

1.19.1.2.5.1 Introduction The sensor uses a global shutter, i.e. light exposure takes place on all pixels in parallel, although subsequent readout is sequential.

Feature	Description
Manufacturer	Sony
Sensor name	IMX531
Max. frames per second	66.0
Device Structure	CMOS image sensor
SNR_{max} [dB][1/]	39,7
DNR (normal / HDR) [dB][1/]	70,4 / -
Image size	1.1
Number of effective pixels	4512 (H) x 4512 (V)
Unit cell size	2.74µm (H) x 2.74µm (V)
ADC resolution / output	12 bit → 8/10/(12)

[1] Measured accord. to EMVA1288 with gray scale version of the camera

1.19.1.2.5.2 Spectral Sensitivity

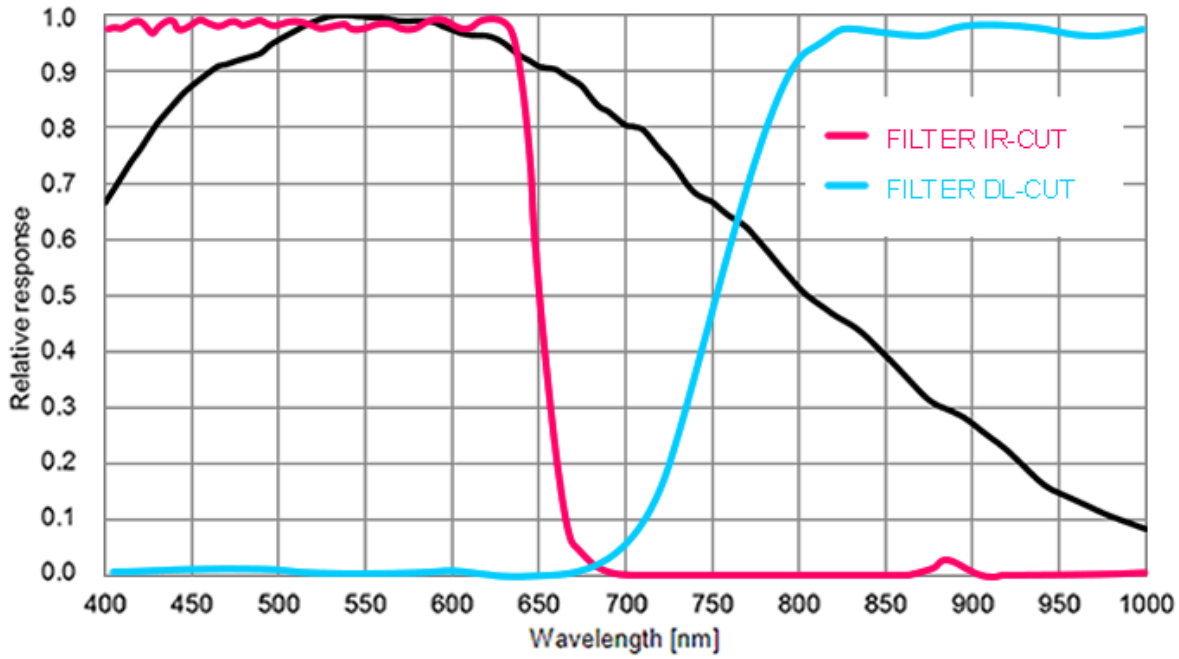


Figure 1: Spectral sensitivity BVS CA-BN4-0204ZG

1.19.1.2.5.3 Timings For calculating the frame rates, please use the frame rate calculator of the online manual:

See also

`https://www.matrix-vision.com/manuals/mvBlueNAOS/mvBlue_models_BVS_CA-BN4-0204ZG_.html#mvBC_subsubsection_sensors_CMOS_models_BVS_CA-BN4-0204ZG_Calc`

The following trigger modes are available:

Setting (GenICam)	Mode / Setting (obsolete "Device Specific")	Description
`"TriggerSelector = FrameStart"` `"TriggerMode = Off"`	Continuous	Free running, no external trigger signal needed.

`"TriggerSelector = FrameStart"`
`"TriggerMode = On"`
`"TriggerSource = Software"`
`"ExposureMode = Timed"`
 	
To trigger one frame execute the *TriggerSoftware@i* command then.
	OnDemand	Image acquisition triggered by command (software trigger).

"TriggerSelector = AcquisitionActive" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = LevelLow" "ExposureMode = Timed"	OnLowLevel	Start an exposure of a frame as long as the trigger input is below the trigger threshold. (No Frame↩ Trigger!)
"TriggerSelector = AcquisitionActive" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = LevelHigh" "ExposureMode = Timed"	OnHighLevel	Start an exposure of a frame as long as the trigger input is above the trigger threshold. (No Frame↩ Trigger!)
"TriggerSelector = FrameStart" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = FallingEdge" "ExposureMode = Timed"	OnFallingEdge	Each falling edge of trigger signal acquires one image.
"TriggerSelector = FrameStart" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = RisingEdge" "ExposureMode = Timed"	OnRisingEdge	Each rising edge of trigger signal acquires one image.
"TriggerSelector = FrameStart" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = AnyEdge" "ExposureMode = Timed"	OnAnyEdge	Start the exposure of a frame when the trigger input level changes from high to low or from low to high.

Device Feature And Property List (p. 240)

1.19.1.2.5.4 Device Feature And Property List

1.19.1.2.6 BVS CA-BN4-0246ZG (24.6 Mpix [5328 x 4608])

1.19.1.2.6.1 Introduction The sensor uses a global shutter, i.e. light exposure takes place on all pixels in parallel, although subsequent readout is sequential.

Feature	Description
Manufacturer	Sony
Sensor name	IMX530
Max. frames per second	55.5
Device Structure	CMOS image sensor
SNR_{max} [dB][1/]	39,8
DNR (normal / HDR) [dB][1/]	70,3 / -
Image size	1.2
Number of effective pixels	5328 (H) x 4608 (V)
Unit cell size	2.74µm (H) x 2.74µm (V)
ADC resolution / output	12 bit → 8/10/(12)

[1] Measured accord. to EMVA1288 with gray scale version of the camera

1.19.1.2.6.2 Spectral Sensitivity

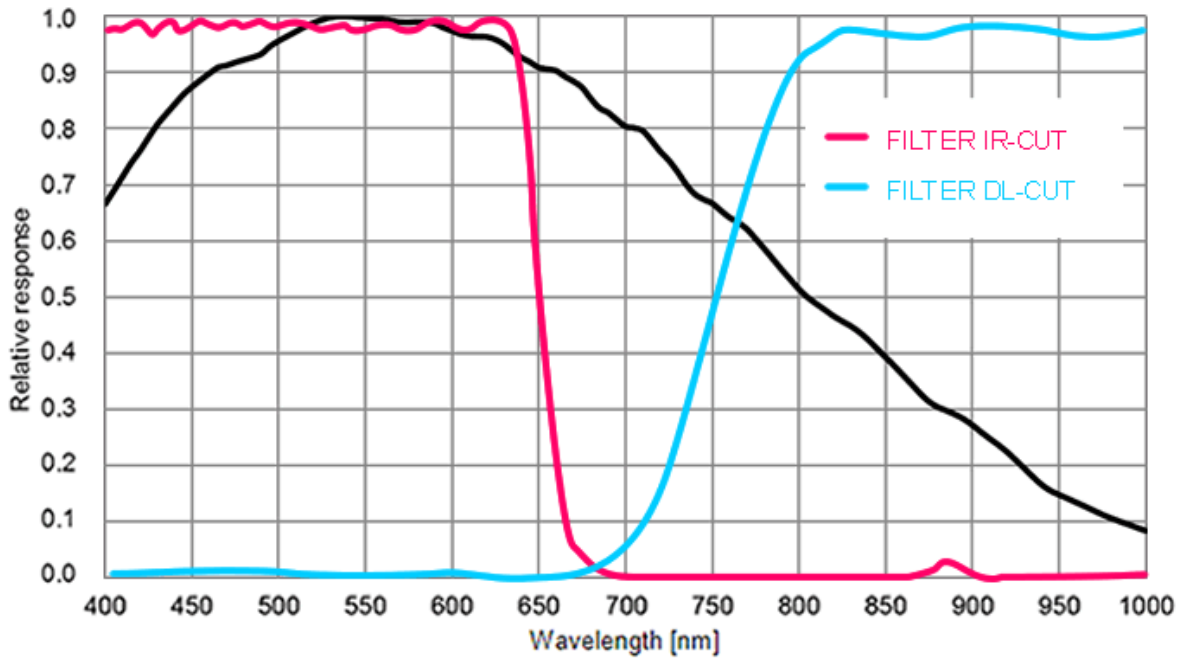


Figure 1: Spectral sensitivity BVS CA-BN4-0246ZG

1.19.1.2.6.3 Timings For calculating the frame rates, please use the frame rate calculator of the online manual:

See also

https://www.matrix-vision.com/manuals/mvBlueNAOS/mvBlue_models_BVS_CA-BN4-0246ZG_.html#mvBC_subsubsection_sensors_CMOS_models_BVS_CA-BN4-0246ZG_Calc

The following trigger modes are available:

Setting (GenICam)	Mode / Setting (obsolete "Device Specific")	Description
"TriggerSelector = FrameStart" "TriggerMode = Off"	Continuous	Free running, no external trigger signal needed.
"TriggerSelector = FrameStart" "TriggerMode = On" "TriggerSource = Software" "ExposureMode = Timed" To trigger one frame execute the *TriggerSoftware@i* command then.	OnDemand	Image acquisition triggered by command (software trigger).
"TriggerSelector = AcquisitionActive" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = LevelLow" "ExposureMode = Timed"	OnLowLevel	Start an exposure of a frame as long as the trigger input is below the trigger threshold. (No Frame↩ Trigger!)
"TriggerSelector = AcquisitionActive" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = LevelHigh" "ExposureMode = Timed"	OnHighLevel	Start an exposure of a frame as long as the trigger input is above the trigger threshold. (No Frame↩ Trigger!)
"TriggerSelector = FrameStart" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = FallingEdge" "ExposureMode = Timed"	OnFallingEdge	Each falling edge of trigger signal acquires one image.
"TriggerSelector = FrameStart" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = RisingEdge" "ExposureMode = Timed"	OnRisingEdge	Each rising edge of trigger signal acquires one image.

"TriggerSelector = FrameStart" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = AnyEdge" "ExposureMode = Timed"	OnAnyEdge	Start the exposure of a frame when the trigger input level changes from high to low or from low to high.

Device Feature And Property List (p. 243)

1.19.1.2.6.4 Device Feature And Property List

1.19.2 A.2 Pregius CMOS (Color)

- **mvBlueNAOS2** (p. 243)
- **mvBlueNAOS4** (p. 258)

1.19.2.1 mvBlueNAOS2

- **BVS CA-BN2-0016ZC (1.6 Mpix [1456 x 1088])** (p. 243)
- **BVS CA-BN2-0032ZC (3.2 Mpix [2064 x 1544])** (p. 246)
- **BVS CA-BN2-0051ZC (5.1 Mpix [2464 x 2056])** (p. 248)
- **BVS CA-BN2-0089ZC (8.9 Mpix [4096 x 2176])** (p. 250)
- **BVS CA-BN2-0124ZC (12.3 Mpix [4096 x 3008])** (p. 253)
- **BVS CA-BN2-0246AC (24.6 Mpix [5328 x 4608])** (p. 255)

1.19.2.1.1 BVS CA-BN2-0016ZC (1.6 Mpix [1456 x 1088])

1.19.2.1.1.1 Introduction The sensor uses a global shutter, i.e. light exposure takes place on all pixels in parallel, although subsequent readout is sequential.

Feature	Description
Manufacturer	Sony
Sensor name	IMX273
Max. frames per second	226.5
Device Structure	CMOS image sensor
SNR_{max} [dB][1/]	40,2
DNR (normal / HDR) [dB][1/]	71,4 / -
Image size	1/2.9
Number of effective pixels	1456 (H) x 1088 (V)
Unit cell size	3.45µm (H) x 3.45µm (V)
ADC resolution / output	12 bit → 8/10/(12)

[1] Measured accord. to EMVA1288 with gray scale version of the camera

1.19.2.1.1.2 Spectral Sensitivity

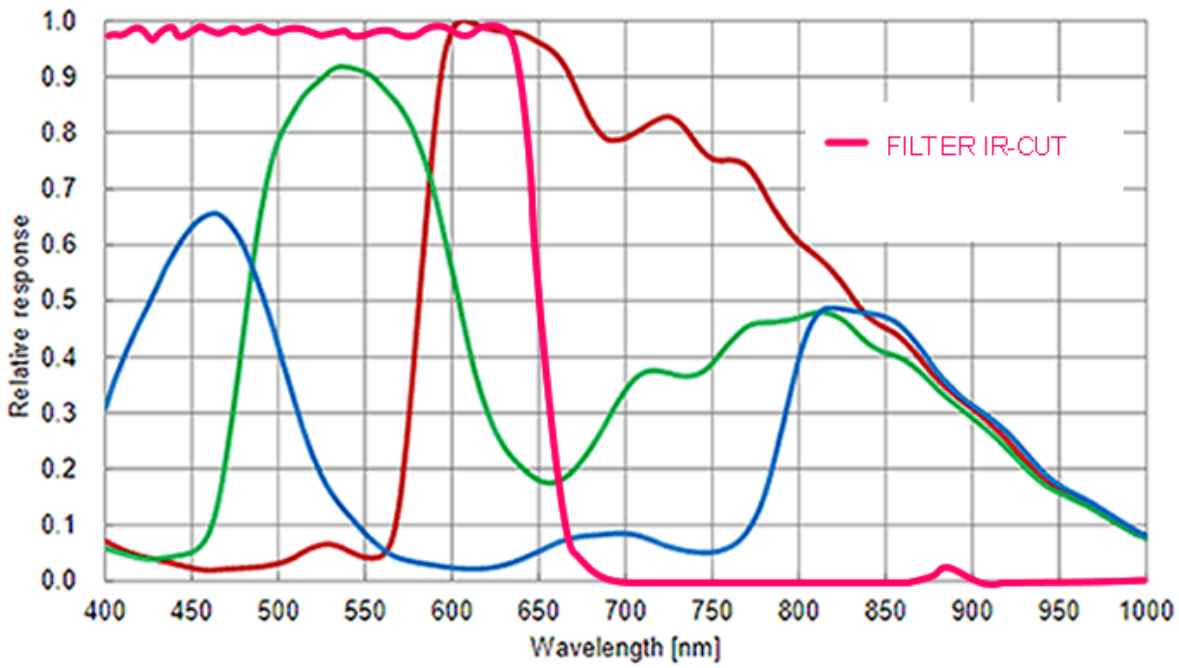


Figure 1: Spectral sensitivity BVS CA-BN2-0016ZC

1.19.2.1.1.3 Timings For calculating the frame rates, please use the frame rate calculator of the online manual:

See also

https://www.matrix-vision.com/manuals/mvBlueNAOS/mvBlue_models_BVS_CA-BN2-0016ZC_.html#mvBC_subsubsection_sensors_CMOS_models_BVS_CA-BN2-0016ZC_Calc

The following trigger modes are available:

Setting (GenICam)	Mode / Setting (obsolete "Device Specific")	Description
"TriggerSelector = FrameStart" "TriggerMode = Off"	Continuous	Free running, no external trigger signal needed.
"TriggerSelector = FrameStart" "TriggerMode = On" "TriggerSource = Software" "ExposureMode = Timed" To trigger one frame execute the *TriggerSoftware@i* command then.	OnDemand	Image acquisition triggered by command (software trigger).
"TriggerSelector = AcquisitionActive" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = LevelLow" "ExposureMode = Timed"	OnLowLevel	Start an exposure of a frame as long as the trigger input is below the trigger threshold. (No Frame↩ Trigger!)
"TriggerSelector = AcquisitionActive" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = LevelHigh" "ExposureMode = Timed"	OnHighLevel	Start an exposure of a frame as long as the trigger input is above the trigger threshold. (No Frame↩ Trigger!)
"TriggerSelector = FrameStart" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = FallingEdge" "ExposureMode = Timed"	OnFallingEdge	Each falling edge of trigger signal acquires one image.
"TriggerSelector = FrameStart" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = RisingEdge" "ExposureMode = Timed"	OnRisingEdge	Each rising edge of trigger signal acquires one image.
"TriggerSelector = FrameStart" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = AnyEdge" "ExposureMode = Timed"	OnAnyEdge	Start the exposure of a frame when the trigger input level changes from high to low or from low to high.

Device Feature And Property List (p. 246)

1.19.2.1.1.4 Device Feature And Property List

1.19.2.1.2 BVS CA-BN2-0032ZC (3.2 Mpix [2064 x 1544])

1.19.2.1.2.1 Introduction The sensor uses a global shutter, i.e. light exposure takes place on all pixels in parallel, although subsequent readout is sequential.

Feature	Description
Manufacturer	Sony
Sensor name	IMX252
Max. frames per second	191.5
Device Structure	CMOS image sensor
SNR_{max} [dB][1/]	40,3
DNR (normal / HDR) [dB][1/]	71,4 / -
Image size	1/1.8
Number of effective pixels	2064 (H) x 1544 (V)
Unit cell size	3.45µm (H) x 3.45µm (V)
ADC resolution / output	12 bit → 8/10/(12)

[1] Measured accord. to EMVA1288 with gray scale version of the camera

1.19.2.1.2.2 Spectral Sensitivity

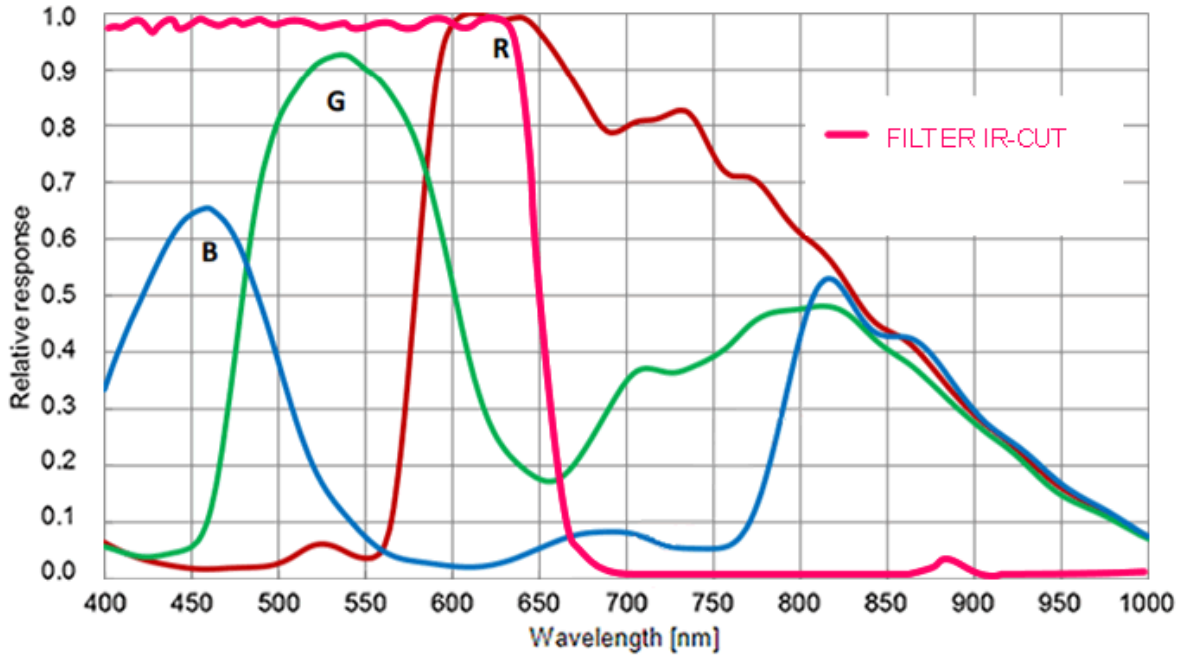


Figure 1: Spectral sensitivity BVS CA-BN2-0032ZC

1.19.2.1.2.3 Timings For calculating the frame rates, please use the frame rate calculator of the online manual:

See also

https://www.matrix-vision.com/manuals/mvBlueNAOS/mvBlue_models_BVS_CA-BN2-0032ZC_.html#mvBC_subsubsection_sensors_CMOS_models_BVS_CA-BN2-0032ZC_Calc

The following trigger modes are available:

Setting (GenICam)	Mode / Setting (obsolete "Device Specific")	Description
"TriggerSelector = FrameStart" "TriggerMode = Off"	Continuous	Free running, no external trigger signal needed.
"TriggerSelector = FrameStart" "TriggerMode = On" "TriggerSource = Software" "ExposureMode = Timed" To trigger one frame execute the *TriggerSoftware@i* command then.	OnDemand	Image acquisition triggered by command (software trigger).
"TriggerSelector = AcquisitionActive" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = LevelLow" "ExposureMode = Timed"	OnLowLevel	Start an exposure of a frame as long as the trigger input is below the trigger threshold. (No FrameTrigger!)
"TriggerSelector = AcquisitionActive" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = LevelHigh" "ExposureMode = Timed"	OnHighLevel	Start an exposure of a frame as long as the trigger input is above the trigger threshold. (No FrameTrigger!)
"TriggerSelector = FrameStart" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = FallingEdge" "ExposureMode = Timed"	OnFallingEdge	Each falling edge of trigger signal acquires one image.

"TriggerSelector = FrameStart" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = RisingEdge" "ExposureMode = Timed"	OnRisingEdge	Each rising edge of trigger signal acquires one image.
"TriggerSelector = FrameStart" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = AnyEdge" "ExposureMode = Timed"	OnAnyEdge	Start the exposure of a frame when the trigger input level changes from high to low or from low to high.

Device Feature And Property List (p. 248)

1.19.2.1.2.4 Device Feature And Property List

1.19.2.1.3 BVS CA-BN2-0051ZC (5.1 Mpix [2464 x 2056])

1.19.2.1.3.1 Introduction The sensor uses a global shutter, i.e. light exposure takes place on all pixels in parallel, although subsequent readout is sequential.

Feature	Description
Manufacturer	Sony
Sensor name	IMX250
Max. frames per second	140
Device Structure	CMOS image sensor
SNR_{max} [dB][1/]	40,2
DNR (normal / HDR) [dB][1/]	71,2 / -
Image size	2/3
Number of effective pixels	2464 (H) x 2056 (V)
Unit cell size	3.45µm (H) x 3.45µm (V)
ADC resolution / output	12 bit → 8/10/(12)

[1] Measured accord. to EMVA1288 with gray scale version of the camera

1.19.2.1.3.2 Spectral Sensitivity

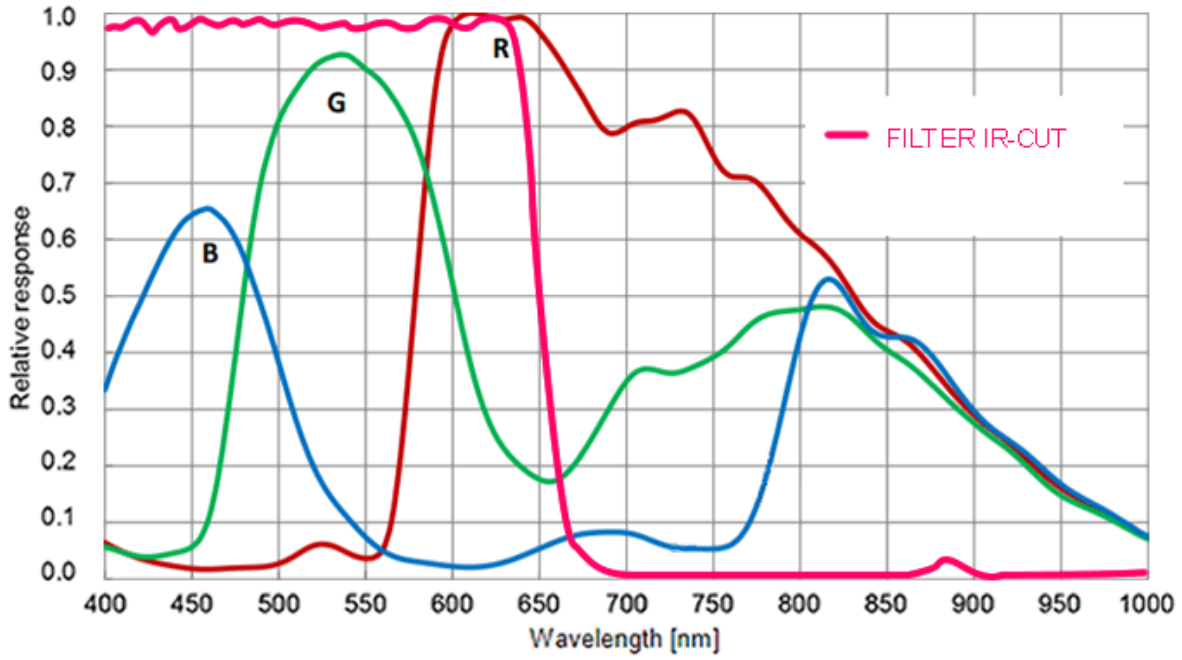


Figure 1: Spectral sensitivity BVS CA-BN2-0051ZC

1.19.2.1.3.3 Timings For calculating the frame rates, please use the frame rate calculator of the online manual:

See also

`https://www.matrix-vision.com/manuals/mvBlueNAOS/mvBlue_models_BVS_CA-BN2-0051ZC_.html#mvBC_subsubsection_sensors_CMOS_models_BVS_CA-BN2-0051ZC_Calc`

The following trigger modes are available:

Setting (GenICam)	Mode / Setting (obsolete "Device Specific")	Description
`"TriggerSelector = FrameStart"` `"TriggerMode = Off"`	Continuous	Free running, no external trigger signal needed.
`"TriggerSelector = FrameStart"` `"TriggerMode = On"` `"TriggerSource = Software"` `"ExposureMode = Timed"` To trigger one frame execute the *TriggerSoftware@i* command then.	OnDemand	Image acquisition triggered by command (software trigger).

"TriggerSelector = AcquisitionActive" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = LevelLow" "ExposureMode = Timed"	OnLowLevel	Start an exposure of a frame as long as the trigger input is below the trigger threshold. (No Frame↩ Trigger!)
"TriggerSelector = AcquisitionActive" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = LevelHigh" "ExposureMode = Timed"	OnHighLevel	Start an exposure of a frame as long as the trigger input is above the trigger threshold. (No Frame↩ Trigger!)
"TriggerSelector = FrameStart" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = FallingEdge" "ExposureMode = Timed"	OnFallingEdge	Each falling edge of trigger signal acquires one image.
"TriggerSelector = FrameStart" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = RisingEdge" "ExposureMode = Timed"	OnRisingEdge	Each rising edge of trigger signal acquires one image.
"TriggerSelector = FrameStart" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = AnyEdge" "ExposureMode = Timed"	OnAnyEdge	Start the exposure of a frame when the trigger input level changes from high to low or from low to high.

Device Feature And Property List (p. 250)

1.19.2.1.3.4 Device Feature And Property List

1.19.2.1.4 BVS CA-BN2-0089ZC (8.9 Mpix [4096 x 2176])

1.19.2.1.4.1 Introduction The sensor uses a global shutter, i.e. light exposure takes place on all pixels in parallel, although subsequent readout is sequential.

Feature	Description
Manufacturer	Sony
Sensor name	IMX255
Max. frames per second	88.7
Device Structure	CMOS image sensor
SNR_{max} [dB][1/]	40,2
DNR (normal / HDR) [dB][1/]	70,9 / -
Image size	1
Number of effective pixels	4096 (H) x 2176 (V)
Unit cell size	3.45µm (H) x 3.45µm (V)
ADC resolution / output	12 bit → 8/10/(12)

[1] Measured accord. to EMVA1288 with gray scale version of the camera

1.19.2.1.4.2 Spectral Sensitivity

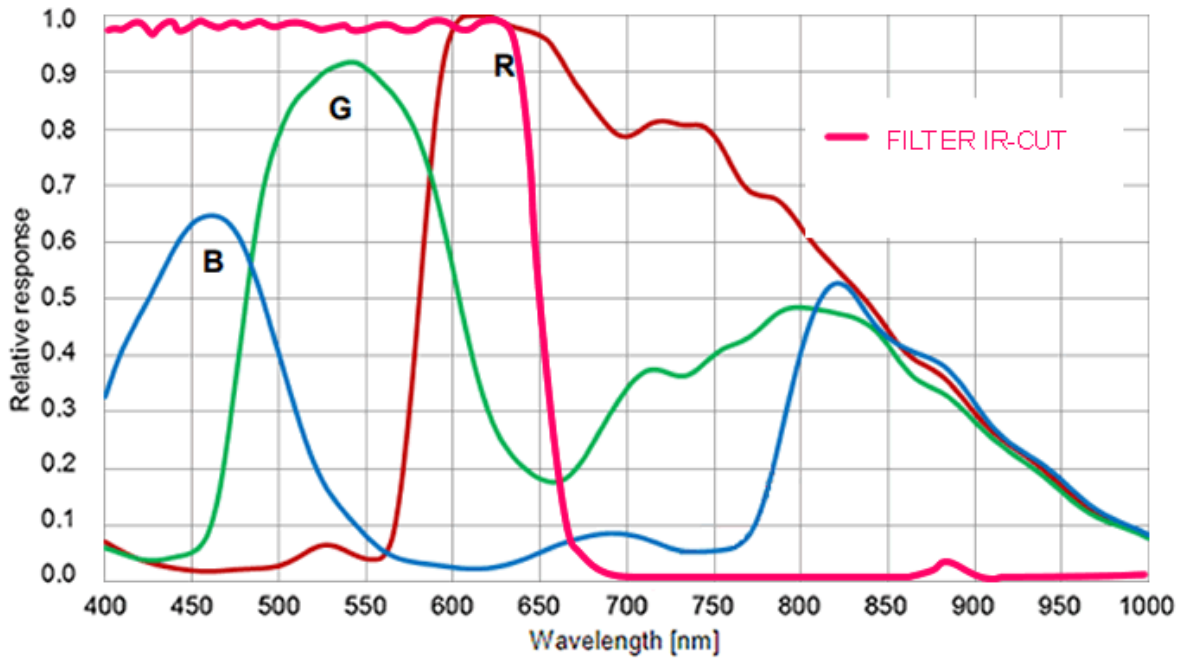


Figure 1: Spectral sensitivity BVS CA-BN2-0089ZC

1.19.2.1.4.3 Timings For calculating the frame rates, please use the frame rate calculator of the online manual:

See also

https://www.matrix-vision.com/manuals/mvBlueNAOS/mvBlue_models_BVS_CA-BN2-0089ZC_.html#mvBC_subsubsection_sensors_CMOS_models_BVS_CA-BN2-0089ZC_Calc

The following trigger modes are available:

Setting (GenICam)	Mode / Setting (obsolete "Device Specific")	Description
"TriggerSelector = FrameStart" "TriggerMode = Off"	Continuous	Free running, no external trigger signal needed.
"TriggerSelector = FrameStart" "TriggerMode = On" "TriggerSource = Software" "ExposureMode = Timed" To trigger one frame execute the *TriggerSoftware@i* command then.	OnDemand	Image acquisition triggered by command (software trigger).
"TriggerSelector = AcquisitionActive" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = LevelLow" "ExposureMode = Timed"	OnLowLevel	Start an exposure of a frame as long as the trigger input is below the trigger threshold. (No Frame↩ Trigger!)
"TriggerSelector = AcquisitionActive" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = LevelHigh" "ExposureMode = Timed"	OnHighLevel	Start an exposure of a frame as long as the trigger input is above the trigger threshold. (No Frame↩ Trigger!)
"TriggerSelector = FrameStart" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = FallingEdge" "ExposureMode = Timed"	OnFallingEdge	Each falling edge of trigger signal acquires one image.
"TriggerSelector = FrameStart" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = RisingEdge" "ExposureMode = Timed"	OnRisingEdge	Each rising edge of trigger signal acquires one image.

"TriggerSelector = FrameStart" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = AnyEdge" "ExposureMode = Timed"	OnAnyEdge	Start the exposure of a frame when the trigger input level changes from high to low or from low to high.

Device Feature And Property List (p. 253)

1.19.2.1.4.4 Device Feature And Property List

1.19.2.1.5 BVS CA-BN2-0124ZC (12.3 Mpix [4096 x 3008])

1.19.2.1.5.1 Introduction The sensor uses a global shutter, i.e. light exposure takes place on all pixels in parallel, although subsequent readout is sequential.

Feature	Description
Manufacturer	Sony
Sensor name	IMX253
Max. frames per second	64.6
Device Structure	CMOS image sensor
SNR_{max} [dB][1/]	40,2
DNR (normal / HDR) [dB][1/]	70,9 / -
Image size	1.1
Number of effective pixels	4096 (H) x 3008 (V)
Unit cell size	3.45µm (H) x 3.45µm (V)
ADC resolution / output	12 bit → 8/10/(12)

[1] Measured accord. to EMVA1288 with gray scale version of the camera

1.19.2.1.5.2 Spectral Sensitivity

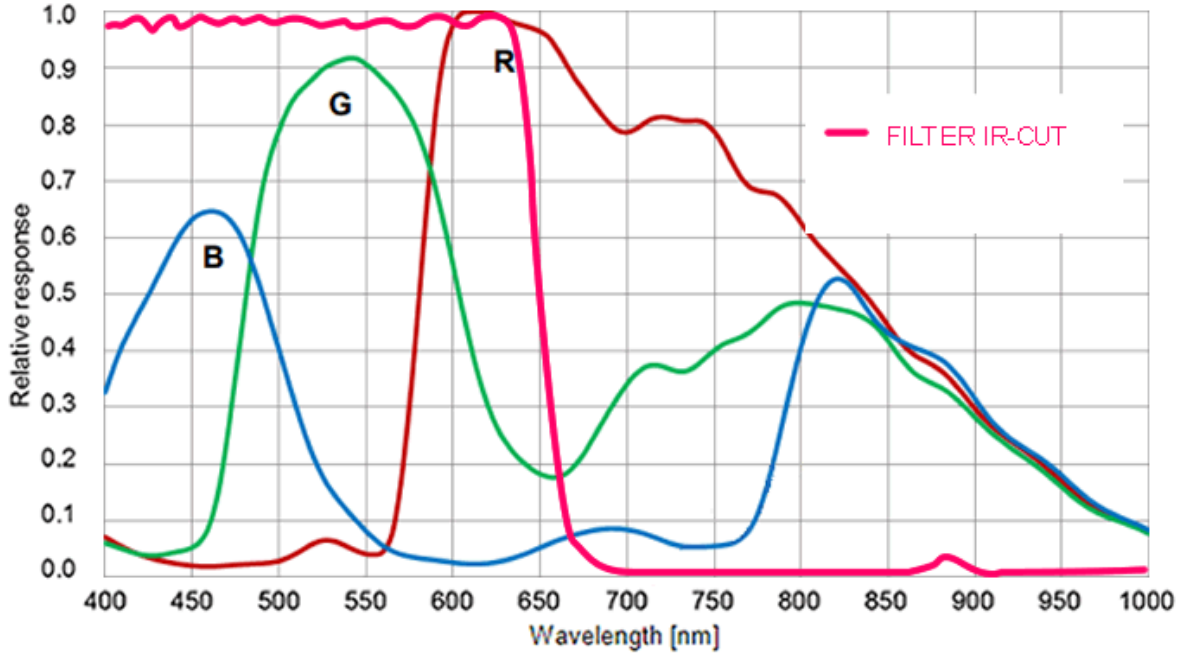


Figure 1: Spectral sensitivity BVS CA-BN2-0124ZC

1.19.2.1.5.3 Timings For calculating the frame rates, please use the frame rate calculator of the online manual:

See also

`https://www.matrix-vision.com/manuals/mvBlueNAOS/mvBlue_models_BVS_CA-BN2-0124ZC_.html#mvBC_subsubsection_sensors_CMOS_models_BVS_CA-BN2-0124ZC_Calc`

The following trigger modes are available:

Setting (GenICam)	Mode / Setting (obsolete "Device Specific")	Description
`"TriggerSelector = FrameStart"` `"TriggerMode = Off"`	Continuous	Free running, no external trigger signal needed.

`"TriggerSelector = FrameStart"`
`"TriggerMode = On"`
`"TriggerSource = Software"`
`"ExposureMode = Timed"`
 	
To trigger one frame execute the *TriggerSoftware@i* command then.
	OnDemand	Image acquisition triggered by command (software trigger).

"TriggerSelector = AcquisitionActive" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = LevelLow" "ExposureMode = Timed"	OnLowLevel	Start an exposure of a frame as long as the trigger input is below the trigger threshold. (No Frame↩ Trigger!)
"TriggerSelector = AcquisitionActive" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = LevelHigh" "ExposureMode = Timed"	OnHighLevel	Start an exposure of a frame as long as the trigger input is above the trigger threshold. (No Frame↩ Trigger!)
"TriggerSelector = FrameStart" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = FallingEdge" "ExposureMode = Timed"	OnFallingEdge	Each falling edge of trigger signal acquires one image.
"TriggerSelector = FrameStart" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = RisingEdge" "ExposureMode = Timed"	OnRisingEdge	Each rising edge of trigger signal acquires one image.
"TriggerSelector = FrameStart" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = AnyEdge" "ExposureMode = Timed"	OnAnyEdge	Start the exposure of a frame when the trigger input level changes from high to low or from low to high.

Device Feature And Property List (p. 255)

1.19.2.1.5.4 Device Feature And Property List

1.19.2.1.6 BVS CA-BN2-0246AC (24.6 Mpix [5328 x 4608])

1.19.2.1.6.1 Introduction The sensor uses a global shutter, i.e. light exposure takes place on all pixels in parallel, although subsequent readout is sequential.

Feature	Description
Manufacturer	Sony
Sensor name	IMX540
Max. frames per second	24.1
Device Structure	CMOS image sensor
SNR_{max} [dB][1/]	39,8
DNR (normal / HDR) [dB][1/]	70,5 / -
Image size	1.2
Number of effective pixels	5328 (H) x 4608 (V)
Unit cell size	2.74µm (H) x 2.74µm (V)
ADC resolution / output	12 bit → 8/10/(12)

[1] Measured accord. to EMVA1288 with gray scale version of the camera

1.19.2.1.6.2 Spectral Sensitivity

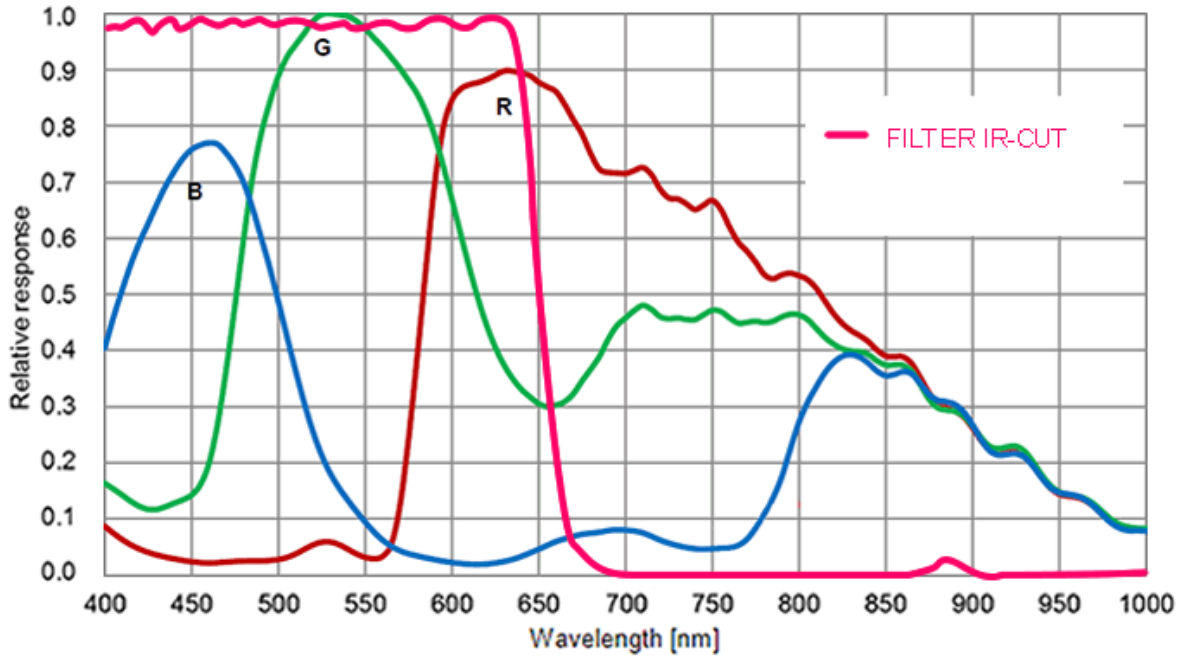


Figure 1: Spectral sensitivity BVS CA-BN2-0246AC

1.19.2.1.6.3 Timings For calculating the frame rates, please use the frame rate calculator of the online manual:

See also

https://www.matrix-vision.com/manuals/mvBlueNAOS/mvBlue_models_BVS_CA-BN2-0246AC_.html#mvBC_subsubsection_sensors_CMOS_models_BVS_CA-BN2-0246AC_Calc

The following trigger modes are available:

Setting (GenICam)	Mode / Setting (obsolete "Device Specific")	Description
"TriggerSelector = FrameStart" "TriggerMode = Off"	Continuous	Free running, no external trigger signal needed.
"TriggerSelector = FrameStart" "TriggerMode = On" "TriggerSource = Software" "ExposureMode = Timed" To trigger one frame execute the *TriggerSoftware@i* command then.	OnDemand	Image acquisition triggered by command (software trigger).
"TriggerSelector = AcquisitionActive" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = LevelLow" "ExposureMode = Timed"	OnLowLevel	Start an exposure of a frame as long as the trigger input is below the trigger threshold. (No Frame↩ Trigger!)
"TriggerSelector = AcquisitionActive" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = LevelHigh" "ExposureMode = Timed"	OnHighLevel	Start an exposure of a frame as long as the trigger input is above the trigger threshold. (No Frame↩ Trigger!)
"TriggerSelector = FrameStart" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = FallingEdge" "ExposureMode = Timed"	OnFallingEdge	Each falling edge of trigger signal acquires one image.
"TriggerSelector = FrameStart" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = RisingEdge" "ExposureMode = Timed"	OnRisingEdge	Each rising edge of trigger signal acquires one image.

"TriggerSelector = FrameStart" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = AnyEdge" "ExposureMode = Timed"	OnAnyEdge	Start the exposure of a frame when the trigger input level changes from high to low or from low to high.

Device Feature And Property List (p. 258)

1.19.2.1.6.4 Device Feature And Property List

1.19.2.2 mvBlueNAOS4

- **BVS CA-BN4-0051BC (5.1 Mpix [2464 x 2064])** (p. 259)
- **BVS CA-BN4-0081ZC (8.1 Mpix [2848 x 2848])** (p. 261)
- **BVS CA-BN4-0124BC (12.4 Mpix [4128 x 3008])** (p. 263)
- **BVS CA-BN4-0162ZC (16.2 Mpix [5328 x 3040])** (p. 266)
- **BVS CA-BN4-0204ZC (20.4 Mpix [4512 x 4512])** (p. 268)
- **BVS CA-BN4-0246ZC (24.6 Mpix [5328 x 4608])** (p. 271)

1.19.2.2.1 BVS CA-BN4-0051BC (5.1 Mpix [2464 x 2064])

1.19.2.2.1.1 Introduction The sensor uses a global shutter, i.e. light exposure takes place on all pixels in parallel, although subsequent readout is sequential.

Feature	Description
Manufacturer	Sony
Sensor name	IMX537
Max. frames per second	234.0
Device Structure	CMOS image sensor
SNR_{max} [dB][1/]	39,8
DNR (normal / HDR) [dB][1/]	69,6 / -
Image size	1/1.8
Number of effective pixels	2464 (H) x 2064 (V)
Unit cell size	2.74µm (H) x 2.74µm (V)
ADC resolution / output	12 bit → 8/10/(12)

[1] Measured accord. to EMVA1288 with gray scale version of the camera

1.19.2.2.1.2 Spectral Sensitivity

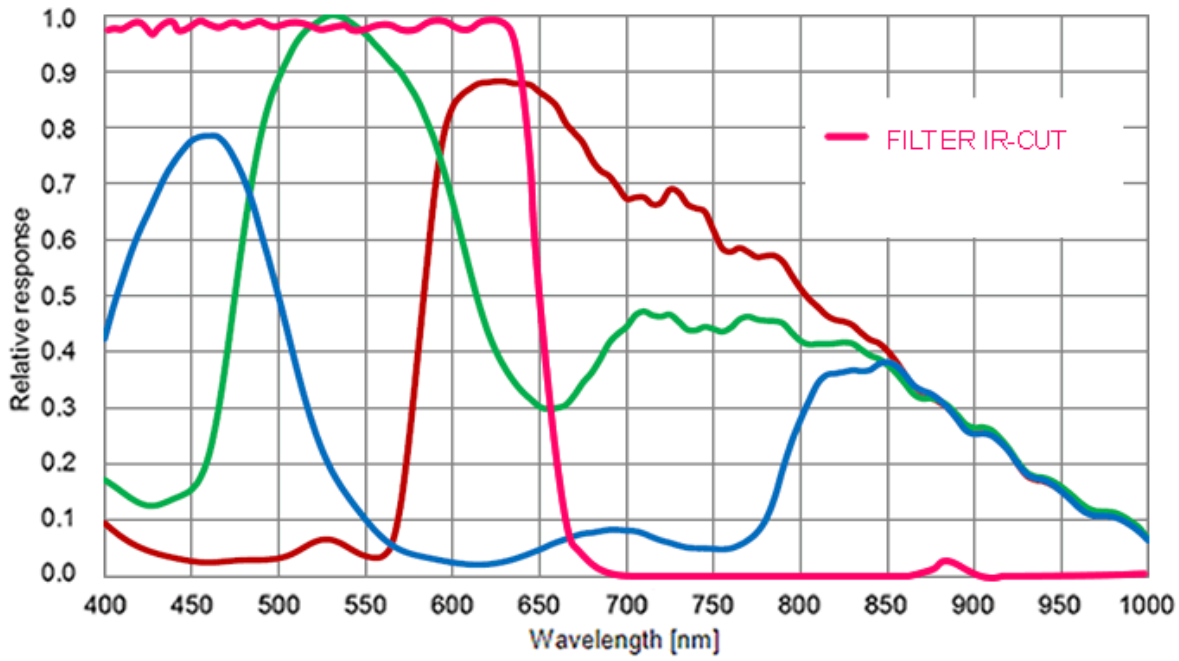


Figure 1: Spectral sensitivity BVS CA-BN4-0051BC

1.19.2.2.1.3 Timings For calculating the frame rates, please use the frame rate calculator of the online manual:

See also

https://www.matrix-vision.com/manuals/mvBlueNAOS/mvBlue_models_BVS_CA-BN4-0051BC_.html#mvBC_subsubsection_sensors_CMOS_models_BVS_CA-BN4-0051BC_Calc

The following trigger modes are available:

Setting (GenICam)	Mode / Setting (obsolete "Device Specific")	Description
"TriggerSelector = FrameStart" "TriggerMode = Off"	Continuous	Free running, no external trigger signal needed.
"TriggerSelector = FrameStart" "TriggerMode = On" "TriggerSource = Software" "ExposureMode = Timed" To trigger one frame execute the *TriggerSoftware@i* command then.	OnDemand	Image acquisition triggered by command (software trigger).
"TriggerSelector = AcquisitionActive" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = LevelLow" "ExposureMode = Timed"	OnLowLevel	Start an exposure of a frame as long as the trigger input is below the trigger threshold. (No Frame↩ Trigger!)
"TriggerSelector = AcquisitionActive" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = LevelHigh" "ExposureMode = Timed"	OnHighLevel	Start an exposure of a frame as long as the trigger input is above the trigger threshold. (No Frame↩ Trigger!)
"TriggerSelector = FrameStart" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = FallingEdge" "ExposureMode = Timed"	OnFallingEdge	Each falling edge of trigger signal acquires one image.
"TriggerSelector = FrameStart" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = RisingEdge" "ExposureMode = Timed"	OnRisingEdge	Each rising edge of trigger signal acquires one image.

"TriggerSelector = FrameStart" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = AnyEdge" "ExposureMode = Timed"	OnAnyEdge	Start the exposure of a frame when the trigger input level changes from high to low or from low to high.

Device Feature And Property List (p. 261)

1.19.2.2.1.4 Device Feature And Property List

1.19.2.2.2 BVS CA-BN4-0081ZC (8.1 Mpix [2848 x 2848])

1.19.2.2.2.1 Introduction The sensor uses a global shutter, i.e. light exposure takes place on all pixels in parallel, although subsequent readout is sequential.

Feature	Description
Manufacturer	Sony
Sensor name	IMX536
Max. frames per second	152.5
Device Structure	CMOS image sensor
SNR_{max} [dB][1/]	39,8
DNR (normal / HDR) [dB][1/]	69,8 / -
Image size	2/3
Number of effective pixels	2848 (H) x 2848 (V)
Unit cell size	2.74µm (H) x 2.74µm (V)
ADC resolution / output	12 bit → 8/10/(12)

[1] Measured accord. to EMVA1288 with gray scale version of the camera

1.19.2.2.2.2 Spectral Sensitivity

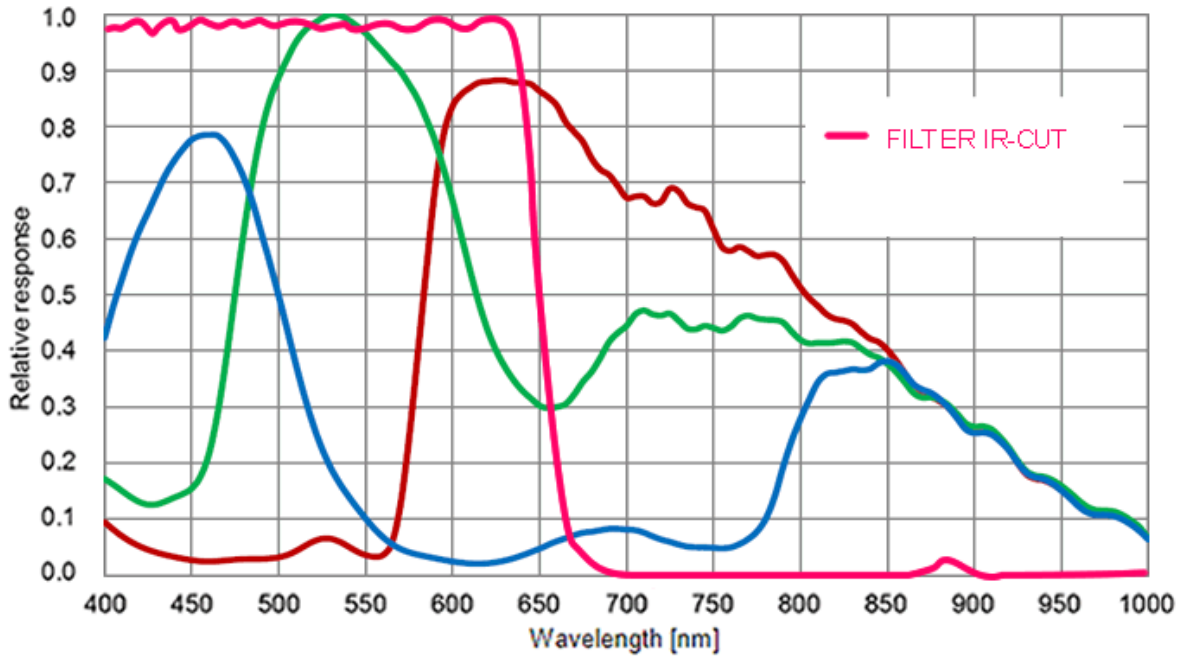


Figure 1: Spectral sensitivity BVS CA-BN4-0081ZC

1.19.2.2.2.3 Timings For calculating the frame rates, please use the frame rate calculator of the online manual:

See also

`https://www.matrix-vision.com/manuals/mvBlueNAOS/mvBlue_models_BVS_CA-BN4-0081ZC_.html#mvBC_subsubsection_sensors_CMOS_models_BVS_CA-BN4-0081ZC_Calc`

The following trigger modes are available:

Setting (GenICam)	Mode / Setting (obsolete "Device Specific")	Description
`"TriggerSelector = FrameStart"` `"TriggerMode = Off"`	Continuous	Free running, no external trigger signal needed.

`"TriggerSelector = FrameStart"`
`"TriggerMode = On"`
`"TriggerSource = Software"`
`"ExposureMode = Timed"`
 	
To trigger one frame execute the *TriggerSoftware@i* command then.
	OnDemand	Image acquisition triggered by command (software trigger).

"TriggerSelector = AcquisitionActive" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = LevelLow" "ExposureMode = Timed"	OnLowLevel	Start an exposure of a frame as long as the trigger input is below the trigger threshold. (No Frame↩ Trigger!)
"TriggerSelector = AcquisitionActive" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = LevelHigh" "ExposureMode = Timed"	OnHighLevel	Start an exposure of a frame as long as the trigger input is above the trigger threshold. (No Frame↩ Trigger!)
"TriggerSelector = FrameStart" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = FallingEdge" "ExposureMode = Timed"	OnFallingEdge	Each falling edge of trigger signal acquires one image.
"TriggerSelector = FrameStart" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = RisingEdge" "ExposureMode = Timed"	OnRisingEdge	Each rising edge of trigger signal acquires one image.
"TriggerSelector = FrameStart" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = AnyEdge" "ExposureMode = Timed"	OnAnyEdge	Start the exposure of a frame when the trigger input level changes from high to low or from low to high.

Device Feature And Property List (p. 263)

1.19.2.2.2.4 Device Feature And Property List

1.19.2.2.3 BVS CA-BN4-0124BC (12.4 Mpix [4128 x 3008])

1.19.2.2.3.1 Introduction The sensor uses a global shutter, i.e. light exposure takes place on all pixels in parallel, although subsequent readout is sequential.

Feature	Description
Manufacturer	Sony
Sensor name	IMX535
Max. frames per second	104.5
Device Structure	CMOS image sensor
SNR_{max} [dB][1/]	39,6
DNR (normal / HDR) [dB][1/]	69,5 / -
Image size	1/1.1
Number of effective pixels	4128 (H) x 3008 (V)
Unit cell size	2.74µm (H) x 2.74µm (V)
ADC resolution / output	12 bit → 8/10/(12)

[1] Measured accord. to EMVA1288 with gray scale version of the camera

1.19.2.2.3.2 Spectral Sensitivity

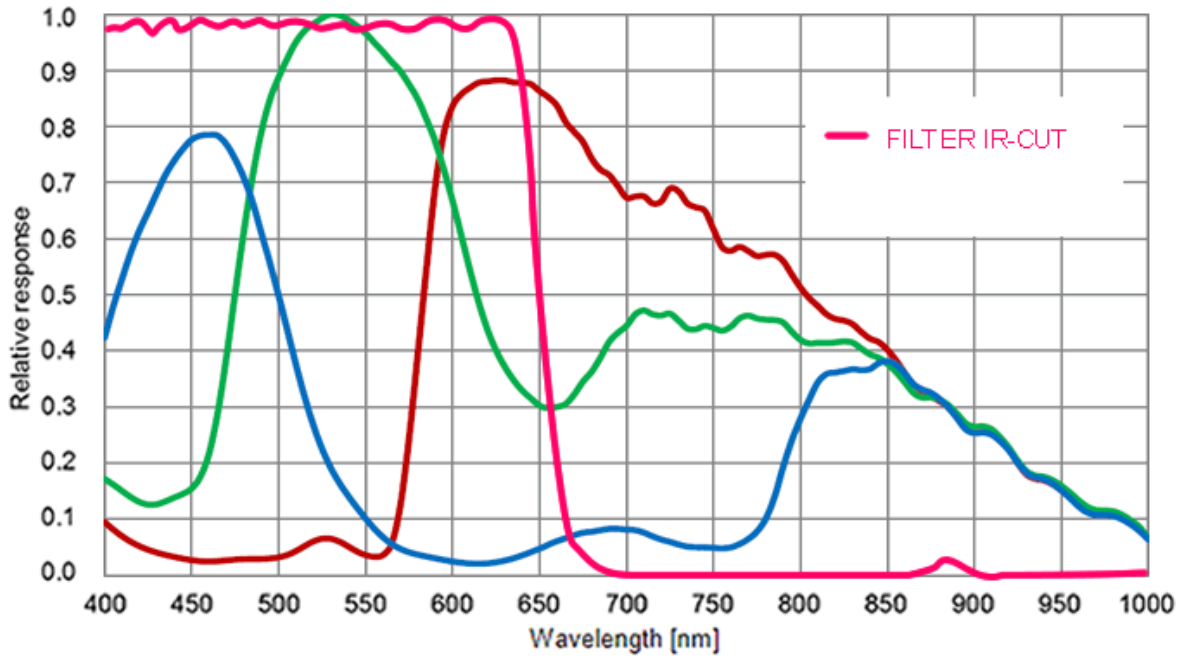


Figure 1: Spectral sensitivity BVS CA-BN4-0124BC

1.19.2.2.3.3 Timings For calculating the frame rates, please use the frame rate calculator of the online manual:

See also

https://www.matrix-vision.com/manuals/mvBlueNAOS/mvBlue_models_BVS_CA-BN4-0124BC_.html#mvBC_subsubsection_sensors_CMOS_models_BVS_CA-BN4-0124BC_Calc

The following trigger modes are available:

Setting (GenICam)	Mode / Setting (obsolete "Device Specific")	Description
"TriggerSelector = FrameStart" "TriggerMode = Off"	Continuous	Free running, no external trigger signal needed.
"TriggerSelector = FrameStart" "TriggerMode = On" "TriggerSource = Software" "ExposureMode = Timed" To trigger one frame execute the *TriggerSoftware@i* command then.	OnDemand	Image acquisition triggered by command (software trigger).
"TriggerSelector = AcquisitionActive" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = LevelLow" "ExposureMode = Timed"	OnLowLevel	Start an exposure of a frame as long as the trigger input is below the trigger threshold. (No Frame↩ Trigger!)
"TriggerSelector = AcquisitionActive" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = LevelHigh" "ExposureMode = Timed"	OnHighLevel	Start an exposure of a frame as long as the trigger input is above the trigger threshold. (No Frame↩ Trigger!)
"TriggerSelector = FrameStart" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = FallingEdge" "ExposureMode = Timed"	OnFallingEdge	Each falling edge of trigger signal acquires one image.
"TriggerSelector = FrameStart" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = RisingEdge" "ExposureMode = Timed"	OnRisingEdge	Each rising edge of trigger signal acquires one image.

"TriggerSelector = FrameStart" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = AnyEdge" "ExposureMode = Timed"	OnAnyEdge	Start the exposure of a frame when the trigger input level changes from high to low or from low to high.

Device Feature And Property List (p. 266)

1.19.2.2.3.4 Device Feature And Property List

1.19.2.2.4 BVS CA-BN4-0162ZC (16.2 Mpix [5328 x 3040])

1.19.2.2.4.1 Introduction The sensor uses a global shutter, i.e. light exposure takes place on all pixels in parallel, although subsequent readout is sequential.

Feature	Description
Manufacturer	Sony
Sensor name	IMX532
Max. frames per second	82.8
Device Structure	CMOS image sensor
SNR_{max} [dB][1/]	39,8
DNR (normal / HDR) [dB][1/]	70,5 / -
Image size	1.1
Number of effective pixels	5328 (H) x 3040 (V)
Unit cell size	2.74µm (H) x 2.74µm (V)
ADC resolution / output	12 bit → 8/10/(12)

[1] Measured accord. to EMVA1288 with gray scale version of the camera

1.19.2.2.4.2 Spectral Sensitivity

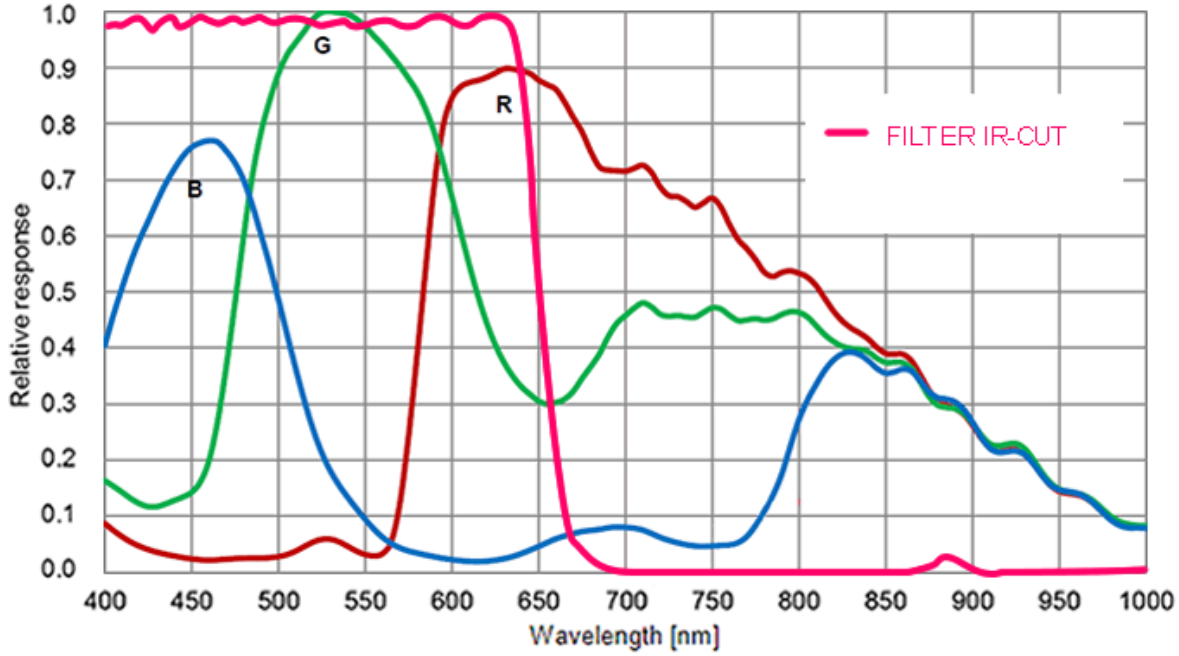


Figure 1: Spectral sensitivity BVS CA-BN4-0162ZC

1.19.2.2.4.3 Timings For calculating the frame rates, please use the frame rate calculator of the online manual:

See also

`https://www.matrix-vision.com/manuals/mvBlueNAOS/mvBlue_models_BVS_CA-BN4-0162ZC_.html#mvBC_subsubsection_sensors_CMOS_models_BVS_CA-BN4-0162ZC_Calc`

The following trigger modes are available:

Setting (GenICam)	Mode / Setting (obsolete "Device Specific")	Description
`"TriggerSelector = FrameStart"` `"TriggerMode = Off"`	Continuous	Free running, no external trigger signal needed.

`"TriggerSelector = FrameStart"`
`"TriggerMode = On"`
`"TriggerSource = Software"`
`"ExposureMode = Timed"`
 	
To trigger one frame execute the *TriggerSoftware@i* command then.
	OnDemand	Image acquisition triggered by command (software trigger).

"TriggerSelector = AcquisitionActive" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = LevelLow" "ExposureMode = Timed"	OnLowLevel	Start an exposure of a frame as long as the trigger input is below the trigger threshold. (No Frame↩ Trigger!)
"TriggerSelector = AcquisitionActive" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = LevelHigh" "ExposureMode = Timed"	OnHighLevel	Start an exposure of a frame as long as the trigger input is above the trigger threshold. (No Frame↩ Trigger!)
"TriggerSelector = FrameStart" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = FallingEdge" "ExposureMode = Timed"	OnFallingEdge	Each falling edge of trigger signal acquires one image.
"TriggerSelector = FrameStart" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = RisingEdge" "ExposureMode = Timed"	OnRisingEdge	Each rising edge of trigger signal acquires one image.
"TriggerSelector = FrameStart" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = AnyEdge" "ExposureMode = Timed"	OnAnyEdge	Start the exposure of a frame when the trigger input level changes from high to low or from low to high.

Device Feature And Property List (p. 268)

1.19.2.2.4.4 Device Feature And Property List

1.19.2.2.5 BVS CA-BN4-0204ZC (20.4 Mpix [4512 x 4512])

1.19.2.2.5.1 Introduction The sensor uses a global shutter, i.e. light exposure takes place on all pixels in parallel, although subsequent readout is sequential.

Feature	Description
Manufacturer	Sony
Sensor name	IMX531
Max. frames per second	66.0
Device Structure	CMOS image sensor
SNR_{max} [dB][1/]	tbd
DNR (normal / HDR) [dB][1/]	tbd / -
Image size	1.1
Number of effective pixels	4512 (H) x 4512 (V)
Unit cell size	2.74µm (H) x 2.74µm (V)
ADC resolution / output	12 bit → 8/10/(12)

[1] Measured accord. to EMVA1288 with gray scale version of the camera

1.19.2.2.5.2 Spectral Sensitivity

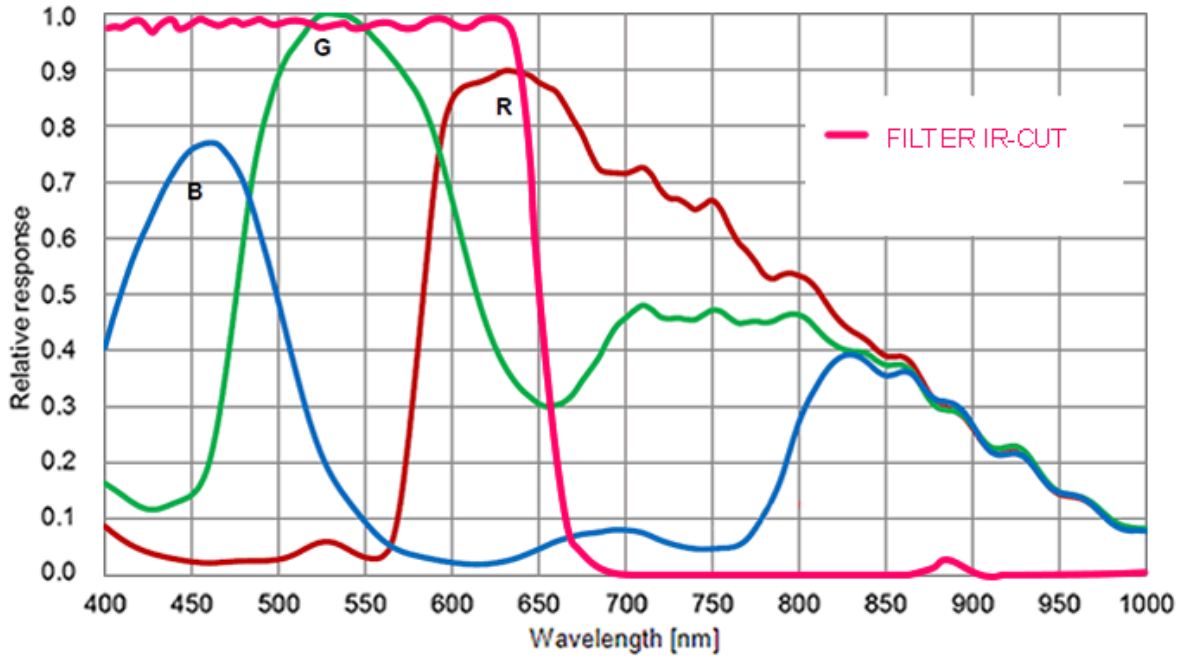


Figure 1: Spectral sensitivity BVS CA-BN4-0204ZC

1.19.2.2.5.3 Timings For calculating the frame rates, please use the frame rate calculator of the online manual:

See also

https://www.matrix-vision.com/manuals/mvBlueNAOS/mvBlue_models_BVS_CA-BN4-0204ZC_.html#mvBC_subsubsection_sensors_CMOS_models_BVS_CA-BN4-0204ZC_Calc

The following trigger modes are available:

Setting (GenICam)	Mode / Setting (obsolete "Device Specific")	Description
"TriggerSelector = FrameStart" "TriggerMode = Off"	Continuous	Free running, no external trigger signal needed.
"TriggerSelector = FrameStart" "TriggerMode = On" "TriggerSource = Software" "ExposureMode = Timed" To trigger one frame execute the *TriggerSoftware@i* command then.	OnDemand	Image acquisition triggered by command (software trigger).
"TriggerSelector = AcquisitionActive" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = LevelLow" "ExposureMode = Timed"	OnLowLevel	Start an exposure of a frame as long as the trigger input is below the trigger threshold. (No Frame↩ Trigger!)
"TriggerSelector = AcquisitionActive" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = LevelHigh" "ExposureMode = Timed"	OnHighLevel	Start an exposure of a frame as long as the trigger input is above the trigger threshold. (No Frame↩ Trigger!)
"TriggerSelector = FrameStart" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = FallingEdge" "ExposureMode = Timed"	OnFallingEdge	Each falling edge of trigger signal acquires one image.
"TriggerSelector = FrameStart" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = RisingEdge" "ExposureMode = Timed"	OnRisingEdge	Each rising edge of trigger signal acquires one image.

"TriggerSelector = FrameStart" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = AnyEdge" "ExposureMode = Timed"	OnAnyEdge	Start the exposure of a frame when the trigger input level changes from high to low or from low to high.

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1.19.2.2.5.4 Device Feature And Property List

1.19.2.2.6 BVS CA-BN4-0246ZC (24.6 Mpix [5328 x 4608])

1.19.2.2.6.1 Introduction The sensor uses a global shutter, i.e. light exposure takes place on all pixels in parallel, although subsequent readout is sequential.

Feature	Description
Manufacturer	Sony
Sensor name	IMX530
Max. frames per second	55.5
Device Structure	CMOS image sensor
SNR_{max} [dB][1/]	39,8
DNR (normal / HDR) [dB][1/]	70,4 / -
Image size	1.2
Number of effective pixels	5328 (H) x 4608 (V)
Unit cell size	2.74µm (H) x 2.74µm (V)
ADC resolution / output	12 bit → 8/10/(12)

[1] Measured accord. to EMVA1288 with gray scale version of the camera

1.19.2.2.6.2 Spectral Sensitivity

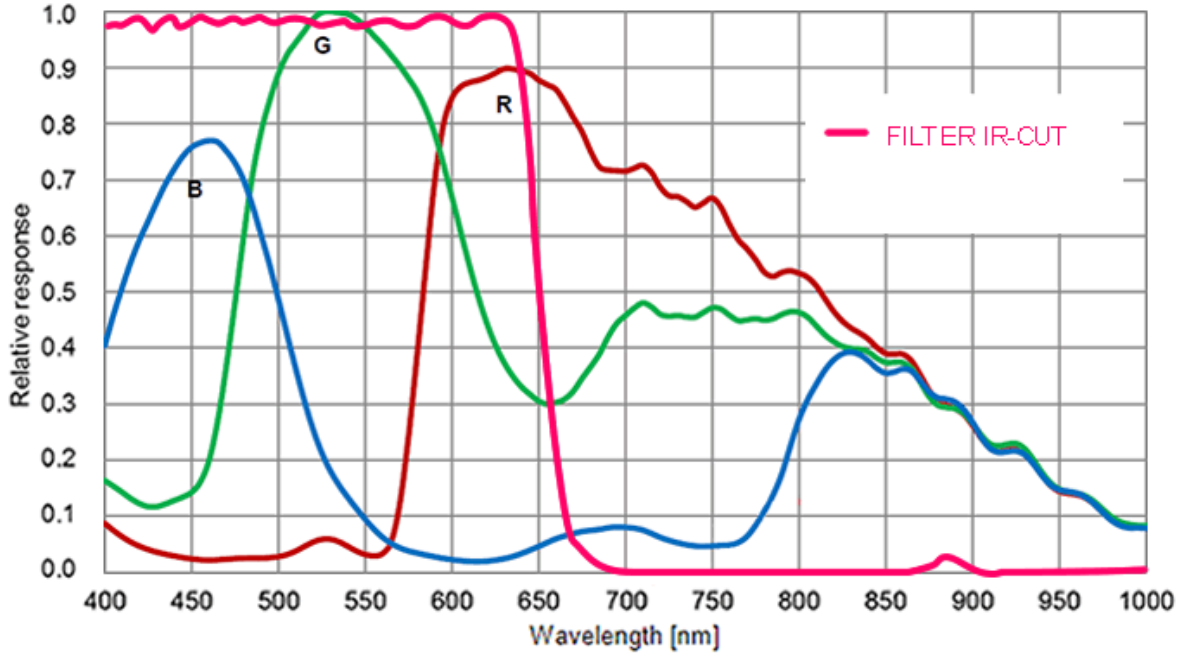


Figure 1: Spectral sensitivity BVS CA-BN4-0246ZC

1.19.2.2.6.3 Timings For calculating the frame rates, please use the frame rate calculator of the online manual:

See also

`https://www.matrix-vision.com/manuals/mvBlueNAOS/mvBlue_models_BVS_CA-BN4-0246ZC_.html#mvBC_subsubsection_sensors_CMOS_models_BVS_CA-BN4-0246ZC_Calc`

The following trigger modes are available:

Setting (GenICam)	Mode / Setting (obsolete "Device Specific")	Description
`"TriggerSelector = FrameStart"` `"TriggerMode = Off"`	Continuous	Free running, no external trigger signal needed.

`"TriggerSelector = FrameStart"`
`"TriggerMode = On"`
`"TriggerSource = Software"`
`"ExposureMode = Timed"`
 	
To trigger one frame execute the *TriggerSoftware@i* command then.
	OnDemand	Image acquisition triggered by command (software trigger).

"TriggerSelector = AcquisitionActive" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = LevelLow" "ExposureMode = Timed"	OnLowLevel	Start an exposure of a frame as long as the trigger input is below the trigger threshold. (No Frame↩ Trigger!)
"TriggerSelector = AcquisitionActive" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = LevelHigh" "ExposureMode = Timed"	OnHighLevel	Start an exposure of a frame as long as the trigger input is above the trigger threshold. (No Frame↩ Trigger!)
"TriggerSelector = FrameStart" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = FallingEdge" "ExposureMode = Timed"	OnFallingEdge	Each falling edge of trigger signal acquires one image.
"TriggerSelector = FrameStart" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = RisingEdge" "ExposureMode = Timed"	OnRisingEdge	Each rising edge of trigger signal acquires one image.
"TriggerSelector = FrameStart" "TriggerMode = On" "TriggerSource = <desired Line>" "TriggerActivation = AnyEdge" "ExposureMode = Timed"	OnAnyEdge	Start the exposure of a frame when the trigger input level changes from high to low or from low to high.

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1.19.2.2.6.4 Device Feature And Property List

1.20 Appendix C. Tested ARM platforms

MATRIX VISION devices can run on ARM-based Linux platforms without limitations regarding available feature sets or API functions. However, each platform may have its own limits in terms of achievable data throughput, RAM or bus speeds. Apart from that, each platform may also come with its own specific set of challenges. Therefore, certain modifications may need to be adapted in order to get your devices run at maximum performance.

This chapter contains test results from different ARM platforms, as well as the specific information on each platform, especially changes that need to be applied to achieve better performance.

The following platforms have been tested by MATRIX VISION:

System	ARM-Architecture	Technology Test Results					Approx. price	Performance	Suitable for	More information
		USB2.0	USB3.0	GigE	10GigE	PCIe				
NVIDIA Jetson AGX Xavier (p. 275)	NVIDIA Carmel ARMv8.2						ca. 650-700$		Demanding Applications	nvidia.com
NVIDIA Jetson Xavier NX (p. 277)	NVIDIA Carmel ARMv8.2						ca. 700$		Demanding Applications	nvidia.com
NVIDIA Jetson Nano	ARM Cortex-A57						90-100$		Mid-Range Applications	nvidia.com
NVIDIA Jetson TX2	NVIDIA Denver 2 and ARM Cortex-A57						ca. 400$		Demanding Applications	nvidia.com
i.MX8M Mini (p. 280)	ARM Cortex-A53								Mid-Range Applications	nxp.com
Raspberry Pi 4	ARM Cortex-A72						35-75$		Price Sensitive Projects	raspberrypi.org
Raspberry Pi Compute Module 4 (p. 281)	ARM Cortex-A72						30-80$		Price Sensitive Projects	raspberrypi.org

The system delivers good performance with this device.

The system doesn't work with this device.

The developer kit doesn't work with this device because it provides no PCI Express interface.

The system delivers limited performance with this device.

The system hasn't been tested yet with this device.

Appendices:

- **C.1 ARM64 based devices** (p. 275)

1.20.1 C.1 ARM64 based devices

- **NVIDIA Jetson AGX Xavier** (p. 275)
- **NVIDIA Jetson Xavier NX** (p. 277)
- **i.MX8M Mini** (p. 280)
- **Raspberry Pi Compute Module 4** (p. 281)

1.20.1.1 NVIDIA Jetson AGX Xavier

CPU	NVIDIA Carmel ARMv8.2 @ 2.26GHz
Cores	8
RAM	32GB
USB2.0 Interfaces	4
USB3.1 Interfaces	3
Ethernet	10/100/1000 MBit
PCIe	1x8 + 1x4 + 1x2 + 2x1 Gen 4.0

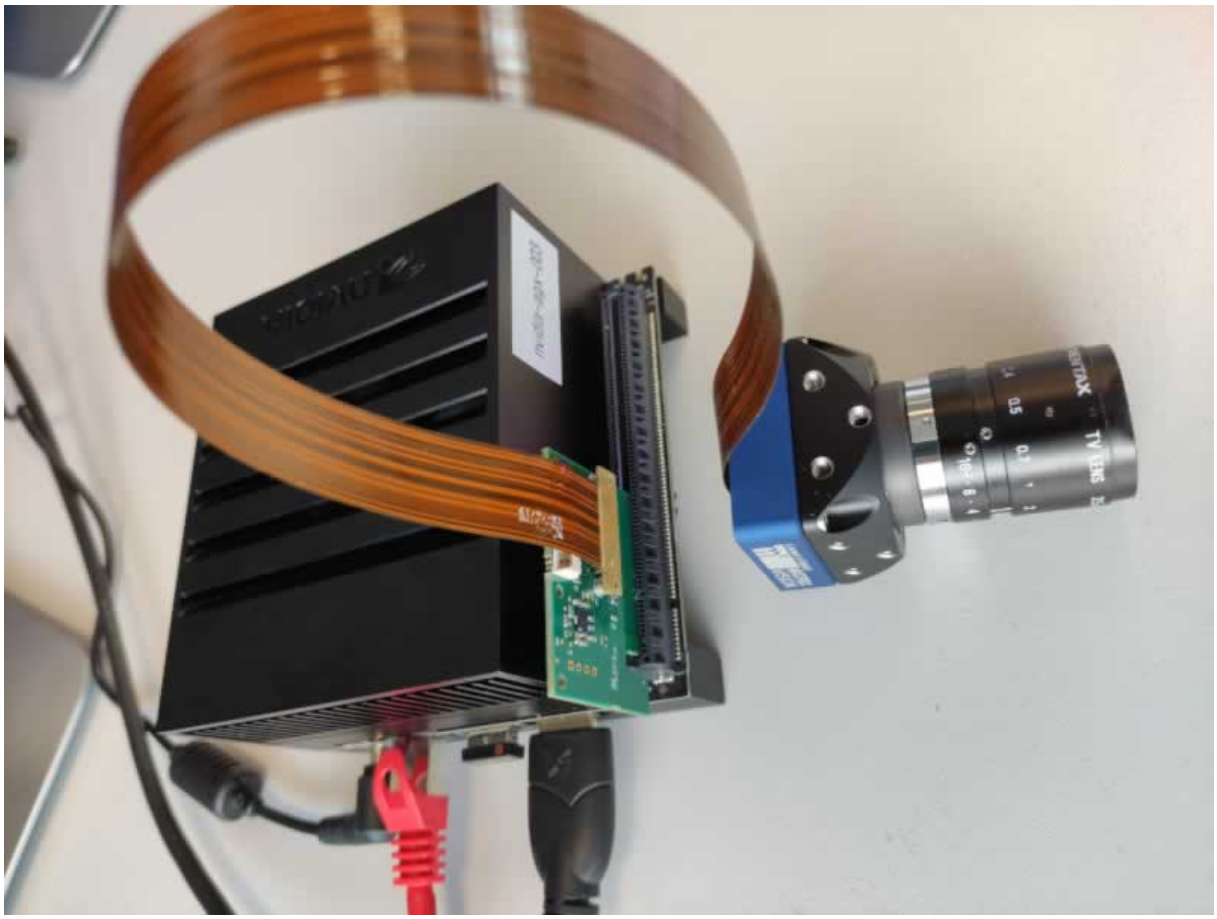
1.20.1.1.1 General

Note

The above table describes the specification of the NVIDIA Jetson AGX Xavier Developer Kit.

1.20.1.1.2 Benchmarks

1.20.1.1.2.1 PCIe Performance



Test setup

> **Note**
>
> The mvBlueNAOS device uses 4-lane PCIe 2.0 on the NVIDIA Jetson AGX Xavier.

1.20.1.1.2.2 Results The following tests have been performed using different de-Bayering scenarios to achieve the max. FPS while maintaining 0 lost images. The CPU load during the acquisition is also documented below.

Scenarios that have been tested are listed as follows:

1. When de-Bayering is carried out on the camera: The camera delivers RGB8 image data to the host system. This setting results in a lower CPU load but a lower frame rate.
2. When de-Bayering is carried out on the host system: The camera delivers Bayer8 image data to the host system. The Bayer8 image data then get de-Bayered to RGB8 format on the host system. This setting results in a higher frame rate but a higher CPU load as well.
3. When no de-Bayering is performed: The camera delivers Bayer8 image data to the host system. No de-↩ Bayering is performed. This settings results in a lower CPU load and a higher frame rate. The behavior is identical to monochrome cameras.

Camera	Resolution	Pixel Format	Frame Rate [Frames/s]	Bandwidth [MB/s]	CPU Load With Image Display	CPU Load Without Image Display
CA-BN2-0051ZC	2464 x 2056	RGB8 (on camera) -> RGB8 (on host)	87	1319.88	∼21%	∼12%
Camera	Resolution	Pixel Format	Frame Rate [Frames/s]	Bandwidth [MB/s]	CPU Load With Image Display	CPU Load Without Image Display
CA-BN2-0051ZC	2464 x 2056	BayerRG8 (on camera) -> RGB8 (on host)	139	703.94	∼42%	∼31%
Camera	Resolution	Pixel Format	Frame Rate [Frames/s]	Bandwidth [MB/s]	CPU Load With Image Display	CPU Load Without Image Display
CA-BN2-0051ZC	2464 x 2056	BayerRG8 (on camera) -> Bayer↩ RG8/Raw (on host)	139	703.97	∼21%	∼14%

1.20.1.2 NVIDIA Jetson Xavier NX

CPU	NVIDIA Carmel ARMv8.2 @ 1.9GHz
Cores	6
RAM	8GB
USB3.1 Interfaces	4
Ethernet	10/100/1000 MBit
PCIe	1x1 + 1x4 Gen 3.0

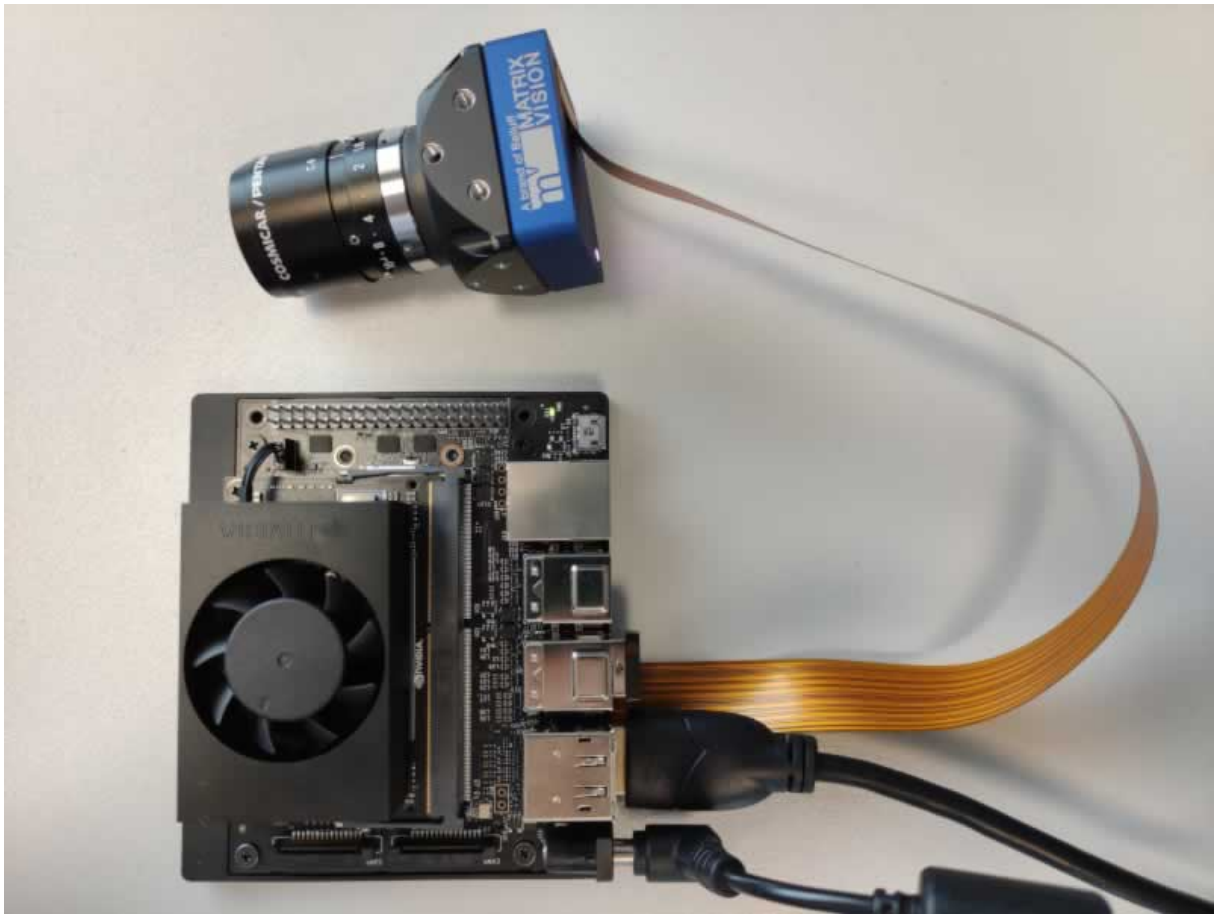
1.20.1.2.1 General

Note

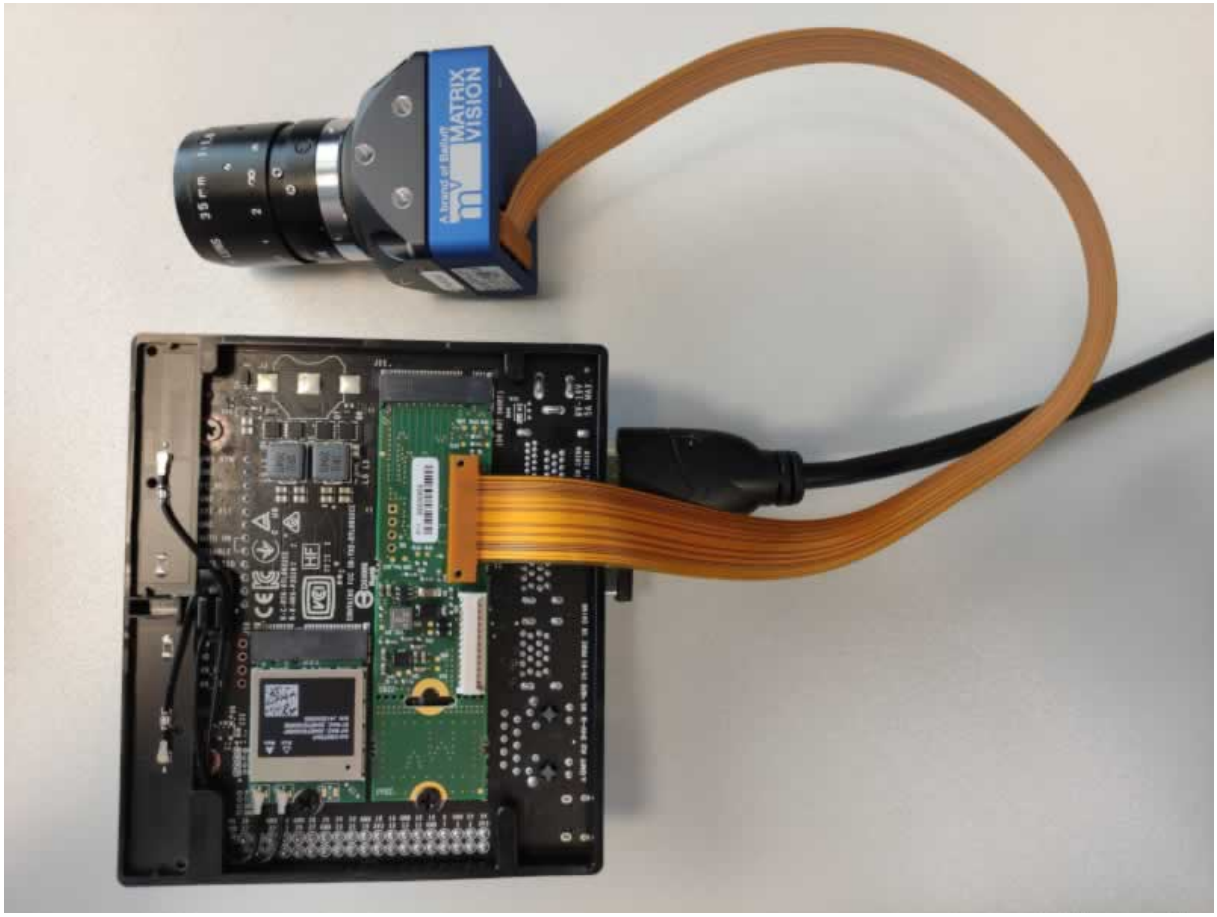
The above table describes the specification of the NVIDIA Jetson Xavier NX Developer Kit.

1.20.1.2.2 Benchmarks

1.20.1.2.2.1 PCIe Performance



Test setup (front side)



Test setup (back side)

Note

The mvBlueNAOS device uses 4-lane PCIe 2.0 on the NVIDIA Jetson Xavier NX.

1.20.1.2.2.2 Results The following tests have been performed using different de-Bayering scenarios to achieve the max. FPS while maintaining 0 lost images. The CPU load during the acquisition is also documented below.

Scenarios that have been tested are listed as follows:

1. When de-Bayering is carried out on the camera: The camera delivers RGB8 image data to the host system. This setting results in a lower CPU load but a lower frame rate.
2. When de-Bayering is carried out on the host system: The camera delivers Bayer8 image data to the host system. The Bayer8 image data then get de-Bayered to RGB8 format on the host system. This setting results in a higher frame rate but a higher CPU load as well.
3. When no de-Bayering is performed: The camera delivers Bayer8 image data to the host system. No de-↩ Bayering is performed. This settings results in a lower CPU load and a higher frame rate. The behavior is identical to monochrome cameras.

Camera	Resolution	Pixel Format	Frame Rate [Frames/s]	Bandwidth [MB/s]	CPU Load With Image Display	CPU Load Without Image Display
CA-BN2-0051ZC	2464 x 2056	RGB8 (on camera) -> RGB8 (on host)	87	1320.03	∼28%	∼10%
Camera	Resolution	Pixel Format	Frame Rate [Frames/s]	Bandwidth [MB/s]	CPU Load With Image Display	CPU Load Without Image Display
CA-BN2-0051ZC	2464 x 2056	BayerRG8 (on camera) -> RGB8 (on host)	139	703.96	∼63%	∼56%
Camera	Resolution	Pixel Format	Frame Rate [Frames/s]	Bandwidth [MB/s]	CPU Load With Image Display	CPU Load Without Image Display
CA-BN2-0051ZC	2464 x 2056	BayerRG8 (on camera) -> Bayer↩ RG8/Raw (on host)	139	703.97	∼29%	∼12%

1.20.1.3 i.MX8M Mini

CPU	ARM Cortex®-A53 @ 1.6GHz
Cores	4
RAM	1 GB
USB2.0 Interfaces	2
USB3.0 Interfaces	None
Ethernet	1000 MBit
PCIe	1 x 1 Lane Gen 2.0

1.20.1.3.1 General The carrier-board used in this test: MBa8Mx from TQ-Systems GmbH

Note

If you are looking for more information and guidance about installing mvIMPACT Acquire driver packages via the Yocto Project, please choose an API-manual suited for your programming language and then go to chapter "Installation From Private Setup Routines -> Embedded Linux -> Yocto Project". All API-manuals can be found under `https://www.matrix-vision.com/manuals/`.

1.20.1.3.2 Benchmarks

1.20.1.3.2.1 PCIe Performance Test

1.20.1.3.2.2 Results Since i.MX8M Mini has only 1 GB memory, it is difficult to receive all the images in time when Debayer is performed on the host system (Error: mvdma_irq_tasklet-0=HUNGER!). Running acquistion without Debayering on the host results in no error images, there are lost frames. Ideally the camera should run at 1 PCIe 2.0 Lane speed (i.e. 500 MB/s).

1.20.1.4 Raspberry Pi Compute Module 4

1.20.1.4.1 General The Raspberry Pi Compute Module 4 is a well priced platform regarding its performance.

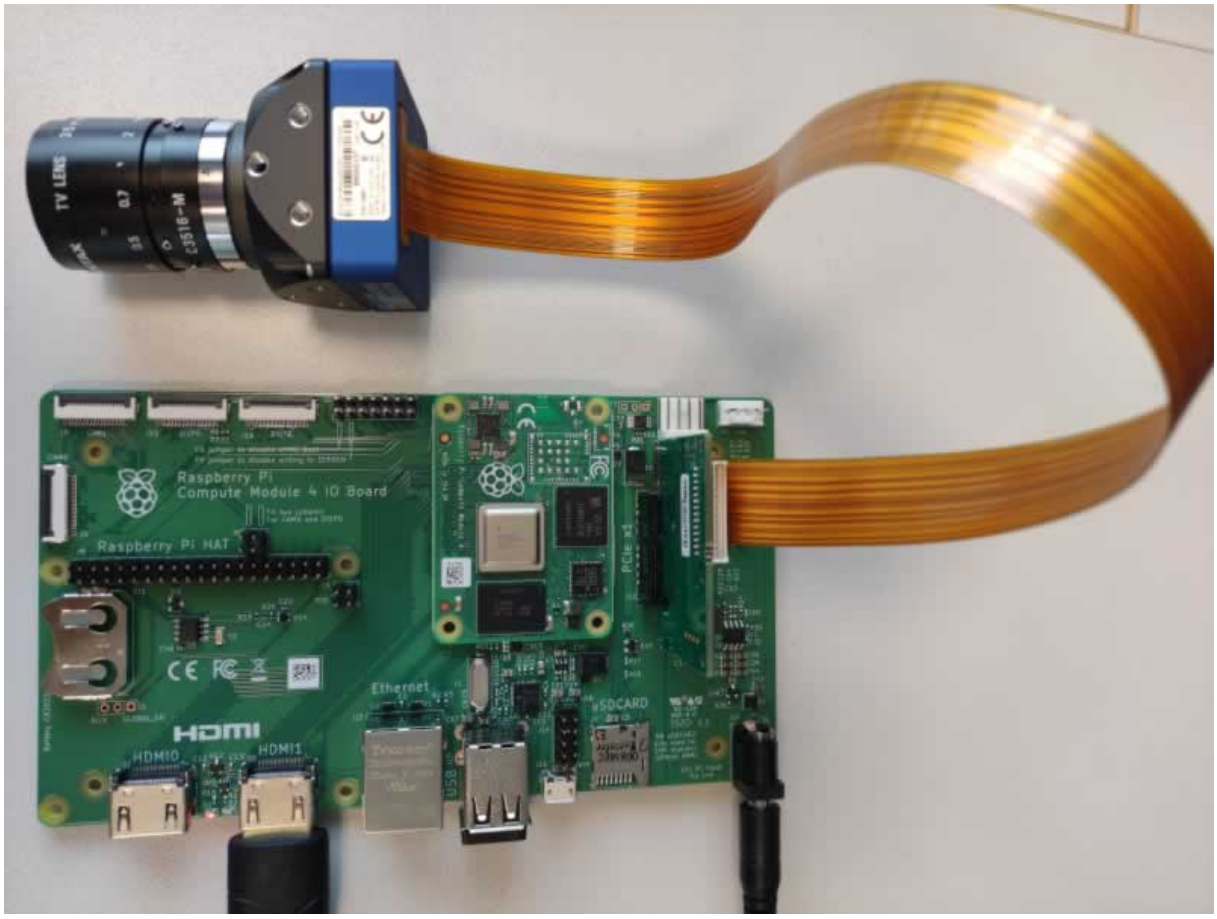
CPU	Cortex-A72 @ 1500MHz
Cores	4
RAM	4 GB
USB2.0 Interfaces	2
Ethernet	10/100/1000 MBit
PCIe	1 x 1 Lane Gen 2.0

Note

The above table describes the specification of the 4GB version Raspberry Pi Compute Module 4 IO Board.

1.20.1.4.2 Benchmarks

1.20.1.4.2.1 PCIe Performance



Test setup

Note

The mvBlueNAOS device uses 1-lane PCIe 2.0 on the Raspberry Pi Compute Module 4. If you are using the Raspberry Pi Compute Module 4 IO Board like us, you may need to cut out the edge of the slot to be able to insert the adapter board (like shown in the test setup above).

Though the following test has been carried out with a mvBlueNAOS2 device, it is recommended to use the Raspberry Pi Compute Module 4 together with a mvBlueNAOS1 device due to the board's limited PCIe speed and CPU processing power.

If you are using Raspberry Pi OS (64-bit) on the Raspberry Pi Compute Module 4, please make sure that the Kernel version is >=5.10.11.

1.20.1.4.2.2 Additional Settings Applied On The System In order for the system's X Server not to auto-detect the mvBlueNAOS as a GPU, the following configuration has to be adapted:

1. Create a directory `/etc/X11/xorg.conf.d`.
2. Create a file named `10-force_no_auto.conf` under the directory `/etc/X11/xorg.conf.d`.
3. Added the following content to the file:

```
Section "ServerFlags"
        Option "AutoAddGPU" "false"
EndSection
```

1.20.1.4.2.3 Results The following tests have been performed using different de-Bayering scenarios to achieve the max. FPS while maintaining 0 lost images. The CPU load during the acquisition is also documented below.

Scenarios that have been tested are listed as follows:

1. When de-Bayering is carried out on the camera: The camera delivers RGB8 image data to the host system. This setting results in a lower CPU load but a lower frame rate.
2. When de-Bayering is carried out on the host system: The camera delivers Bayer8 image data to the host system. The Bayer8 image data then get de-Bayered to RGB8 format on the host system. This setting results in a higher frame rate but a higher CPU load as well.
3. When no de-Bayering is performed: The camera delivers Bayer8 image data to the host system. No de-↩ Bayering is performed. This settings results in a lower CPU load and a higher frame rate. The behavior is identical to monochrome cameras.

Camera	Resolution	Pixel Format	Frame Rate [Frames/s]	Bandwidth [MB/s]	CPU Load With Image Display	CPU Load Without Image Display
CA-BN2-0051ZC	2464 x 2056	RGB8 (on camera) -> RGB8 (on host)	25	375.47	∼8%	∼4%
Camera	Resolution	Pixel Format	Frame Rate [Frames/s]	Bandwidth [MB/s]	CPU Load With Image Display	CPU Load Without Image Display

CA-BN2-0051ZC	2464 x 2056	BayerRG8 (on camera) -> RGB8 (on host)	57	288.29	∼72%	∼67%
Camera	Resolution	Pixel Format	Frame Rate [Frames/s]	Bandwidth [MB/s]	CPU Load With Image Display	CPU Load Without Image Display
CA-BN2-0051ZC	2464 x 2056	BayerRG8 (on camera) -> BayerRG8/Raw (on host)	74	375.94	∼14%	∼8%

Note

Since performing De-Bayering on the host (e.g. the 2.scenario) requires a lot more CPU processing power, the brutto transmission bandwidth (i.e. `DeviceLinkThroughput`) has been limited to 330MB/s to avoid frame loss during this test.